

41. Serial Communications Interface (SCIj, SCLi, SCIlh)

This MCU has 13 independent serial communications interface (SCI) channels. The SCI consists of the SCIj module (SCI0 to SCI6), the SCLi module (SCI7 to SCI11), and the SCIlh module (SCI12).

The SCIj module (SCI0 to SCI6) and SCLi module (SCI7 to SCI11) can handle both asynchronous and clock synchronous serial communications.

Asynchronous serial data communications can be carried out with standard asynchronous communications chips such as a Universal Asynchronous Receiver/Transmitter (UART) or Asynchronous Communications Interface Adapter (ACIA). As an extended function in asynchronous communications mode, the SCI also supports smart card (IC card) interfaces conforming to ISO/IEC 7816-3 (standard for Identification Cards). The SCI also supports simple SPI interfaces, and simple I²C-bus interfaces when configured for single-master systems.

The SCIlh module includes the functions of the SCIg module, and supports an extended serial communication protocol formed of Start Frames and Information Frames.

In this section, “PCLK” for SCI0 to SCI6 and SCI12 is used to refer to PCLKB and “PCLK” for SCI7 to SCI11 is used to refer to PCLKA.

41.1 Overview

Table 41.1 lists the specifications of the SCIj module, Table 41.2 lists the specifications of the SCLi module, Table 41.3 lists the specifications of the SCIlh module, and Table 41.4 lists the specifications of the individual SCI channels. Figure 41.1 shows the block diagram of SCI0 to SCI4, Figure 41.2 shows the block diagram of SCI5 and SCI6, Figure 41.3 shows the block diagram of SCI7 to SCI11, and Figure 41.4 shows the block diagram of SCI12 (SCIlh).

Table 41.1 SCIj Specifications (1/2)

Item	Description
Serial communication modes	- Asynchronous - Clock synchronous - Smart card interface - Simple I²C-bus - Simple SPI bus
Transfer speed	Bit rate specifiable with the on-chip baud rate generator.
Full-duplex communications	Transmitter: Continuous transmission possible using double-buffer structure. Receiver: Continuous reception possible using double-buffer structure.
I/O pins	Refer to Table 41.5 to Table 41.7.
Data transfer	Selectable as LSB first or MSB first transfer*1
Interrupt sources	Transmit end, transmit data empty, receive data full, receive error, and data match Completion of generation of a start condition, restart condition, or stop condition (for simple I ² C mode)
Low power consumption function	Module stop state can be set for each channel.

Table 41.1 SCIj Specifications (2/2)

Item	Description	
Asynchronous mode	Data length	7, 8, or 9 bits
	Transmission stop bit	1 or 2 bits
	Parity	Even parity, odd parity, or no parity
	Receive error detection	Parity, overrun, and framing errors
	Hardware flow control	CTSn# and RTSn# pins can be used in controlling transmission/reception.
	Data match detection	Compares receive data and comparison data, and generates interrupt when they are matched.
	Start-bit detection	Low level or falling edge is selectable.
	Break detection	When a framing error occurs, a break can be detected by reading the RXDn pin level directly or reading the SPTR.RXDMON flag.
	Clock source	An internal or external clock can be selected. Transfer rate clock input from the TMR can be used. (SCI5 and SCI6)
	Double-speed mode	Baud rate generator double-speed mode is selectable.
	Multi-processor communications function	Serial communication among multiple processors
	Noise cancellation	The signal paths from input on the RXDn pins incorporate digital noise filters.
Clock synchronous mode	Data length	8 bits
	Receive error detection	Overrun error
	Hardware flow control	CTSn# and RTSn# pins can be used in controlling transmission/reception.
Smart card interface mode	Error processing	An error signal can be automatically transmitted when detecting a parity error during reception
		Data can be automatically retransmitted when receiving an error signal during transmission
	Data type	Both direct convention and inverse convention are supported.
Simple I ² C mode	Transfer format	I ² C-bus format
	Operating mode	Master (single-master operation only)
	Transfer rate	Fast mode is supported (refer to section 41.2.13, Bit Rate Register (BRR) to set the transfer rate).
	Noise cancellation	The signal paths from input on the SSCLn and SSDAn pins incorporate digital noise filters, and the interval for noise cancellation is adjustable.
Simple SPI bus	Data length	8 bits
	Detection of errors	Overrun error
	SS input pin function	Applying the high level to the SSn# pin can cause the output pins to enter the high-impedance state.
	Clock settings	Four kinds of settings for clock phase and clock polarity are selectable.
Bit rate modulation function		Correction of outputs from the on-chip baud rate generator can reduce errors.
Event link function (supported by SCI5 only)		Error (receive error or error signal detection) event output
		Receive data full event output
		Transmit data empty event output
		Transmit end event output

Note 1. In simple I²C mode, only MSB first is available.

Table 41.2 SCli Specifications (1/2)

Item	Description
Serial communication modes	- Asynchronous - Clock synchronous - Smart card interface - Simple I²C-bus - Simple SPI bus
Transfer speed	Bit rate specifiable with the on-chip baud rate generator.

Table 41.2 SCli Specifications (2/2)

Item		Description
Full-duplex communications		Transmitter: Continuous transmission possible using double-buffer structure. Receiver: Continuous reception possible using double-buffer structure.
I/O pins		Refer to Table 41.5 to Table 41.7.
Data transfer		Selectable as LSB first or MSB first transfer*1
Interrupt sources		Transmit end, transmit data empty, receive data full, receive error, receive data ready, and data match Completion of generation of a start condition, restart condition, or stop condition (for simple I ² C mode)
Low power consumption function		Module stop state can be set for each channel.
Asynchronous mode	Data length	7, 8, or 9 bits
	Transmission stop bit	1 or 2 bits
	Parity	Even parity, odd parity, or no parity
	Receive error detection	Parity, overrun, and framing errors
	Hardware flow control	CTS _n # and RTS _n # pins can be used in controlling transmission/reception.
	Transmit/receive FIFO	16-stage FIFOs for transmit and receive buffers
	Data match detection	Compares receive data and comparison data, and generates interrupt when they are matched
	Start-bit detection	Low level or falling edge is selectable.
	Break detection	When a framing error occurs, a break can be detected by reading the RXD _n pin level directly or reading the SPTR.RXDMON flag.
	Clock source	An internal or external clock can be selected.
	Double-speed mode	Baud rate generator double-speed mode is selectable.
	Multi-processor communications function	Serial communication among multiple processors
Clock synchronous mode	Noise cancellation	The signal paths from input on the RXD _n pins incorporate digital noise filters.
	Data length	8 bits
	Receive error detection	Overrun error
	Hardware flow control	CTS _n # and RTS _n # pins can be used in controlling transmission/reception.
Smart card interface mode	Transmit/receive FIFO	16-stage FIFOs for transmit and receive buffers
	Error processing	An error signal can be automatically transmitted when detecting a parity error during reception Data can be automatically retransmitted when receiving an error signal during transmission
	Data type	Both direct convention and inverse convention are supported.
Simple I ² C mode	Transfer format	I ² C-bus format
	Operating mode	Master (single-master operation only)
	Transfer rate	Fast mode is supported (refer to section 41.2.13, Bit Rate Register (BRR) to set the transfer rate).
	Noise cancellation	The signal paths from input on the SSCL _n and SSDA _n pins incorporate digital noise filters, and the interval for noise cancellation is adjustable.
Simple SPI bus	Data length	8 bits
	Detection of errors	Overrun error
	SS input pin function	Applying the high level to the SS _n # pin can cause the output pins to enter the high-impedance state.
	Clock settings	Four kinds of settings for clock phase and clock polarity are selectable.
Bit rate modulation function		Correction of outputs from the on-chip baud rate generator can reduce errors.

Note 1. In simple I²C mode, only MSB first is available.

Table 41.3 SClh Specifications (1/2)

Item	Description	
Serial communication modes	• Asynchronous • Clock synchronous • Smart card interface • Simple I²C-bus • Simple SPI bus	
Transfer speed	Bit rate specifiable with the on-chip baud rate generator.	
Full-duplex communications	Transmitter: Continuous transmission possible using double-buffer structure. Receiver: Continuous reception possible using double-buffer structure.	
I/O pins	Refer to Table 41.5 to Table 41.8.	
Data transfer	Selectable as LSB first or MSB first transfer* ¹	
Interrupt sources	Transmit end, transmit data empty, receive data full, and receive error Completion of generation of a start condition, restart condition, or stop condition (for simple I ² C mode)	
Low power consumption function	Module stop state can be set.	
Asynchronous mode	Data length	7, 8, or 9 bits
	Transmission stop bit	1 or 2 bits
	Parity	Even parity, odd parity, or no parity
	Receive error detection	Parity, overrun, and framing errors
	Hardware flow control	CTS _n # and RTS _n # pins can be used in controlling transmission/reception.
	Start-bit detection	Low level or falling edge is selectable.
	Break detection	When a framing error occurs, a break can be detected by reading the RXD _n pin level directly.
	Clock source	An internal or external clock can be selected. Transfer rate clock input from the TMR can be used.
	Double-speed mode	Baud rate generator double-speed mode is selectable.
	Multi-processor communications function	Serial communication among multiple processors
	Noise cancellation	The signal paths from input on the RXD _n pins incorporate digital noise filters.
Clock synchronous mode	Data length	8 bits
	Receive error detection	Overrun error
	Hardware flow control	CTS _n # and RTS _n # pins can be used in controlling transmission/reception.
Smart card interface mode	Error processing	An error signal can be automatically transmitted when detecting a parity error during reception Data can be automatically retransmitted when receiving an error signal during transmission
	Data type	Both direct convention and inverse convention are supported.
Simple I ² C mode	Transfer format	I ² C-bus format
	Operating mode	Master (single-master operation only)
	Transfer rate	Fast mode is supported (refer to section 41.2.13, Bit Rate Register (BRR) to set the transfer rate).
	Noise cancellation	The signal paths from input on the SSCL _n and SSDA _n pins incorporate digital noise filters, and the interval for noise cancellation is adjustable.
Simple SPI bus	Data length	8 bits
	Detection of errors	Overrun error
	SS input pin function	Applying the high level to the SS _n # pin can cause the output pins to enter the high-impedance state.
	Clock settings	Four kinds of settings for clock phase and clock polarity are selectable.

Table 41.3 SClh Specifications (2/2)

Item	Description	
Extended serial mode	Start Frame transmission	• Output of a low level as the Break Field over a specified width and generation of interrupts on completion • Detection of bus collisions and the generation of interrupts on detection
	Start Frame reception	• Detection of the Break Field low width and generation of an interrupt on detection • Comparison of Control Fields 0 and 1 and generation of an interrupt when the two match • Two kinds of data for comparison (primary and secondary) can be set in Control Field 1. • A priority interrupt bit can be set in Control Field 1. • Handling of Start Frames that do not include a Break Field • Handling of Start Frames that do not include a Control Field 0 • Function for measuring bit rates
	I/O control function	• Selectable polarity for TXDX12 and RXDX12 signals • Selection of a digital filter for the RXDX12 signal • Half-duplex operation employing RXDX12 and TXDX12 signals multiplexed on the same pin • Selectable timing for the sampling of data received through RXDX12
	Timer function	• Usable as a reload timer
Bit rate modulation function		Correction of outputs from the on-chip baud rate generator can reduce errors.

Note 1. In simple I²C mode, only MSB first is available.

Table 41.4 Functions of SCI Channels

Item	SCI0 to SCI4	SCI5	SCI6	SCI7 to SCI11	SCI12
Asynchronous mode	Available	Available	Available	Available	Available
Clock synchronous mode	Available	Available	Available	Available	Available
Smart card interface mode	Available	Available	Available	Available	Available
Simple I ² C mode	Available	Available	Available	Available	Available
Simple SPI mode	Available	Available	Available	Available	Available
FIFO mode	Not available	Not available	Not available	Available	Not available
Data match detection	Available	Available	Available	Available	Not available
Extended serial mode	Not available	Not available	Not available	Not available	Available
TMR clock input	Not available	Available	Available	Not available	Available
Event link function	Not available	Available	Not available	Not available	Not available
Peripheral module clock	PCLKB	PCLKB	PCLKB	PCLKA	PCLKB

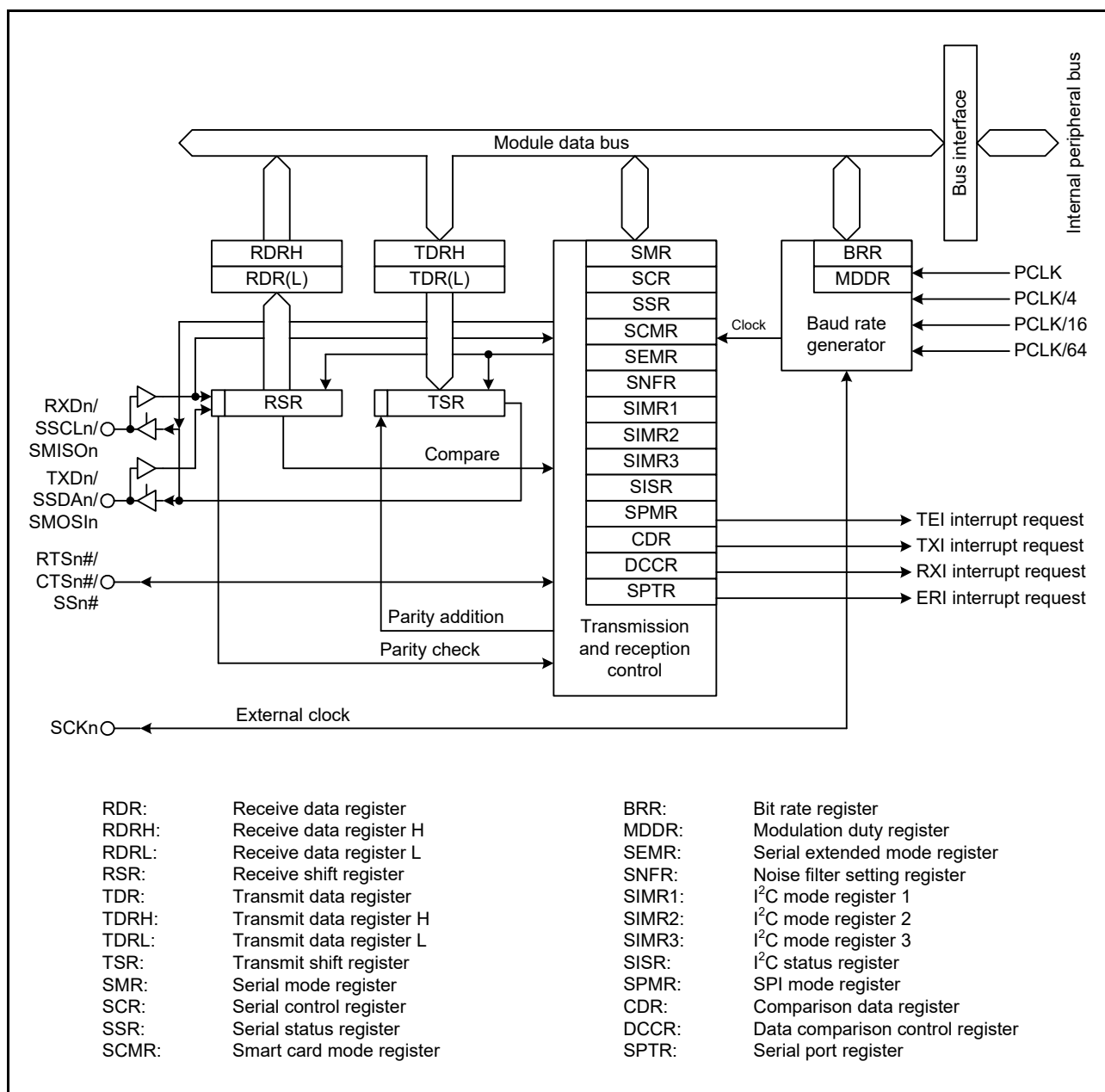


Figure 41.1 Block Diagram of SCIj (SCI0 to SCI4)

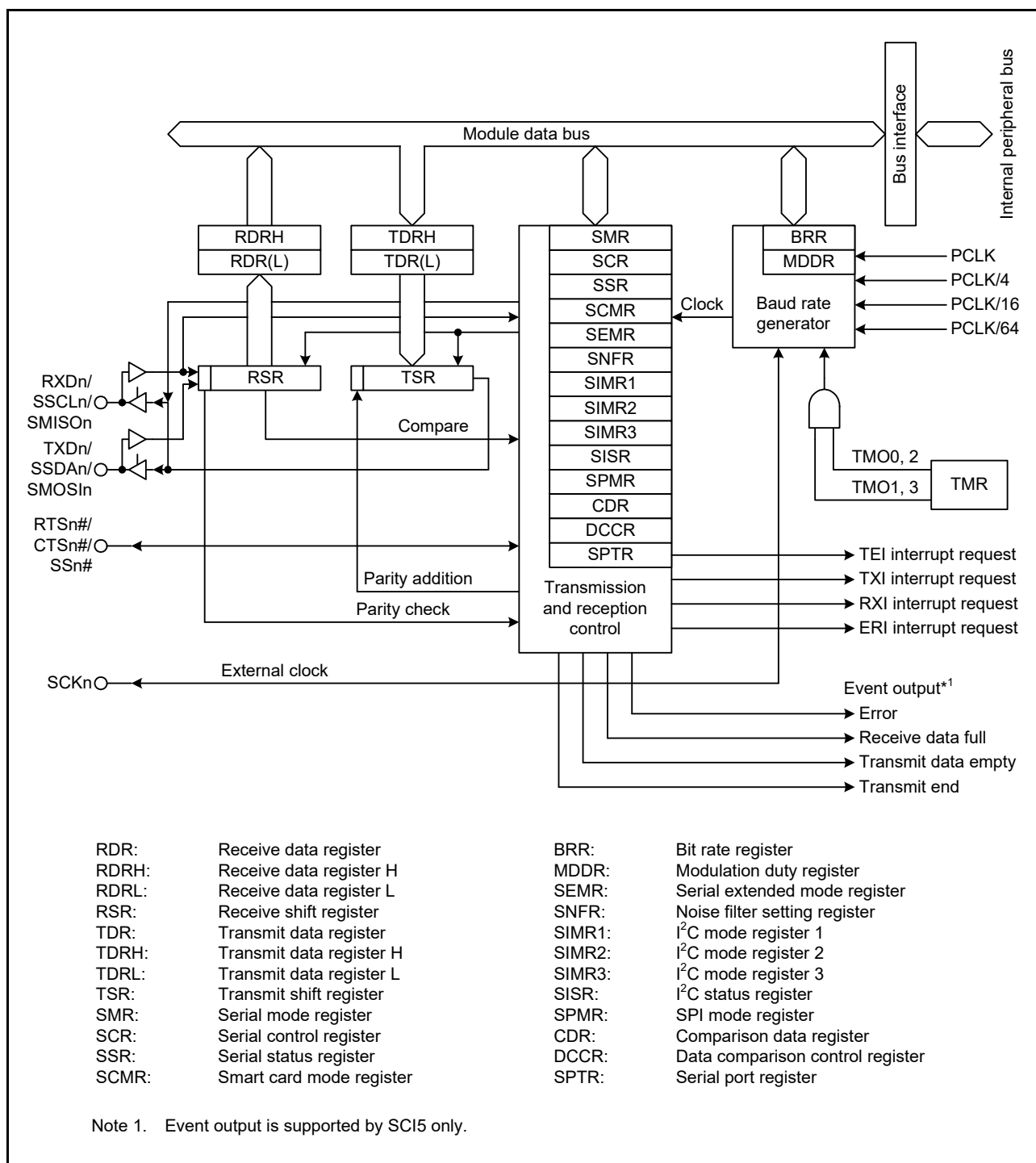


Figure 41.2 **Block Diagram of SCIj (SCI5 and SCI6)**

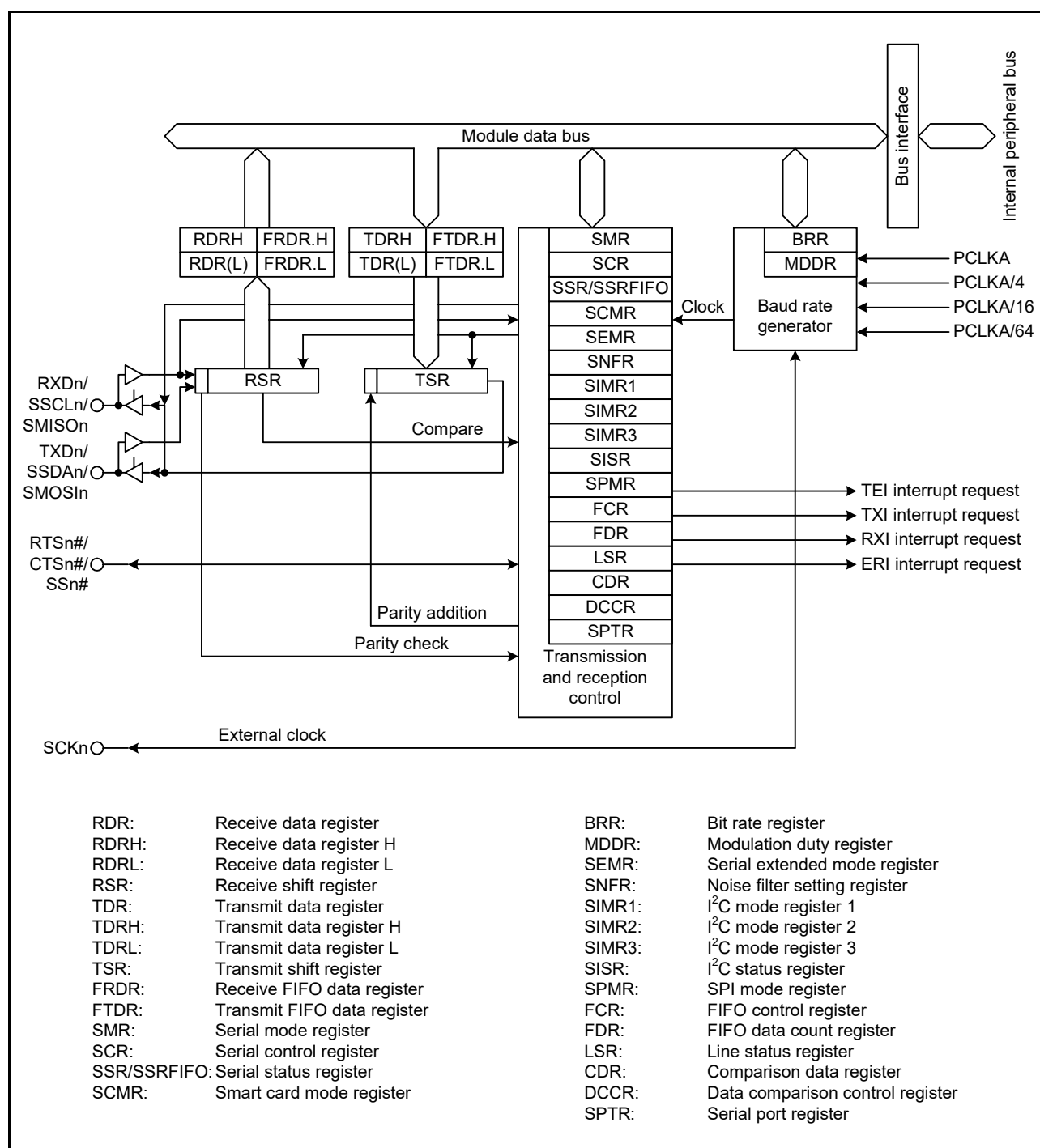


Figure 41.3 Block Diagram of SCli (SCI7 to SCI11)

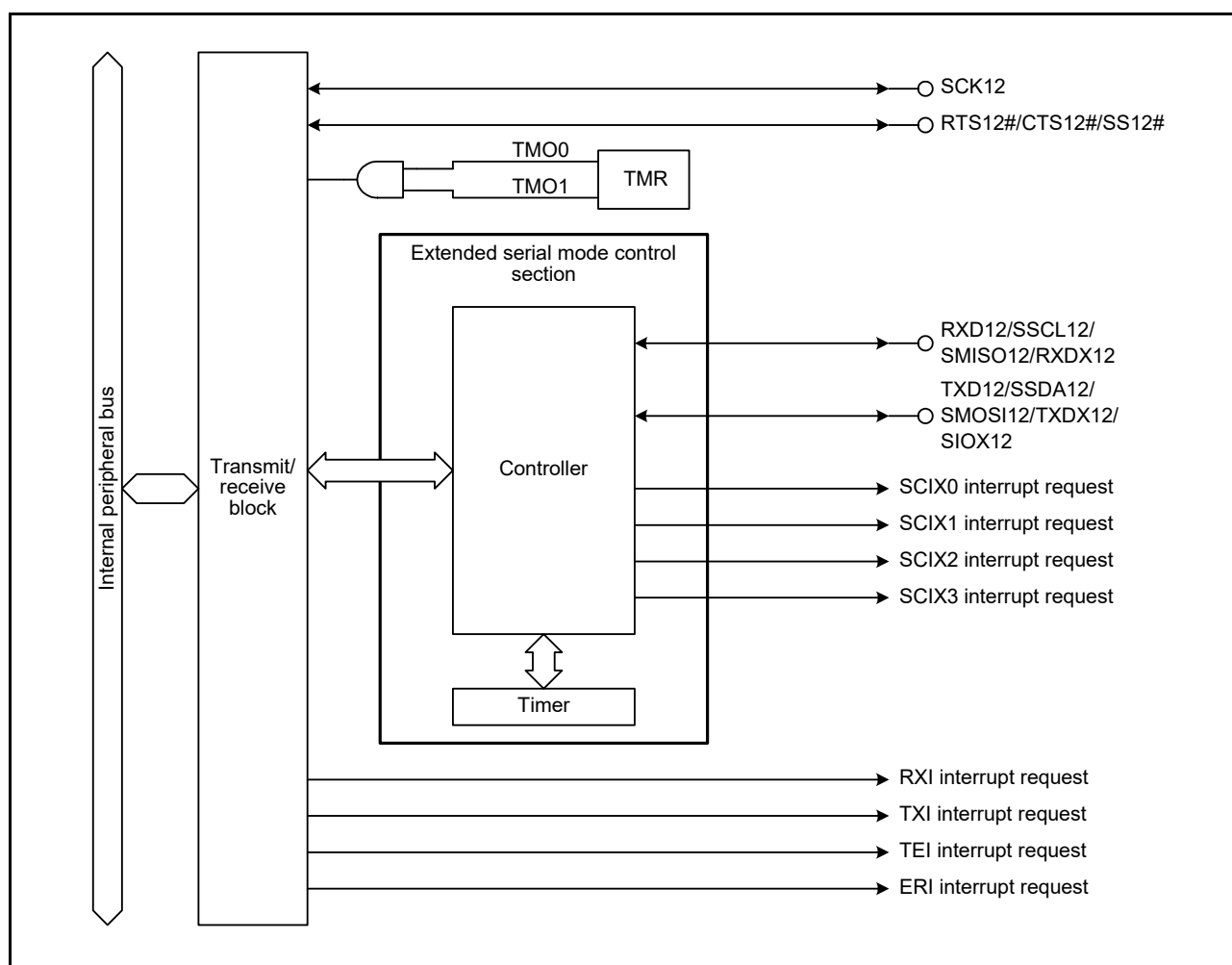


Figure 41.4 Block Diagram of SCli (SCI12)

Table 41.5 to Table 41.8 list the pin configuration of the SCIs for the individual modes.

Table 41.5 SCI Pin Configuration in Asynchronous Mode and Clock Synchronous Mode

Channel	Pin Name	I/O	Function
SCI0	SCK0	I/O	SCI0 clock input/output
	RXD0	Input	SCI0 receive data input
	TXD0	Output	SCI0 transmit data output
	CTS0#/RTS0#	I/O	SCI0 transfer start control input/output
SCI1	SCK1	I/O	SCI1 clock input/output
	RXD1	Input	SCI1 receive data input
	TXD1	Output	SCI1 transmit data output
	CTS1#/RTS1#	I/O	SCI1 transfer start control input/output
SCI2	SCK2	I/O	SCI2 clock input/output
	RXD2	Input	SCI2 receive data input
	TXD2	Output	SCI2 transmit data output
	CTS2#/RTS2#	I/O	SCI2 transfer start control input/output
SCI3	SCK3	I/O	SCI3 clock input/output
	RXD3	Input	SCI3 receive data input
	TXD3	Output	SCI3 transmit data output
	CTS3#/RTS3#	I/O	SCI3 transfer start control input/output
SCI4	SCK4	I/O	SCI4 clock input/output
	RXD4	Input	SCI4 receive data input
	TXD4	Output	SCI4 transmit data output
	CTS4#/RTS4#	I/O	SCI4 transfer start control input/output
SCI5	SCK5	I/O	SCI5 clock input/output
	RXD5	Input	SCI5 receive data input
	TXD5	Output	SCI5 transmit data output
	CTS5#/RTS5#	I/O	SCI5 transfer start control input/output
SCI6	SCK6	I/O	SCI6 clock input/output
	RXD6	Input	SCI6 receive data input
	TXD6	Output	SCI6 transmit data output
	CTS6#/RTS6#	I/O	SCI6 transfer start control input/output
SCI7	SCK7	I/O	SCI7 clock input/output
	RXD7	Input	SCI7 receive data input
	TXD7	Output	SCI7 transmit data output
	CTS7#/RTS7#	I/O	SCI7 transfer start control input/output
SCI8	SCK8	I/O	SCI8 clock input/output
	RXD8	Input	SCI8 receive data input
	TXD8	Output	SCI8 transmit data output
	CTS8#/RTS8#	I/O	SCI8 transfer start control input/output
SCI9	SCK9	I/O	SCI9 clock input/output
	RXD9	Input	SCI9 receive data input
	TXD9	Output	SCI9 transmit data output
	CTS9#/RTS9#	I/O	SCI9 transfer start control input/output

Table 41.5 SCI Pin Configuration in Asynchronous Mode and Clock Synchronous Mode

Channel	Pin Name	I/O	Function
SCI10	SCK10	I/O	SCI10 clock input/output
	RXD10	Input	SCI10 receive data input
	TXD10	Output	SCI10 transmit data output
	CTS10#/RTS10#	I/O	SCI10 transfer start control input/output
SCI11	SCK11	I/O	SCI11 clock input/output
	RXD11	Input	SCI11 receive data input
	TXD11	Output	SCI11 transmit data output
	CTS11#/RTS11#	I/O	SCI11 transfer start control input/output
SCI12	SCK12	I/O	SCI12 clock input/output
	RXD12	Input	SCI12 receive data input
	TXD12	Output	SCI12 transmit data output
	CTS12#/RTS12#	I/O	SCI12 transfer start control input/output

Table 41.6 SCI Pin Configuration in Simple I²C Mode

Channel	Pin Name	I/O	Function
SCI0	SSCL0	I/O	SCI0 I ² C clock input/output
	SSDA0	I/O	SCI0 I ² C data input/output
SCI1	SSCL1	I/O	SCI1 I ² C clock input/output
	SSDA1	I/O	SCI1 I ² C data input/output
SCI2	SSCL2	I/O	SCI2 I ² C clock input/output
	SSDA2	I/O	SCI2 I ² C data input/output
SCI3	SSCL3	I/O	SCI3 I ² C clock input/output
	SSDA3	I/O	SCI3 I ² C data input/output
SCI4	SSCL4	I/O	SCI4 I ² C clock input/output
	SSDA4	I/O	SCI4 I ² C data input/output
SCI5	SSCL5	I/O	SCI5 I ² C clock input/output
	SSDA5	I/O	SCI5 I ² C data input/output
SCI6	SSCL6	I/O	SCI6 I ² C clock input/output
	SSDA6	I/O	SCI6 I ² C data input/output
SCI7	SSCL7	I/O	SCI7 I ² C clock input/output
	SSDA7	I/O	SCI7 I ² C data input/output
SCI8	SSCL8	I/O	SCI8 I ² C clock input/output
	SSDA8	I/O	SCI8 I ² C data input/output
SCI9	SSCL9	I/O	SCI9 I ² C clock input/output
	SSDA9	I/O	SCI9 I ² C data input/output
SCI10	SSCL10	I/O	SCI10 I ² C clock input/output
	SSDA10	I/O	SCI10 I ² C data input/output
SCI11	SSCL11	I/O	SCI11 I ² C clock input/output
	SSDA11	I/O	SCI11 I ² C data input/output
SCI12	SSCL12	I/O	SCI12 I ² C clock input/output
	SSDA12	I/O	SCI12 I ² C data input/output

Table 41.7 SCI Pin Configuration in Simple SPI Mode (1/2)

Channel	Pin Name	I/O	Function
SCI0	SCK0	I/O	SCI0 clock input/output
	SMISO0	I/O	SCI0 slave transmit data input/output
	SMOSI0	I/O	SCI0 master transmit data input/output
	SS0#	Input	SCI0 chip select input
SCI1	SCK1	I/O	SCI1 clock input/output
	SMISO1	I/O	SCI1 slave transmit data input/output
	SMOSI1	I/O	SCI1 master transmit data input/output
	SS1#	Input	SCI1 chip select input
SCI2	SCK2	I/O	SCI2 clock input/output
	SMISO2	I/O	SCI2 slave transmit data input/output
	SMOSI2	I/O	SCI2 master transmit data input/output
	SS2#	Input	SCI2 chip select input
SCI3	SCK3	I/O	SCI3 clock input/output
	SMISO3	I/O	SCI3 slave transmit data input/output
	SMOSI3	I/O	SCI3 master transmit data input/output
	SS3#	Input	SCI3 chip select input
SCI4	SCK4	I/O	SCI4 clock input/output
	SMISO4	I/O	SCI4 slave transmit data input/output
	SMOSI4	I/O	SCI4 master transmit data input/output
	SS4#	Input	SCI4 chip select input
SCI5	SCK5	I/O	SCI5 clock input/output
	SMISO5	I/O	SCI5 slave transmit data input/output
	SMOSI5	I/O	SCI5 master transmit data input/output
	SS5#	Input	SCI5 chip select input
SCI6	SCK6	I/O	SCI6 clock input/output
	SMISO6	I/O	SCI6 slave transmit data input/output
	SMOSI6	I/O	SCI6 master transmit data input/output
	SS6#	Input	SCI6 chip select input
SCI7	SCK7	I/O	SCI7 clock input/output
	SMISO7	I/O	SCI7 slave transmit data input/output
	SMOSI7	I/O	SCI7 master transmit data
	SS7#	Input	SCI7 chip select input
SCI8	SCK8	I/O	SCI8 clock input/output
	SMISO8	I/O	SCI8 slave transmit data input/output
	SMOSI8	I/O	SCI8 master transmit data input/output
	SS8#	Input	SCI8 chip select input
SCI9	SCK9	I/O	SCI9 clock input/output
	SMISO9	I/O	SCI9 slave transmit data input/output
	SMOSI9	I/O	SCI9 master transmit data input/output
	SS9#	Input	SCI9 chip select input
SCI10	SCK10	I/O	SCI10 clock input/output
	SMISO10	I/O	SCI10 slave transmit data input/output
	SMOSI10	I/O	SCI10 master transmit data input/output
	SS10#	Input	SCI10 chip select input

Table 41.7 SCI Pin Configuration in Simple SPI Mode (2/2)

Channel	Pin Name	I/O	Function
SCI11	SCK11	I/O	SCI11 clock input/output
	SMISO11	I/O	SCI11 slave transmit data input/output
	SMOSI11	I/O	SCI11 master transmit data input/output
	SS11#	Input	SCI11 chip select input
SCI12	SCK12	I/O	SCI12 clock input/output
	SMISO12	I/O	SCI12 slave transmit data input/output
	SMOSI12	I/O	SCI12 master transmit data input/output
	SS12#	Input	SCI12 chip select input

Table 41.8 SCI Pin Configuration in Extended Serial Mode

Channel	Pin Name	I/O	Function
SCI12	RDX12	Input	SCI12 receive data input
	TXDX12	Output	SCI12 transmit data output
	SIOX12	I/O	SCI12 transfer data input/output

41.2 Register Descriptions

41.2.1 Receive Shift Register (RSR)

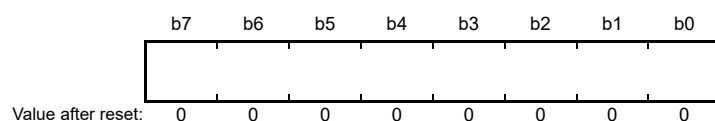
The RSR register is a shift register which is used to receive serial data input from the RXDn pin and converts it into parallel data.

When one frame of data has been received, it is automatically transferred to the RDR register.

The RSR register cannot be directly accessed by the CPU.

41.2.2 Receive Data Register (RDR)

Address(es): SCI0.RDR 0008 A005h, SCI1.RDR 0008 A025h, SCI2.RDR 0008 A045h, SCI3.RDR 0008 A065h, SCI4.RDR 0008 A085h, SCI5.RDR 0008 A0A5h, SCI6.RDR 0008 A0C5h, SCI7.RDR 000D 00E5h, SCI8.RDR 000D 0005h, SCI9.RDR 000D 0025h, SCI10.RDR 000D 0045h, SCI11.RDR 000D 0065h, SCI12.RDR 0008 B305h



The RDR register is an 8-bit register that stores receive data.

When one frame of serial data has been received, the received serial data is transferred from the RSR register to the RDR register. Then the RSR register can receive the next data.

Since the RSR and RDR registers function as a double buffer in this way, continuous receive operations can be performed.

Read the RDR register only once after a receive data full interrupt (RXI) has occurred. Note that if next one frame of data is received before reading receive data from the RDR register, an overrun error occurs.

The RDR register cannot be written to by the CPU.

41.2.3 Receive Data Register H, L, HL (RDRH, RDRL, RDRHL)

- Receive Data Register H (RDRH)

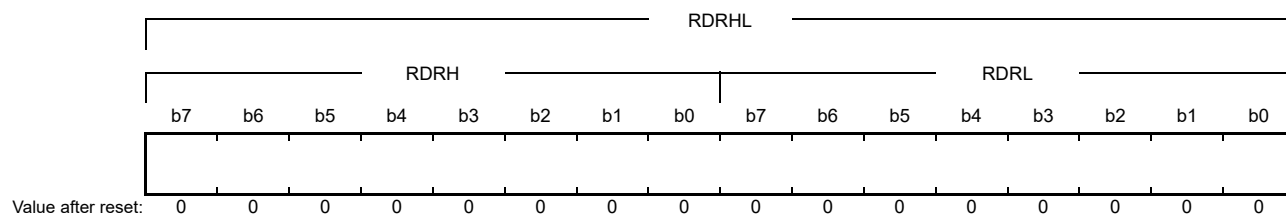
Address(es): SCI0.RDRH 0008 A010h, SCI1.RDRH 0008 A030h, SCI2.RDRH 0008 A050h, SCI3.RDRH 0008 A070h, SCI4.RDRH 0008 A090h, SCI5.RDRH 0008 A0B0h, SCI6.RDRH 0008 A0D0h, SCI7.RDRH 000D 00F0h, SCI8.RDRH 000D 0010h, SCI9.RDRH 000D 0030h, SCI10.RDRH 000D 0050h, SCI11.RDRH 000D 0070h, SCI12.RDRH 0008 B310h

- Receive Data Register L (RDRL)

Address(es): SCI0.RDRL 0008 A011h, SCI1.RDRL 0008 A031h, SCI2.RDRL 0008 A051h, SCI3.RDRL 0008 A071h, SCI4.RDRL 0008 A091h, SCI5.RDRL 0008 A0B1h, SCI6.RDRL 0008 A0D1h, SCI7.RDRL 000D 00F1h, SCI8.RDRL 000D 0011h, SCI9.RDRL 000D 0031h, SCI10.RDRL 000D 0051h, SCI11.RDRL 000D 0071h, SCI12.RDRL 0008 B311h

- Receive Data Register HL (RDRHL)

Address(es): SCI0.RDRHL 0008 A010h, SCI1.RDRHL 0008 A030h, SCI2.RDRHL 0008 A050h, SCI3.RDRHL 0008 A070h, SCI4.RDRHL 0008 A090h, SCI5.RDRHL 0008 A0B0h, SCI6.RDRHL 0008 A0D0h, SCI7.RDRHL 000D 00F0h, SCI8.RDRHL 000D 0010h, SCI9.RDRHL 000D 0030h, SCI10.RDRHL 000D 0050h, SCI11.RDRHL 000D 0070h, SCI12.RDRHL 0008 B310h



The RDRH and RDRL registers are 8-bit registers that store receive data. Use these registers when asynchronous mode and 9-bit data length are selected.

The RDRL register is the shadow register of the RDR register; i.e. access to the RDRL register is equivalent to access to the RDR register.

After one frame of data is received, the received data is transferred from the RSR register to these registers, thus allowing the RSR register to receive the next data.

The RSR, RDRH and RDRL registers have a double-buffered construction to enable continuous reception.

Read the RDRH and RDRL registers should be performed only once in the order from the RDRH register to the RDRL register when a receive data full interrupt (RXI) request is issued. Note that an overrun error occurs when the next frame of data is received before the received data has been read from the RDRL register.

The CPU cannot write to the RDRH and RDRL registers. Bits 0 to 7 in the RDRH register are fixed to 0. These bits are read as 0.

The RDRHL register can be accessed in 16-bit units.

41.2.4 Receive FIFO Data Register (FRDR)

Address(es): SCI7.FRDR 000D 00F0h, SCI8.FRDR 000D 0010h, SCI9.FRDR 000D 0030h, SCI10.FRDR 000D 0050h, SCI11.FRDR 000D 0070h,
 SCI7.FRDR.H 000D 00F0h, SCI8.FRDR.H 000D 0010h, SCI9.FRDR.H 000D 0030h, SCI10.FRDR.H 000D 0050h,
 SCI11.FRDR.H 000D 0070h,
 SCI7.FRDR.L 000D 00F1h, SCI8.FRDR.L 000D 0011h, SCI9.FRDR.L 000D 0031h, SCI10.FRDR.L 000D 0051h,
 SCI11.FRDR.L 000D 0071h



Bit	Symbol	Bit Name	Description	R/W
b8 to b0	RDAT[8:0]	Receive Data	Received data can be read	R
b9	MPB	Multi-Processor Bit Monitor Flag	Value of the multi-processor bit in the reception frame 0: Data transmission cycles 1: ID transmission cycles	R
b10	DR	Receive Data Ready Flag* ¹	0: The FRDR register does not contain valid data 1: The FRDR register contains valid data	R
b11	PER	Parity Error Flag	0: No parity error occurred 1: A parity error has occurred	R
b12	FER	Framing Error Flag	0: No framing error occurred 1: A framing error has occurred	R
b13	ORER	Overflow Error Flag* ¹	0: No overflow error occurred 1: An overflow error has occurred	R
b14	RDF	Receive FIFO Full Flag* ¹	0: The number of data stored in the receive FIFO is less than the threshold value 1: The number of data stored in the receive FIFO is equal to or greater than the threshold value	R
b15	—	Reserved	The read value is undefined	R

Note 1. These flags are the same as the flags of the same name in the SSRFIFO register. To clear these flags, clear the corresponding flag in the SSRFIFO register.

This register is used to read the data at the top of the sixteen-stage receive FIFO. This register is enabled when the FCR.FM bit is 1 (FIFO mode).

Reading the FRDR register when the receive FIFO is empty results in an undefined value.

When reading the lower 8 bits (FRDR.L), the value of the FRDR register is updated with the next data in the receive FIFO. When reading the upper 8 bits (FRDR.H), the FRDR register is not updated. When reading the FRDR register in 8-bit units, read first from the FRDR.H register and then the FRDR.L register.

When the character length is 8 bits 0 is stored in the RDAT[8] bit and when the character length is 7 bits 00b is stored in the RDAT[8:7] bits.

The MPB flag holds the value of the multi-processor bit that was added to the data at the top of the receive FIFO. The SSR.MPB flag is not used when the FCR.FM bit is 1 (FIFO mode).

The FER and PER flags indicate whether the relevant error exists in the data at the top of the receive FIFO. Its value is updated each time the FRDR register is read.

The RDF, ORER, and DR flags are the same as the flags of the same name in the SSRFIFO register. When these flags are read with a value of 1, the flag can be cleared simply by writing 0 to the corresponding flag in the SSRFIFO register.

41.2.5 Transmit Data Register (TDR)

Address(es): SCI0.TDR 0008 A003h, SCI1.TDR 0008 A023h, SCI2.TDR 0008 A043h, SCI3.TDR 0008 A063h, SCI4.TDR 0008 A083h, SCI5.TDR 0008 A0A3h, SCI6.TDR 0008 A0C3h, SCI7.TDR 000D 00E3h, SCI8.TDR 000D 0003h, SCI9.TDR 000D 0023h, SCI10.TDR 000D 0043h, SCI11.TDR 000D 0063h, SCI12.TDR 0008 B303h



The TDR register is an 8-bit register that stores transmit data.

When the SCI detects that the TSR register is empty, it transfers the transmit data written in the TDR register to the TSR register and starts transmission.

The double-buffered structures of the TDR register and the TSR register enable continuous serial transmission. If the next transmit data has already been written to the TDR register when one frame of data is transmitted, the SCI transfers the written data to the TSR register to continue transmission.

The CPU is able to read from or write to the TDR register at any time. Only write transmit data to the TDR register once after each instance of the transmit data empty interrupt (TXI).

41.2.6 Transmit Data Register H, L, HL (TDRH, TDRL, TDRHL)

- Transmit Data Register H (TDRH)

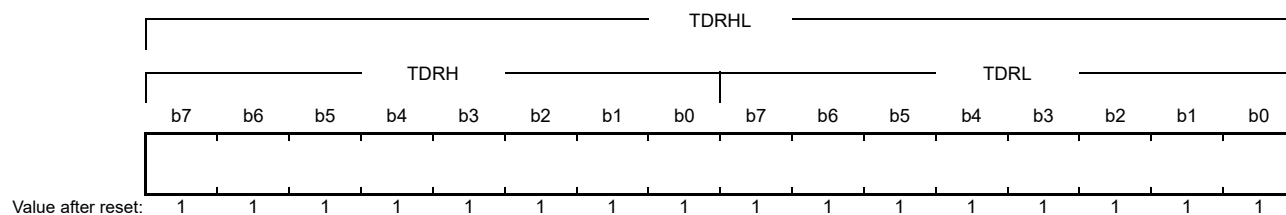
Address(es): SCI0.TDRH 0008 A00Eh, SCI1.TDRH 0008 A02Eh, SCI2.TDRH 0008 A04Eh, SCI3.TDRH 0008 A06Eh, SCI4.TDRH 0008 A08Eh, SCI5.TDRH 0008 A0AEh, SCI6.TDRH 0008 A0CEh, SCI7.TDRH 000D 00EEh, SCI8.TDRH 000D 000Eh, SCI9.TDRH 000D 002Eh, SCI10.TDRH 000D 004Eh, SCI11.TDRH 000D 006Eh, SCI12.TDRH 0008 B30Eh

- Transmit Data Register L (TDRL)

Address(es): SCI0.TDRL 0008 A00Fh, SCI1.TDRL 0008 A02Fh, SCI2.TDRL 0008 A04Fh, SCI3.TDRL 0008 A06Fh, SCI4.TDRL 0008 A08Fh, SCI5.TDRL 0008 A0AFh, SCI6.TDRL 0008 A0CFh, SCI7.TDRL 000D 00EFh, SCI8.TDRL 000D 000Fh, SCI9.TDRL 000D 002Fh, SCI10.TDRL 000D 004Fh, SCI11.TDRL 000D 006Fh, SCI12.TDRL 0008 B30Fh

- Transmit Data Register HL (TDRHL)

Address(es): SCI0.TDRHL 0008 A00Eh, SCI1.TDRHL 0008 A02Eh, SCI2.TDRHL 0008 A04Eh, SCI3.TDRHL 0008 A06Eh, SCI4.TDRHL 0008 A08Eh, SCI5.TDRHL 0008 A0AEh, SCI6.TDRHL 0008 A0CEh, SCI7.TDRHL 000D 00EEh, SCI8.TDRHL 000D 000Eh, SCI9.TDRHL 000D 002Eh, SCI10.TDRHL 000D 004Eh, SCI11.TDRHL 000D 006Eh, SCI12.TDRHL 0008 B30Eh



The TDRH and TDRL registers are 8-bit registers that store transmit data. Use these registers when asynchronous mode and 9-bit data length are selected.

The TDRL register is the shadow register of the TDR register; i.e. access to the TDRL register is equivalent to access to the TDR register.

When empty space is detected in the TSR register, the transmit data stored in the TDRH and TDRL registers is transferred to the TSR register; i.e., transmitting is started.

The TSR, TDRH and TDRL registers have a double-buffered construction to realize continuous reception. When the next data to be transmitted is stored in the TDRL register after one frame of data has been transmitted, the transmitting operation is continued by transfer to the TSR register.

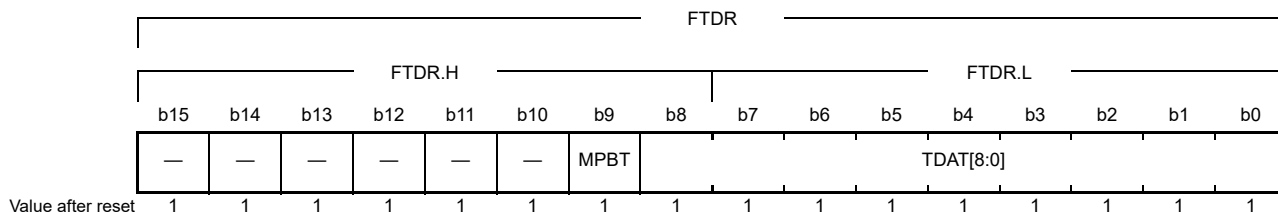
The CPU can read and write to the TDRH and TDRL registers. Bits 0 to 7 in the RDRH register are fixed to 1. These bits are read as 1. The write value should be 1.

Writing transmit data to the TDRH and TDRL registers should be performed only once in the order from the TDRH register to the TDRL register when a transmit data empty interrupt (TXI) request is issued.

The TDRHL register can be accessed in 16-bit units.

41.2.7 Transmit FIFO Data Register (FTDR)

Address(es): SCI7.FTDR 000D 00EEh, SCI8.FTDR 000D 000Eh, SCI9.FTDR 000D 002Eh, SCI10.FTDR 000D 004Eh, SCI11.FTDR 000D 006Eh,
 SCI7.FTDR.H 000D 00EEh, SCI8.FTDR.H 000D 000Eh, SCI9.FTDR.H 000D 002Eh, SCI10.FTDR.H 000D 004Eh,
 SCI11.FTDR.H 000D 006Eh,
 SCI7.FTDR.L 000D 00EFh, SCI8.FTDR.L 000D 000Fh, SCI9.FTDR.L 000D 002Fh, SCI10.FTDR.L 000D 004Fh,
 SCI11.FTDR.L 000D 006Fh



Bit	Symbol	Bit Name	Description	R/W
b8 to b0	TDAT[8:0]	Transmit Data	Specifies the data to be transmitted	W
b9	MPBT	Transmit Multi-Processor	Specifies the value of the multi-processor bit in the transmit frame 0: Data transmission cycles 1: ID transmission cycles	W
b15 to b10	—	Reserved	The write value should be 1	W

This register is used to write data to the 16-stage transmit FIFO. This register is enabled when the FCR.FM bit is 1 (FIFO mode) and the SCR.TE bit is 1.

When the transmit FIFO contains 16 frames of data, no transmit data can be set in the FTDR register.

When writing a value to the lower 8 bits (FTDR.L), the data in the FTDR register is transferred to the transmit FIFO.

When writing a value to the upper 8 bits (FTDR.H), no data is transferred to the transmit FIFO. When writing a 16-bit data to the FTDR register in 8-bit units, write first to the FTDR.H register and then the FTDR.L register.

MPBT Bit (Transmit Multi-Processor)

This bit is used to set the value of the multi-processor bit to add to the transmit frame. The SSR.MPBT bit is not used when the FCR.FM bit is 1 (FIFO mode).

41.2.8 Transmit Shift Register (TSR)

The TSR register is a shift register that transmits serial data.

To perform serial data transmission, the SCI first automatically transfers transmit data from the TDR register to the TSR register, and then sends the data to the TXDn pin.

The TSR register cannot be directly accessed by the CPU.

41.2.9 Serial Mode Register (SMR)

Some bits in the SMR register have different functions in smart card interface mode and non-smart card interface mode.

(1) Non-Smart Card Interface Mode (SCMR.SMIF = 0)

Address(es): SCI0.SMR 0008 A000h, SCI1.SMR 0008 A020h, SCI2.SMR 0008 A040h, SCI3.SMR 0008 A060h, SCI4.SMR 0008 A080h, SCI5.SMR 0008 A0A0h, SCI6.SMR 0008 A0C0h, SCI7.SMR 000D 00E0h, SCI8.SMR 000D 0000h, SCI9.SMR 000D 0020h, SCI10.SMR 000D 0040h, SCI11.SMR 000D 0060h, SCI12.SMR 0008 B300h

b7	b6	b5	b4	b3	b2	b1	b0
CM	CHR	PE	PM	STOP	MP	CKS[1:0]	
Value after reset:	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b1, b0	CKS[1:0]	Clock Select	b1 b0 0 0: PCLK (n = 0)*1 0 1: PCLK/4 (n = 1)*1 1 0: PCLK/16 (n = 2)*1 1 1: PCLK/64 (n = 3)*1	R/W*4
b2	MP	Multi-Processor Mode	(Valid only in asynchronous mode) 0: Multi-processor communications function is disabled 1: Multi-processor communications function is enabled	R/W*4
b3	STOP	Stop Bit Length	(Valid only in asynchronous mode) 0: 1 stop bit 1: 2 stop bits	R/W*4
b4	PM	Parity Mode	(Valid only when the PE bit is 1) 0: Selects even parity 1: Selects odd parity	R/W*4
b5	PE	Parity Enable	(Valid only in asynchronous mode) • When transmitting 0: Parity bit addition is not performed 1: The parity bit is added • When receiving 0: Parity bit checking is not performed 1: The parity bit is checked	R/W*4
b6	CHR	Character Length	(Valid only in asynchronous mode*2) Selects in combination with the SCMR.CHR1 bit. CHR1 CHR 0 0: Transmit/receive in 9-bit data length 0 1: Transmit/receive in 9-bit data length 1 0: Transmit/receive in 8-bit data length (initial value) 1 1: Transmit/receive in 7-bit data length*3	R/W*4
b7	CM	Communications Mode	0: Asynchronous mode or simple I2C mode 1: Clock synchronous mode or simple SPI mode	R/W*4

Note 1. n is the decimal notation of the value of n in the BRR register (refer to section 41.2.13, Bit Rate Register (BRR)).

Note 2. In other than asynchronous mode, this bit setting is invalid and a fixed data length of 8 bits is used.

Note 3. LSB first is fixed and the MSB (bit 7) in the TDR register is not transmitted in transmission.

Note 4. Writable only when the SCR.TE bit is 0 and the SCR.RE bit is 0 (both serial transmission and reception are disabled).

CKS[1:0] Bits (Clock Select)

These bits select the clock source for the on-chip baud rate generator.

For the relation between the settings of these bits and the baud rate, refer to section 41.2.13, Bit Rate Register (BRR).

MP Bit (Multi-Processor Mode)

Disables/enables the multi-processor communications function. The settings of the PE bit and PM bit are invalid in multi-processor mode.

STOP Bit (Stop Bit Length)

Selects the stop bit length in transmission.

In reception, only the first stop bit is checked regardless of this bit setting. If the second stop bit is 0, it is treated as the start bit of the next transmit frame.

PM Bit (Parity Mode)

Selects the parity mode (even or odd) for transmission and reception.

The setting of the PM bit is invalid in multi-processor mode.

PE Bit (Parity Enable)

When this bit is set to 1, the parity bit is added to transmit data, and the parity bit is checked in reception.

Irrespective of the setting of the PE bit, the parity bit is not added or checked in multi-processor format.

CHR Bit (Character Length)

Selects the data length for transmission and reception.

Selects in combination with the SCMR.CHR1 bit.

In other than asynchronous mode, a fixed data length of 8 bits is used.

(2) Smart Card Interface Mode (SCMR.SMIF = 1)

Address(es): SMC10.SMR 0008 A000h, SMC11.SMR 0008 A020h, SMC12.SMR 0008 A040h, SMC13.SMR 0008 A060h, SMC14.SMR 0008 A080h, SMC15.SMR 0008 A0A0h, SMC16.SMR 0008 A0C0h, SMC17.SMR 000D 00E0h, SMC18.SMR 000D 0000h, SMC19.SMR 000D 0020h, SMC110.SMR 000D 0040h, SMC111.SMR 000D 0060h, SMC112.SMR 0008 B300h

b7	b6	b5	b4	b3	b2	b1	b0
GM	BLK	PE	PM	BCP[1:0]		CKS[1:0]	
Value after reset:	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b1, b0	CKS[1:0]	Clock Select	b1 b0 0 0: PCLK (n = 0)*1 0 1: PCLK/4 (n = 1)*1 1 0: PCLK/16 (n = 2)*1 1 1: PCLK/64 (n = 3)*1	R/W*2
b3, b2	BCP[1:0]	Base Clock Pulse	Selects the number of base clock cycles in combination with the SCMR.BCP2 bit. Table 41.9 lists the combinations of the SCMR.BCP2 bit and SMR.BCP[1:0] bits.	R/W*2
b4	PM	Parity Mode	(Valid only when the PE bit is 1) 0: Selects even parity 1: Selects odd parity	R/W*2
b5	PE	Parity Enable	When this bit is set to 1, a parity bit is added to transmit data, and the parity of received data is checked. Set this bit to 1 in smart card interface mode.	R/W*2
b6	BLK	Block Transfer Mode	0: Non-block transfer mode operation 1: Block transfer mode operation	R/W*2
b7	GM	GSM Mode	0: Non-GSM mode operation 1: GSM mode operation	R/W*2

Note 1. n is the decimal notation of the value of n in the BRR register (refer to section 41.2.13, Bit Rate Register (BRR)).

Note 2. Writable only when the SCR.TE bit is 0 and the SCR.RE bit is 0 (both serial transmission and reception are disabled).

CKS[1:0] Bits (Clock Select)

These bits select the clock source for the on-chip baud rate generator.

For the relationship between the settings of these bits and the baud rate, refer to section 41.2.13, Bit Rate Register (BRR).

BCP[1:0] Bits (Base Clock Pulse)

These bits select the number of base clock cycles in a 1-bit data transfer time in smart card interface mode.

Set these bits in combination with the SCMR.BCP2 bit.

For details, refer to section 41.6.4, Receive Data Sampling Timing and Reception Margin.

Table 41.9 Combinations of the SCMR.BCP2 Bit and SMR.BCP[1:0] Bits

SCMR.BCP2 Bit	SMR.BCP[1:0] Bits		Number of Base Clock Cycles for 1-Bit Transfer Period
0	0	0	93 clock cycles (S = 93)*1
0	0	1	128 clock cycles (S = 128)*1
0	1	0	186 clock cycles (S = 186)*1
0	1	1	512 clock cycles (S = 512)*1
1	0	0	32 clock cycles (S = 32)*1 (Initial Value)
1	0	1	64 clock cycles (S = 64)*1
1	1	0	372 clock cycles (S = 372)*1
1	1	1	256 clock cycles (S = 256)*1

Note 1. S is the value of S in the BRR register (refer to section 41.2.13, Bit Rate Register (BRR)).

PM Bit (Parity Mode)

Selects the parity mode for transmission and reception (even or odd).

For details on the usage of this bit in smart card interface mode, refer to [section 41.6.2, Data Format \(Except in Block Transfer Mode\)](#).

PE Bit (Parity Enable)

Set the PE bit to 1.

The parity bit is added to transmit data before transmission, and the parity bit is checked in reception.

BLK Bit (Block Transfer Mode)

Setting this bit to 1 allows block transfer mode operation.

For details, refer to [section 41.6.3, Block Transfer Mode](#).

GM Bit (GSM Mode)

Setting this bit to 1 allows GSM mode operation.

In GSM mode, the SSR.TEND flag set timing is put forward to 11.0 etu (elementary time unit = 1-bit transfer time) from the start and the clock output control function is appended. For details, refer to [section 41.6.6, Serial Data Transmission \(Except in Block Transfer Mode\)](#) and [section 41.6.8, Clock Output Control](#).

41.2.10 Serial Control Register (SCR)

Some bits in the SCR register have different functions in smart card interface mode and non-smart card interface mode.

(1) Non-Smart Card Interface Mode (SCMR.SMIF = 0)

Address(es): SCI0.SCR 0008 A002h, SCI1.SCR 0008 A022h, SCI2.SCR 0008 A042h, SCI3.SCR 0008 A062h, SCI4.SCR 0008 A082h, SCI5.SCR 0008 A0A2h, SCI6.SCR 0008 A0C2h, SCI7.SCR 000D 00E2h, SCI8.SCR 000D 0002h, SCI9.SCR 000D 0022h, SCI10.SCR 000D 0042h, SCI11.SCR 000D 0062h, SCI12.SCR 0008 B302h

b7	b6	b5	b4	b3	b2	b1	b0
TIE	RIE	TE	RE	MPIE	TEIE	CKE[1:0]	
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b1, b0	CKE[1:0]	Clock Enable	(Asynchronous mode) b1 b0 0 0: On-chip baud rate generator The SCKn pin becomes high-impedance. 0 1: On-chip baud rate generator The clock with the same frequency as the bit rate is output from the SCKn pin. 1 x: External clock or TMR clock*2 • The clock with a frequency 16 times the bit rate should be input from the SCKn pin. Input a clock signal with a frequency eight times the bit rate when the SEMR.ABCS bit is 1. • The SCKn pin becomes high-impedance when the TMR clock*2 is used. (Clock synchronous mode) b1 b0 0 x: Internal clock The SCKn pin functions as the clock output pin. 1 x: External clock The SCKn pin functions as the clock input pin.	R/W*1
b2	TEIE	Transmit End Interrupt Enable	0: A TEI interrupt request is disabled 1: A TEI interrupt request is enabled	R/W
b3	MPIE	Multi-Processor Interrupt Enable	(Valid in asynchronous mode when the SMR.MP bit is 1) 0: Normal reception 1: When the data with the multi-processor bit set to 0 is received, the data is not read, and setting the status flags RDRF, ORER, and FER in the SSR register to 1 is disabled. When the data with the multi-processor bit set to 1 is received, the MPIE bit is automatically cleared to 0, and normal reception is resumed.	R/W
b4	RE	Receive Enable	0: Serial reception is disabled 1: Serial reception is enabled	R/W*3
b5	TE	Transmit Enable	0: Serial transmission is disabled 1: Serial transmission is enabled	R/W*3
b6	RIE	Receive Interrupt Enable	0: RXI and ERI interrupt requests are disabled 1: RXI and ERI interrupt requests are enabled	R/W
b7	TIE	Transmit Interrupt Enable	0: A TXI interrupt request is disabled 1: A TXI interrupt request is enabled	R/W

x: Don't care

Note 1. Writable only when TE = 0 and RE = 0.

Note 2. TMR clock is selectable for SCI5, SCI6, and SCI12.

Note 3. 1 can be written only when TE = 0 and RE = 0, while the SMR.CM bit is 1. After setting TE or RE to 1, only 0 can be written to TE and RE. While the SMR.CM bit is 0 and the SIMR1.IICM bit is 0, writing is enabled under any condition.

CKE[1:0] Bits (Clock Enable)

These bits select the clock source and SCKn pin function.

The combination of the settings of these bits and of the SEMR.ACS0 bit sets the internal TMR clock.

TEIE Bit (Transmit End Interrupt Enable)

Enables or disables a TEI interrupt request.

A TEI interrupt request is disabled by setting the TEIE bit to 0.

In simple I²C mode, the TEI is allocated to the interrupt on completion of issuing a start, restart, or stop condition (STI).

In this case, the TEIE bit can be used to enable or disable the STI.

MPIE Bit (Multi-Processor Interrupt Enable)

When this bit is set to 1 and the data with the multi-processor bit set to 0 is received, the data is not read and setting the status flags RDRF, ORER, and FER in the SSR register (and SSRFIFO.DR flag for SCI7 to SCI11) to 1 is disabled.

When the data with the multi-processor bit set to 1 is received, the MPIE bit is automatically cleared to 0, and normal reception is resumed. For details, refer to section 41.4, Multi-Processor Communications Function.

When the data with the multi-processor bit set to 0 is received, the receive data is not transferred from the RSR register to the RDR register, a receive error is not detected, and setting the flags RDRF, ORER, and FER (and SSRFIFO.DR flag for SCI7 to SCI11) to 1 is disabled.

When the data with the multi-processor bit set to 1 is received, the MPB bit is set to 1, the MPIE bit is automatically cleared to 0, the RXI and ERI interrupt requests are enabled (if the SCR.RIE bit is set to 1), and setting the flags RDRF, ORER, and FER (and SSRFIFO.DR flag for SCI7 to SCI11) to 1 is enabled.

Set the MPIE bit to 0 if multi-processor communications function is not to be used.

RE Bit (Receive Enable)

Enables or disables serial reception.

When this bit is set to 1, serial reception is started by detecting the start bit in asynchronous mode or the synchronous clock input in clock synchronous mode. Note that the SMR register should be set prior to setting the RE bit to 1 in order to designate the reception format.

Even if reception is halted by setting the RE bit to 0 while the FCR.FM bit is 0 (non-FIFO mode), the ORER, FER, PER, and RDRF flags in the SSR register are not affected and the previous value is retained.

Even if reception is halted by setting the RE bit to 0 while the FCR.FM bit is 1 (FIFO mode), the RDF, ORER, FER, PER, DR flags in the SSRFIFO register are not affected and the previous value is retained.

TE Bit (Transmit Enable)

Enables or disables serial transmission.

When this bit is set to 1, serial transmission is started by writing transmit data to the TDR register. Note that the SMR register should be set prior to setting the TE bit to 1 in order to designate the transmission format.

RIE Bit (Receive Interrupt Enable)

Enables or disables RXI and ERI interrupt requests.

An RXI interrupt request is disabled by setting the RIE bit to 0.

An ERI interrupt request can be canceled by reading 1 from the ORER, FER, or PER flag in the SSR register and then setting the flag to 0, or setting the RIE bit to 0.

TIE Bit (Transmit Interrupt Enable)

Enables or disables TXI interrupt request.

A TXI interrupt request is disabled by setting the TIE bit to 0.

(2) Smart Card Interface Mode (SCMR.SMIF = 1)

Address(es): SMC10.SCR 0008 A002h, SMC11.SCR 0008 A022h, SMC12.SCR 0008 A042h, SMC13.SCR 0008 A062h, SMC14.SCR 0008 A082h, SMC15.SCR 0008 A0A2h, SMC16.SCR 0008 A0C2h, SMC17.SCR 000D 00E2h, SMC18.SCR 000D 0002h, SMC19.SCR 000D 0022h, SMC110.SCR 000D 0042h, SMC111.SCR 000D 0062h, SMC112.SCR 0008 B302h

b7	b6	b5	b4	b3	b2	b1	b0
TIE	RIE	TE	RE	MPIE	TEIE	CKE[1:0]	
Value after reset:	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b1, b0	CKE[1:0]	Clock Enable	- When SMR.GM = 0 - b1 b0 0 0: Output disabled The SCKn pin becomes high-impedance. 0 1: Clock output 1 x: Setting prohibited - When SMR.GM = 1 - b1 b0 0 0: Output fixed low - x 1: Clock output 1 0: Output fixed high	R/W*1
b2	TEIE	Transmit End Interrupt Enable	This bit should be 0 in smart card interface mode.	R/W
b3	MPIE	Multi-Processor Interrupt Enable	This bit should be 0 in smart card interface mode.	R/W
b4	RE	Receive Enable	0: Serial reception is disabled 1: Serial reception is enabled	R/W*2
b5	TE	Transmit Enable	0: Serial transmission is disabled 1: Serial transmission is enabled	R/W*2
b6	RIE	Receive Interrupt Enable	0: RXI and ERI interrupt requests are disabled 1: RXI and ERI interrupt requests are enabled	R/W
b7	TIE	Transmit Interrupt Enable	0: A TXI interrupt request is disabled 1: A TXI interrupt request is enabled	R/W

x: Don't care

Note 1. Writable only when TE = 0 and RE = 0.

Note 2. 1 can be written only when TE = 0 and RE = 0. After setting TE or RE to 1, only 0 can be written in TE and RE.

For details on interrupt requests, refer to section 41.12, Interrupt Sources.

CKE[1:0] Bits (Clock Enable)

These bits control the clock output from the SCKn pin.

In GSM mode, clock output can be dynamically switched. For details, refer to section 41.6.8, Clock Output Control.

TEIE Bit (Transmit End Interrupt Enable)

This bit should be 0 in smart card interface mode.

MPIE Bit (Multi-Processor Interrupt Enable)

This bit should be 0 in smart card interface mode.

RE Bit (Receive Enable)

Enables or disables serial reception.

When this bit is set to 1, serial reception is started by detecting the start bit. Note that the SMR register should be set prior to setting the RE bit to 1 in order to designate the reception format.

Even if reception is halted by setting the RE bit to 0, the ORER, FER, and PER flags in the SSR register are not affected and the previous value is retained.

TE Bit (Transmit Enable)

Enables or disables serial transmission.

When this bit is set to 1, serial transmission is started by writing transmit data to the TDR register. Note that the SMR register should be set prior to setting the TE bit to 1 in order to designate the transmission format.

RIE Bit (Receive Interrupt Enable)

Enables or disables RXI and ERI interrupt requests.

An RXI interrupt request is disabled by setting the RIE bit to 0.

An ERI interrupt request can be canceled by reading 1 from the ORER, FER, or PER flag in the SSR register and then setting the flag to 0, or setting the RIE bit to 0.

TIE Bit (Transmit Interrupt Enable)

Enables or disables TXI interrupt request.

A TXI interrupt request is disabled by setting the TIE bit to 0.

41.2.11 Serial Status Register (SSR/SSRFIFO)

Some bits in the SSR register have different functions in smart card interface mode and non-smart card interface mode or FIFO mode and non-FIFO mode.

(1) Non-Smart Card Interface Mode and non-FIFO Mode (SCMR.SMIF = 0 and FCR.FM = 0)

Address(es): SCI0.SSR 0008 A004h, SCI1.SSR 0008 A024h, SCI2.SSR 0008 A044h, SCI3.SSR 0008 A064h, SCI4.SSR 0008 A084h, SCI5.SSR 0008 A0A4h, SCI6.SSR 0008 A0C4h, SCI7.SSR 000D 00E4h, SCI8.SSR 000D 0004h, SCI9.SSR 000D 0024h, SCI10.SSR 000D 0044h, SCI11.SSR 000D 0064h, SCI12.SSR 0008 B304h

b7	b6	b5	b4	b3	b2	b1	b0
TDRE	RDRF	ORER	FER	PER	TEND	MPB	MPBT
Value after reset: 1	0	0	0	0	1	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	MPBT	Multi-Processor Bit Transfer	Sets the multi-processor bit for adding to the transmission frame 0: Data transmission cycles 1: ID transmission cycles	R/W
b1	MPB	Multi-Processor	Value of the multi-processor bit in the reception frame 0: Data transmission cycles 1: ID transmission cycles	R
b2	TEND	Transmit End Flag	0: A character is being transmitted. 1: Character transfer has been completed.	R
b3	PER	Parity Error Flag	0: No parity error occurred 1: A parity error has occurred	R/(W) *1
b4	FER	Framing Error Flag	0: No framing error occurred 1: A framing error has occurred	R/(W) *1
b5	ORER	Overrun Error Flag	0: No overrun error occurred 1: An overrun error has occurred	R/(W) *1
b6	RDRF	Receive Data Full Flag	0: No valid data is held in the RDR register 1: Received data is held in the RDR register	R/(W) *2
b7	TDRE	Transmit Data Empty Flag	0: Data to be transmitted is held in the TDR register 1: No data is held in the TDR register	R/(W) *2

Note 1. Only 0 can be written to this bit, to clear the flag. To clear this flag, confirm that the flag is 1 and then set it to 0.

Note 2. Write 1 when writing is necessary.

MPB Bit (Multi-Processor)

Holds the value of the multi-processor bit in the reception frame. This bit does not change when the SCR.RE bit is 0.

TEND Flag (Transmit End Flag)

Indicates completion of transmission.

[Setting conditions]

- When the SCR.TE bit is set to 0 (serial transmission is disabled)
When the SCR.TE bit is changed from 0 to 1, the TEND flag is not affected and retains the value 1.
- When the TDR register is not updated at the time of transmission of the tail-end bit of a character being transmitted

[Clearing condition]

- When transmit data are written to the TDR register while the SCR.TE bit is 1
When setting the TEND flag to 0 to complete the interrupt handling, refer to section 15.5.2, Level Detection.

PER Flag (Parity Error Flag)

Indicates that a parity error has occurred during reception in asynchronous mode and the reception ends abnormally.

[Setting condition]

- When a parity error is detected during reception while data match detection is disabled (for SCI0 to SCI11)
- When a parity error is detected during reception (for SCI12)
Although receive data when the parity error occurs is transferred to the RDR register, no RXI interrupt request occurs. Note that when the PER flag is being set to 1, the subsequent receive data is not transferred to the RDR register.

[Clearing condition]

- When 0 is written to the PER flag after reading PER = 1
When setting the PER flag to 0 to complete the interrupt handling, refer to **section 15.5.2, Level Detection**. Even when the SCR.RE bit is set to 0 (serial reception is disabled), the PER flag is not affected and retains its previous value.

FER Flag (Framing Error Flag)

Indicates that a framing error has occurred during reception in asynchronous mode and the reception ends abnormally.

[Setting condition]

- When the stop bit is 0 while data match detection is disabled (for SCI0 to SCI11)
- When the stop bit is 0 (for SCI12)
In 2-stop-bit mode, only the first stop bit is checked whether it is 1 but the second stop bit is not checked. Note that although receive data when the framing error occurs is transferred to the RDR register, no RXI interrupt request occurs. In addition, when the FER flag is being set to 1, the subsequent receive data is not transferred to the RDR register.

[Clearing condition]

- When 0 is written to the FER flag after reading FER = 1
When setting the FER flag to 0 to complete the interrupt handling, refer to **section 15.5.2, Level Detection**. Even when the SCR.RE bit is set to 0, the FER flag is not affected and retains its previous value.

ORER Flag (Overrun Error Flag)

Indicates that an overrun error has occurred during reception and the reception ends abnormally.

[Setting condition]

- When the next data is received before receive data is read from the RDR register
In the RDR register, receive data prior to an overrun error occurrence is retained, but data received after the overrun error occurrence is lost. When the ORER flag is set to 1, subsequent serial reception cannot be performed. Note that, in clock synchronous mode, serial transmission also cannot continue.

[Clearing condition]

- When 0 is written to the ORER flag after reading ORER = 1
When setting the ORER flag to 0 to complete the interrupt handling, refer to **section 15.5.2, Level Detection**. Even when the SCR.RE bit is set to 0, the ORER flag is not affected and retains its previous value.

RDRF Flag (Receive Data Full Flag)

Indicates whether the RDR register has received data.

[Setting condition]

- When data has been received normally, and transferred from the RSR register to the RDR register

[Clearing condition]

- When data is read from the RDR register

TDRE Flag (Transmit Data Empty Flag)

Indicates whether the TDR register has data to be transmitted.

[Setting condition]

- When data is transferred from the TDR register to the TSR register

[Clearing condition]

- When data is written to the TDR register

(2) Smart Card Interface Mode (SCMR.SMIF = 1)

Address(es): SMCi0.SSR 0008 A004h, SMCi1.SSR 0008 A024h, SMCi2.SSR 0008 A044h, SMCi3.SSR 0008 A064h, SMCi4.SSR 0008 A084h, SMCi5.SSR 0008 A0A4h, SMCi6.SSR 0008 A0C4h, SMCi7.SSR 000D 00E4h, SMCi8.SSR 000D 0004h, SMCi9.SSR 000D 0024h, SMCi10.SSR 000D 0044h, SMCi11.SSR 000D 0064h, SMCi12.SSR 0008 B304h

b7	b6	b5	b4	b3	b2	b1	b0
TDRE	RDRF	ORER	ERS	PER	TEND	MPB	MPBT

Value after reset: 1 0 0 0 0 1 0 0

Bit	Symbol	Bit Name	Description	R/W
b0	MPBT	Multi-Processor Bit Transfer	This bit should be set to 0 in smart card interface mode.	R/W
b1	MPB	Multi-Processor	This bit is not used in smart card interface mode. It should be set to 0.	R
b2	TEND	Transmit End Flag	0: A character is being transmitted. 1: Character transfer has been completed.	R
b3	PER	Parity Error Flag	0: No parity error occurred 1: A parity error has occurred	R/(W) *1
b4	ERS	Error Signal Status Flag	0: Low error signal not responded 1: Low error signal responded	R/(W) *1
b5	ORER	Overrun Error Flag	0: No overrun error occurred 1: An overrun error has occurred	R/(W) *1
b6	RDRF	Receive Data Full Flag	0: No valid data is held in the RDR register 1: Received data is held in the RDR register	R/(W) *2
b7	TDRE	Transmit Data Empty Flag	0: Data to be transmitted is held in the TDR register 1: No data is held in the TDR register	R/(W) *2

Note 1. Only 0 can be written to this bit, to clear the flag. To clear this flag, confirm that the flag is 1 and then set it to 0.

Note 2. Write 1 when writing is necessary.

TEND Flag (Transmit End Flag)

With no error signal from the receiving side, this bit is set to 1 when further data for transfer is ready to be transferred to the TDR register.

[Setting conditions]

- When the SCR.TE bit is 0 (serial transmission is disabled)
When the SCR.TE bit is changed from 0 to 1, the TEND flag is not affected and retains the value 1.
- When a specified period has elapsed after the latest transmission of 1 byte, the ERS flag is 0, and the TDR register is not updated
The set timing is determined by register settings as listed below.
When SMR.GM = 0 and SMR.BLK = 0, 12.5 etu after the start of transmission
When SMR.GM = 0 and SMR.BLK = 1, 11.5 etu after the start of transmission
When SMR.GM = 1 and SMR.BLK = 0, 11.0 etu after the start of transmission
When SMR.GM = 1 and SMR.BLK = 1, 11.0 etu after the start of transmission

[Clearing condition]

- When transmit data are written to the TDR register while the SCR.TE bit is 1
When setting the TEND flag to 0 to complete the interrupt handling, refer to **section 15.5.2, Level Detection**.

PER Flag (Parity Error Flag)

Indicates that a parity error has occurred during reception in asynchronous mode and the reception ends abnormally.

[Setting condition]

- When a parity error is detected during reception
Although receive data when the parity error occurs is transferred to the RDR register, no RXI interrupt request occurs. Note that when the PER flag is being set to 1, the subsequent receive data is not transferred to the RDR register.

[Clearing condition]

- When 0 is written to the PER flag after reading PER = 1
When setting the PER flag to 0 to complete the interrupt handling, refer to **section 15.5.2, Level Detection**.
Even when the SCR.RE bit is set to 0 (serial reception is disabled), the PER flag is not affected and retains its previous value.

ERS Flag (Error Signal Status Flag)

[Setting condition]

- When a low error signal is sampled

[Clearing condition]

- When 0 is written to the ERS flag after reading ERS = 1
When setting the ERS flag to 0 to complete the interrupt handling, refer to **section 15.5.2, Level Detection**.
Even when the SCR.RE bit is set to 0, the ERS flag is not affected and retains its previous value.

ORER Flag (Overrun Error Flag)

Indicates that an overrun error has occurred during reception and the reception ends abnormally.

[Setting condition]

- When the next data is received before receive data is read from the RDR register
In the RDR register, the receive data prior to an overrun error occurrence is retained, but data received following the overrun error occurrence is lost. When the ORER flag is set to 1, subsequent serial reception cannot be performed.

[Clearing condition]

- When 0 is written to the ORER flag after reading ORER = 1
When setting the ORER flag to 0 to complete the interrupt handling, refer to **section 15.5.2, Level Detection**.
Even when the SCR.RE bit is set to 0, the ORER flag is not affected and retains its previous value.

RDRF Flag (Receive Data Full Flag)

Indicates whether the RDR register has received data.

[Setting condition]

- When data has been received normally, and transferred from the RSR register to the RDR register

[Clearing condition]

- When data is read from the RDR register

TDRE Flag (Transmit Data Empty Flag)

Indicates whether the TDR register has data to be transmitted.

[Setting condition]

- When data is transferred from the TDR register to the TSR register

[Clearing condition]

- When data is written to the TDR register

(3) Non-Smart Card Interface Mode and FIFO Mode (SCMR.SMIF = 0 and FCR.FM = 1)

Address(es): SCI7.SSRFIFO 000D 00E4h, SCI8.SSRFIFO 000D 0004h, SCI9.SSRFIFO 000D 0024h, SCI10.SSRFIFO 000D 0044h, SCI11.SSRFIFO 000D 0064h

	b7	b6	b5	b4	b3	b2	b1	b0
	TDFE	RDF	ORER	FER	PER	TEND	—	DR
Value after reset	1	0	0	0	0	1	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	DR	Receive Data Ready Flag*1	0: Reception in progress, or receive FIFO is empty 1: Reception completed, and number of data in the receive FIFO is less than the threshold value	R/(W) *2
b1	—	Reserved	The read value is undefined. The write value should be 1	R/W
b2	TEND	Transmit End Flag	0: A character is being transmitted 1: Character transfer has been completed	R/(W) *2
b3	PER	Parity Error Flag	0: No parity error occurred 1: A parity error has occurred	R/(W) *2
b4	FER	Framing Error Flag	0: No framing error occurred 1: A framing error has occurred	R/(W) *2
b5	ORER	Overrun Error Flag	0: No overrun error occurred 1: An overrun error has occurred	R/(W) *2
b6	RDF	Receive FIFO Full Flag	0: The number of data stored in the receive FIFO is less than the threshold value 1: The number of data stored in the receive FIFO is equal to or greater than the threshold value	R/(W) *2
b7	TDFE	Transmit FIFO Empty Flag	0: The number of data stored in the transmit FIFO is greater than the threshold value 1: The number of data stored in the transmit FIFO is equal to or less than the threshold value	R/(W) *2

Note 1. This flag is only enabled in asynchronous mode. This flag does not become 1 in clock synchronous mode.

Note 2. This flag can only be written to 0 to clear the flag. To clear this flag, confirm that the flag is 1 and then set it to 0.

DR Flag (Receive Data Ready Flag)

After receiving data, this flag indicates when 15 etus (Elementary Time Unit: time required to transmit a single bit) have elapsed since the number of data stored in the receive FIFO has dropped to less than the threshold value (FCR.RTRG[3:0]).

[Setting condition]

- When the last received data has neither framing nor parity error, and the number of data in the receive FIFO is less than the threshold value at the time that data is transferred from the RSR register, and the next data is not yet fully received even after 15 etus have elapsed since the last stop bit.

[Clearing condition]

- When 0 is written to the DR flag, after all data in the receive FIFO is read and the DR flag is confirmed to be 1.
- When the FCR.FM bit is changed from 0 to 1.

When the FCR.DRES bit is 1 (ERI interrupt), refer to **section 15.5.2, Level Detection** when setting the DR flag to 0 to exit from the interrupt handling routine.

TEND Flag (Transmit End Flag)

This flag indicates that transmission has completed.

[Setting condition]

- When transmitting the last bit of a transmit character, and there is no transmit data in the FTDR register.

[Clearing condition]

- When the SCR.TE bit is 1 and transmit data is written to the FTDR register.
- When the SCR.TE bit is 1, and 0 is written to the TEND flag after it is confirmed to be 1.
- When the FCR.FM bit is changed from 0 to 1.

PER Flag (Parity Error Flag)

In asynchronous mode when data match detection is disabled, this flag indicates that a parity error was detected for data in the receive FIFO.

[Setting condition]

- When data match detection is disabled, and a parity error is detected in the received data.

[Clearing condition]

- When 0 is written to the PER flag after it is confirmed to be 1.

When setting the PER flag to 0 to exit from the interrupt handling routine, refer to **section 15.5.2, Level Detection**.

Unlike in non-FIFO mode, the receive operation continues even after a parity error is detected in the receive data.

Even when the SCR.RE bit is set to 0 (serial reception is disabled), the PER flag is not affected and retains its previous value.

FER Flag (Framing Error Flag)

In asynchronous mode when data match detection is disabled, this flag indicates that a framing error was detected for data in the receive FIFO.

[Setting condition]

- When the stop bit of the received data is 0 while data match detection is disabled.

[Clearing condition]

- When 0 is written to the FER flag after it is confirmed to be 1.

When setting the FER flag to 0 to exit from the interrupt handling routine, refer to **section 15.5.2, Level Detection**.

Unlike in non-FIFO mode, the receive operation continues even after a framing error is detected in the receive data.

Even when the SCR.RE bit is set to 0, the FER flag is not affected and retains its previous value.

ORER Flag (Overrun Error Flag)

This flag indicates that an overrun error has occurred during reception and the reception ends abnormally

[Setting condition]

- When the receive FIFO has 16 frames of data, and the next data reception has completed.

[Clearing condition]

- When 0 is written to the ORER flag after it is confirmed to be 1.

When setting the ORER flag to 0 to exit from the interrupt handling routine, refer to **section 15.5.2, Level Detection**.

Even when the SCR.RE bit is set to 0, the ORER flag is not affected and retains its previous value.

RDF Flag (Receive FIFO Full Flag)

This flag indicates when the number of data stored in the receive FIFO is equal to or greater than the threshold value (FCR.RTRG[3:0]).

[Setting condition]

- When the data stored in the receive FIFO becomes equal to or greater than the threshold value.

[Clearing condition]

- When 0 is written to the RDF flag after it is confirmed to be 1.
- When the receive FIFO contains fewer receive data than the receive FIFO threshold value on completion of DTC or DMA transfer to read receive data from the FRDR register.

When the conditions for this flag to become 0 and 1 occur simultaneously, the RDF flag initially becomes 0. At this time, if the data stored in the receive FIFO is equal to or greater than the threshold value, the RDF flag becomes 1 after 1 PCLK.

TDFE Flag (Transmit FIFO Empty Flag)

This flag indicates that data has been transferred from the transmit FIFO to the TSR register, and the number of data remaining in the transmit FIFO is equal to or less than the threshold value (FCR.TTRG[3:0]).

[Setting condition]

- When the SCR.TE bit is 0.
- When the data stored in the transmit FIFO becomes equal to or less than the threshold value.

[Clearing condition]

- When the transmit FIFO contains more transmit data than the transmit FIFO threshold value on completion of DTC or DMA transfer to write transmit data to the FTDR register.
- When 0 is written to the TDFE flag after it is confirmed to be 1.

When the SCR.TE bit is set to 0, the TDFE flag becomes 1 regardless of any other condition. In all other cases, when the conditions for this flag to become 1 and 0 occur simultaneously, the TDFE flag initially becomes 0. At this time, if the data stored in the transmit FIFO is equal to or less than the threshold value, the TDFE flag becomes 1 after 1 PCLK.

To enable use of DMA or DTC transfer, do not write 0 to the TDFE flag.

41.2.12 Smart Card Mode Register (SCMR)

Address(es): SCIO.SCMR 0008 A006h, SCI1.SCMR 0008 A026h, SCI2.SCMR 0008 A046h, SCI3.SCMR 0008 A066h, SCI4.SCMR 0008 A086h, SCI5.SCMR 0008 A0A6h, SCI6.SCMR 0008 A0C6h, SCI7.SCMR 000D 00E6h, SCI8.SCMR 000D 0006h, SCI9.SCMR 000D 0026h, SCI10.SCMR 000D 0046h, SCI11.SCMR 000D 0066h, SCI12.SCMR 0008 B306h, SMCIO.SCMR 0008 A006h, SMC11.SCMR 0008 A026h, SMC12.SCMR 0008 A046h, SMC13.SCMR 0008 A066h, SMC14.SCMR 0008 A086h, SMC15.SCMR 0008 A0A6h, SMC16.SCMR 0008 A0C6h, SMC17.SCMR 000D 00E6h, SMC18.SCMR 000D 0006h, SMC19.SCMR 000D 0026h, SMC110.SCMR 000D 0046h, SMC111.SCMR 000D 0066h, SMC112.SCMR 0008 B306h

b7	b6	b5	b4	b3	b2	b1	b0
BCP2	—	—	CHR1	SDIR	SINV	—	SMIF

Value after reset: 1 1 1 1 0 0 1 0

Bit	Symbol	Bit Name	Description	R/W
b0	SMIF	Smart Card Interface Mode Select	0: Non-smart card interface mode (Asynchronous mode, clock synchronous mode, simple SPI mode, or simple I ² C mode) 1: Smart card interface mode	R/W*1
b1	—	Reserved	This bit is read as 1. The write value should be 1.	R/W
b2	SINV	Transmitted/Received Data Invert*2, *3	0: Data bits in the TDR register are transferred to the TSR register as they are. Data bits in the RSR register are transferred to the RDR register as they are. 1: Data bits in the TDR register are transferred to the TSR register with inverting. Data bits in the RSR register are transferred to the RDR register with inverting.	R/W*1
b3	SDIR	Transmitted/Received Data Transfer Direction*2, *4	0: Transfer with LSB first 1: Transfer with MSB first	R/W*1
b4	CHR1	Character Length 1*5	Selects in combination with the SMR.CHR bit. CHR1 CHR 0 0: Transmit/receive in 9-bit data length 0 1: Transmit/receive in 9-bit data length 1 0: Transmit/receive in 8-bit data length (initial value) 1 1: Transmit/receive in 7-bit data length*6	R/W*1
b6, b5	—	Reserved	These bits are read as 1. The write value should be 1.	R/W
b7	BCP2	Base Clock Pulse 2	Selects the number of base clock cycles in combination with the SMR.BCP[1:0] bits. Table 41.10 lists the combinations of the SCMR.BCP2 bit and SMR.BCP[1:0] bits.	R/W*1

Note 1. Writable only when the SCR.TE bit is 0 and the SCR.RE bit is 0 (both serial transmission and reception are disabled).

Note 2. This bit can be used in the smart card interface mode, asynchronous mode (multi-processor mode), clock synchronous mode, and simple SPI mode.

Note 3. Set this bit to 0 if operation is to be in simple I²C mode.

Note 4. Set this bit to 1 if operation is to be in simple I²C mode.

Note 5. This bit is only valid in asynchronous mode. The setting is invalid and a fixed data length of 8 bits is used in modes other than asynchronous mode.

Note 6. LSB first should be selected and the value of MSB (b7) in the TDR register cannot be transmitted.

SMIF Bit (Smart Card Interface Mode Select)

When this bit is set to 1, smart card interface mode is selected.

When this bit is set to 0, non-smart card interface mode, i.e., asynchronous mode (including multi-processor mode), clock synchronous mode, simple SPI mode, or simple I²C mode is selected.

SINV Bit (Transmitted/Received Data Invert)

This bit is used to invert the logic level of the data bits when the data is transferred between data register and shift register. This bit does not affect the logic level of the parity bit. To invert the parity bit, invert the SMR.PM bit.

CHR1 Bit (Character Length 1)

Selects the data length of transmit/receive data.

Selects in combination with the SMR.CHR bit.

A fixed data length of 8 bits is used in modes other than asynchronous mode.

BCP2 Bit (Base Clock Pulse 2)

Selects the number of base clock cycles in a 1-bit data transfer time in smart card interface mode. Set this bit in combination with the SMR.BCP[1:0] bits.

Table 41.10 Combinations of the SCMR.BCP2 Bit and SMR.BCP[1:0] Bits

SCMR.BCP2 Bit	SMR.BCP[1:0] Bits		Number of Base Clock Cycles for 1-Bit Transfer Period
0	0	0	93 clock cycles ($S = 93$)* ¹
0	0	1	128 clock cycles ($S = 128$)* ¹
0	1	0	186 clock cycles ($S = 186$)* ¹
0	1	1	512 clock cycles ($S = 512$)* ¹
1	0	0	32 clock cycles ($S = 32$)* ¹ (Initial Value)
1	0	1	64 clock cycles ($S = 64$)* ¹
1	1	0	372 clock cycles ($S = 372$)* ¹
1	1	1	256 clock cycles ($S = 256$)* ¹

Note 1. S is the value of S in the BRR register (refer to section 41.2.13, Bit Rate Register (BRR)).

41.2.13 Bit Rate Register (BRR)

Address(es): SCI0.BRR 0008 A001h, SCI1.BRR 0008 A021h, SCI2.BRR 0008 A041h, SCI3.BRR 0008 A061h, SCI4.BRR 0008 A081h, SCI5.BRR 0008 A0A1h, SCI6.BRR 0008 A0C1h, SCI7.BRR 000D 00E1h, SCI8.BRR 000D 0001h, SCI9.BRR 000D 0021h, SCI10.BRR 000D 0041h, SCI11.BRR 000D 0061h, SCI12.BRR 0008 B301h



The BRR register is an 8-bit register that adjusts the bit rate.

As each SCI channel has independent baud rate generator control, different bit rates can be set for each. Table 41.11 and Table 41.12 shows the relationship between the setting (N) in the BRR register and the bit rate (B) for normal asynchronous mode, multi-processor communication, clock synchronous mode, smart card interface mode, simple SPI mode, and simple I²C mode.

For SCI7 to SCI11, do not set the BRR register to 00h when the SMR.CM bit is 1 (clock synchronous mode or simple SPI mode), the FCR.FM bit is 1 (FIFO mode), and the SMR.CKS[1:0] bits are 00b (PCLK).

The BRR register is writable only when the TE and RE bits in the SCR register are 0.

Table 41.11 Relationship between N Setting in the BRR Register and Bit Rate B (for SCI0 to SCI11)

Mode	SEMR Settings			BRR Setting	Error (%)
	BGDM bit	ABCS bit	ABCSE bit		
Asynchronous, multi-processor communication	0	0	0	$N = \frac{PCLK \times 10^6}{64 \times 2^{2n-1} \times B} - 1$	$Error = \left\{ \frac{PCLK \times 10^6}{B \times 64 \times 2^{2n-1} \times (N+1)} - 1 \right\} \times 100$
	0	1	0	$N = \frac{PCLK \times 10^6}{32 \times 2^{2n-1} \times B} - 1$	$Error = \left\{ \frac{PCLK \times 10^6}{B \times 32 \times 2^{2n-1} \times (N+1)} - 1 \right\} \times 100$
	1	0	0		
	1	1	0	$N = \frac{PCLK \times 10^6}{16 \times 2^{2n-1} \times B} - 1$	$Error = \left\{ \frac{PCLK \times 10^6}{B \times 16 \times 2^{2n-1} \times (N+1)} - 1 \right\} \times 100$
	0 or 1	0 or 1	1	$N = \frac{PCLK \times 10^6}{12 \times 2^{2n-1} \times B} - 1$	$Error = \left\{ \frac{PCLK \times 10^6}{B \times 12 \times 2^{2n-1} \times (N+1)} - 1 \right\} \times 100$
Clock synchronous, simple SPI				$N = \frac{PCLK \times 10^6}{8 \times 2^{2n-1} \times B} - 1$	
Smart card interface				$N = \frac{PCLK \times 10^6}{S \times 2^{2n+1} \times B} - 1$	$Error = \left\{ \frac{PCLK \times 10^6}{B \times S \times 2^{2n+1} \times (N+1)} - 1 \right\} \times 100$
Simple I ² C*1				$N = \frac{PCLK \times 10^6}{64 \times 2^{2n-1} \times B} - 1$	

B: Bit rate (bps)

N: BRR setting for on-chip baud rate generator ($0 \leq N \leq 255$)

PCLK: Operating frequency (MHz)

n and S: Determined by the settings of the SMR and SCMR registers as listed in Table 41.14 and Table 41.15.

Note 1. Adjust the bit rate so that the widths at high and low level of the SCL output in simple I²C mode satisfy the I²C-bus standard.

Table 41.12 Relationship between N Setting in the BRR Register and Bit Rate B (for SCI12)

Mode	SEMR Settings		BRR Setting	Error (%)
	BGDM bit	ABCS bit		
Asynchronous, multi-processor communication	0	0	$N = \frac{PCLK \times 10^6}{64 \times 2^{2n-1} \times B} - 1$	$Error = \left\{ \frac{PCLK \times 10^6}{B \times 64 \times 2^{2n-1} \times (N+1)} - 1 \right\} \times 100$
	0	1	$N = \frac{PCLK \times 10^6}{32 \times 2^{2n-1} \times B} - 1$	$Error = \left\{ \frac{PCLK \times 10^6}{B \times 32 \times 2^{2n-1} \times (N+1)} - 1 \right\} \times 100$
	1	0		
	1	1	$N = \frac{PCLK \times 10^6}{16 \times 2^{2n-1} \times B} - 1$	$Error = \left\{ \frac{PCLK \times 10^6}{B \times 16 \times 2^{2n-1} \times (N+1)} - 1 \right\} \times 100$
Clock synchronous, simple SPI			$N = \frac{PCLK \times 10^6}{8 \times 2^{2n-1} \times B} - 1$	
Smart card interface			$N = \frac{PCLK \times 10^6}{S \times 2^{2n+1} \times B} - 1$	$Error = \left\{ \frac{PCLK \times 10^6}{B \times S \times 2^{2n+1} \times (N+1)} - 1 \right\} \times 100$
Simple I ² C*1			$N = \frac{PCLK \times 10^6}{64 \times 2^{2n-1} \times B} - 1$	

B: Bit rate (bps)

N: BRR setting for on-chip baud rate generator ($0 \leq N \leq 255$)

PCLK: Operating frequency (MHz)

n and S: Determined by the settings of the SMR and SCMR registers as listed in Table 41.14 and Table 41.15.

Note 1. Adjust the bit rate so that the widths at high and low level of the SCL output in simple I²C mode satisfy the I²C-bus standard.**Table 41.13 Calculating Widths at High and Low Level for SCL**

Mode	SCL	Formula (Result in Seconds)
I ² C	High period (minimum value)	$(N+1) \times 4 \times 2^{2n-1} \times 7 \times \frac{1}{PCLK \times 10^6}$
	Low period (minimum value)	$(N+1) \times 4 \times 2^{2n-1} \times 8 \times \frac{1}{PCLK \times 10^6}$

Table 41.14 Clock Source Settings

SMR.CKS[1:0] Bit Setting	Clock Source	n
0 0	PCLK	0
0 1	PCLK/4	1
1 0	PCLK/16	2
1 1	PCLK/64	3

Table 41.15 Base Clock Settings in Smart Card Interface Mode

SCMR.BCP2 Bit Setting	SMR.BCP[1:0] Bit Setting	Base Clock Cycles for 1-bit Period	S
0	0 0	93 clock cycles	93
0	0 1	128 clock cycles	128
0	1 0	186 clock cycles	186
0	1 1	512 clock cycles	512
1	0 0	32 clock cycles	32
1	0 1	64 clock cycles	64
1	1 0	372 clock cycles	372
1	1 1	256 clock cycles	256

Table 41.16 lists examples of N settings in the BRR register in normal asynchronous mode. Table 41.18 lists the maximum bit rate settable for each operating frequency. Examples of BRR (N) settings in clock synchronous mode and simple SPI mode are listed in Table 41.21. Examples of BRR (N) settings in smart card interface mode are listed in Table 41.23. Examples of BRR (N) settings in simple I²C mode are listed in Table 41.25. In smart card interface mode, the number of base clock cycles S in a 1-bit data transfer time can be selected. For details, refer to section 41.6.4, Receive Data Sampling Timing and Reception Margin. Table 41.19 and Table 41.22 list the maximum bit rates with external clock input.

When either the SEMR.ABCS or BGDM bit is set to 1 in asynchronous mode, the bit rate becomes twice that listed in Table 41.16. When both of those bits are set to 1, the bit rate becomes four times the listed value.

Table 41.16 Examples of BRR Settings for Various Bit Rates (Asynchronous Mode)

Bit Rate (bps)	Operating Frequency PCLK (MHz)														
	8			9.8304			10			12			12.288		
	n	N	Error (%)	n	N	Error (%)	n	N	Error (%)	n	N	Error (%)	n	N	Error (%)
110	2	141	0.03	2	174	-0.26	2	177	-0.25	2	212	0.03	2	217	0.08
150	2	103	0.16	2	127	0.00	2	129	0.16	2	155	0.16	2	159	0.00
300	1	207	0.16	1	255	0.00	2	64	0.16	2	77	0.16	2	79	0.00
600	1	103	0.16	1	127	0.00	1	129	0.16	1	155	0.16	1	159	0.00
1200	0	207	0.16	0	255	0.00	1	64	0.16	1	77	0.16	1	79	0.00
2400	0	103	0.16	0	127	0.00	0	129	0.16	0	155	0.16	0	159	0.00
4800	0	51	0.16	0	63	0.00	0	64	0.16	0	77	0.16	0	79	0.00
9600	0	25	0.16	0	31	0.00	0	32	-1.36	0	38	0.16	0	39	0.00
19200	0	12	0.16	0	15	0.00	0	15	1.73	0	19	-2.34	0	19	0.00
31250	0	7	0.00	0	9	-1.70	0	9	0.00	0	11	0.00	0	11	2.40
38400	—	—	—	0	7	0.00	0	7	1.73	0	9	-2.34	0	9	0.00

Bit Rate (bps)	Operating Frequency PCLK (MHz)														
	14			16			17.2032			18			19.6608		
	n	N	Error (%)	n	N	Error (%)	n	N	Error (%)	n	N	Error (%)	n	N	Error (%)
110	2	248	-0.17	3	70	0.03	3	75	0.48	3	79	-0.12	3	86	0.31
150	2	181	0.16	2	207	0.16	2	223	0.00	2	233	0.16	2	255	0.00
300	2	90	0.16	2	103	0.16	2	111	0.00	2	116	0.16	2	127	0.00
600	1	181	0.16	1	207	0.16	1	223	0.00	1	233	0.16	1	255	0.00
1200	1	90	0.16	1	103	0.16	1	111	0.00	1	116	0.16	1	127	0.00
2400	0	181	0.16	0	207	0.16	0	223	0.00	0	233	0.16	0	255	0.00
4800	0	90	0.16	0	103	0.16	0	111	0.00	0	116	0.16	0	127	0.00
9600	0	45	-0.93	0	51	0.16	0	55	0.00	0	58	-0.69	0	63	0.00
19200	0	22	-0.93	0	25	0.16	0	27	0.00	0	28	1.02	0	31	0.00
31250	0	13	0.00	0	15	0.00	0	16	1.20	0	17	0.00	0	19	-1.70
38400	—	—	—	0	12	0.16	0	13	0.00	0	14	-2.34	0	15	0.00

Bit Rate (bps)	Operating Frequency PCLK (MHz)														
	20			25			30			33			40		
	n	N	Error (%)	n	N	Error (%)	n	N	Error (%)	n	N	Error (%)	n	N	Error (%)
110	3	88	-0.25	3	110	-0.02	3	132	0.13	3	145	0.33	3	177	-0.25
150	3	64	0.16	3	80	0.47	3	97	-0.35	3	106	0.39	3	129	0.16
300	2	129	0.16	2	162	-0.15	2	194	0.16	2	214	-0.07	3	64	0.16
600	2	64	0.16	2	80	0.47	2	97	-0.35	2	106	0.39	2	129	0.16
1200	1	129	0.16	1	162	-0.15	1	194	0.16	1	214	-0.07	2	64	0.16
2400	1	64	0.16	1	80	0.47	1	97	-0.35	1	106	0.39	1	129	0.16
4800	0	129	0.16	0	162	-0.15	0	194	0.16	0	214	-0.07	1	64	0.16
9600	0	64	0.16	0	80	0.47	0	97	-0.35	0	106	0.39	0	129	0.16
19200	0	32	-1.36	0	40	-0.76	0	48	-0.35	0	53	-0.54	0	64	0.16
31250	0	19	0.00	0	24	0.00	0	29	0.00	0	32	0.00	0	39	0.00
38400	0	15	1.73	0	19	1.73	0	23	1.73	0	26	-0.54	0	32	-1.36

Bit Rate (bps)	Operating Frequency PCLK (MHz)								
	50			60			120*1		
	n	N	Error (%)	n	N	Error (%)	n	N	Error (%)
110	3	221	-0.02						
150	3	162	-0.15	3	194	0.16			
300	3	80	0.47	3	97	-0.35	3	194	0.16
600	2	162	-0.15	3	48	-0.35	3	97	-0.35
1200	2	80	0.47	2	97	-0.35	3	48	-0.35
2400	1	162	-0.15	2	48	-0.35	2	97	-0.35
4800	1	80	0.47	1	97	-0.35	2	48	-0.35
9600	0	162	-0.15	1	48	-0.35	1	97	-0.35
19200	0	80	0.47	0	97	-0.35	1	48	-0.35
31250	0	49	0.00	0	59	0.00	0	119	0.00
38400	0	40	-0.76	0	48	-0.35	0	97	-0.35

Note: This is an example when the ABCS, ABCSE, and BGDM bits in the SEMR register are 0.
 When either the ABCS bit or BGDM bit is set to 1, the bit rate doubles.
 When both ABCS and BGDM bits in the SEMR register are set to 1, the bit rate increases four times.
 When the ABCSE bit is set to 1, the bit rate increases 16/3 times.

Note 1. Supported by SCI7 to SCI11 only.

Table 41.17 Maximum Bit Rate for Each Operating Frequency (Asynchronous Mode) (for SCI0 to SCI11)

PCLK (MHz)	SEMR Settings					Maximum Bit Rate (bps)	PCLK (MHz)	SEMR Settings					Maximum Bit Rate (bps)
	BGDM Bit	ABCS Bit	ABCSE Bit	n	N			BGDM Bit	ABCS Bit	ABCSE Bit	n	N	
8	0	0	0	0	0	250000	19.6608	0	0	0	0	0	614400
		1	0	0	0	500000			1	0	0	0	1228800
	1	0	0	0	0	1000000		1	0	0	0	0	2457600
		1	0	0	0				0	0	0	0	
	0 or 1	0 or 1	1	0	0	1333333		0 or 1	0 or 1	1	0	0	3276800
9.8304	0	0	0	0	0	307200	20	0	0	0	0	0	625000
		1	0	0	0	614400			1	0	0	0	1250000
	1	0	0	0	0	1228800		1	0	0	0	0	2500000
		1	0	0	0				0	0	0	0	
	0 or 1	0 or 1	1	0	0	1638400		0 or 1	0 or 1	1	0	0	3333333
10	0	0	0	0	0	312500	25	0	0	0	0	0	781250
		1	0	0	0	625000			1	0	0	0	1562500
	1	0	0	0	0	1250000		1	0	0	0	0	3125000
		1	0	0	0				0	0	0	0	
	0 or 1	0 or 1	1	0	0	1666667		0 or 1	0 or 1	1	0	0	4166667
12	0	0	0	0	0	375000	30	0	0	0	0	0	937500
		1	0	0	0	750000			1	0	0	0	1875000
	1	0	0	0	0	1500000		1	0	0	0	0	3750000
		1	0	0	0				0	0	0	0	
	0 or 1	0 or 1	1	0	0	2000000		0 or 1	0 or 1	1	0	0	5000000
12.288	0	0	0	0	0	384000	33	0	0	0	0	0	1031250
		1	0	0	0	768000			1	0	0	0	2062500
	1	0	0	0	0	1536000		1	0	0	0	0	4125000
		1	0	0	0				0	0	0	0	
	0 or 1	0 or 1	1	0	0	2048000		0 or 1	0 or 1	1	0	0	5500000
14	0	0	0	0	0	437500	40	0	0	0	0	0	1250000
		1	0	0	0	875000			1	0	0	0	2500000
	1	0	0	0	0	1750000		1	0	0	0	0	5000000
		1	0	0	0				0	0	0	0	
	0 or 1	0 or 1	1	0	0	2333333		0 or 1	0 or 1	1	0	0	6666667
16	0	0	0	0	0	500000	50	0	0	0	0	0	1562500
		1	0	0	0	1000000			1	0	0	0	3125000
	1	0	0	0	0	2000000		1	0	0	0	0	6250000
		1	0	0	0				0	0	0	0	
	0 or 1	0 or 1	1	0	0	2666667		0 or 1	0 or 1	1	0	0	8333333
17.2032	0	0	0	0	0	537600	60	0	0	0	0	0	1875000
		1	0	0	0	1075200			1	0	0	0	3750000
	1	0	0	0	0	2150400		1	0	0	0	0	7500000
		1	0	0	0				0	0	0	0	
	0 or 1	0 or 1	1	0	0	2867200		0 or 1	0 or 1	1	0	0	10000000
18	0	0	0	0	0	562500	120*1	0	0	0	0	0	3750000
		1	0	0	0	1125000			1	0	0	0	7500000
	1	0	0	0	0	2250000		1	0	0	0	0	15000000
		1	0	0	0				0	0	0	0	
	0 or 1	0 or 1	1	0	0	3000000		0 or 1	0 or 1	1	0	0	20000000

Note 1. Supported by SCI7 to SCI11 only.

Table 41.18 Maximum Bit Rate for Each Operating Frequency (Asynchronous Mode) (for SCI12)

PCLK (MHz)	SEMR Settings				Maximum Bit Rate (bps)	PCLK (MHz)	SEMR Settings				Maximum Bit Rate (bps)
	BGDM Bit	ABCS Bit	n	N			BGDM Bit	ABCS Bit	n	N	
8	0	0	0	0	250000	19.6608	0	0	0	0	614400
		1	0	0	500000			1	0	0	1228800
	1	0	0	0			1	0	0	0	
		1	0	0	1000000			1	0	0	2457600
9.8304	0	0	0	0	307200	20	0	0	0	0	625000
		1	0	0	614400			1	0	0	1250000
	1	0	0	0			1	0	0	0	
		1	0	0	1228800			1	0	0	2500000
10	0	0	0	0	312500	25	0	0	0	0	781250
		1	0	0	625000			1	0	0	1562500
	1	0	0	0			1	0	0	0	
		1	0	0	1250000			1	0	0	3125000
12	0	0	0	0	375000	30	0	0	0	0	937500
		1	0	0	750000			1	0	0	1875000
	1	0	0	0			1	0	0	0	
		1	0	0	1500000			1	0	0	3750000
12.288	0	0	0	0	384000	33	0	0	0	0	1031250
		1	0	0	768000			1	0	0	2062500
	1	0	0	0			1	0	0	0	
		1	0	0	1536000			1	0	0	4125000
14	0	0	0	0	437500	40	0	0	0	0	1250000
		1	0	0	875000			1	0	0	2500000
	1	0	0	0			1	0	0	0	
		1	0	0	1750000			1	0	0	5000000
16	0	0	0	0	500000	50	0	0	0	0	1562500
		1	0	0	1000000			1	0	0	3125000
	1	0	0	0			1	0	0	0	
		1	0	0	2000000			1	0	0	6250000
17.2032	0	0	0	0	537600	60	0	0	0	0	1875000
		1	0	0	1075200			1	0	0	3750000
	1	0	0	0			1	0	0	0	
		1	0	0	2150400			1	0	0	7500000
18	0	0	0	0	562500						
		1	0	0	1125000						
	1	0	0	0							
		1	0	0	2250000						

Table 41.19 Maximum Bit Rate with External Clock Input (Asynchronous Mode)

PCLK (MHz)	External Input Clock (MHz)	Maximum Bit Rate (bps)	
		SEMR.ABCS Bit = 0	SEMR.ABCS Bit = 1
8	2.0000	125000	250000
9.8304	2.4576	153600	307200
10	2.5000	156250	312500
12	3.0000	187500	375000
12.288	3.0720	192000	384000
14	3.5000	218750	437500
16	4.0000	250000	500000
17.2032	4.3008	268800	537600
18	4.5000	281250	562500
19.6608	4.9152	307200	614400
20	5.0000	312500	625000
25	6.2500	390625	781250
30	7.5000	468750	937500
33	8.2500	515625	1031250
40	10.0000	625000	1250000
50	12.5000	781250	1562500
60	15.0000	937500	1875000
120*1	30.0000	1875000	3750000

Note 1. Supported by SCI7 to SCI11 only.

Table 41.20 Maximum Bit Rate with TMR Clock Input (Asynchronous Mode)

PCLK (MHz)	TMR Clock (MHz)	Maximum Bit Rate (bps)	
		SEMR.ABCS Bit = 0	SEMR.ABCS Bit = 1
8	4	250000	500000
9.8304	4.9152	307200	614400
10	5	312500	625000
12	6	375000	750000
12.288	6.144	384000	768000
14	7	437500	875000
16	8	500000	1000000
17.2032	8.6016	537600	1075200
18	9	562500	1125000
19.6608	9.8304	614400	1228800
20	10	625000	1250000
25	12.5	781250	1562500
30	15	937500	1875000
33	16.5	1031250	2062500
40	20	1250000	2500000
50	25	1562500	3125000
60	30	1875000	3750000

Table 41.21 BRR Settings for Various Bit Rates (Clock Synchronous Mode, Simple SPI Mode)

Bit Rate (bps)	Operating Frequency PCLK (MHz)																					
	8		10		16		20		25		30		33		40		50		60		120*1	
	n	N	n	N	n	N	n	N	n	N	n	N	n	N	n	N	n	N	n	N	n	N
110																						
250	3	124	3	155	3	249																
500	2	249	3	77	3	124	3	155	3	194	3	233	3	255								
1 k	2	124	2	155	2	249	3	77	3	97	3	116	3	128	3	155	3	194	3	233		
2.5 k	1	199	1	249	2	99	2	124	2	155	2	187	2	205	2	249	3	77	3	93	3	187
5 k	1	99	1	124	1	199	1	249	2	77	2	93	2	102	2	124	2	155	3	46	3	93
10 k	0	199	0	249	1	99	1	124	1	155	1	187	1	205	1	249	2	77	2	93	3	46
25 k	0	79	0	99	0	159	0	199	0	249	1	74	1	82	1	99	1	124	1	149	2	74
50 k	0	39	0	49	0	79	0	99	0	124	0	149	0	164	1	49	1	61	1	74	1	149
100 k	0	19	0	24	0	39	0	49	0	62	0	74	0	82	0	99	0	124	0	149	1	74
250 k	0	7	0	9	0	15	0	19	0	24	0	29	0	32	0	39	0	49	0	59	1	29
500 k	0	3	0	4	0	7	0	9	—	—	0	14	—	—	0	19	0	24	0	29	1	14
1 M	0	1			0	3	0	4	—	—			—	—	0	9	—	—	0	14	0	29
2.5 M			0	0*2			0	1			0	2			0	3	0	4	0	5	0	11
5 M							0	0*2							0	1			0	2	0	5
7.5 M											0	0*2							0	1	0	3

Blank cell: Cannot be set since the bit rate error exceeds 5%.

—: Can be set, but a bit rate error of 1 to 5% will occur.

Note 1. Supported by SCI7 to SCI11 only.

Note 2. For SCI7 to SCI11, this setting cannot be used when the FCR.FM bit is 1 (FIFO mode).

When this setting is used while the FCR.FM bit is 0 (non-FIFO mode) or with other channel, continuous transmission or reception is not possible. After transmitting/receiving one frame of data, there is an interval of a 1-bit period before starting transmitting/receiving the next frame of data. The output of the synchronization clock is stopped for a 1-bit period. For this reason, it takes 9 bits worth of time to transfer one frame (8 bits) of data, and the average transfer rate is 8/9 times the bit rate.

Table 41.22 Maximum Bit Rate with External Clock Input (Clock Synchronous Mode, Simple SPI Mode)

PCLK (MHz)	External Input Clock (MHz)	Maximum Bit Rate (Mbps)
8	1.3333	1.3333
10	1.6667	1.6667
12	2.0000	2.0000
14	2.3333	2.3333
16	2.6667	2.6667
18	3.0000	3.0000
20	3.3333	3.3333
25	4.1667	4.1667
30	5.0000	5.0000
33	5.5000	5.5000
40	6.6667	6.6667
50	8.3333	8.3333
60	10.0000	10.0000
120*1	20.0000	20.0000

Note 1. Supported by SCI7 to SCI11 only.

Table 41.23 BRR Settings for Various Bit Rates (Smart Card Interface Mode, n = 0, S = 372)

Bit Rate (bps)	PCLK (MHz)	n	N	Error (%)
9600	7.1424	0	0	0.00
	10.00	0	1	-30.00
	10.7136	0	1	-25.00
	13.00	0	1	-8.99
	14.2848	0	1	0.00
	16.00	0	1	12.01
	18.00	0	2	-15.99
	20.00	0	2	-6.66
	25.00	0	3	-12.49
	30.00	0	3	5.01
	33.00	0	4	-7.59
	40.00	0	5	-6.66
	50.00	0	6	0.01
	60.00	0	7	5.01
	120.00*1	0	16	-1.17

Note 1. Supported by SCI7 to SCI11 only.

Table 41.24 Maximum Bit Rate for Each Operating Frequency (Smart Card Interface Mode, S = 32)

PCLK (MHz)	Maximum Bit Rate (bps)	n	N
10.00	156250	0	0
10.7136	167400	0	0
13.00	203125	0	0
16.00	250000	0	0
18.00	281250	0	0
20.00	312500	0	0
25.00	390625	0	0
30.00	468750	0	0
33.00	515625	0	0
40.00	625000	0	0
50.00	781250	0	0
60.00	937500	0	0
120.00*1	1875000	0	0

Note 1. Supported by SCI7 to SCI11 only.

Table 41.25 BRR Settings for Various Bit Rates (Simple I²C Mode)

Bit Rate (bps)	Operating Frequency PCLK (MHz)														
	8			10			16			20			25		
	n	N	Error (%)	n	N	Error (%)	n	N	Error (%)	n	N	Error (%)	n	N	Error (%)
10 k	0	24	0.0	0	31	-2.3	1	12	-3.8	1	15	-2.3	1	19	-2.3
25 k	0	9	0.0	0	12	-3.8	1	4	0.0	1	6	-10.7	1	7	-2.3
50 k	0	4	0.0	0	6	-10.7	1	2	-16.7	1	3	-21.9	1	3	-2.3
100 k	0	2	-16.7	0	3	-21.9	0	4	0.0	0	6	-10.7	1	1	-2.3
250 k	0	0	0.0	0	1	-37.5	0	1	0.0	0	2	-16.7	0	3	-21.9
350 k										0	1	-10.7	0	2	-25.6

Bit Rate (bps)	Operating Frequency PCLK (MHz)														
	30			33			40			50			60		
	n	N	Error (%)	n	N	Error (%)	n	N	Error (%)	n	N	Error (%)	n	N	Error (%)
10 k	1	23	-2.3	1	25	-0.8	0	124	0.00	2	9	-2.3	1	46	-0.27
25 k	1	9	-6.3	1	10	-6.3	0	40	0.00	2	3	-2.3	0	74	0.00
50 k	1	4	-6.3	1	5	-14.1	0	24	0.00	2	1	-2.3	0	37	-1.32
100 k	1	2	-21.9	1	2	-14.1	0	12	-3.85	1	3	-2.3	0	18	-1.32
250 k	0	3	-6.3	0	4	-17.5	0	4	0.00	0	6	-10.7	0	7	-6.25
350 k	0	2	-10.7	0	2	-1.8	0	3	-10.71	0	4	-10.7	0	4	7.14

Bit Rate (bps)	Operating Frequency PCLK (MHz)		
	120*1		
	n	N	Error (%)
10 k	1	93	-0.27
25 k	0	149	0.00
50 k	0	74	0.00
100 k	0	37	-1.31
250 k	0	14	0.00
350 k	0	10	-2.60

Note 1. Supported by SCI7 to SCI11 only.

Table 41.26 Minimum Widths at High and Low Level for SCL at Various Bit Rates (Simple I²C Mode)

Bit Rate (bps)	Operating Frequency PCLK (MHz)											
	8			10			16			20		
	n	N	Min. Widths at High/Low Level for SCL (μs)	n	N	Min. Widths at High/Low Level for SCL (μs)	n	N	Min. Widths at High/Low Level for SCL (μs)	n	N	Min. Widths at High/Low Level for SCL (μs)
10 k	0	24	43.75/50.00	0	31	44.80/51.20	1	12	45.50/52.00	1	15	44.80/51.20
25 k	0	9	17.50/20.00	0	12	18.20/20.80	1	4	17.50/20.00	1	6	19.60/22.40
50 k	0	4	8.75/10.00	0	6	9.80/11.20	1	2	10.50/12.00	1	3	11.20/12.80
100 k	0	2	5.25/6.00	0	3	5.60/6.40	0	4	4.37/5.00	0	6	4.90/5.60
250 k	0	0	1.75/2.00	0	1	2.80/3.20	0	1	1.75/2.00	0	2	2.10/2.40
350 k										0	1	1.40/1.60

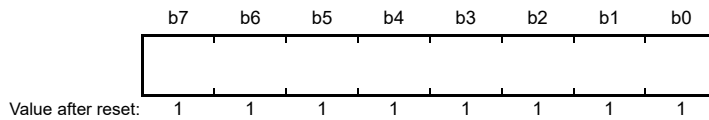
Bit Rate (bps)	Operating Frequency PCLK (MHz)											
	25			30			33			40		
	n	N	Min. Widths at High/Low Level for SCL (μs)	n	N	Min. Widths at High/Low Level for SCL (μs)	n	N	Min. Widths at High/Low Level for SCL (μs)	n	N	Min. Widths at High/Low Level for SCL (μs)
10 k	1	19	44.80/51.20	1	23	44.80/51.20	1	25	44.12/50.42	1	32	46.20/52.80
25 k	1	7	17.92/20.48	1	9	18.66/21.33	1	10	18.66/21.33	1	12	18.20/20.80
50 k	1	3	8.96/10.24	1	4	9.33/10.66	1	5	10.18/11.63	1	6	9.80/11.20
100 k	1	1	4.48/5.12	1	2	5.60/6.40	1	2	5.09/5.81	0	13	4.90/5.60
250 k	0	3	2.24/2.56	0	3	1.86/2.13	0	4	2.12/2.42	0	4	1.75/2.00
350 k	0	2	1.68/1.92	0	2	1.40/1.60	0	2	1.27/1.45	0	3	1.40/1.60

Bit Rate (bps)	Operating Frequency PCLK (MHz)								
	50			60			120*1		
	n	N	Min. Widths at High/Low Level for SCL (μs)	n	N	Min. Widths at High/Low Level for SCL (μs)	n	N	Min. Widths at High/Low Level for SCL (μs)
10 k	2	9	44.80/51.20	1	47	44.80/51.20	1	93	43.87/50.13
25 k	2	3	17.92/20.48	0	74	17.50/20.00	0	149	17.50/20.00
50 k	2	1	8.96/10.24	0	37	8.87/10.13	0	74	8.75/10.00
100 k	1	3	4.48/5.12	0	18	4.43/5.07	0	37	4.43/5.07
250 k	0	6	1.96/2.24	0	7	1.87/2.13	0	15	1.87/2.13
350 k	0	4	1.40/1.60	0	5	1.40/1.60	0	10	1.28/1.47

Note 1. Supported by SCI7 to SCI11 only.

41.2.14 Modulation Duty Register (MDDR)

Address(es): SCI0.MDDR 0008 A012h, SCI1.MDDR 0008 A032h, SCI2.MDDR 0008 A052h, SCI3.MDDR 0008 A072h, SCI4.MDDR 0008 A092h, SCI5.MDDR 0008 A0B2h, SCI6.MDDR 0008 A0D2h, SCI7.MDDR 000D 00F2h, SCI8.MDDR 000D 0012h, SCI9.MDDR 000D 0032h, SCI10.MDDR 000D 0052h, SCI11.MDDR 000D 0072h, SCI12.MDDR 0008 B312h



The MDDR register corrects the bit rate adjusted by the BRR register.

When the SEMR.BRME bit is set to 1, the bit rate generated by the on-chip baud rate generator is evenly corrected according to the settings of the MDDR register ($M/256$). The relationship between the MDDR register setting (M) and the bit rate (B) is given in Table 41.27 and Table 41.28.

The range of the value that can be set in the MDDR register is from 80h to FFh. A value other than these cannot be set. The MDDR register is writable only when the TE and RE bits in the SCR register are 0.

Table 41.27 Relationship between MDDR Setting (M) and Bit Rate (B) When Bit Rate Modulation Function is Used (for SCI0 to SCI11)

Mode	SEMR Settings			BRR Setting	Error (%)
	BGDM Bit	ABCS Bit	ABCSE Bit		
Asynchronous, multi-processor communication	0	0	0	$N = \frac{PCLK \times 10^6}{64 \times 2^{2n-1} \times \frac{256}{M} \times B} - 1$	$\text{Error} = \left\{ \frac{PCLK \times 10^6}{B \times 64 \times 2^{2n-1} \times \frac{256}{M} \times (N+1)} - 1 \right\} \times 100$
	1	0	0	$N = \frac{PCLK \times 10^6}{32 \times 2^{2n-1} \times \frac{256}{M} \times B} - 1$	$\text{Error} = \left\{ \frac{PCLK \times 10^6}{B \times 32 \times 2^{2n-1} \times \frac{256}{M} \times (N+1)} - 1 \right\} \times 100$
	0	1	0		
	1	1	0	$N = \frac{PCLK \times 10^6}{16 \times 2^{2n-1} \times \frac{256}{M} \times B} - 1$	$\text{Error} = \left\{ \frac{PCLK \times 10^6}{B \times 16 \times 2^{2n-1} \times \frac{256}{M} \times (N+1)} - 1 \right\} \times 100$
	0 or 1	0 or 1	1	$N = \frac{PCLK \times 10^6}{12 \times 2^{2n-1} \times \frac{256}{M} \times B} - 1$	$\text{Error} = \left\{ \frac{PCLK \times 10^6}{B \times 12 \times 2^{2n-1} \times \frac{256}{M} \times (N+1)} - 1 \right\} \times 100$
Clock synchronous, simple SPI*1				$N = \frac{PCLK \times 10^6}{8 \times 2^{2n-1} \times \frac{256}{M} \times B} - 1$	
Smart card interface				$N = \frac{PCLK \times 10^6}{S \times 2^{2n+1} \times \frac{256}{M} \times B} - 1$	$\text{Error} = \left\{ \frac{PCLK \times 10^6}{B \times S \times 2^{2n+1} \times \frac{256}{M} \times (N+1)} - 1 \right\} \times 100$
Simple I ² C*2				$N = \frac{PCLK \times 10^6}{64 \times 2^{2n-1} \times \frac{256}{M} \times B} - 1$	

B: Bit rate (bps)

M: MDDR setting ($128 \leq MDDR \leq 256$)

N: BRR setting for baud rate generator ($0 \leq N \leq 255$)

PCLK: Operating frequency (MHz)

n and S: Determined by the settings of the SMR and SCMR registers as listed in Table 41.14 and Table 41.15, section 41.2.13, Bit Rate Register (BRR).

Note 1. Do not use this function in clock synchronous mode and in the highest speed settings in simple SPI mode (SMR.CKS[1:0] = 00b, SCR.CKE[1] = 0, and BRR = 0).

Note 2. Adjust the bit rate so that the widths at high and low level of the SCL output in simple I²C mode satisfy the I²C-bus standard.

Table 41.28 Relationship between MDDR Setting (M) and Bit Rate (B) When Bit Rate Modulation Function is Used (for SCI12)

Mode	SEMR Settings		BRR Setting	Error (%)
	BGDM Bit	ABCS Bit		
Asynchronous, multi-processor communication	0	0	$N = \frac{PCLK \times 10^6}{64 \times 2^{2n-1} \times \frac{256}{M} \times B} - 1$	$\text{Error} = \left\{ \frac{PCLK \times 10^6}{B \times 64 \times 2^{2n-1} \times \frac{256}{M} \times (N+1)} - 1 \right\} \times 100$
	1	0	$N = \frac{PCLK \times 10^6}{32 \times 2^{2n-1} \times \frac{256}{M} \times B} - 1$	$\text{Error} = \left\{ \frac{PCLK \times 10^6}{B \times 32 \times 2^{2n-1} \times \frac{256}{M} \times (N+1)} - 1 \right\} \times 100$
	0	1		
	1	1	$N = \frac{PCLK \times 10^6}{16 \times 2^{2n-1} \times \frac{256}{M} \times B} - 1$	$\text{Error} = \left\{ \frac{PCLK \times 10^6}{B \times 16 \times 2^{2n-1} \times \frac{256}{M} \times (N+1)} - 1 \right\} \times 100$
Clock synchronous, simple SPI*1			$N = \frac{PCLK \times 10^6}{8 \times 2^{2n-1} \times \frac{256}{M} \times B} - 1$	
Smart card interface			$N = \frac{PCLK \times 10^6}{S \times 2^{2n+1} \times \frac{256}{M} \times B} - 1$	$\text{Error} = \left\{ \frac{PCLK \times 10^6}{B \times S \times 2^{2n+1} \times \frac{256}{M} \times (N+1)} - 1 \right\} \times 100$
Simple I ² C*2			$N = \frac{PCLK \times 10^6}{64 \times 2^{2n-1} \times \frac{256}{M} \times B} - 1$	

B: Bit rate (bps)

M: MDDR setting ($128 \leq MDDR \leq 256$)

N: BRR setting for baud rate generator ($0 \leq N \leq 255$)

PCLK: Operating frequency (MHz)

n and S: Determined by the settings of the SMR and SCMR registers as listed in Table 41.14 and Table 41.15, section 41.2.13, Bit Rate Register (BRR).

Note 1. Do not use this function in clock synchronous mode and in the highest speed settings in simple SPI mode (SMR.CKS[1:0] = 00b, SCR.CKE[1] = 0, and BRR = 0).

Note 2. Adjust the bit rate so that the widths at high and low level of the SCL output in simple I²C mode satisfy the I²C-bus standard.

Smaller settings of the SMR.CKS[1:0] bits and larger settings of the BRR register reduce difference in the length of the 1-bit period.

41.2.15 Serial Extended Mode Register (SEMR)

Address(es): SCI0.SEMR 0008 A007h, SCI1.SEMR 0008 A027h, SCI2.SEMR 0008 A047h, SCI3.SEMR 0008 A067h, SCI4.SEMR 0008 A087h, SCI5.SEMR 0008 A0A7h, SCI6.SEMR 0008 A0C7h, SCI7.SEMR 000D 00E7h, SCI8.SEMR 000D 0007h, SCI9.SEMR 000D 0027h, SCI10.SEMR 000D 0047h, SCI11.SEMR 000D 0067h, SCI12.SEMR 0008 B307h

b7	b6	b5	b4	b3	b2	b1	b0
RXDESEL	BGDM	NFEN	ABCS	ABCSE	BRME	—	ACS0

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Bit Name	Description	R/W
b0	ACS0	Asynchronous Mode Clock Source Select	(Valid only in asynchronous mode) 0: External clock input 1: Logical AND of two compare matches output from TMR (valid for SCI5, SCI6, and SCI12 only) Available compare match output varies per SCI channel.	R/W*1
b1	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b2	BRME	Bit Rate Modulation Enable	0: Bit rate modulation function is disabled. 1: Bit rate modulation function is enabled.	R/W*1
b3	ABCSE	Asynchronous Mode Base Clock Select Extended*2	(Valid only when using a on-chip baud rate generator in asynchronous mode) 0: The number of clock cycles for 1-bit period depends on the BGDM and ABCS bits setting. 1: Selects 6 base clock cycles for 1-bit period.	R/W*1
b4	ABCS	Asynchronous Mode Base Clock Select	(Valid only in asynchronous mode) 0: Selects 16 base clock cycles for 1-bit period. 1: Selects 8 base clock cycles for 1-bit period.	R/W*1
b5	NFEN	Digital Noise Filter Function Enable	(In asynchronous mode) 0: Noise cancellation function for the RXDn input signal is disabled. 1: Noise cancellation function for the RXDn input signal is enabled. (in simple I ² C mode) 0: Noise cancellation function for the SSCLn and SSDAn input signals is disabled. 1: Noise cancellation function for the SSCLn and SSDAn input signals is enabled. The NFEN bit should be 0 in any mode other than above.	R/W*1
b6	BGDM	Baud Rate Generator Double-Speed Mode Select	(Only valid the SCR.CKE[1] bit is 0 in asynchronous mode) 0: Baud rate generator outputs the clock with normal frequency. 1: Baud rate generator outputs the clock with doubled frequency.	R/W*1
b7	RXDESEL	Asynchronous Start Bit Edge Detection Select	(Valid only in asynchronous mode) 0: The low level on the RXDn pin is detected as the start bit. 1: A falling edge on the RXDn pin is detected as the start bit.	R/W*1

Note 1. Writable only when the SCR.TE bit is 0 and the SCR.RE bit is 0 (both serial transmission and reception are disabled).

Note 2. This bit is reserved for SCI12. It is read as 0. The write value should be 0.

The SEMR register is used to select a clock source for 1-bit period in asynchronous mode or a detection method of the start bit.

ACS0 Bit (Asynchronous Mode Clock Source Select)

Selects the clock source in the asynchronous mode.

The ACS0 bit is valid in asynchronous mode (SMR.CM bit = 0) and when an external clock input is selected (SCR.CKE[1:0] bits = 10b or 11b). This bit is used to select an external clock input or the logical AND of compare matches output from the internal TMR.

Set the ACS0 bit to 0 in other than asynchronous mode.

For SCI5, SCI6, and SCI12, the TMO_n output (n = 0 to 3) of TMR units 0 and 1 can be set as the base clock source. Refer to Table 41.29 for details.

The ACS0 bits for SCI0 to SCI4 and SCI7 to SCI11 are reserved. The write values to these bits for SCI0 to SCI4 and SCI7 to SCI11 should be 0.

Table 41.29 Correspondence between SCI Channels and Compare Match Outputs

SCI	TMR	Compare Match Output
SCI5	Unit 0	TMO0, TMO1
SCI6	Unit 1	TMO2, TMO3
SCI12	Unit 0	TMO0, TMO1

Figure 41.5 shows a setting example of when TMO0 and TMO1 in the TMR unit 0 are selected for output.

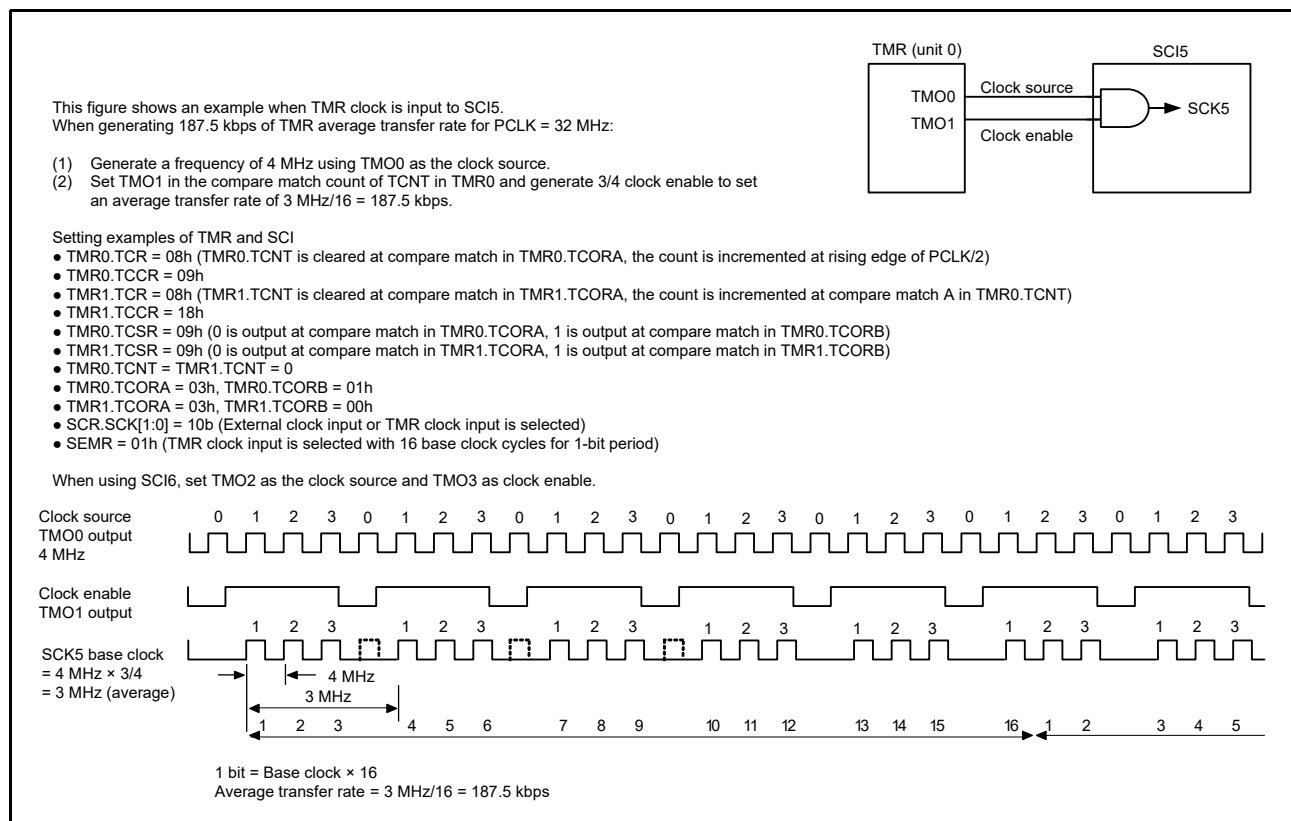


Figure 41.5 Example of Average Transfer Rate Setting When TMR Clock is Input

BRME Bit (Bit Rate Modulation Enable)

This bit enables and disables the bit rate modulation function. The bit rate generated by on-chip baud rate generator is evenly corrected when this function is enabled.

ABCSE Bit (Asynchronous Mode Base Clock Select Extended)

When setting this bit to 1, the 1-bit period becomes 6 cycles of the base clock and the clock of doubled frequency is output from the baud rate generator.

This bit is valid when the on-chip baud rate generator is selected as the clock source (`SCR.CKE[1] = 0`) in asynchronous mode (`SMR.CM = 0`).

When the bit rate is made to be 1/6 of PCLK frequency, set this bit to 1, the `SMR.CKS[1:0]` bits to 00b, and the `BRR` register to 00h.

NFEN Bit (Digital Noise Filter Function Enable)

This bit enables or disables the digital noise filter function.

When the function is enabled, noise cancellation is applied to the `RXDn` input signal in asynchronous mode, and noise cancellation is applied to the `SSDAn` and `SSCLn` input signals in simple I²C mode.

In any mode other than above, set the `NFEN` bit to 0 to disable the digital noise filter function.

When the function is disabled, input signals are transferred as is, as internal signals.

BGDM Bit (Baud Rate Generator Double-Speed Mode Select)

Selects the cycle of output clock for the baud rate generator.

This bit is valid when the on-chip baud rate generator is selected as the clock source (`SCR.CKE[1] = 0`) in asynchronous mode (`SMR.CM = 0`). For the clock output from the baud rate generator, either normal or doubled frequency can be selected. The base clock is generated by the clock output from the baud rate generator. When the `BGDM` bit is set to 1, the base clock cycle is halved and the bit rate is doubled.

Set this bit to 0 in modes other than asynchronous mode.

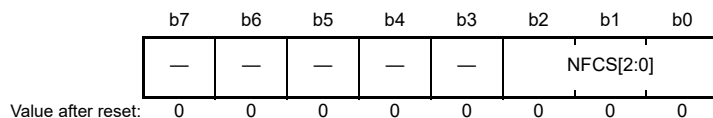
RXDESEL Bit (Asynchronous Start Bit Edge Detection Select)

Selects the detection method of the start bit for reception in asynchronous mode. When a break occurs, data receiving operation depends on the settings of this bit. Set this bit to 1 when reception should be stopped while a break occurs or when reception should be started without retaining the `RXDn` pin input at high level for the period of one data frame or longer after completion of the break.

Set this bit to 0 in modes other than asynchronous mode.

41.2.16 Noise Filter Setting Register (SNFR)

Address(es): SCI0.SNFR 0008 A008h, SCI1.SNFR 0008 A028h, SCI2.SNFR 0008 A048h, SCI3.SNFR 0008 A068h, SCI4.SNFR 0008 A088h, SCI5.SNFR 0008 A0A8h, SCI6.SNFR 0008 A0C8h, SCI7.SNFR 000D 00E8h, SCI8.SNFR 000D 0008h, SCI9.SNFR 000D 0028h, SCI10.SNFR 000D 0048h, SCI11.SNFR 000D 0068h, SCI12.SNFR 0008 B308h



Bit	Symbol	Bit Name	Description	R/W
b2 to b0	NFCS[2:0]	Noise Filter Clock Select	In asynchronous mode, the standard setting for the base clock is as follows. b2 b0 0 0 0: The clock signal divided by 1 is used with the noise filter. In simple I²C mode, the standard settings for the clock source of the on-chip baud rate generator selected by the SMR.CKS[1:0] bits are given below. b2 b0 0 0 1: The clock signal divided by 1 is used with the noise filter. 0 1 0: The clock signal divided by 2 is used with the noise filter. 0 1 1: The clock signal divided by 4 is used with the noise filter. 1 0 0: The clock signal divided by 8 is used with the noise filter. Settings other than above are prohibited.	R/W*1
b7 to b3	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

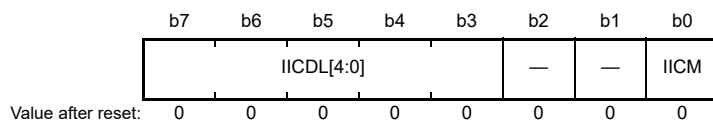
Note 1. Writing to these bits is only possible when the RE and TE bits in the SCR register are 0 (serial reception and transmission disabled).

NFCS[2:0] Bits (Noise Filter Clock Select)

These bits select the sampling clock for the digital noise filter. To use the noise filter in asynchronous mode, set these bits to 000b. In simple I²C mode, set the bits to a value in the range from 001b to 100b.

41.2.17 I²C Mode Register 1 (SIMR1)

Address(es): SCi0.SIMR1 0008 A009h, SCi1.SIMR1 0008 A029h, SCi2.SIMR1 0008 A049h, SCi3.SIMR1 0008 A069h, SCi4.SIMR1 0008 A089h, SCi5.SIMR1 0008 A0A9h, SCi6.SIMR1 0008 A0C9h, SCi7.SIMR1 000D 00E9h, SCi8.SIMR1 000D 0009h, SCi9.SIMR1 000D 0029h, SCi10.SIMR1 000D 0049h, SCi11.SIMR1 000D 0069h, SCi12.SIMR1 0008 B309h



Bit	Symbol	Bit Name	Description	R/W
b0	IICM	Simple I ² C Mode Select	SMIF IICM 0 0: Asynchronous mode, Multi-processor mode, Clock synchronous mode (in asynchronous mode, synchronous, or simple SPI mode) 0 1: Simple I ² C mode 1 0: Smart card interface mode 1 1: Setting prohibited.	R/W*1
b2, b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b7 to b3	IICDL[4:0]	SSDA Output Delay Select	(Cycles below are of the clock signal from the on-chip baud rate generator.) b7 b3 0 0 0 0 0: No output delay 0 0 0 0 1: 0 to 1 cycle 0 0 0 1 0: 1 to 2 cycles 0 0 0 1 1: 2 to 3 cycles 0 0 1 0 0: 3 to 4 cycles 0 0 1 0 1: 4 to 5 cycles : : 1 1 1 1 0: 29 to 30 cycles 1 1 1 1 1: 30 to 31 cycles	R/W*1

Note 1. Writing to these bits is only possible when the RE and TE bits in the SCR register are 0 (both serial transmission and reception are disabled).

SIMR1 is used to select simple I²C mode and the number of delay stages for the SSDA output.

IICM Bit (Simple I²C Mode Select)

In conjunction with the SCMR.SMIF bit, this bit selects the operating mode.

IICDL[4:0] Bits (SSDA Output Delay Select)

These bits are used to set a delay for output on the SSDAn pin relative to the falling edge of the output on the SSCLn pin. The available delay settings range from no delay to 31 cycles, with the clock signal from the on-chip baud rate generator as the base. The signal obtained by frequency-dividing PCLK by the divisor set in the SMR.CKS[1:0] bits is supplied as the clock signal from the on-chip baud rate generator. Set these bits to 00000b unless operation is in simple I²C mode. In simple I²C mode, set the bits to a value in the range from 00001b to 11111b.

41.2.18 I²C Mode Register 2 (SIMR2)

Address(es): SCi0.SIMR2 0008 A00Ah, SCi1.SIMR2 0008 A02Ah, SCi2.SIMR2 0008 A04Ah, SCi3.SIMR2 0008 A06Ah, SCi4.SIMR2 0008 A08Ah, SCi5.SIMR2 0008 A0AAh, SCi6.SIMR2 0008 A0CAh, SCi7.SIMR2 000D 00EAh, SCi8.SIMR2 000D 000Ah, SCi9.SIMR2 000D 002Ah, SCi10.SIMR2 000D 004Ah, SCi11.SIMR2 000D 006Ah, SCi12.SIMR2 0008 B30Ah

b7	b6	b5	b4	b3	b2	b1	b0
—	—	IICACK T	—	—	—	IICCSC	IICINT M

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Bit Name	Description	R/W
b0	IICINTM	I ² C Interrupt Mode Select	0: Use ACK/NACK interrupts. 1: Use reception and transmission interrupts.	R/W*1
b1	IICCSC	Clock Synchronization	0: No synchronization with the clock signal 1: Synchronization with the clock signal	R/W*1
b4 to b2	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b5	IICACKT	ACK Transmission Data	0: ACK transmission 1: NACK transmission and reception of ACK/NACK	R/W
b7, b6	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. Writing to these bits is only possible when the RE and TE bits in the SCR register are 0 (serial reception and transmission disabled).

SIMR2 is used to select how reception and transmission are controlled in simple I²C mode.

IICINTM Bit (I²C Interrupt Mode Select)

This bit selects the sources of interrupt requests in simple I²C mode.

IICCSC Bit (Clock Synchronization)

Set the IICCSC bit to 1 if the internally generated SSCLn clock signal is to be synchronized when the SSCLn pin has been placed at the low level in the case of a wait inserted by the other device, etc.

The SSCLn clock signal is not synchronized if the IICCSC bit is 0. The SSCLn clock signal is generated in accord with the rate selected in the BRR regardless of the level being input on the SSCLn pin.

Set the IICCSC bit to 1 except during debugging.

IICACKT Bit (ACK Transmission Data)

Transmitted data contains ACK bits. Set this bit to 1 when ACK and NACK bits are received.

41.2.19 I²C Mode Register 3 (SIMR3)

Address(es): SCI0.SIMR3 0008 A00Bh, SCI1.SIMR3 0008 A02Bh, SCI2.SIMR3 0008 A04Bh, SCI3.SIMR3 0008 A06Bh, SCI4.SIMR3 0008 A08Bh, SCI5.SIMR3 0008 A0ABh, SCI6.SIMR3 0008 A0CBh, SCI7.SIMR3 000D 00EBh, SCI8.SIMR3 000D 000Bh, SCI9.SIMR3 000D 002Bh, SCI10.SIMR3 000D 004Bh, SCI11.SIMR3 000D 006Bh, SCI12.SIMR3 0008 B30Bh

b7	b6	b5	b4	b3	b2	b1	b0
IICSCLS[1:0]	IICSDAS[1:0]	IICSTIF	IICSTPREQ	IICRSTAREQ	IICSTAREQ		
Value after reset:	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	IICSTAREQ	Start Condition Generation	0: A start condition is not generated. 1: A start condition is generated.*1, *3, *4, *5	R/W
b1	IICRSTAREQ	Restart Condition Generation	0: A restart condition is not generated. 1: A restart condition is generated.*2, *3, *4, *5	R/W
b2	IICSTPREQ	Stop Condition Generation	0: A stop condition is not generated. 1: A stop condition is generated.*2, *3, *4, *5	R/W
b3	IICSTIF	Issuing of Start, Restart, or Stop Condition Completed Flag	0: There are no requests for generating conditions or a condition is being generated. 1: A start, restart, or stop condition is completely generated.	R/W
b5, b4	IICSDAS[1:0]	SSDA Output Select	b5 b4 0 0: Serial data output 0 1: Generate a start, restart, or stop condition. 1 0: Output the low level on the SSDAn pin. 1 1: Place the SSDAn pin in the high-impedance state.	R/W
b7, b6	IICSCLS[1:0]	SSCL Output Select	b7 b6 0 0: Serial clock output 0 1: Generate a start, restart, or stop condition. 1 0: Output the low level on the SSCLn pin. 1 1: Place the SSCLn pin in the high-impedance state.	R/W

Note 1. Generate a start condition only when the SSCLn and SSDAn pins are both high (the corresponding bits in the corresponding PIDR registers are 1).

Note 2. Generate a restart or stop condition only when the SSCLn pin is low (the corresponding bit in the PIDR register is 0).

Note 3. Do not set more than one from among the IICSTAREQ, IICRSTAREQ, and IICSTPREQ bits to 1 at a given time.

Note 4. Execute the generation of a condition after the value of the IICSTIF flag is 0.

Note 5. Do not write 0 to this bit while it is 1. Generation of a condition is suspended by writing 0 to this bit while it is 1.

SIMR3 is used to control the simple I²C mode start and stop conditions, and to hold the SSDAn and SSCLn pins at fixed levels.

IICSTAREQ Bit (Start Condition Generation)

When a start condition is to be generated, set both the IICSDAS[1:0] and IICSCLS[1:0] bits to 01b as well as setting the IICSTAREQ bit to 1.

[Setting condition]

- Writing 1 to the bit

[Clearing condition]

- Completion of generation of the start condition

IICRSTAREQ Bit (Restart Condition Generation)

When a restart condition is to be generated, set both the IICSDAS[1:0] and IICSCLS[1:0] bits to 01b as well as setting the IICRSTAREQ bit to 1.

[Setting condition]

- Writing 1 to the bit

[Clearing condition]

- Completion of generation of the restart condition

IICSTPREQ Bit (Stop Condition Generation)

When a stop condition is to be generated, set both the IICSDAS[1:0] and IICSCLS[1:0] bits to 01b as well as setting the IICSTPREQ bit to 1.

[Setting condition]

- Writing 1 to the bit

[Clearing condition]

- Completion of generation of the stop condition

IICSTIF Flag (Issuing of Start, Restart, or Stop Condition Completed Flag)

After generating a condition, this bit indicates that the generation is completed. When using the IICSTAREQ, IICRSTAREQ, or IICSTPREQ bit to cause generation of a condition, do so after setting the IICSTIF flag to 0. When the IICSTIF flag is 1 while an interrupt request is enabled by setting the SCR.TEIE bit, an STI request is output.

[Setting condition]

- Completion of the generation of a start, restart, or stop condition (however, in cases where this conflicts with any of the conditions for the flag becoming 0 listed below, the other condition takes precedence)

[Clearing conditions]

- Writing 0 to the bit (confirm that the IICSTIF flag is 0 before doing so)
- Writing 0 to the SIMR1.IICM bit (when operation is not in simple I²C mode)
- Writing 0 to the SCR.TE bit

IICSDAS[1:0] Bits (SSDA Output Select)

These bits control output from the SSDAn pin.

Set the IICSDAS[1:0] and IICSCLS[1:0] bits to the same value during normal operations.

IICSCLS[1:0] Bits (SSCL Output Select)

These bits control output from the SSCLn pin.

Set the IICSCLS[1:0] and IICSDAS[1:0] bits to the same value during normal operations.

41.2.20 I²C Status Register (SISR)

Address(es): SCI0.SISR 0008 A00Ch, SCI1.SISR 0008 A02Ch, SCI2.SISR 0008 A04Ch, SCI3.SISR 0008 A06Ch, SCI4.SISR 0008 A08Ch, SCI5.SISR 0008 A0ACh, SCI6.SISR 0008 A0CCh, SCI7.SISR 000D 00ECh, SCI8.SISR 000D 000Ch, SCI9.SISR 000D 002Ch, SCI10.SISR 000D 004Ch, SCI11.SISR 000D 006Ch, SCI12.SISR 0008 B30Ch

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	—	IICACK R

Value after reset: 0 0 x x 0 x 0 0

x: Undefined

Bit	Symbol	Bit Name	Description	R/W
b0	IICACKR	ACK Reception Data Flag	0: ACK received 1: NACK received	R/W*1
b1	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b2	—	Reserved	The read value is undefined.	R
b3	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b5, b4	—	Reserved	The read value is undefined.	R
b7, b6	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. Only 0 can be written to this bit, to clear the flag.

SISR is used to monitor state in relation to simple I²C mode.

IICACKR Flag (ACK Reception Data Flag)

Received ACK and NACK bits can be read from this bit.

The IICACKR flag is updated at the rising of SSCLn clock for the ACK/NACK receiving bit.

41.2.21 SPI Mode Register (SPMR)

Address(es): SCI0.SPMR 0008 A00Dh, SCI1.SPMR 0008 A02Dh, SCI2.SPMR 0008 A04Dh, SCI3.SPMR 0008 A06Dh, SCI4.SPMR 0008 A08Dh, SCI5.SPMR 0008 A0ADh, SCI6.SPMR 0008 A0CDh, SCI7.SPMR 000D 00EDh, SCI8.SPMR 000D 000Dh, SCI9.SPMR 000D 002Dh, SCI10.SPMR 000D 004Dh, SCI11.SPMR 000D 006Dh, SCI12.SPMR 0008 B30Dh

b7	b6	b5	b4	b3	b2	b1	b0
CKPH	CKPOL	—	MFF	—	MSS	CTSE	SSE

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Bit Name	Description	R/W
b0	SSE	SSn# Pin Function Enable	0: SSn# pin function is disabled. 1: SSn# pin function is enabled.	R/W*1
b1	CTSE	CTS Enable	0: CTS function is disabled (RTS output function is enabled). 1: CTS function is enabled.	R/W*1
b2	MSS	Master Slave Select	0: Transmission is through the SMOSIn pin and reception is through the SMISOn pin (master mode). 1: Reception is through the SMOSIn pin and transmission is through the SMISOn pin (slave mode).	R/W*1
b3	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b4	MFF	Mode Fault Flag	0: No mode fault error 1: Mode fault error	R/W*2
b5	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b6	CKPOL	Clock Polarity Select	0: Clock polarity is not inverted. 1: Clock polarity is inverted.	R/W*1
b7	CKPH	Clock Phase Select	0: Clock is not delayed. 1: Clock is delayed.	R/W*1

Note 1. Writing to these bits is only possible when the RE and TE bits in the SCR register are 0 (both serial transmission and reception are disabled).

Note 2. Only 0 can be written to these bits, which clears the flag.

SPMR is used to select the extension settings in asynchronous and clock synchronous modes.

SSE Bit (SSn# Pin Function Enable)

Set this bit to 1 if the SSn# pin is to be used in control of transmission and reception (in simple SPI mode). Set this bit to 0 in any other mode. Furthermore, even for usage in simple SPI mode, the SSn# pin on the master side is not required to control reception and transmission when master mode (SCR.CKE[1:0] = 00b and MSS = 0) is selected and there is a single master, so the setting for the SSE bit is 0. Do not set both the SSE and CTSE bits to enabled (even if this setting is made, operation is the same as that when these bits are set to 0).

CTSE Bit (CTS Enable)

Set this bit to 1 if the SSn# pin is to be used for inputting of the CTS control signal to control of transmission and reception. The RTS signal is output when this bit is set to 0. Set this bit to 0 in smart card interface mode, simple SPI mode, and simple I²C mode. Do not set both the CTSE and SSE bits to enabled (even if this setting is made, operation is the same as that when these bits are set to 0).

MSS Bit (Master Slave Select)

This bit selects between master and slave operation in simple SPI mode. When the MSS bit is set to 1, data is received through the SMOSIn pin and transmitted through the SMISOn pin.

Set this bit to 0 in modes other than simple SPI mode.

MFF Flag (Mode Fault Flag)

This bit indicates mode fault errors.

In a multi-master configuration, determine the mode fault error occurrence by reading the MFF flag.

[Setting condition]

- Input on the SSn# pin being at the low level during master operation in simple SPI mode (SSE bit = 1 and MSS bit = 0)

[Clearing condition]

- Writing 0 to the bit after it was read as 1

CKPOL Bit (Clock Polarity Select)

This bit selects the polarity of the clock signal output through the SCKn pin. Refer to Figure 41.62 for details.

Set the bit to 0 in other than simple SPI mode and clock synchronous mode.

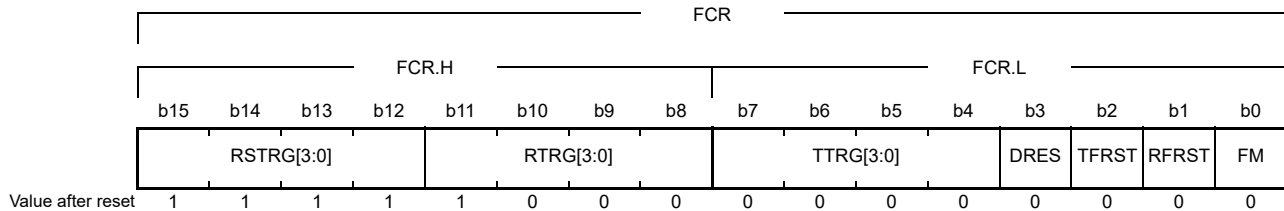
CKPH Bit (Clock Phase Select)

This bit selects the phase of the clock signal output through the SCKn pin. Refer to Figure 41.62 for details.

Set the bit to 0 in other than simple SPI mode and clock synchronous mode.

41.2.22 FIFO Control Register (FCR)

Address(es): SCI7.FCR 000D 00F4h, SCI8.FCR 000D 0014h, SCI9.FCR 000D 0034h, SCI10.FCR 000D 0054h, SCI11.FCR 000D 0074h,
 SCI7.FCR.H 000D 00F4h, SCI8.FCR.H 000D 0014h, SCI9.FCR.H 000D 0034h, SCI10.FCR.H 000D 0054h, SCI11.FCR.H 000D 0074h,
 SCI7.FCR.L 000D 00F5h, SCI8.FCR.L 000D 0015h, SCI9.FCR.L 000D 0035h, SCI10.FCR.L 000D 0055h, SCI11.FCR.L 000D 0075h



Bit	Symbol	Bit Name	Description	R/W
b0	FM	FIFO Mode Select*1	0: Non-FIFO mode (TDR and RDR registers are used for transmit and receive) 1: FIFO mode (FTDR and FRDR registers are used for transmit and receive)	R/W*2
b1	RFRST	Receive FIFO Reset*3	Writing 1 clears the number of data stored in the receive FIFO to 0	R/W
b2	TFRST	Transmit FIFO Reset*3	Writing 1 clears the number of data stored in the transmit FIFO to 0	R/W
b3	DRES	Receive Data Ready Interrupt Select	0: Receive data full interrupt (RXI) 1: Error interrupt (ERI)	R/W*2
b7 to b4	TTRG[3:0]	Transmit FIFO Threshold Setting*1	Specifies the threshold value for the SSRFIFO.TDFE flag to become 1 b7 b4 0 0 0 0: Transmit FIFO threshold value is 0 0 0 0 1: Transmit FIFO threshold value is 1 : 1 1 1 1: Transmit FIFO threshold value is 15	R/W
b11 to b8	RTRG[3:0]	Receive FIFO Threshold Setting*1	Specifies the threshold value for the SSRFIFO.RDF flag to become 1 b11 b8 0 0 0 0: Setting prohibited 0 0 0 1: Receive FIFO threshold value is 1 : 1 1 1 1: Receive FIFO threshold value is 15	R/W
b15 to b12	RSTRG[3:0]	RTS# Output Threshold Setting*4	Specifies the threshold value for the RTSn# pin to be driven high b15 b12 0 0 0 0: Setting prohibited 0 0 0 1: Receive FIFO threshold value is 1 : 1 1 1 1: Receive FIFO threshold value is 15	R/W

Note 1. These bits are enabled in asynchronous mode and clock synchronous mode only.

Note 2. These bits can be rewritten only when the TE and RE bits in the SCR register are both 0.

Note 3. These bits are enabled only when the FM bit is 1.

Note 4. These bits are enabled when the FM bit is 1, and the CTSE and SSE bits in the SPMR register are both 0 (RTS# output is enabled).

FM Bit (FIFO Mode Select)

This bit is used to enable FIFO mode. Setting this bit to 1 enables FIFO, and causes FTDR and FRDR to be used as the data register for transmit and receive, respectively.

Setting this bit to 0 disables FIFO, and causes the TDR and RDR registers, or the TDRHL, TDRH, TDRL, RDRHL, RDRH, and RDRL registers to be used as the data register for transmit and receive.

This bit is enabled in asynchronous mode and clock synchronous mode only. This bit should be set to 0 in any other mode. When rewriting this bit, transmit and receive should be disabled.

RFRST Bit (Receive FIFO Reset)

Setting the RFRST bit to 1, causes the FDR.R[4:0] bits to become 0. The value of the RFRST bit automatically returns to 0 after 1 PCLK.

TFRST Bit (Transmit FIFO Reset)

Setting the TFRST bit to 1, causes the FDR.T[4:0] bits to become 0. The value of the TFRST bit automatically returns to 0 after 1 PCLK.

DRES Bit (Receive Data Ready Interrupt Select)

This bit is used to select the interrupt request that is generated when the SSRFIFO.DR flag becomes 1. When this bit is 0, a DR flag of 1 generates a receive data full interrupt (RXI) request. When this bit is 1, a DR flag of 1 generates a receive error interrupt (ERI) request.

TTRG[3:0] Bits (Transmit FIFO Threshold Setting)

The SSRFIFO.TDFE flag becomes 1 when the value of the FDR.T[4:0] bits becomes equal to or less than the value set in the TTRG[3:0] bits. At this time, if the SCR.TIE bit is 1, a transmit data empty interrupt (TXI) request is generated.

RTRG[3:0] Bits (Receive FIFO Threshold Setting)

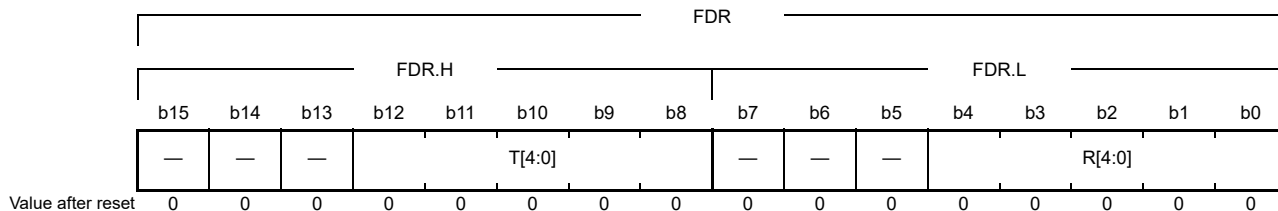
The SSRFIFO.RDF flag becomes 1 when the values of the FDR.R[4:0] bits are equal to or greater than the values set in the RTRG[3:0] bits. At this time, if the SCR.RIE bit is 1, a receive data full interrupt (RXI) request is generated.

RSTRG[3:0] Bits (RTS# Output Threshold Setting)

When the values of the FDR.R[4:0] bits becomes equal to or greater than the values set in the RSTRG[3:0] bits, the RTS# pin is driven high. When using this function, set both the CTSE and SSE bits in the SPMR register to 0 to enable RTS# output.

41.2.23 FIFO Data Count Register (FDR)

Address(es): SCI7.FDR 000D 00F6h, SCI8.FDR 000D 0016h, SCI9.FDR 000D 0036h, SCI10.FDR 000D 0056h, SCI11.FDR 000D 0076h,
 SCI7.FDR.H 000D 00F6h, SCI8.FDR.H 000D 0016h, SCI9.FDR.H 000D 0036h, SCI10.FDR.H 000D 0056h, SCI11.FDR.H 000D 0076h,
 SCI7.FDR.L 000D 00F7h, SCI8.FDR.L 000D 0017h, SCI9.FDR.L 000D 0037h, SCI10.FDR.L 000D 0057h, SCI11.FDR.L 000D 0077h



Bit	Symbol	Bit Name	Description	R/W
b4 to b0	R[4:0]	Receive FIFO Data Count*1	These bits indicate the number of data stored in the receive FIFO	R
b7 to b5	—	Reserved	These bits are read as 0. The write value should be 0.	R
b12 to b8	T[4:0]	Transmit FIFO Data Count*1	These bits indicate the number of data stored in the transmit FIFO	R
b15 to b13	—	Reserved	These bits are read as 0. The write value should be 0.	R

Note 1. These bits are valid only in asynchronous mode or clock synchronous mode.

R[4:0] Bits (Receive FIFO Data Count)

These bits indicate the number of data stored in the receive FIFO. 00h indicates that no received data is in FIFO, and 10h indicates that 16 frame data received have been stored in FIFO.

These bits never have values in the range from 11h to 1Fh.

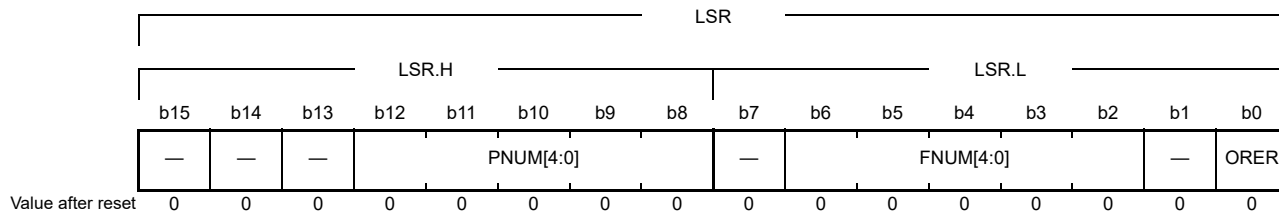
T[4:0] Bits (Transmit FIFO Data Count)

These bits indicate the number of untransmitted data in the transmit FIFO. 00h indicates that no untransmitted data is in FIFO, and 10h indicates that 16 frames of untransmitted data are stored in FIFO.

These bits never have values in the range from 11h to 1Fh.

41.2.24 Line Status Register (LSR)

Address(es): SCI7.LSR 000D 00F8h, SCI8.LSR 000D 0018h, SCI9.LSR 000D 0038h, SCI10.LSR 000D 0058h, SCI11.LSR 000D 0078h,
SCI7.LSR.H 000D 00F8h, SCI8.LSR.H 000D 0018h, SCI9.LSR.H 000D 0038h, SCI10.LSR.H 000D 0058h, SCI11.LSR.H 000D 0078h,
SCI7.LSR.L 000D 00F9h, SCI8.LSR.L 000D 0019h, SCI9.LSR.L 000D 0039h, SCI10.LSR.L 000D 0059h, SCI11.LSR.L 000D 0079h



Bit	Symbol	Bit Name	Description	R/W
b0	ORER	Overrun Error Flag ^{*1}	The same value as that in the SSRFIFO.ORER flag is read. 0: An overrun error has not occurred. 1: An overrun error has occurred.	R ^{*2}
b1	—	Reserved	This bit are read as 0. The write value should be 0.	R
b6 to b2	FNUM[4:0]	Framing Error Count	These bits indicate the number of data with a framing error among the data stored in the receive FIFO.	R
b7	—	Reserved	This bit are read as 0. The write value should be 0.	R
b12 to b8	PNUM[4:0]	Parity Error Count	These bits indicate the number of data with a parity error among the data stored in the receive FIFO.	R
b15 to b13	—	Reserved	This bit are read as 0. The write value should be 0.	R

Note 1. The bit can be read only in FIFO mode, and asynchronous mode or clock synchronous mode.

Note 2. To clear this flag, confirm that the SSRFIFO.ORER flag is 1 first, and then write 0 to the SSRFIFO.ORER flag.

ORER Flag (Overrun Error Flag)

This bit becomes 1 when an overrun error occurs. The value of the SSRFIFO.ORER flag is reflected to this bit. To clear this flag, clear the SSRFIFO.ORER flag.

FNUM[4:0] Bits (Framing Error Count)

These bits indicate the number of data in which a framing error has occurred among the received data in the receive FIFO.

PNUM[4:0] Bits (Parity Error Count)

These bits indicate the number of data in which a parity error has occurred among the received data in the receive FIFO.

41.2.25 Comparison Data Register (CDR)

Address(es): SCI0.CDR 0008 A01Ah, SCI1.CDR 0008 A03Ah, SCI2.CDR 0008 A05Ah, SCI3.CDR 0008 A07Ah, SCI4.CDR 0008 A09Ah, SCI5.CDR 0008 A0BAh, SCI6.CDR 0008 A0DAh, SCI7.CDR 000D 00FAh, SCI8.CDR 000D 001Ah, SCI9.CDR 000D 003Ah, SCI10.CDR 000D 005Ah, SCI11.CDR 000D 007Ah, SCI10.CDR.H 0008 A01Ah, SCI1.CDR.H 0008 A03Ah, SCI2.CDR.H 0008 A05Ah, SCI3.CDR.H 0008 A07Ah, SCI4.CDR.H 0008 A09Ah, SCI5.CDR.H 0008 A0BAh, SCI6.CDR.H 0008 A0DAh, SCI7.CDR.H 000D 00FAh, SCI8.CDR.H 000D 001Ah, SCI9.CDR.H 000D 003Ah, SCI10.CDR.H 000D 005Ah, SCI11.CDR.H 000D 007Ah, SCI10.CDR.L 0008 A01Bh, SCI1.CDR.L 0008 A03Bh, SCI2.CDR.L 0008 A05Bh, SCI3.CDR.L 0008 A07Bh, SCI4.CDR.L 0008 A09Bh, SCI5.CDR.L 0008 A0BBh, SCI6.CDR.L 0008 A0DBh, SCI7.CDR.L 000D 00FBh, SCI8.CDR.L 000D 001Bh, SCI9.CDR.L 000D 003Bh, SCI10.CDR.L 000D 005Bh, SCI11.CDR.L 000D 007Bh



Bit	Symbol	Bit Name	Description	R/W
b8 to b0	CMPD[8:0]	Comparison Data	These bits specify the data of comparison source when using the data match detection function.	R/W
b15 to b9	—	Reserved	These bits are read as 0. The write value should be 0.	R

CMPD[8:0] Bits (Comparison Data)

These bits are used for detecting a data match. The length of the valid bit is the same as the character length set in the SMR.CHR and SCMR.CHR1 bits.

The DCCR.DCMF flag becomes 1 when the received data matches with the value of these bits.

41.2.26 Data Comparison Control Register (DCCR)

Address(es): SCI0.DCCR 0008 A013h, SCI1.DCCR 0008 A033h, SCI2.DCCR 0008 A053h, SCI3.DCCR 0008 A073h, SCI4.DCCR 0008 A093h, SCI5.DCCR 0008 A0B3h, SCI6.DCCR 0008 A0D3h, SCI7.DCCR 000D 00F3h, SCI8.DCCR 000D 0013h, SCI9.DCCR 000D 0033h, SCI10.DCCR 000D 0053h, SCI11.DCCR 000D 0073h

b7	b6	b5	b4	b3	b2	b1	b0
DCME	IDSEL	—	DFER	DPER	—	—	DCMF
Value after reset	0	1	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	DCMF	Data Match Flag	0: Data is not matched 1: Data is matched	R/(W) *1
b2, b1	—	Reserved	These bits are read as 0. The write value should be 0	R/W
b3	DPER	Match Data Parity Error Flag	0: A parity error is not found in the matched data. 1: A parity error is found in the matched data.	R/(W) *1
b4	DFER	Match Data Framing Error Flag	0: A framing error is not found in the matched data. 1: A framing error is found in the matched data.	R/(W) *1
b5	—	Reserved	This bit is read as 0. The write value should be 0	R/W
b6	IDSEL	ID Frame Select*2	0: All data is to be compared 1: The data with the multi-processor bit set to 1 is to be compared	R/W
b7	DCME	Data Match Detection Enable*2	0: Data match detection is disabled 1: Data match detection is enabled	R/W

Note 1. Only 0 can be written to this bit, which clears the flag. To clear this flag, confirm that the flag is 1, and then write 0 to the flag.

Note 2. This bit is only valid in asynchronous mode.

DCMF Flag (Data Match Flag)

This flag indicates the comparison result of the received data and the value of the CDR register.

[Setting condition]

- When the received data matches with the value in the CDR register while the DCME bit is 1

[Clearing condition]

- When 0 is written to the flag after reading this flag as 1

Even when the SCR.RE bit is set to 0, the DCMF flag has no effect and retains the previous state.

DPER Flag (Match Data Parity Error Flag)

This flag indicates if a parity error has occurred in the matched data.

[Setting condition]

- When a parity error is found in the received data in which a data match is detected.

[Clearing condition]

- When 0 is written to the flag after reading the flag as 1.

Refer to section 15.5.2, Level Detection to exit from the interrupt handling routine after setting the DPER flag to 0.

Even when the SCR.RE bit is set to 0, the DPER flag has no effect and retains the previous state.

DFER Flag (Match Data Framing Error Flag)

This flag indicates if a framing error has occurred in the matched data.

[Setting condition]

- When the stop bit of the received frame in which a data match is detected is 0

[Clearing condition]

- When 0 is written to the flag after reading the flag as 1.

Refer to section 15.5.2, Level Detection to exit from the interrupt handling routine after setting the DFER flag to 0.

Even when the SCR.RE bit is set to 0, the DFER flag has no effect and retains the previous state.

IDSEL Bit (ID Frame Select)

This bit specifies the condition of the received data to be compared. The bit is valid only when the DCME bit is 1.

When setting this bit to 1, only data in received frames (ID frames) in which the multi-processor bit has the value 1 are compared.

When this bit is set to 0, all received data is compared.

DCME Bit (Data Match Detection Enable)

This bit enables or disables the data match detection function. The function is valid only in asynchronous mode. In other modes, set this bit to 0.

This bit automatically becomes 0 when a data match is detected.

41.2.27 Serial Port Register (SPTR)

Address(es): SCI0.SPTR 0008 A01Ch, SCI1.SPTR 0008 A03Ch, SCI2.SPTR 0008 A05Ch, SCI3.SPTR 0008 A07Ch, SCI4.SPTR 0008 A09Ch, SCI5.SPTR 0008 A0BCh, SCI6.SPTR 0008 A0DCh, SCI7.SPTR 000D 00FCh, SCI8.SPTR 000D 001Ch, SCI9.SPTR 000D 003Ch, SCI10.SPTR 000D 005Ch, SCI11.SPTR 000D 007Ch

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	SPB2IO	SPB2DT	RXDMON
0	0	0	0	0	0	1	1

Value after reset

Bit	Symbol	Bit Name	Description	R/W
b0	RXDMON	RXD Line Monitoring Flag	0: The RXDn pin level is low 1: The RXDn pin level is high	R
b1	SPB2DT	Serial Port Break Data*1	Combine bits SPB2DT, SPB2IO, and SCR.TE to control the TXDn pin. Refer to Table 41.30 for details.	R/W
b2	SPB2IO	Serial Port Break I/O*1		R/W
b7 to b3	—	Reserved	These bits are read as 0. The write value should be 0.	R

Note 1. This bit is only valid in asynchronous mode.

RXDMON Flag (RXD Line Monitoring Flag)

This flag is used for monitoring the level of the RXDn pin.

SPB2DT Bit (Serial Port Break Data)

This bit specifies the output level of the TXDn pin when the SCR.TE bit is 0. Refer to Table 41.30 for details.

SPB2IO Bit (Serial Port Break I/O)

This bit specifies input or output of the TXDn pin when the SCR.TE bit is 0. Set this bit to 1 (output) when controlling the TXDn pin by software.

Table 41.30 Controlling the TXDn pin

SCR.TE Bit Setting	SPB2IO Bit Setting	SPB2DT Bit Setting	TXDn Pin Status
0 (Transmission disabled)	0 (Input)	0 or 1	Hi-Z
		0 (Low)	Low is output
		1 (High)	High is output
1 (Transmission enabled)	0 or 1	0 or 1	Transmit data output pin

41.2.28 Extended Serial Module Enable Register (ESMER)

Address(es): SCI12.ESMER 0008 B320h

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	—	ESME
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	ESME	Extended Serial Mode Enable	0: The extended serial mode is disabled. 1: The extended serial mode is enabled.	R/W
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

ESME Bit (Extended Serial Mode Enable)

When the ESME bit is 1, the facilities of the extended serial mode control section are enabled.

When the ESME bit is 0, the extended serial mode control section is initialized.

Table 41.31 Settings of the ESME Bit and Timer Operation Mode

ESME Bit	Timer Mode	Break Field Low Width Determination Mode	Break Field Low Width Output Mode
0	Available*1	Not available	Not available
1	Available	Available	Available

Note 1. Operation is only possible with PCLK selected.

41.2.29 Control Register 0 (CR0)

Address(es): SCI12.CR0 0008 B321h

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	BRME	RXDSF	SFSF	—
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b1	SFSF	Start Frame Status Flag	0: Start Frame detection function is disabled. 1: Start Frame detection function is enabled.	R
b2	RXDSF	RXDX12 Input Status Flag	0: RXDX12 input is enabled. 1: RXDX12 input is disabled.	R
b3	BRME	Bit Rate Measurement Enable	0: Measurement of bit rate is disabled. 1: Measurement of bit rate is enabled.	R/W
b7 to b4	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

41.2.30 Control Register 1 (CR1)

Address(es): SCI12.CR1 0008 B322h

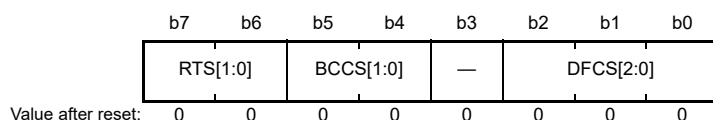
b7	b6	b5	b4	b3	b2	b1	b0
PIBS[2:0]			PIBE	CF1DS[1:0]		CF0RE	BFE
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	BFE	Break Field Enable	0: Break Field detection is disabled. 1: Break Field detection is enabled.	R/W
b1	CF0RE	Control Field 0 Reception Enable	0: Reception of Control Field 0 is disabled. 1: Reception of Control Field 0 is enabled.	R/W
b3, b2	CF1DS[1:0]	Control Field 1 Data Register Select	b3 b2 0 0: Selects comparison with the value in the PCF1DR register. 0 1: Selects comparison with the value in the SCF1DR register. 1 0: Selects comparison with the values in the PCF1DR and SCF1DR registers. 1 1: Setting prohibited.	R/W
b4	PIBE	Priority Interrupt Bit Enable	0: The priority interrupt bit is disabled. 1: The priority interrupt bit is enabled.	R/W
b7 to b5	PIBS[2:0]	Priority Interrupt Bit Select	b7 b5 0 0 0: 0th bit of Control Field 1 0 0 1: 1st bit of Control Field 1 0 1 0: 2nd bit of Control Field 1 0 1 1: 3rd bit of Control Field 1 1 0 0: 4th bit of Control Field 1 1 0 1: 5th bit of Control Field 1 1 1 0: 6th bit of Control Field 1 1 1 1: 7th bit of Control Field 1	R/W

41.2.31 Control Register 2 (CR2)

Address(es): SCI12.CR2 0008 B323h



Bit	Symbol	Bit Name	Description	R/W
b2 to b0	DFCS[2:0]	RXDX12 Signal Digital Filter Clock Select	b2 b0 0 0 0: Filter is disabled. 0 0 1: Filter clock is base clock*1, *2 0 1 0: Filter clock is PCLK/8 0 1 1: Filter clock is PCLK/16 1 0 0: Filter clock is PCLK/32 1 0 1: Filter clock is PCLK/64 1 1 0: Filter clock is PCLK/128 1 1 1: Setting prohibited	R/W
b3	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b5, b4	BCCS[1:0]	Bus Collision Detection Clock Select	• When SEMR.BGDM = 0 or SEMR.BGDM = 1 and SMR.CKS[1:0] = a value other than 00b b5 b4 0 0: Base clock 0 1: Base clock frequency divided by 2 1 0: Base clock frequency divided by 4 1 1: Setting prohibited • When SEMR.BGDM = 1 and SMR.CKS[1:0] = 00b b5 b4 0 0: Base clock frequency divided by 2 0 1: Base clock frequency divided by 4 1 0: Setting prohibited 1 1: Setting prohibited	R/W
b7, b6	RTS[1:0]	RXDX12 Reception Sampling Timing Select	• When SCI12.SEMR.ABCS = 0 b7 b6 0 0: Rising edge of the 8th cycle of base clock 0 1: Rising edge of the 10th cycle of base clock 1 0: Rising edge of the 12th cycle of base clock 1 1: Rising edge of the 14th cycle of base clock • When SCI12.SEMR.ABCS = 1 b7 b6 0 0: Rising edge of the 4th cycle of base clock 0 1: Rising edge of the 5th cycle of base clock 1 0: Rising edge of the 6th cycle of base clock 1 1: Rising edge of the 7th cycle of base clock	R/W

Note: The period of the base clock is 1/16 of a single bit period when the SCI12.SEMR.ABCS is 0, and 1/8 of a single bit period when the SCI12.SEMR.ABCS is 1.

Note 1. To use the base clock, set the SCI12.SCR.TE bit to 1.

Note 2. The base clock divided by 2 is the filter clock when the SEMR.BGDM bit is 1 and the SMR.CKS[1:0] bits are 00b.

41.2.32 Control Register 3 (CR3)

Address(es): SCI12.CR3 0008 B324h

	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	SDST
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	SDST	Start Frame Detection Start	0: Detection of Start Frame is not performed. 1: Detection of Start Frame is performed.	R/W
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

SDST Bit (Start Frame Detection Start)

Detection of a Start Frame begins when this bit is set to 1. The bit is read as 0.

41.2.33 Port Control Register (PCR)

Address(es): SCI12.PCR 0008 B325h

	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	SHARPS	—	—	RDXPS	TXDXPS
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	TXDXPS	TXDX12 Signal Polarity Select	0: The polarity of TXDX12 signal is not inverted for output. 1: The polarity of TXDX12 signal is inverted for output.	R/W
b1	RDXPS	RDX12 Signal Polarity Select	0: The polarity of RDX12 signal is not inverted for input. 1: The polarity of RDX12 signal is inverted for input.	R/W
b3, b2	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b4	SHARPS	TXDX12/RDX12 Pin Multiplexing Select	0: The TXDX12 and RDX12 pins are independent. 1: The TXDX12 and RDX12 signals are multiplexed on the same pin.	R/W
b7 to b5	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

SHARPS Bit (TXDX12/RDX12 Pin Multiplexing Select)

When this bit is set to 1, the TXDX12 and RDX12 signals are multiplexed on the same pin so that half-duplex communications become possible.

41.2.34 Interrupt Control Register (ICR)

Address(es): SCI12.ICR 0008 B326h

b7	b6	b5	b4	b3	b2	b1	b0
—	—	AEDIE	BCDIE	PIBDIE	CF1MIE	CF0MIE	BFDIE
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	BFDIE	Break Field Low Width Detected Interrupt Enable	0: Interrupts on detection of the low width for a Break Field are disabled. 1: Interrupts on detection of the low width for a Break Field are enabled.	R/W
b1	CF0MIE	Control Field 0 Match Detected Interrupt Enable	0: Interrupts on detection of a match with Control Field 0 are disabled. 1: Interrupts on detection of a match with Control Field 0 are enabled.	R/W
b2	CF1MIE	Control Field 1 Match Detected Interrupt Enable	0: Interrupts on detection of a match with Control Field 1 are disabled. 1: Interrupts on detection of a match with Control Field 1 are enabled.	R/W
b3	PIBDIE	Priority Interrupt Bit Detected Interrupt Enable	0: Interrupts on detection of the priority interrupt bit are disabled. 1: Interrupts on detection of the priority interrupt bit are enabled.	R/W
b4	BCDIE	Bus Collision Detected Interrupt Enable	0: Interrupts on detection of a bus collision are disabled. 1: Interrupts on detection of a bus collision are enabled.	R/W
b5	AEDIE	Valid Edge Detected Interrupt Enable	0: Interrupts on detection of a valid edge are disabled. 1: Interrupts on detection of a valid edge are enabled.	R/W
b7, b6	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

41.2.35 Status Register (STR)

Address(es): SCI12.STR 0008 B327h

b7	b6	b5	b4	b3	b2	b1	b0
—	—	AEDF	BCDF	PIBDF	CF1MF	CF0MF	BFDF
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	BFDF	Break Field Low Width Detection Flag	[Setting conditions] • Detection of the low width for a Break Field • Completion of the output of the low width for a Break Field • Underflow of the timer [Clearing condition] • Writing 1 to the STCR.BFDCL bit	R
b1	CF0MF	Control Field 0 Match Flag	[Setting condition] • A match between the value received in Control Field 0 and the set value. [Clearing condition] • Writing 1 to the STCR.CF0MCL bit	R
b2	CF1MF	Control Field 1 Match Flag	[Setting condition] • A match between the data received in Control Field 1 and the set values. [Clearing condition] • Writing 1 to the STCR.CF1MCL bit	R
b3	PIBDF	Priority Interrupt Bit Detection Flag	[Setting condition] • Detection of the priority interrupt bit [Clearing condition] • Writing 1 to the STCR.PIBDCL bit	R
b4	BCDF	Bus Collision Detected Flag	[Setting condition] • Detection of the bus collision [Clearing condition] • Writing 1 to the STCR.BCDCL bit	R
b5	AEDF	Valid Edge Detection Flag	[Setting condition] • Detection of a valid edge [Clearing condition] • Writing 1 to the STCR.AEDCL bit	R
b7, b6	—	Reserved	These bits are read as 0. The write value should be 0.	R

41.2.36 Status Clear Register (STCR)

Address(es): SCI12.STCR 0008 B328h

b7	b6	b5	b4	b3	b2	b1	b0
—	—	AEDCL	BCDCL	PIBDC L	CF1MC L	CF0MC L	BFDCL
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	BFDCL	BFDF Clear	Setting this bit to 1 clears the STR.BFDF flag. This bit is read as 0.	R/W
b1	CF0MCL	CF0MF Clear	Setting this bit to 1 clears the STR.CF0MF flag. This bit is read as 0.	R/W
b2	CF1MCL	CF1MF Clear	Setting this bit to 1 clears the STR.CF1MF flag. This bit is read as 0.	R/W
b3	PIBDCL	PIBDF Clear	Setting this bit to 1 clears the STR.PIBDF flag. This bit is read as 0.	R/W
b4	BCDCL	BCDF Clear	Setting this bit to 1 clears the STR.BCDF flag. This bit is read as 0.	R/W
b5	AEDCL	AEDF Clear	Setting this bit to 1 clears the STR.AEDF flag. This bit is read as 0.	R/W
b7, b6	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

41.2.37 Control Field 0 Data Register (CF0DR)

Address(es): SCI12.CF0DR 0008 B329h

b7	b6	b5	b4	b3	b2	b1	b0
0	0	0	0	0	0	0	0

Value after reset:

The CF0DR register is an 8-bit readable and writable register that holds a value for comparison with Control Field 0.

41.2.38 Control Field 0 Compare Enable Register (CF0CR)

Address(es): SCI12.CF0CR 0008 B32Ah

b7	b6	b5	b4	b3	b2	b1	b0
CF0CE7	CF0CE6	CF0CE5	CF0CE4	CF0CE3	CF0CE2	CF0CE1	CF0CE0
Value after reset: 0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	CF0CE0	Control Field 0 Bit 0 Compare Enable	0: Comparison with bit 0 of Control Field 0 is disabled. 1: Comparison with bit 0 of Control Field 0 is enabled.	R/W
b1	CF0CE1	Control Field 0 Bit 1 Compare Enable	0: Comparison with bit 1 of Control Field 0 is disabled. 1: Comparison with bit 1 of Control Field 0 is enabled.	R/W
b2	CF0CE2	Control Field 0 Bit 2 Compare Enable	0: Comparison with bit 2 of Control Field 0 is disabled. 1: Comparison with bit 2 of Control Field 0 is enabled.	R/W
b3	CF0CE3	Control Field 0 Bit 3 Compare Enable	0: Comparison with bit 3 of Control Field 0 is disabled. 1: Comparison with bit 3 of Control Field 0 is enabled.	R/W
b4	CF0CE4	Control Field 0 Bit 4 Compare Enable	0: Comparison with bit 4 of Control Field 0 is disabled. 1: Comparison with bit 4 of Control Field 0 is enabled.	R/W
b5	CF0CE5	Control Field 0 Bit 5 Compare Enable	0: Comparison with bit 5 of Control Field 0 is disabled. 1: Comparison with bit 5 of Control Field 0 is enabled.	R/W
b6	CF0CE6	Control Field 0 Bit 6 Compare Enable	0: Comparison with bit 6 of Control Field 0 is disabled. 1: Comparison with bit 6 of Control Field 0 is enabled.	R/W
b7	CF0CE7	Control Field 0 Bit 7 Compare Enable	0: Comparison with bit 7 of Control Field 0 is disabled. 1: Comparison with bit 7 of Control Field 0 is enabled.	R/W

41.2.39 Control Field 0 Receive Data Register (CF0RR)

Address(es): SCI12.CF0RR 0008 B32Bh

b7	b6	b5	b4	b3	b2	b1	b0
Value after reset: 0	0	0	0	0	0	0	0

CF0RR is a readable register that holds the value received in Control Field 0.

41.2.40 Primary Control Field 1 Data Register (PCF1DR)

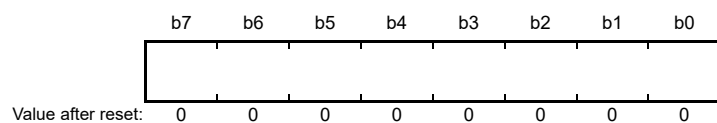
Address(es): SCI12.PCF1DR 0008 B32Ch

b7	b6	b5	b4	b3	b2	b1	b0
Value after reset: 0	0	0	0	0	0	0	0

PCF1DR is an 8-bit readable and writable register that holds the 8-bit primary value for comparison with Control Field 1.

41.2.41 Secondary Control Field 1 Data Register (SCF1DR)

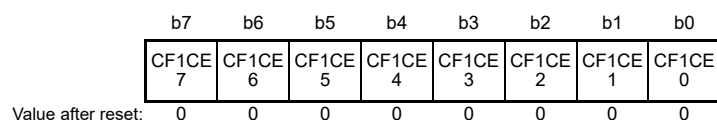
Address(es): SCI12.SCF1DR 0008 B32Dh



PCF1DR is an 8-bit readable and writable register that holds the 8-bit secondary value for comparison with Control Field 1.

41.2.42 Control Field 1 Compare Enable Register (CF1CR)

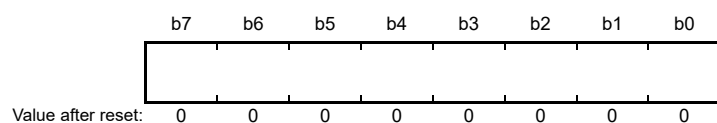
Address(es): SCI12.CF1CR 0008 B32Eh



Bit	Symbol	Bit Name	Description	R/W
b0	CF1CE0	Control Field 1 Bit 0 Compare Enable	0: Comparison with bit 0 of Control Field 1 is disabled. 1: Comparison with bit 0 of Control Field 1 is enabled.	R/W
b1	CF1CE1	Control Field 1 Bit 1 Compare Enable	0: Comparison with bit 1 of Control Field 1 is disabled. 1: Comparison with bit 1 of Control Field 1 is enabled.	R/W
b2	CF1CE2	Control Field 1 Bit 2 Compare Enable	0: Comparison with bit 2 of Control Field 1 is disabled. 1: Comparison with bit 2 of Control Field 1 is enabled.	R/W
b3	CF1CE3	Control Field 1 Bit 3 Compare Enable	0: Comparison with bit 3 of Control Field 1 is disabled. 1: Comparison with bit 3 of Control Field 1 is enabled.	R/W
b4	CF1CE4	Control Field 1 Bit 4 Compare Enable	0: Comparison with bit 4 of Control Field 1 is disabled. 1: Comparison with bit 4 of Control Field 1 is enabled.	R/W
b5	CF1CE5	Control Field 1 Bit 5 Compare Enable	0: Comparison with bit 5 of Control Field 1 is disabled. 1: Comparison with bit 5 of Control Field 1 is enabled.	R/W
b6	CF1CE6	Control Field 1 Bit 6 Compare Enable	0: Comparison with bit 6 of Control Field 1 is disabled. 1: Comparison with bit 6 of Control Field 1 is enabled.	R/W
b7	CF1CE7	Control Field 1 Bit 7 Compare Enable	0: Comparison with bit 7 of Control Field 1 is disabled. 1: Comparison with bit 7 of Control Field 1 is enabled.	R/W

41.2.43 Control Field 1 Receive Data Register (CF1RR)

Address(es): SCI12.CF1RR 0008 B32Fh



CF1RR is a readable register that holds the value received in Control Field 1.

41.2.44 Timer Control Register (TCR)

Address(es): SCI12.TCR 0008 B330h

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	—	TCST
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	TCST	Timer Count Start	0: Stops the timer counting 1: Starts the timer counting	R/W
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

41.2.45 Timer Mode Register (TMR)

Address(es): SCI12.TMR 0008 B331h

b7	b6	b5	b4	b3	b2	b1	b0
—	TCSS[2:0]			TWRC	—	TOMS[1:0]	
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b1, b0	TOMS[1:0]	Timer Operating Mode Select*1	b1 b0 0 0: Timer mode 0 1: Break Field low width determination mode 1 0: Break Field low width output mode 1 1: Setting prohibited	R/W
b2	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b3	TWRC	Counter Write Control	0: Data is written to the reload register and counter 1: Data is written to the reload register only	R/W
b6 to b4	TCSS[2:0]	Timer Count Clock Source Select*1	b6 b4 0 0 0: PCLK 0 0 1: PCLK/2 0 1 0: PCLK/4 0 1 1: PCLK/8 1 0 0: PCLK/16 1 0 1: PCLK/32 1 1 0: PCLK/64 1 1 1: PCLK/128	R/W
b7	—	Reserved	This bit is read as 0. The write value should be 0.	R/W

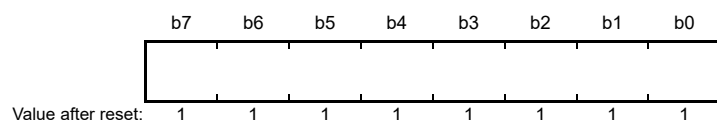
Note 1. Rewrite the TOMS[1:0] and TCSS[2:0] bits only when the timer is stopped (TCST = 0).

TWRC Bit (Counter Write Control)

This bit determines whether a value written to the TPRES or TCNT register is written to the reload register only or is written to both the reload register and the counter.

41.2.46 Timer Prescaler Register (TPRE)

Address(es): SCI12.TPRE 0008 B332h

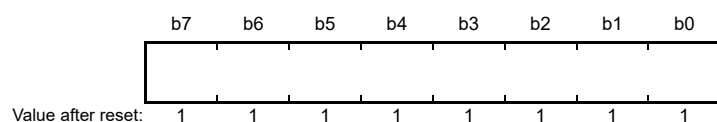


TPRE consists of an 8-bit reload register, a read buffer, and a counter, each of which has FFh as its initial value. The counter counts down in synchronization with the counter clock selected by the TMR.TCSS[2:0] bits, and is reloaded with the value in the reload register when it underflows. Underflows of this register provide the clock source to drive counting by the TCNT register. The reload register and read buffer are allocated to the same address. Data is written to the reload register in writing, and the counter value that has been transferred to the read buffer is returned in reading.

It takes one PCLK cycle to load a value from the reload register to the counter.

41.2.47 Timer Count Register (TCNT)

Address(es): SCI12.TCNT 0008 B333h



TCNT consists of an 8-bit reload register, a read buffer, and a counter, each of which has FFh as its initial value. This down-counter counts underflows of the TPRE register until the TCNT register underflows, and is then reloaded with the value from the reload register. The reload register and read buffer are allocated to the same address. Data is written to the reload register in writing, and the counter value that has been transferred to the read buffer is returned in reading.

It takes one PCLK cycle to load a value from the reload register to the counter.

41.3 Operation in Asynchronous Mode

Figure 41.6 shows the general format for asynchronous serial communications.

One frame consists of a start bit (low level), transmit/receive data, a parity bit, and stop bits (high level).

In asynchronous serial communications, the communications line is usually held in the mark state (high level).

The SCI monitors the communications line. When the SCI detects a low, it regards that as a start bit and starts serial communication.

Inside the SCI, the transmitter and receiver are independent units, enabling full-duplex communications. Both the transmitter and the receiver also have a double-buffered structure, so that data can be read or written during transmission or reception, enabling continuous data transmission and reception. As SCI7 to SCI11 incorporate a FIFO, data can be received and transmitted more efficiently.

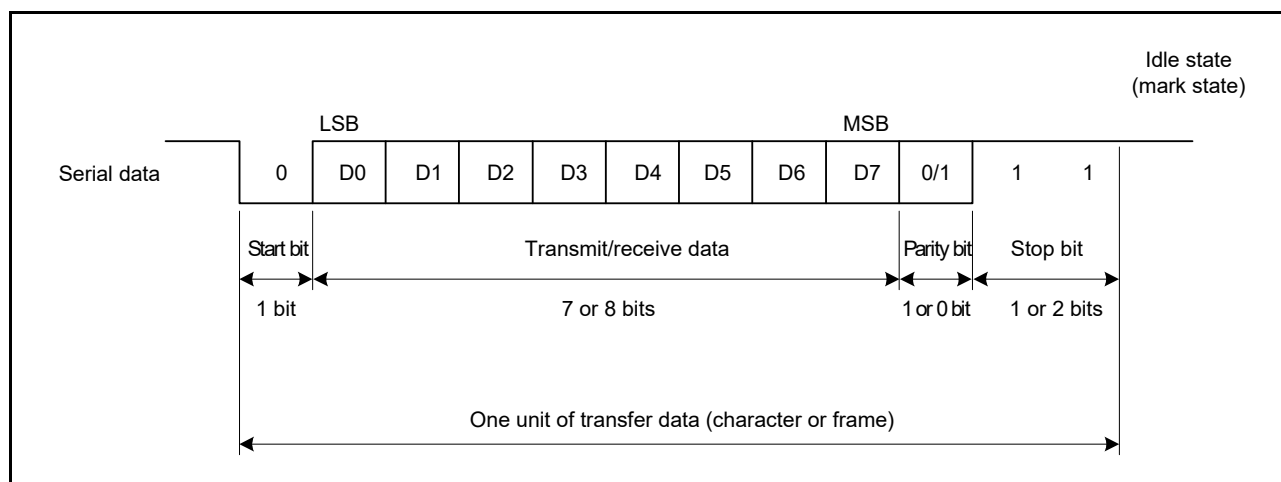


Figure 41.6 Data Format in Asynchronous Serial Communications
(Example with 8-Bit Data, Parity, 2 Stop Bits)

41.3.1 Serial Data Transfer Format

Table 41.32 lists the serial data transfer formats that can be used in asynchronous mode.

Any of 18 transfer formats can be selected according to the SMR and SCMR setting. For details of multi-processor function, refer to section 41.4, Multi-Processor Communications Function.

Table 41.32 Serial Transfer Formats (Asynchronous Mode)

SCMR Setting SMR Setting					Serial Transfer Format and Frame Length												
CHR1	CHR	PE	MP	STOP	1	2	3	4	5	6	7	8	9	10	11	12	13
0	0	0	0	0	S	9-bit data									STOP		
0	0	0	0	1	S	9-bit data									STOP	STOP	
0	0	1	0	0	S	9-bit data									P	STOP	
0	0	1	0	1	S	9-bit data									P	STOP	STOP
1	0	0	0	0	S	8-bit data								STOP			
1	0	0	0	1	S	8-bit data								STOP	STOP		
1	0	1	0	0	S	8-bit data								P	STOP		
1	0	1	0	1	S	8-bit data								P	STOP	STOP	
1	1	0	0	0	S	7-bit data							STOP				
1	1	0	0	1	S	7-bit data							STOP	STOP			
1	1	1	0	0	S	7-bit data							P	STOP			
1	1	1	0	1	S	7-bit data							P	STOP	STOP		
0	0	—	1	0	S	9-bit data									MPB	STOP	
0	0	—	1	1	S	9-bit data									MPB	STOP	STOP
1	0	—	1	0	S	8-bit data								MPB	STOP		
1	0	—	1	1	S	8-bit data								MPB	STOP	STOP	
1	1	—	1	0	S	7-bit data							MPB	STOP			
1	1	—	1	1	S	7-bit data							MPB	STOP	STOP		

S: Start bit
 STOP: Stop bit
 P: Parity bit
 MPB: Multi-processor bit

41.3.2 Receive Data Sampling Timing and Reception Margin in Asynchronous Mode

In asynchronous mode, the SCI operates on a base clock with a frequency of 16 times*1 the bit rate.

In reception, the SCI samples the falling edge of the start bit using the base clock, and performs internal synchronization. Since receive data is sampled at the rising edge of the 8th pulse*1 of the base clock, data is latched at the middle of each bit, as shown in Figure 41.7. Thus the reception margin in asynchronous mode is determined by formula (1) below.

$$M = \left| \left(0.5 - \frac{1}{2N} \right) - (L - 0.5)F - \frac{|D - 0.5|}{N}(1 + F) \right| \times 100 (\%) \quad \cdots \text{Formula (1)}$$

M: Reception margin

N: Ratio of bit rate to clock

(N = 16 when SEMR.ABCSE = 0 and SEMR.ABCS = 0, N = 8 when SEMR.ABCSE = 0 and SEMR.ABCS = 1, N = 6 when SEMR.ABCSE = 1)

D: Duty cycle of clock (D = 0.5 to 1.0)

L: Frame length (L = 9 to 13)

F: Absolute value of clock frequency deviation

Assuming values of F = 0 and D = 0.5 in formula (1), the reception margin is determined by the formula below.

$$M = \{0.5 - 1/(2 \times 16)\} \times 100 (\%) = 46.875 (\%)$$

However, this is only the computed value, and a margin of 20% to 30% should be allowed in system design.

Note 1. These are values when the ABCSE and ABCS bits in the SEMR register are 0. When the ABCSE bit is 0 and the ABCS bit is 1, a frequency of 8 times the bit rate is used as a base clock and receive data is sampled at the rising edge of the 4th pulse of the base clock. When the ABCSE bit is 1, a frequency of 6 times the bit rate is used as a base clock and receive data is sampled at the rising edge of the 3rd pulse of the base clock.

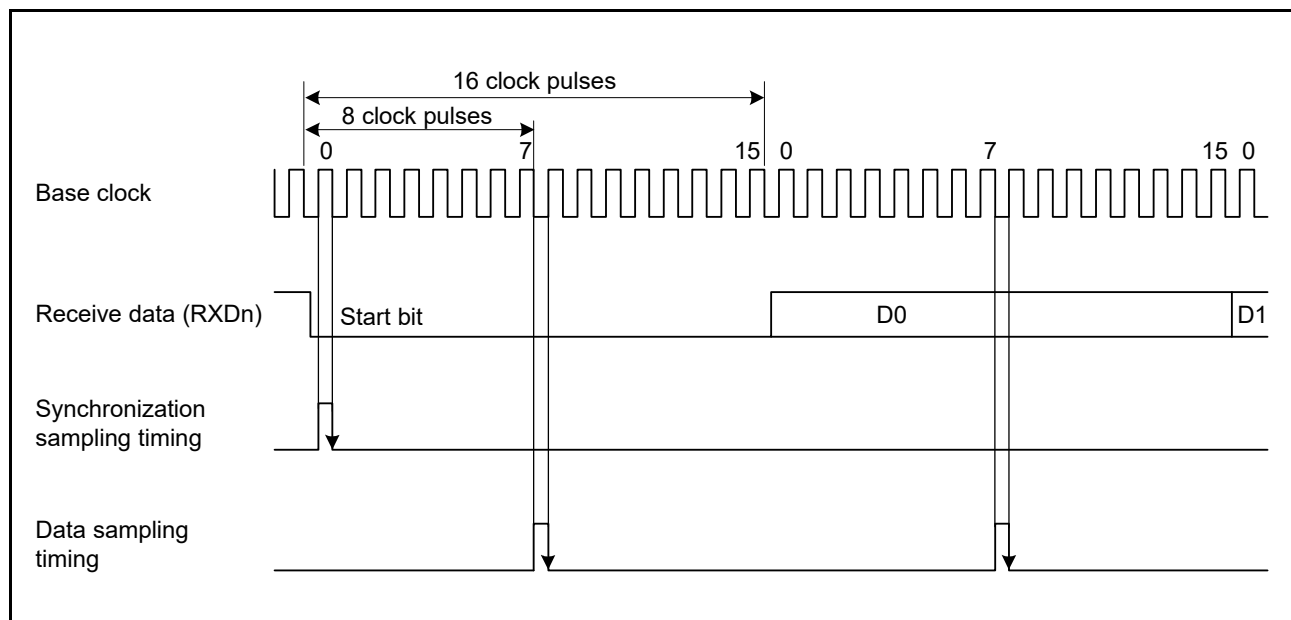


Figure 41.7 Receive Data Sampling Timing in Asynchronous Mode

41.3.3 Clock

Either an internal clock generated by the on-chip baud rate generator or an external clock input to the SCKn pin can be selected as the SCI's transfer clock, according to the setting of the SMR.CM bit and the SCR.CKE[1:0] bits.

When an external clock is input to the SCKn pin, the clock frequency should be 16 times the bit rate (when SEMR.ABCS bit = 0) and 8 times the bit rate (when SEMR.ABCS bit = 1). In addition, when an external clock is specified, the base clock of TMR0 and TMR1 can be selected by the SCIn.SEMR.ACS0 bit (n = 5, 6, 12).

When the SCI is operated on an internal clock, the clock can be output from the SCKn pin. The frequency of the clock output in this case is equal to the bit rate, and the phase is such that the rising edge of the clock is in the middle of the transmit data, as shown in Figure 41.8.

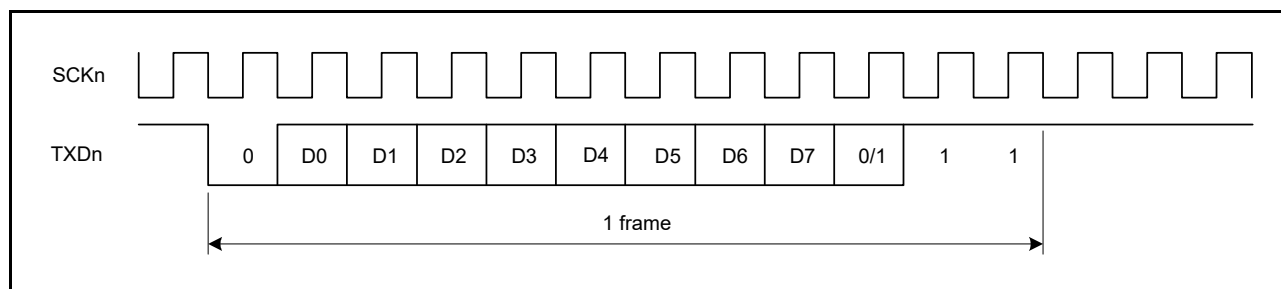


Figure 41.8 Phase Relationship between Output Clock and Transmit Data
(Asynchronous Mode: SMR.CHR = 0, PE = 1, MP = 0, STOP = 1)

41.3.4 Double-Speed Mode and Divide-by-6 Mode

The output clock frequency of the on-chip baud rate generator is doubled by setting the SEMR.BGDM bit to 1, enabling high-speed communication at a doubled bit rate. If the SEMR.ABCS bit is set to 1 under the above condition, the number of base clock cycles changes from 16 to 8, so the bit rate becomes four times faster than the initial state.

When the SEMR.ABCSE bit is set to 1, the number of base clock pulses in 1-bit period becomes 6 and the period of the base clock becomes 1/2, where the bit rate becomes 16/3 times faster compared to a case that all of the ABCS, BDGM, and ABCSE bits are set to 0.

As shown by Formula (1) in section 41.3.2, Receive Data Sampling Timing and Reception Margin in Asynchronous Mode, setting the SEMR.ABCS bit to 1 changes the number of cycles to 8, and the sampling interval becomes longer. This causes the reception margin to decrease. Therefore, setting the SEMR.BGDM bit to 1 and the SEMR.ABCS bit to 0 is recommended instead of setting the SEMR.BGDM bit to 0 and the SEMR.ABCS bit to 1 for high-speed operation at a doubled bit rate.

41.3.5 CTS and RTS Functions

The CTS function is the use of input on the CTSn# pin in transmission control. Setting the SPMR.CTSE bit to 1 enables the CTS function. When the CTS function is enabled, placing the low level on the CTSn# pin causes transmission to start.

Applying the high level to the CTS# pin while transmission is in progress does not affect transmission of the current frame, which continues.

In the RTS function, by using the function of output on the RTSn# pin, a low level is output when reception becomes possible. Conditions for output of the low and high level are shown below.

(a) SCI0 to SCI6 and SCI12, and SCI7 to SCI11 when FIFO is Disabled

[Conditions for low-level output]

When the following conditions are all satisfied.

- The SCR.RE bit is 1.
- Reception is not in progress.
- There are no received data yet to be read.
- The ORER, FER, and PER flags in the SSR register are all 0.

[Condition for high-level output]

When the conditions for low-level output are not satisfied.

(b) SCI7 to SCI11 When FIFO is Enabled

[Conditions for low-level output]

When the following conditions are all satisfied.

- The SCR.RE bit is 1.
- The number of data stored in the receive FIFO is less than the threshold (FCR.RTRG[3:0]).
- The SSRFIFO.ORER flag is 0.

[Condition for high-level output]

When the conditions for low-level output are not satisfied.

Note that either one of CTS and RTS can be selected.

41.3.6 Data Match Detection

The data match detection function is available in asynchronous mode for SCI0 to SCI11.

When the DCCR.DCME bit is set to 1, the received data is compared with the contents of the CDR.CMPD[8:0] bits*1.

Upon a match of the value, a receive data full interrupt (RXI) request is generated.

When the SMR.MP bit is 0, all received data is compared.

When the SMR.MP bit is set to 1 and if the DCCR.IDSEL bit is 1, only the data in which the multiprocessor bit is 1 is compared and data in which the bit is 0 is ignored. When the DCCR.IDSEL bit is 0, all of the received data is compared regardless of the value of the multiprocessor bit.

The received data is not stored or the flag is not updated until the received data matches with the value of the CDR.CMPD[8:0] bits. When the data is matched, the DCCR.DCME bit is automatically set to 0 and the DCMF flag becomes 1. At this time, if the DCCR.IDSEL bit is 1, the SCR.MPIE bit is automatically set to 0. In addition, an receive data full interrupt (RXI) request is generated when the SCR.RIE bit is 1.

When a framing error is found in the matched data, the DCCR.DFER flag becomes 1. When a parity error is found in the matched data, the DCCR.DPER flag becomes 1. The received data matched with the value of the CDR.CMPD[8:0] bits is not stored in the received buffer or the SSR.RDRF flag (the SSRFIFO.RDF flag when the FCR.FM bit is 1) does not become 1.

After a data match is detected and the DCCR.DCME bit is set to 0, data is normally received.

When the DCCR.DFER or DCCR.DPER flag is 1, a data match is not detected. These flags should be set to 0 before enabling the data match detection function.

Note 1. Only the portion that corresponds to the character length specified by the SMR.CHR and SCMR.CHR1 bits is compared.

Figure 41.9 and Figure 41.10 show examples of data match detection.

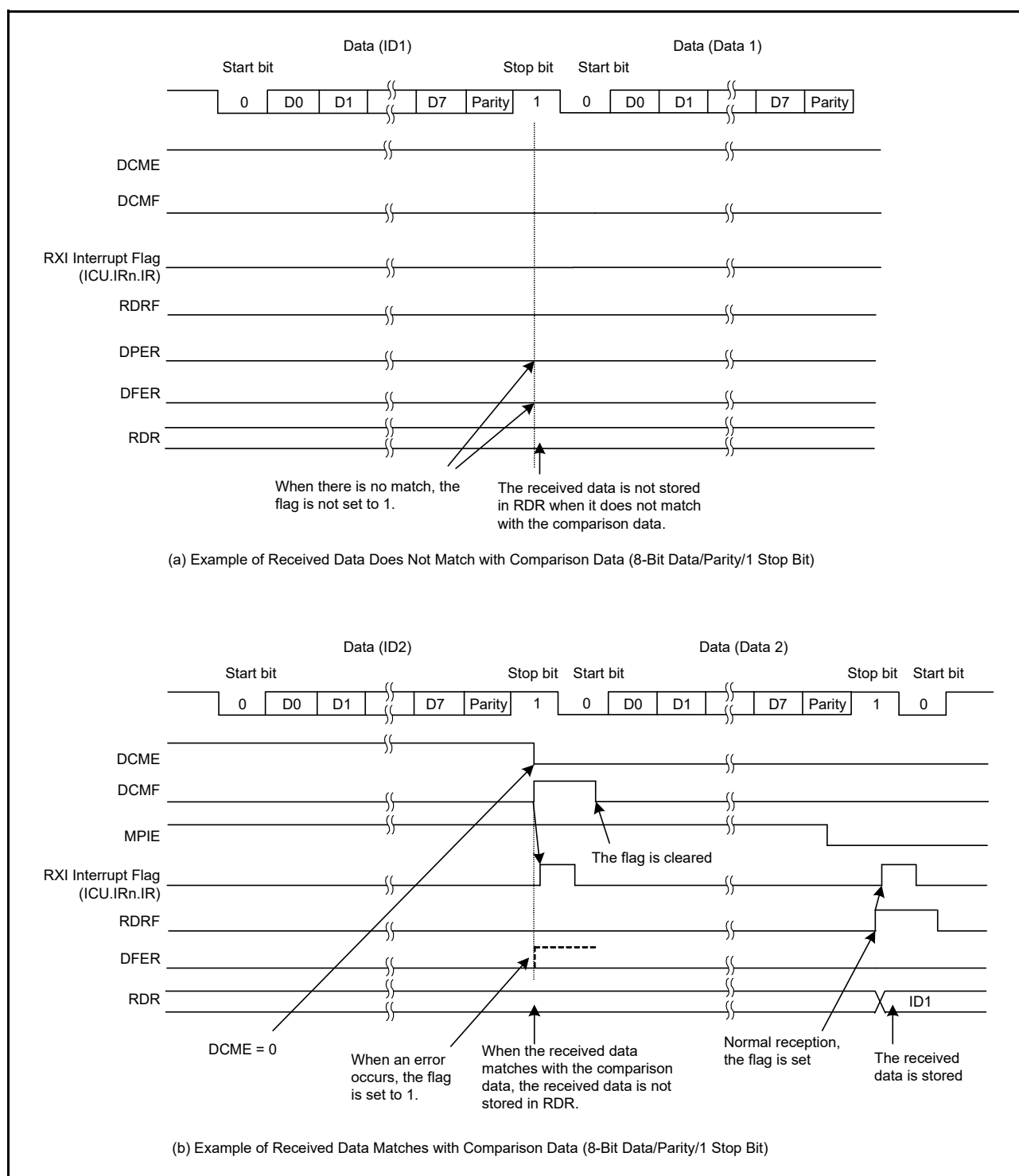


Figure 41.9 Example of Data Match Detection (1) Non Multi-Processor Mode

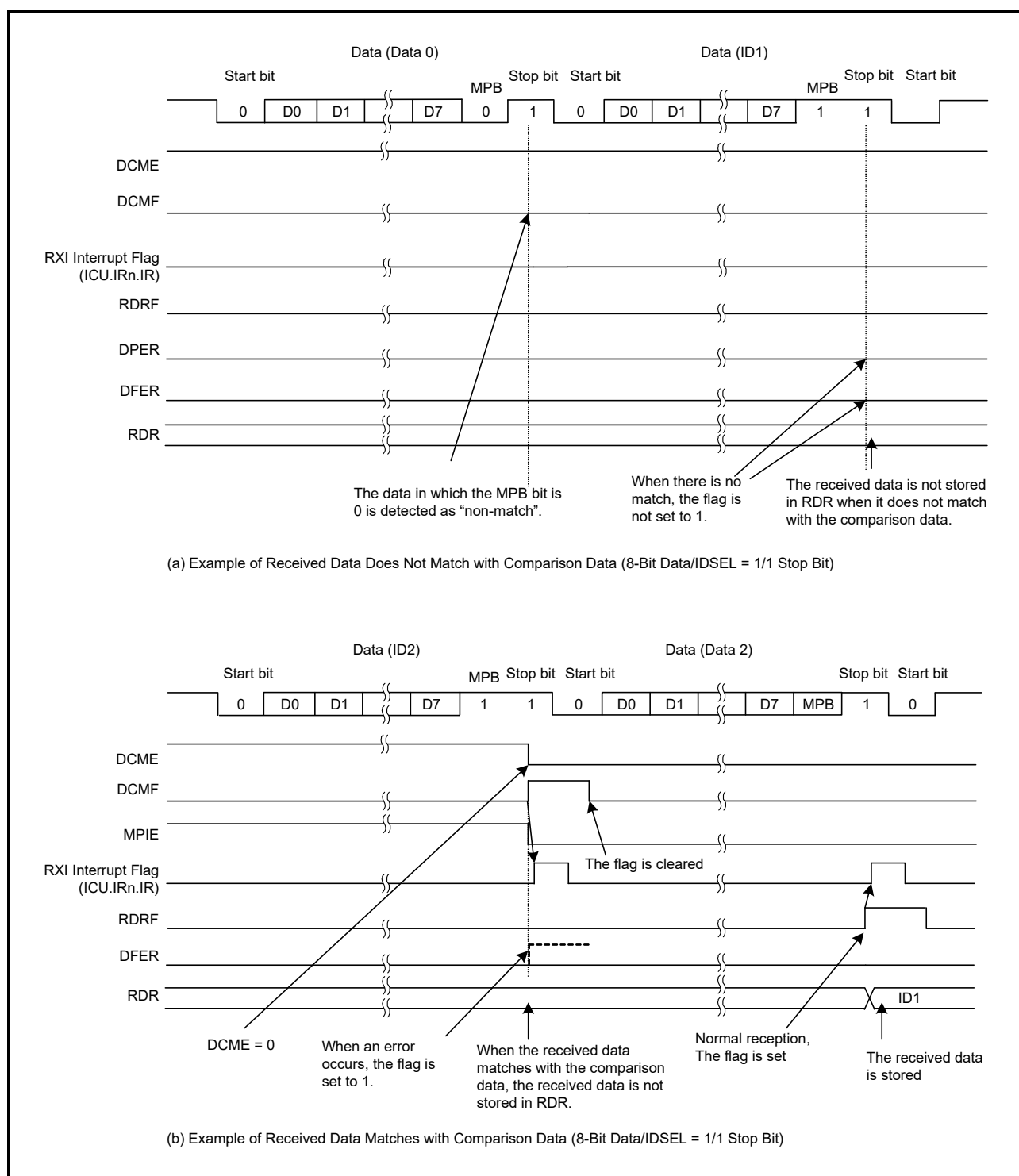


Figure 41.10 Example of Data Match Detection (2) Non Multi-Processor Mode

41.3.7 SCI Initialization (Asynchronous Mode)

Before transmitting and receiving data, start by writing the initial value 00h to the SCR register and then continue through the procedure for SCI given in Figure 41.11 or Figure 41.12. Whenever the operating mode or transfer format is changed, the SCR register must be initialized before the change is made.

When the external clock is used in asynchronous mode, ensure that the clock signal is supplied even during initialization. Note that setting the SCR.RE bit to 0 initializes neither the ORER, FER, PER, and RDRF flags in the SSR register nor registers RDR, RDRH, and RDRL. Note also that the SSRFIFO.TEND flag does not become 1 even when the SCR.TE bit is set to 0.

In addition, note that setting bits TIE, TE, and TEIE in the SCR register to 1 simultaneously leads to the generation of a transmit end interrupt (TEI) request before the generation of a transmit data empty interrupt (TXI) request.

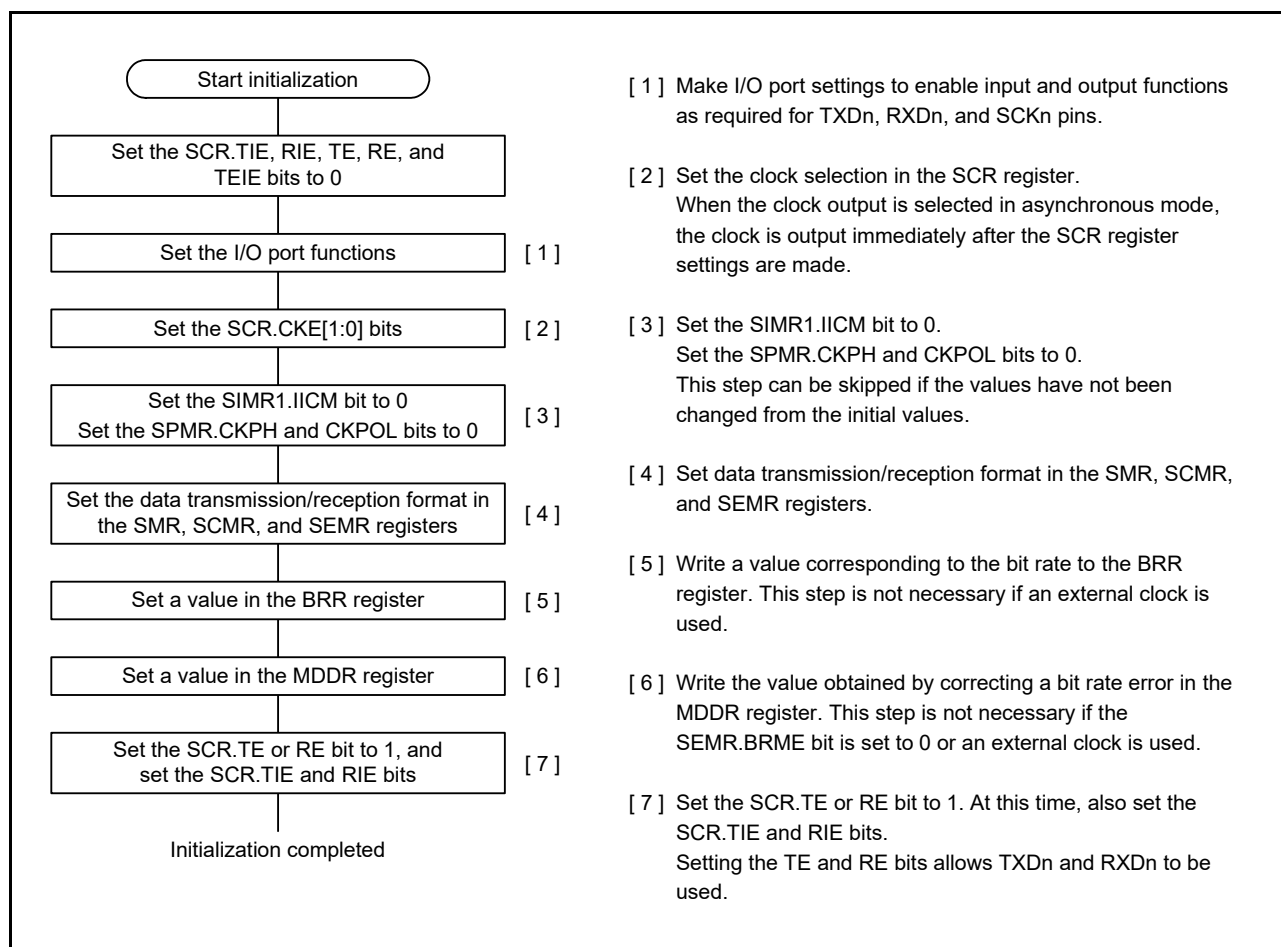


Figure 41.11 Sample SCI Initialization Flowchart (Asynchronous Mode) (for SCI0 to SCI6 and SCI12)

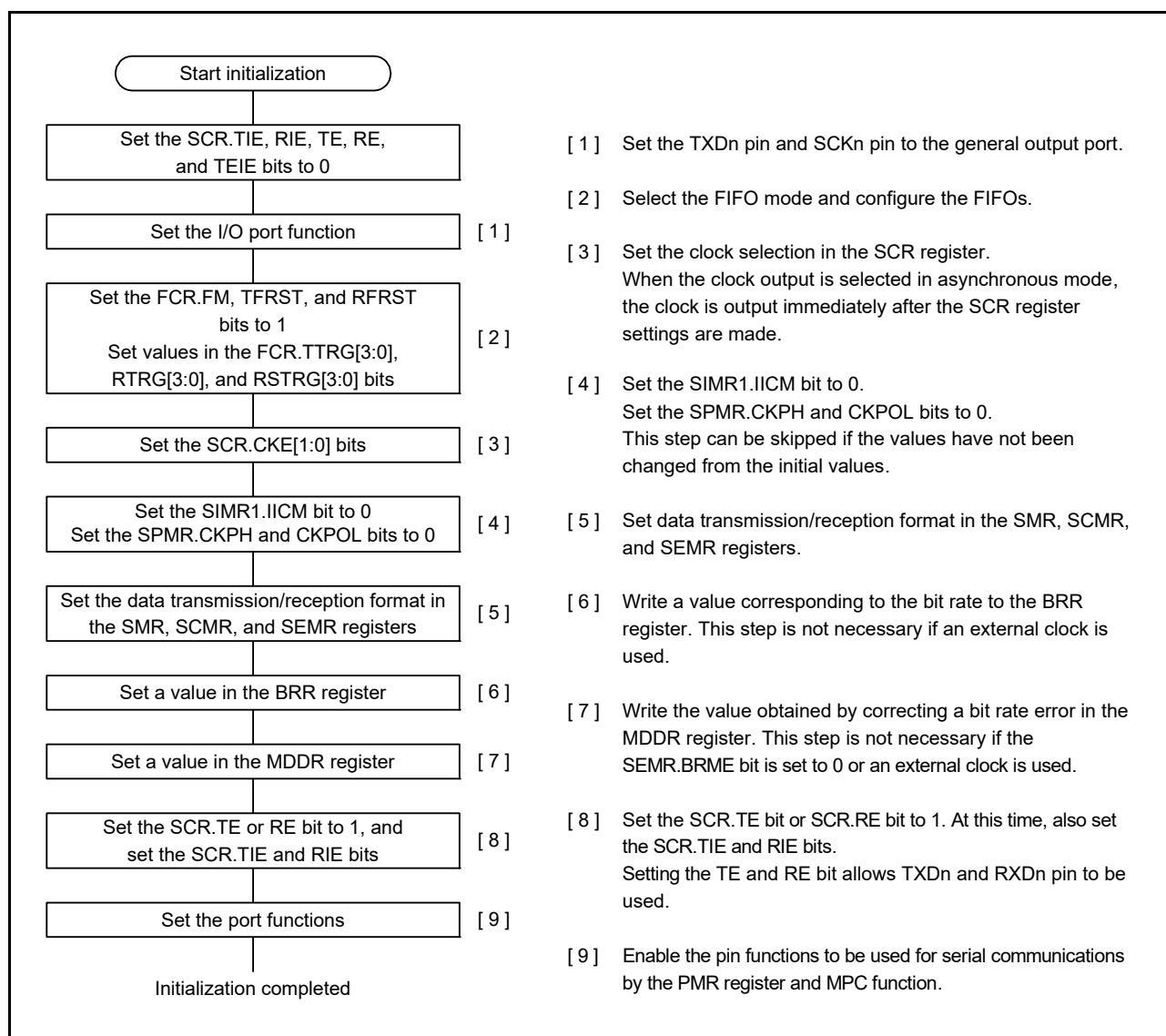


Figure 41.12 Sample SCI Initialization Flowchart (Asynchronous Mode, FIFO enabled) (for SCI7 to SCI11)

Figure 41.13 shows an example of data transmission when the SCI is set to asynchronous mode according to the flow described in Figure 41.11 after a reset. When the pin function is set to the TXD pin, it is still high-impedance because the SCR.TE bit is 0. When the transmit data is written after setting the TE bit to 1, a data transmission starts. After the TE bit is set to 1, one frame of high is output from TXD pin (internal wait time) and then the data transmission starts.

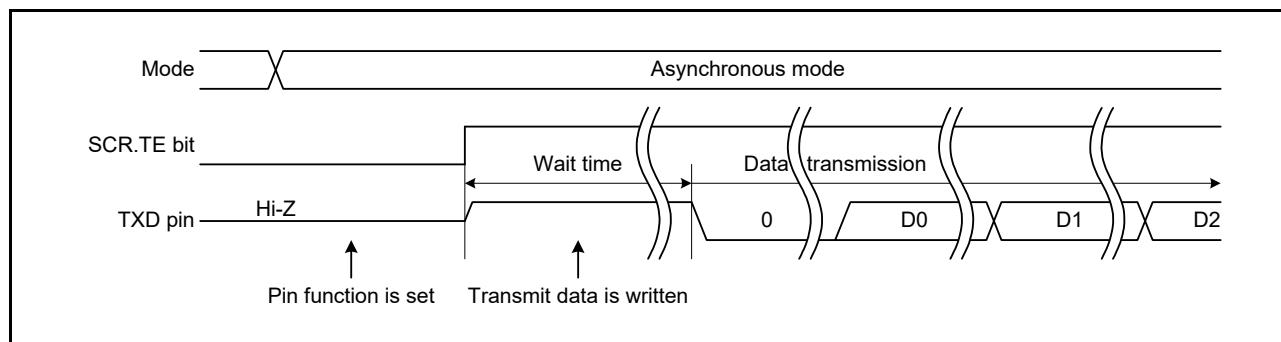


Figure 41.13 Example of Data Transmission Timing in Asynchronous Mode

41.3.8 Serial Data Transmission (Asynchronous Mode)

(1) SCI0 to SCI6 and SCI12, and SCI7 to SCI11 when FIFO is Disabled

Figure 41.14 to Figure 41.16 show an example of the operation for serial transmission in asynchronous mode. In serial transmission, the SCI operates as described below.

1. The SCI transfers data from the TDR register*¹ to the TSR register when data is written to the TDR register*¹ in the TXI interrupt handling routine. The TXI interrupt request at the beginning of transmission is generated when the SCR.TE bit is set to 1 after the SCR.TIE bit is set to 1 or when these 2 bits are set to 1 simultaneously by a single instruction.
2. Transmission starts after the SPMR.CTSE bit is set to 0 (CTS function is disabled) and a low level on the CTSn# pin causes data transfer from the TDR register*¹ to the TSR register. If the SCR.TIE bit is 1 at this time, a TXI interrupt request is generated. Continuous transmission is obtainable by writing the next transmit data to the TDR register*¹ in the TXI interrupt handling routine before transmission of the current transmit data is completed. When TEI interrupt requests are in use, set the SCR.TIE bit to 0 (a TXI interrupt request is disabled) and the SCR.TEIE bit to 1 (a TEI interrupt request is enabled) after the last of the data to be transmitted are written to the TDR register*¹, *² from the handling routine for TXI requests.
3. Data is sent from the TXDn pin in the following order: start bit, transmit data, parity bit or multi-processor bit (may be omitted depending on the format), and stop bit.
4. The SCI checks for updating of (writing to) the TDR register*³ at the time of stop bit output.
5. When the TDR register*³ is updated, setting of the SPMR.CTSE bit to 0 (CTS function is disabled) or a low level input on the CTSn# pin cause the next transfer of the next transmit data from the TDR register*¹ to the TSR register and sending of the stop bit, after which serial transmission of the next frame starts.
6. If the TDR register*³ is not updated, the SCI sets the SSR.TEND flag to 1, sends the stop bit, and then outputs high to put the line in the mark state. If the SCR.TEIE bit is 1 at this time, the SSR.TEND flag is set to 1 and a TEI interrupt request is generated.

Note 1. Write data not to the TDR register but to the TDRH and TDRL registers when 9-bit data length is selected.

Note 2. Write data in the order from the TDRH register to the TDRL register when 9-bit data length is selected.

Note 3. The SCI checks for updating of the TDRL register only and does not check for updating of the TDRH register when 9-bit data length is selected.

Figure 41.17 shows a sample flowchart for serial transmission in asynchronous mode.

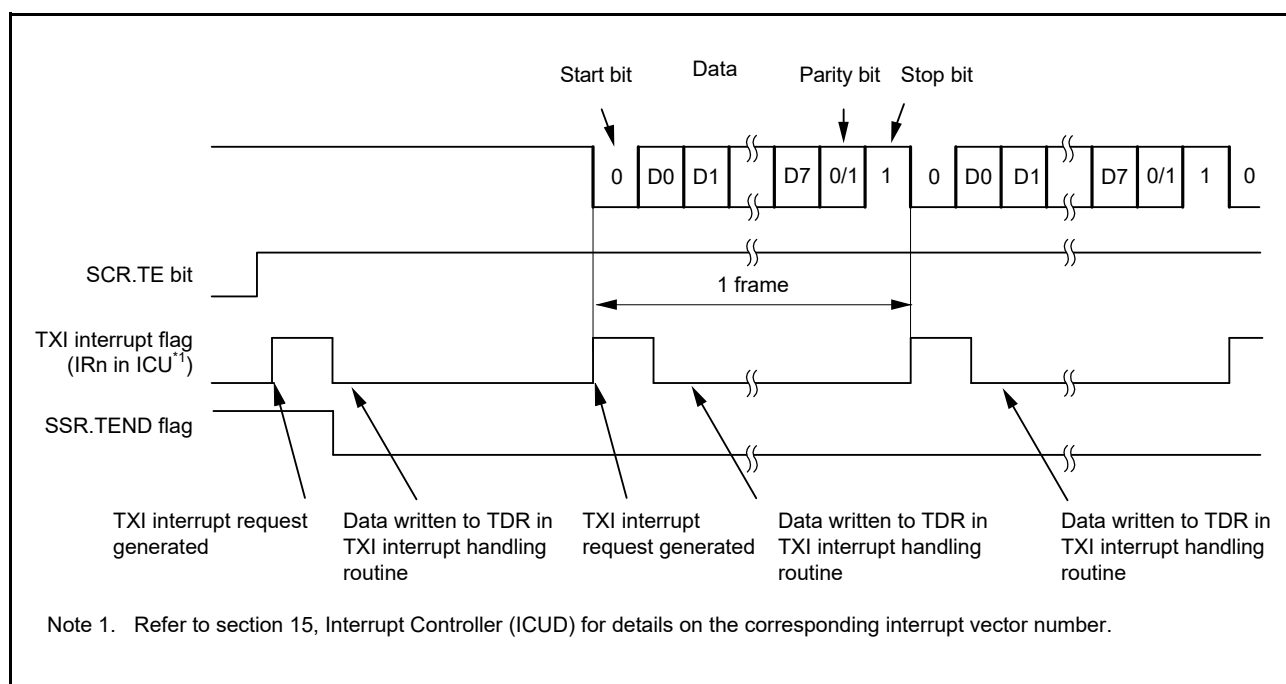


Figure 41.14 Example of Operation for Serial Transmission in Asynchronous Mode (1)
(with 8-Bit Data, Parity, 1 Stop Bit, CTS Function Not Used, at the Beginning of Transmission)

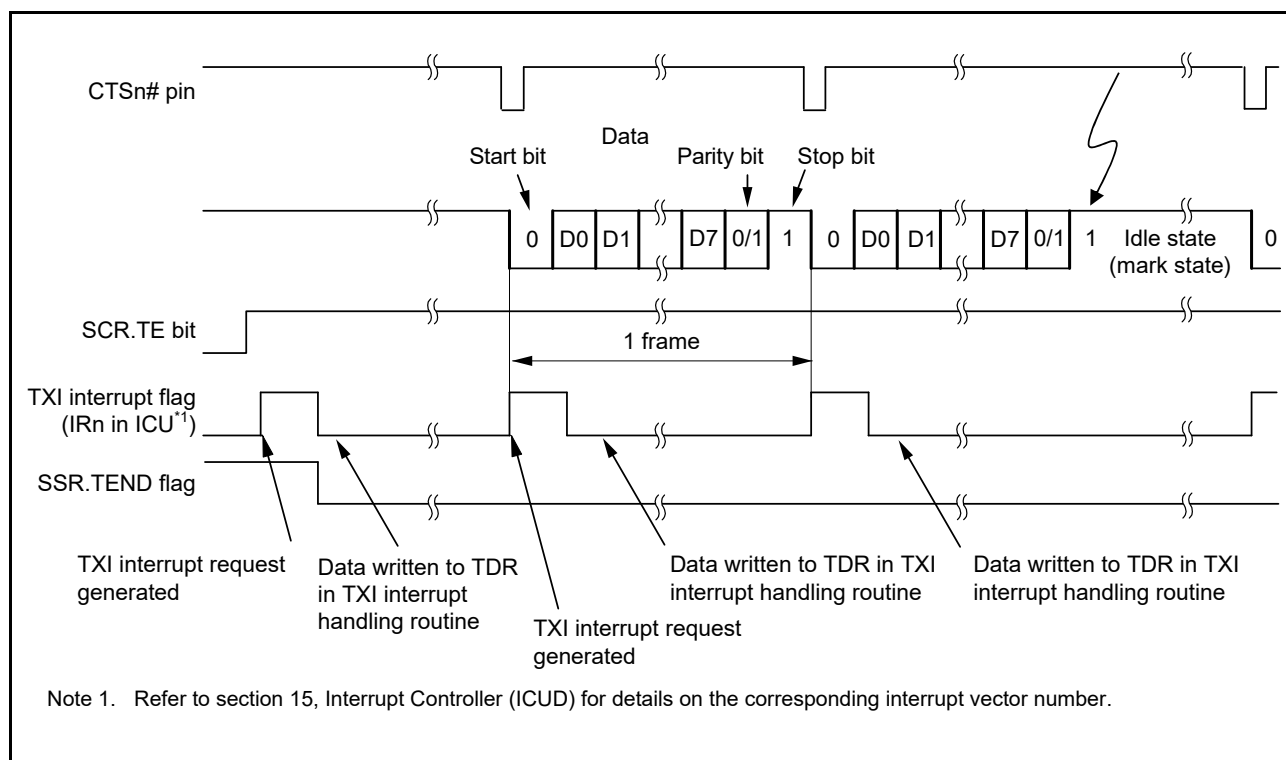


Figure 41.15 Example of Operation for Serial Transmission in Asynchronous Mode (2)
(with 8-Bit Data, Parity, 1 Stop Bit, CTS Function Used, at the Beginning of Transmission)

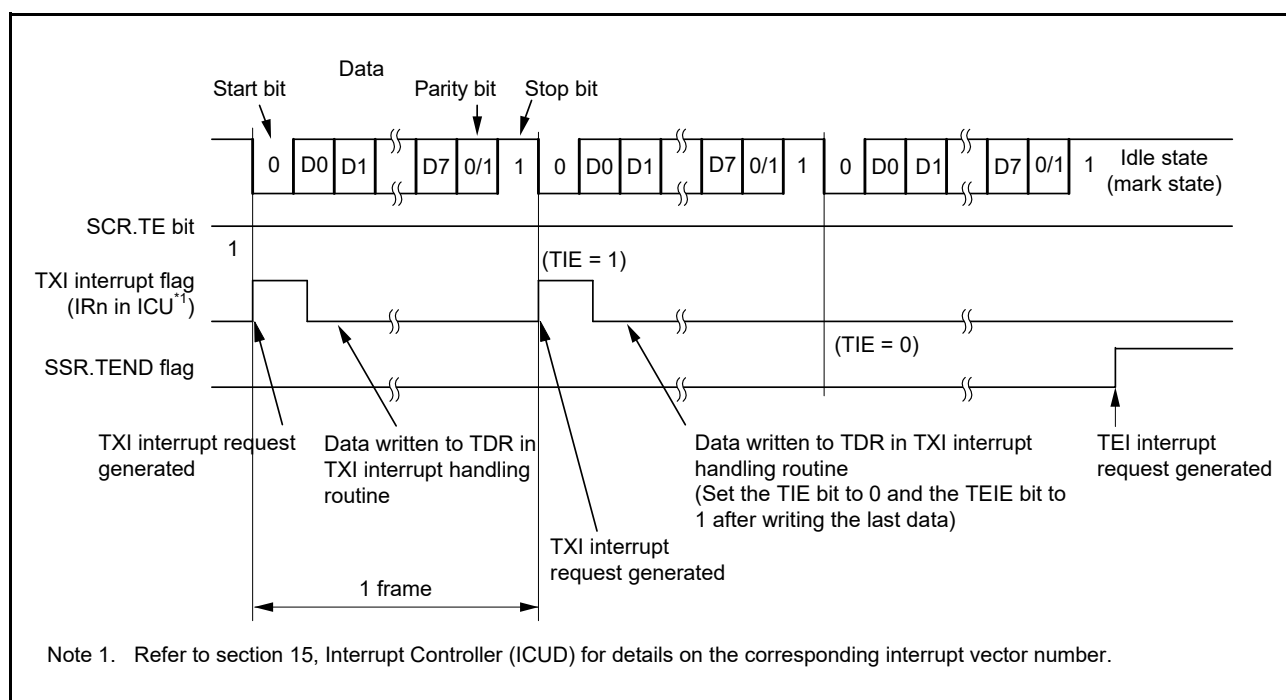


Figure 41.16 Example of Operation for Serial Transmission in Asynchronous Mode (3)
(with 8-Bit Data, Parity, 1 Stop Bit, CTS Function Not Used, from the Middle of Transmission until
Transmission Completion)

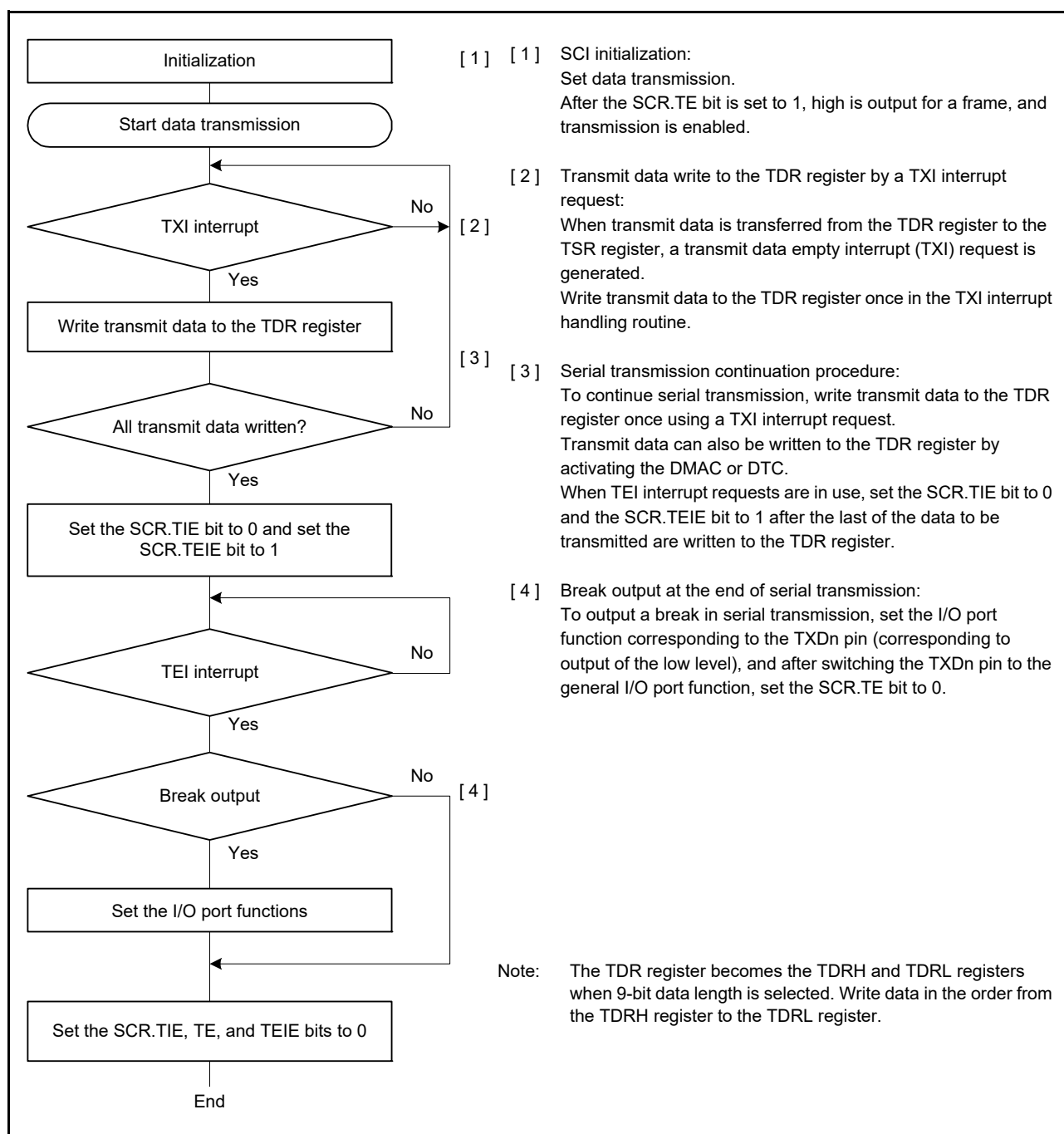


Figure 41.17 Example of Serial Transmission Flowchart in Asynchronous Mode

(2) SCI7 to SCI11 When FIFO is Enabled

The TXI interrupt request at the beginning of transmission is generated when the SCR.TE and SCR.TIE bits are set to 1 simultaneously.

Set the transmit data to the FTDR register (the FTDR.L register for 7-bit and 8-bit) instead of the TDR register. When data is transferred from the transmit FIFO to the TSR register and if the number of data stored in the transmit FIFO is less than or equal to the threshold (FCR.TTRG[3:0] bits), a transmit data empty interrupt (TXI) request is generated.

Up to 16 minus FDR.T[4:0] frames of transmit data can be set within the TXI interrupt processing routine. When the settings of all transmit data are completed, set the SSRFIFO.TDFE flag to 0.

When the transmit data is set by using the DMAC or DTC, the TDFE flag is automatically set to 0.

When sending a break, use the SPB2IO and SPB2DT bits in the SPTR register. Upon completion of the setting, a break is sent by setting the SCR.TE bit to 0.

41.3.9 Serial Data Reception (Asynchronous Mode)

(1) SCI0 to SCI6 and SCI12, and SCI7 to SCI11 when FIFO is Disabled

Figure 41.18 and Figure 41.19 show an example of the operation for serial data reception in asynchronous mode. In serial data reception, the SCI operates as described below.

1. When the value of the SCR.RE bit becomes 1, the output signal on the RTSn# pin goes to the low level.
2. When the SCI monitors the communications line and detects a start bit, it performs internal synchronization, stores receive data in the RSR register, and checks the parity bit and stop bit.
3. If an overrun error occurs, the SSR.ORER flag is set to 1. If the SCR.RIE bit is 1 at this time, an ERI interrupt request is generated. Receive data is not transferred to the RDR register*1.
4. If a parity error is detected, the SSR.PER flag is set to 1 and receive data is transferred to the RDR register*1. If the SCR.RIE bit is 1 at this time, an ERI interrupt request is generated.
5. If a framing error (when the stop bit is 0) is detected, the SSR.FER flag is set to 1 and receive data is transferred to the RDR register*1. If the SCR.RIE bit is 1 at this time, an ERI interrupt request is generated.
6. When reception finishes successfully, receive data is transferred to the RDR register*1. If the SCR.RIE bit is 1 at this time, an RXI interrupt request is generated. Continuous reception is enabled by reading the receive data transferred to the RDR register*1 in this RXI interrupt handling routine before reception of the next receive data is completed. Reading the received data that have been transferred to the RDR register*1 causes the RTSn# pin to output the low level.

Note 1. Read data not in the RDR register but in the RDRH and RDRL registers when 9-bit data length is selected.

Note 2. The SCI checks for reading of the RDRL register only and does not check for reading of the RDRH register when 9-bit data length is selected.

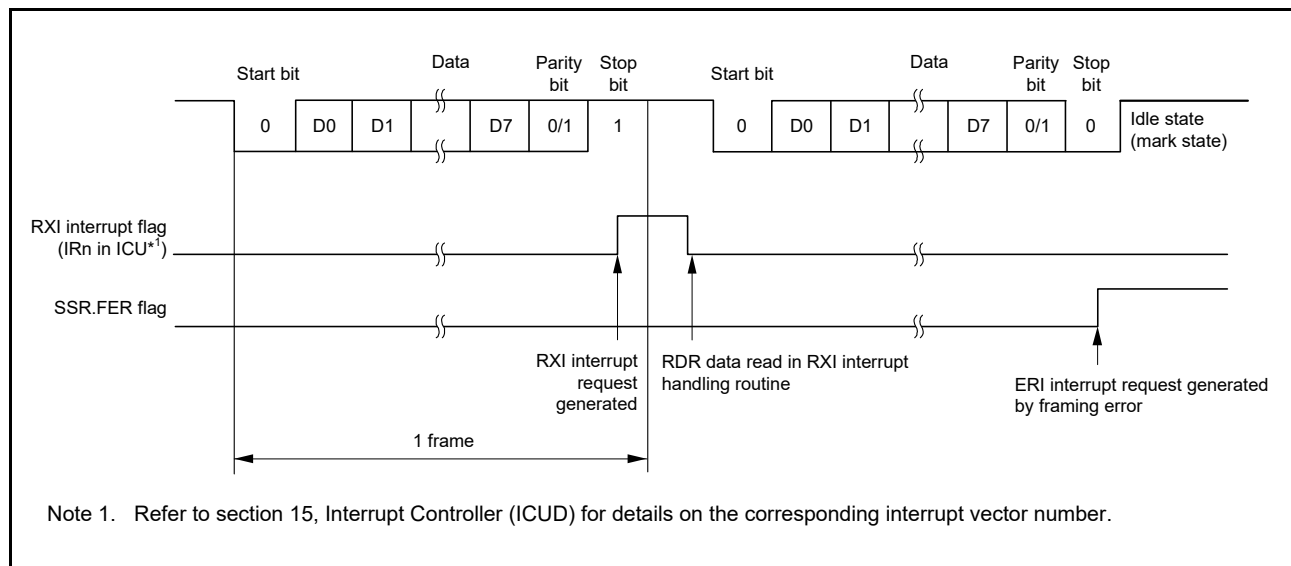
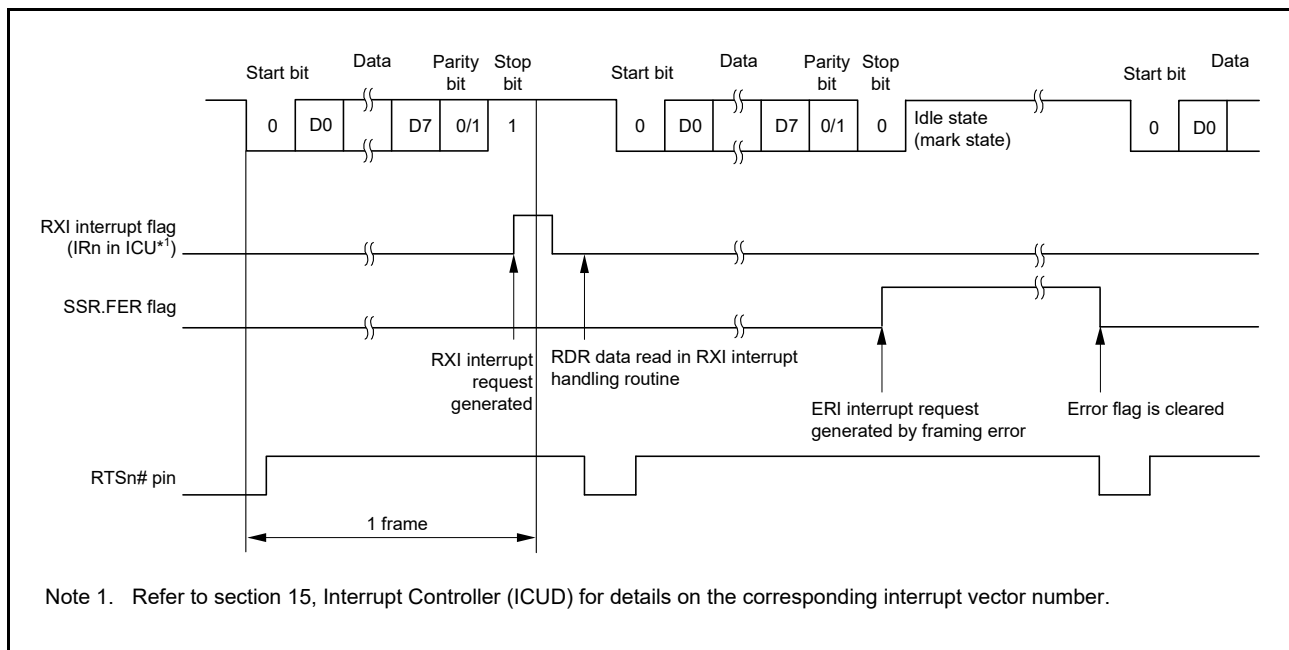


Figure 41.18 Example of SCI Operation for Serial Reception in Asynchronous Mode (1) (When RTS Function is Not Used) (Example with 8-Bit Data, Parity, 1 Stop Bit)



**Figure 41.19 Example of SCI Operation for Serial Reception in Asynchronous Mode (2)
(When RTS Function is Used) (Example with 8-Bit Data, Parity, 1 Stop Bit)**

Table 41.33 lists the states of the status flags in the SSR register and receive data handling when a receive error is detected.

If a receive error is detected, an ERI interrupt request is generated but an RXI interrupt request is not generated. Data reception cannot be resumed while the receive error flag is 1. Accordingly, set the ORER, FER, and PER flags to 0 before resuming reception. Moreover, be sure to read the RDR (or the RDRL) register during overrun error processing. When a reception is forcibly terminated by setting the SCR.RE bit to 0 during operation, read the RDR (or the RDRL) register because received data which has not yet been read may be left in the RDR (or the RDRL) register.

Figure 41.20 and Figure 41.21 show samples of flowcharts for serial data reception.

Table 41.33 Status Flags in the SSR Register and Receive Data Handling

Status Flags in the SSR Register			Receive Data	Receive Error Type
ORER	FER	PER		
1	0	0	Lost	Overrun error
0	1	0	Transferred to the RDR register*1	Framing error
0	0	1	Transferred to the RDR register*1	Parity error
1	1	0	Lost	Overrun error + framing error
1	0	1	Lost	Overrun error + parity error
0	1	1	Transferred to the RDR register*1	Framing error + parity error
1	1	1	Lost	Overrun error + framing error + parity error

Note 1. Read data not in the RDR register but in the RDRH and RDRL registers when 9-bit data length is selected.

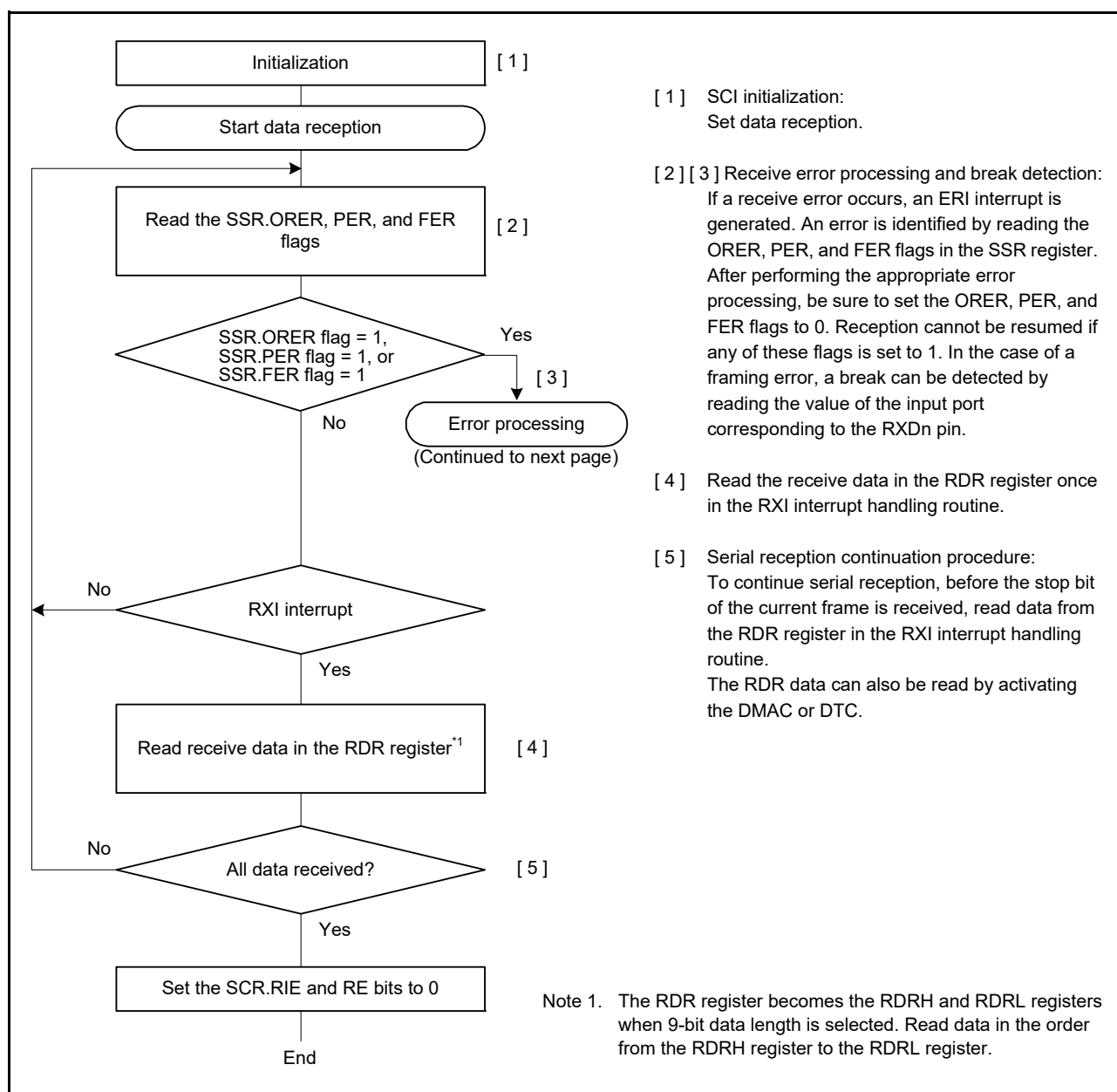


Figure 41.20 Example Flowchart of Serial Reception in Asynchronous Mode (1)

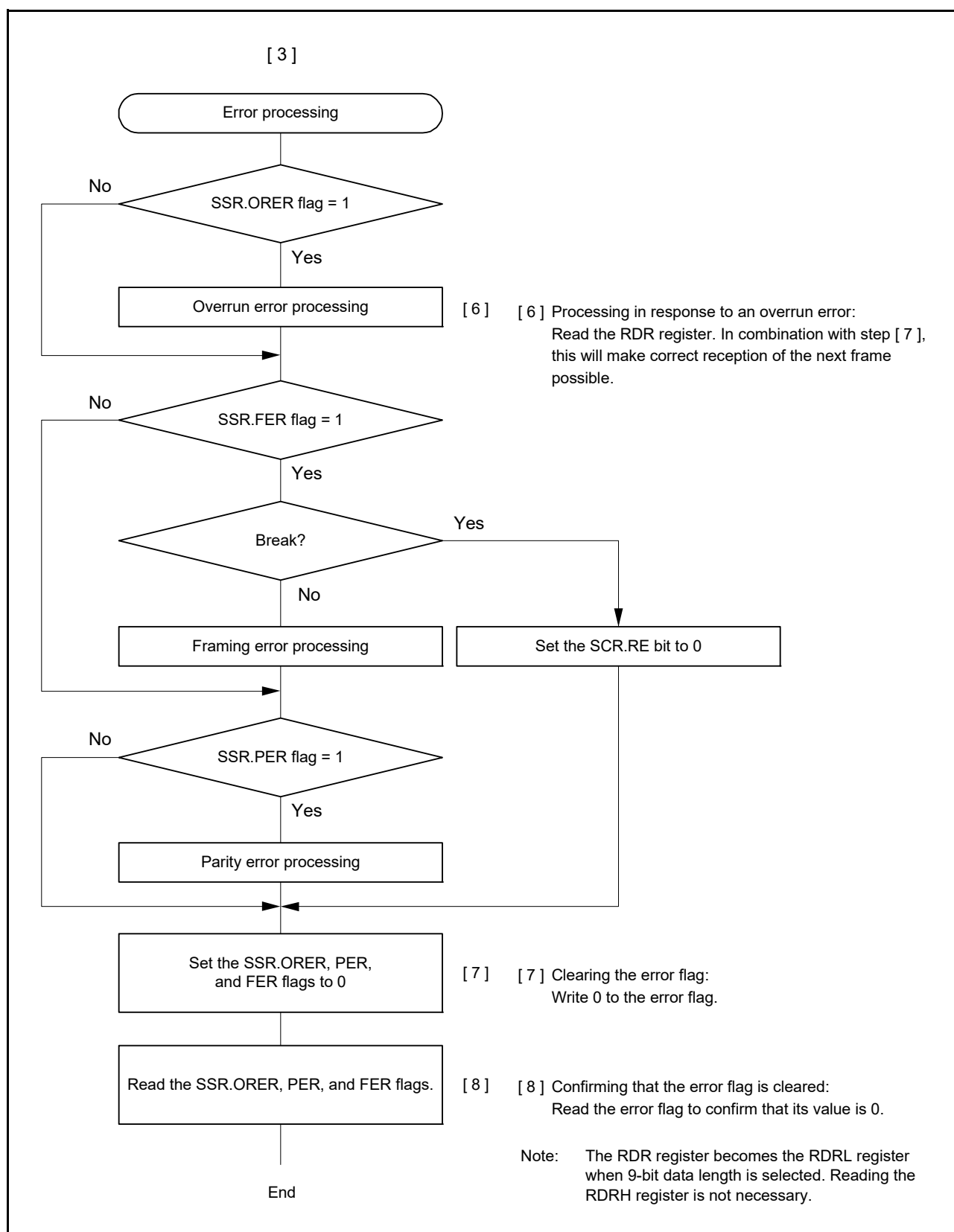


Figure 41.21 Example Flowchart of Serial Reception in Asynchronous Mode (2)

(2) SCI7 to SCI11 When FIFO is Enabled

The received data and status flag are read from the FRDR register instead of the RDR register. When reading in byte units, read the FRDR.H register and then the FRDR.L register in this order. Reading the FRDR.L register updates the FER, PER, and RDATA[8:0] bits in the FRDR register. The value of the RDF, ORER, and DR flags in the FRDR register is the same as that of the SSRFIFO register.

When receiving serial data, the SCI operates as follows.

1. If the SCR.RE bit is set to 1 and the RTSn# function is in use, the SCI drives the RTSn# pin low.
2. When the SCI monitors the communications line and detects a start bit, it performs internal synchronization, stores the received data to the RSR register, and checks the parity and stop bits.
3. When there is no space in the receive FIFO, an overrun error occurs. When an overrun error occurs, the SCI sets the SSRFIFO.ORER flag to 1. At this time, if the SCR.RIE bit is 1, an ERI interrupt request is generated. The received data is not transferred to the receive FIFO.
4. When a parity error is detected, the SCI transfers the received data to the receive FIFO and sets the PER flag in the receive FIFO to 1. At this time, if the SCR.RIE bit is 1, an ERI interrupt request is generated.
5. When a framing error (the stop bit is 0) is detected, the SCI transfers the received data to the receive FIFO and sets the FER flag in the receive FIFO to 1. At this time, if the SCR.RIE bit is 1, an ERI interrupt request is generated.
6. When a framing error is detected and the following one-frame data received is all 0s, the SCI stops receiving operation.
7. If the number of data stored in the receive FIFO is less than the threshold (FCR.RTRG[3:0] bits) and reception of the next frame is not completed even after 15 μ s have elapsed from the stop bit of the previously received frame, the SSRFIFO.DR flag is set to 1. At this time, if the SCR.RIE bit is 1, an RXI interrupt request (when the FCR.DRES bit is 0) or an ERI interrupt request (when the FCR.DRES bit is 1) is generated.
8. When the frame is successfully received, the SCI transfers the received data to the receive FIFO. If the number of data stored in the receive FIFO is equal to or greater than the threshold (FCR.RTRG[3:0] bits), the SCI sets the SSRFIFO.RDF flag to 1. At this time, if the SCR.RIE bit is 1, an RXI interrupt request is generated. Using the RXI interrupt processing routine allows reading of the received data in the RDRF register before an overrun error occurs, thus enabling the sequence reception of data. When the received data that have been transferred to the receive FIFO are read and the number of unread data is less than the value of the FCR.RSTRG[3:0] bits, the SCI drives the RTSn# pin low.

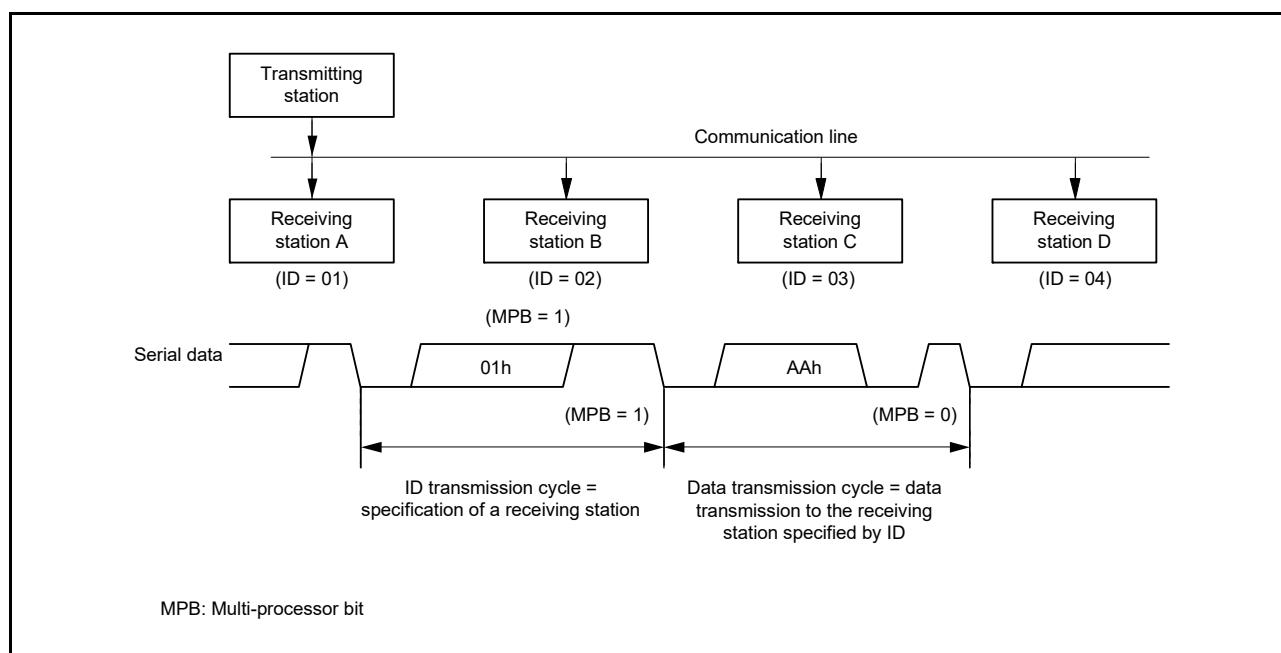
41.4 Multi-Processor Communications Function

Using the multi-processor communication functions enables to transmit and receive data by sharing a communication line between multiple processors by using asynchronous serial communication in which the multi-processor bit is added. In multi-processor communication, a unique ID code is allocated to each receiving station. Serial communication cycles consist of an ID transmission cycle to specify the receiving station and a data transmission cycle to transmit data to the specified receiving station. The multi-processor bit is used to distinguish between the ID transmission cycle and the data transmission cycle. When the multi-processor bit is set to 1, it indicates the ID transmission cycle and when the multi-processor bit is set to 0, it indicates the data transmission cycle. Figure 41.22 shows an example of communication between processors by using a multi-processor format. First, a transmitting station transmits communication data in which the multi-processor bit set to 1 is added to the ID code of the receiving station. Next, the transmitting station transmits the communication data in which the multi-processor bit set to 0 is added to the transmit data. Upon receiving the communication data in which the multi-processor bit is set to 1, the receiving station compares the received ID with the ID of the receiving station itself and if the two match, receives the communication data that is subsequently transmitted. If the received ID does not match with the ID of the receiving station, the receiving station skips the communication data until again receiving the communication data in which the multi-processor bit is set to 1.

(1) SCI0 to SCI6 and SCI12, and SCI7 to SCI11 when FIFO is Disabled

For supporting this function, the SCI provides the SCR.MPIE bit. When the MPIE bit is set to 1, transfer of receive data from the RSR register to the RDR register (the RDRH and RDRL registers when 9-bit data length is selected), detection of a receive error, and setting the respective status flags RDRF, ORER, and FER in the SSR register are disabled until reception of data in which the multi-processor bit is set to 1. Upon receiving a reception character in which the multi-processor bit is set to 1, the SSR.MPB bit is set to 1 and the SCR.MPIE bit is automatically cleared, thus returning to a normal reception operation. During this time, an RXI interrupt is generated if the SCR.RIE bit is 1.

When the multi-processor format is specified, specification of the parity bit is disabled. Apart from this, there is no difference from the operation in the normal asynchronous mode. A clock which is used for the multi-processor communication is also the same as the clock used in the normal asynchronous mode.



**Figure 41.22 An Example of Communication using the Multi-Processor Format
(Example of Transmission of Data AAh to Receiving Station A)**

(2) SCI7 to SCI11 When FIFO is Enabled

In transmitting data, the SCI uses the FTDR.MPBT bit instead of the SSR.MPBT bit. Set a value and the transmit data to the FTDR register at the same timing.

In receiving data, the SCI uses the FRDR.MPB flag instead of the SSR.MPB flag. Upon completion of data reception, when the data in the RSR register is stored to the receive FIFO, the value of the multi-processor bit is stored at the same time.

Setting the SCR.MPIE bit to 1 disables transfer of the received data from the RSR register to the receive FIFO, detection of receive errors, and settings of the RDF, ORER, or FER flag in the SSRFIFO register until the data in which the multi-processor bit has the value 1 is received.

When the reception character in which the multi-processor bit has the value 1 is received, the multi-processor bit and received data are stored in the receive FIFO and the SCR.MPIE bit is set to 0, and then the SCI returns to the normal reception operation. At this time, if the SCR.RIE bit is 1, an RXI interrupt is generated.

When the multi-processor format is specified, setting of the parity bit is disabled. Specifying a format other than that, the serial transfer format is the same as that in normal asynchronous mode.

41.4.1 Multi-Processor Serial Data Transmission

Figure 41.23 is a sample flowchart of multi-processor data transmission. In the ID transmission cycle, the ID should be transmitted with the SSR.MPBT bit set to 1. In the data transmission cycle, the data should be transmitted with the MPBT bit set to 0. The other operations are the same as the operations in asynchronous mode.

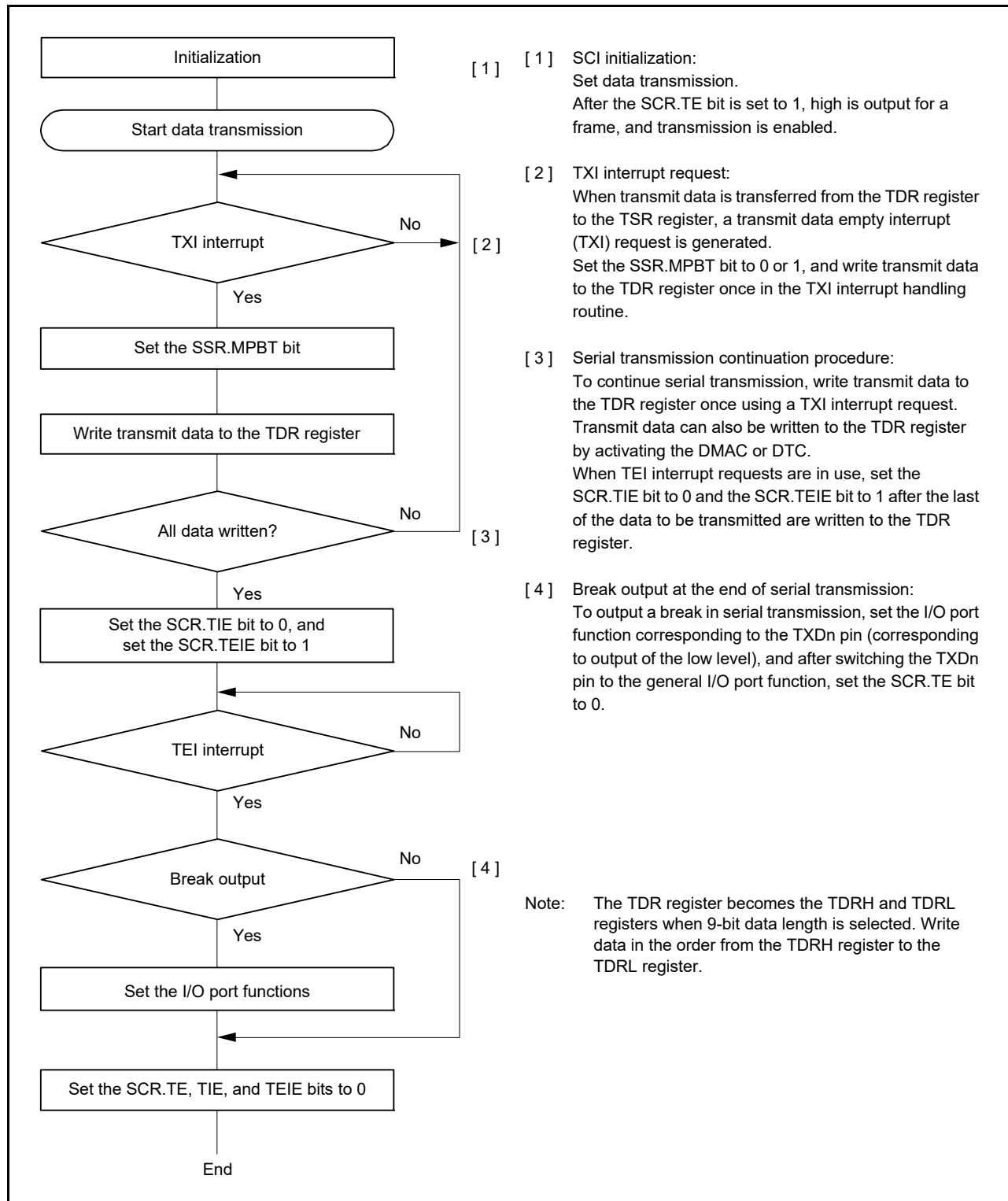


Figure 41.23 Example of Multi-Processor Serial Transmission Flowchart

41.4.2 Multi-Processor Serial Data Reception

Figure 41.25 and Figure 41.26 are sample flowcharts of multi-processor data reception. When the SCR.MPIE bit is set to 1, reading the communication data is skipped until reception of the communication data in which the multi-processor bit is set to 1. When the communication data in which the multi-processor bit is set to 1 is received, the received data is transferred to the RDR register (the RDRH and RDRL registers when 9-bit data length is selected). During this time, the RXI interrupt request is generated. The other operations are the same as the operations in asynchronous mode. Figure 41.24 is the example of operation for reception.

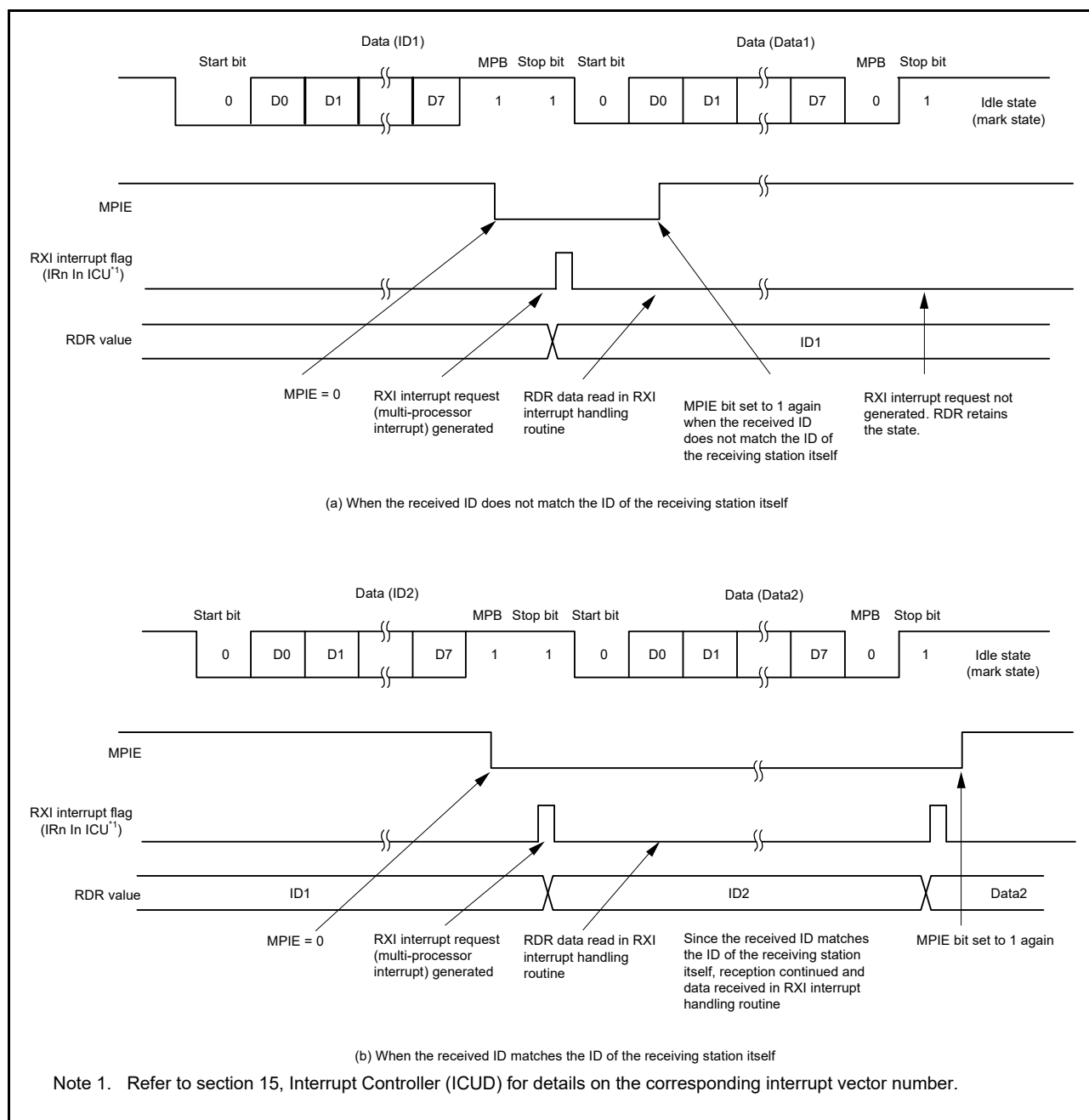


Figure 41.24 Example of SCI Reception (8-Bit Data/Multi-Processor Bit/1 Stop Bit)

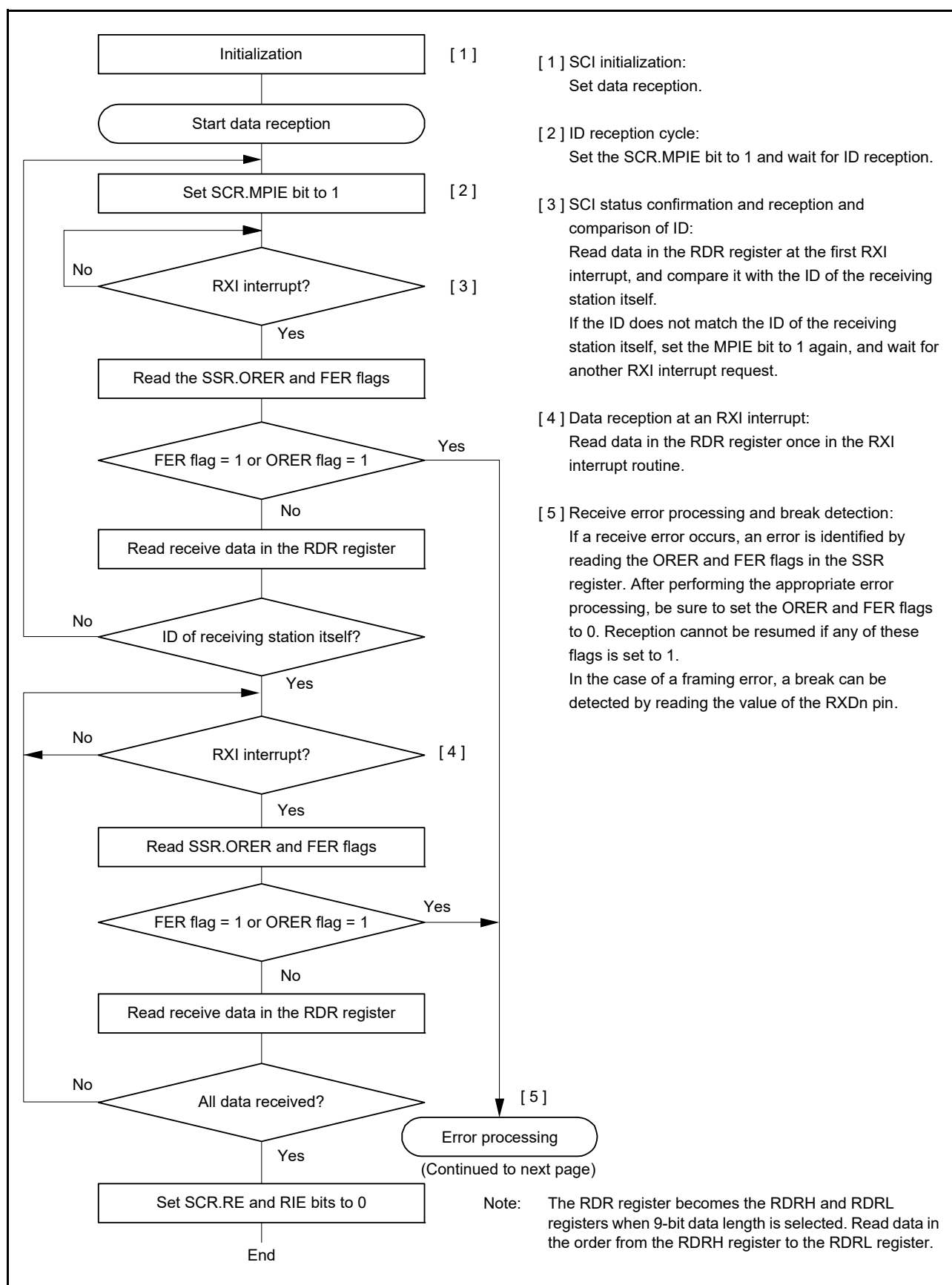


Figure 41.25 Example of Multi-Processor Serial Reception Flowchart (1)

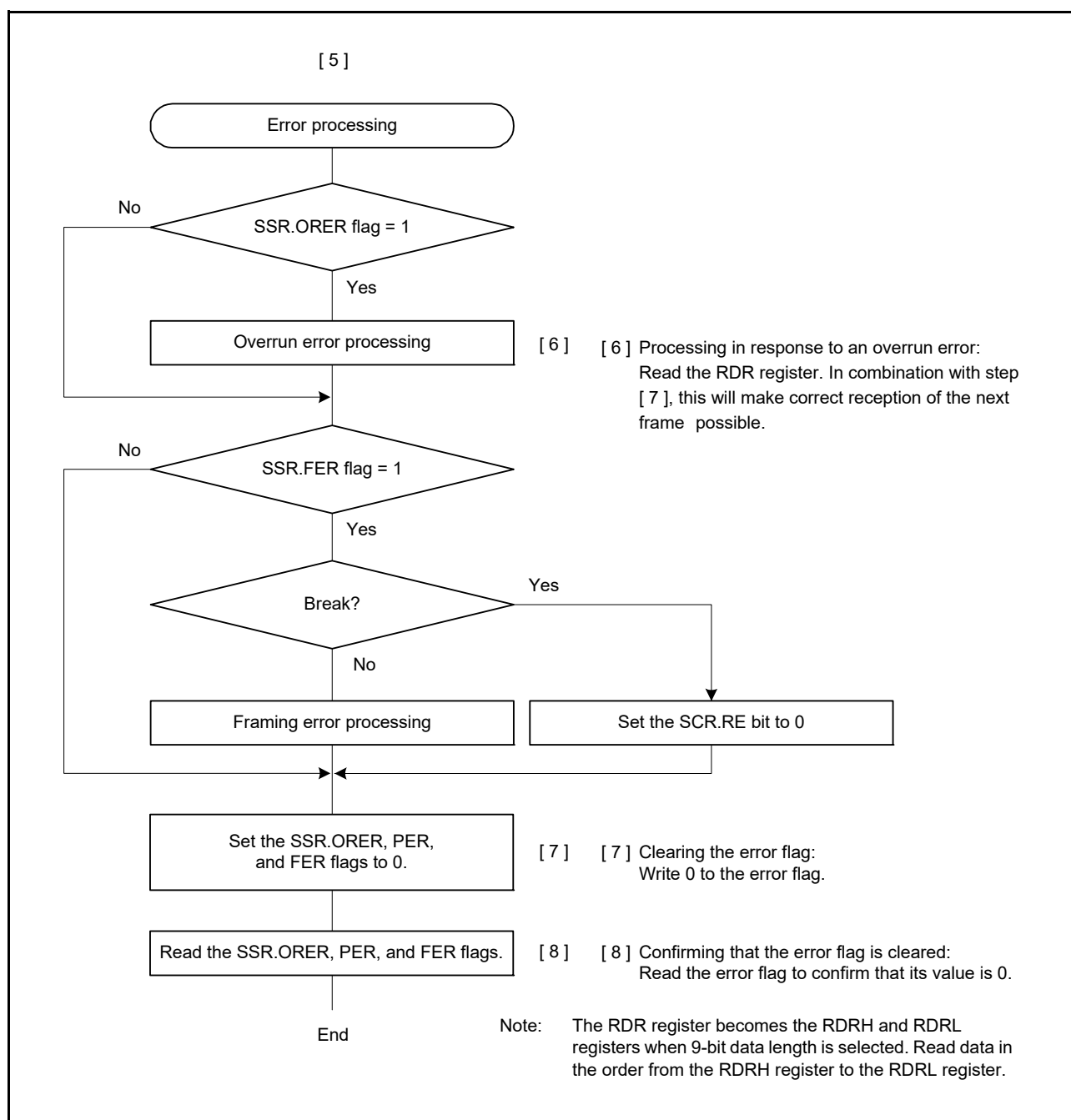


Figure 41.26 Example of Multi-Processor Serial Reception Flowchart (2)

41.5 Operation in Clock Synchronous Mode

Figure 41.27 shows the data format for clock synchronous serial data communications.

In clock synchronous mode, data is transmitted or received in synchronization with clock pulses. One character in transfer data consists of 8-bit data. In clock synchronous mode, no parity bit can be added.

In data transmission, the SCI outputs data from one falling edge of the synchronization clock to the next. In data reception, the SCI receives data in synchronization with the rising edge of the synchronization clock. After 8-bit data is output, the communication line holds the last bit output state.

Inside the SCI, the transmitter and receiver are independent units, enabling full-duplex communications by use of a common clock. Both the transmitter and the receiver also have a double-buffered structure, so that the next transmit data can be written during transmission or the previous receive data can be read during reception, enabling continuous data transfer.

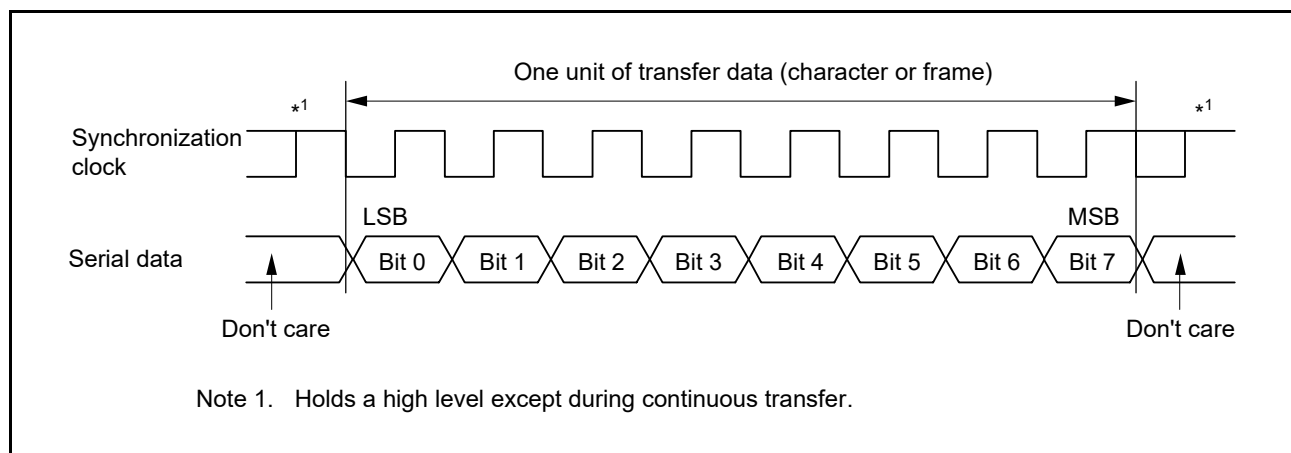


Figure 41.27 Data Format in Clock Synchronous Serial Communications (LSB First)

41.5.1 Clock

Either an internal clock generated by the on-chip baud rate generator or an external synchronization clock input at the SCKn pin can be selected, according to the setting of the SCR.CKE[1:0] bits.

When the SCI is operated on an internal clock, the synchronization clock is output from the SCKn pin. Eight synchronization clock pulses are output in the transfer of one character, and when no transfer is performed the clock is held high. However, when only data reception is performed while the CTS function is disabled, the synchronization clock output is started at the same time when the SCR.RE bit set to 1. The synchronization clock is stopped at the high level when an overrun error occurs or the SCR.RE bit is set to 0.

When only data reception is performed and the CTS function is enabled, the clock output is not started even when the SCR.RE bit set to 1 if the CTSn# pin input is high when the SCR.RE bit is 0. The synchronization clock output is started when the SCR.RE bit is set to 1 and the CTSn# pin input is low. After that, if the CTSn# pin input is high on completion of the frame reception, the synchronization clock output is stopped at the high level. If the CTSn# pin input continues to be low, the synchronization clock is stopped at the high level when an overrun error occurs or the SCR.RE bit is set to 0.

41.5.2 CTS and RTS Functions

In the CTS function, CTSn# pin input is used to control reception/transmission start when the clock source is the internal clock. Setting the SPMR.CTSE bit to 1 enables the CTS function.

When the CTS function is enabled, placing the low level on the CTSn# pin causes reception/transmission to start.

Applying the high level to the CTS# pin while reception/transmission are in progress does not affect reception/transmission of the current frame, which continues.

In the RTS function, RTSn# pin output is used to request reception/transmission start when the clock source is an external synchronizing clock. A low level is output when serial communications become possible. Conditions for output of the low and high level are shown below.

(a) SCI0 to SCI6 and SCI12, and SCI7 to SCI11 when FIFO is Disabled

[Conditions for low-level output]

When the following conditions are all satisfied.

- The SCR.RE or SCR.TE bit is 1.
- Transmission or reception is not in progress.
- There are no received data yet to be read (when the SCR.RE bit is 1).
- Untransmitted data is present (when the SCR.TE bit is 1).
- The SSR.ORER flag is 0.

[Condition for high-level output]

The conditions for low-level output have not been satisfied.

(b) SCI7 to SCI11 When FIFO is Enabled

[Conditions for low-level output]

When the following conditions are all satisfied.

- The SCR.RE or SCR.TE bit is 1.
- Transmission or reception is not in progress.
- The number of data stored in the receive FIFO is less than the threshold (FCR.RTRG[3:0] bits) (when the SCR.RE bit is 1).
- Untransmitted data is present (when the SCR.TE bit is 1).
- The SSRFIFO.ORER flag is 0.

[Condition for high-level output]

When the conditions for low-level output are not satisfied.

41.5.3 SCI Initialization (Clock Synchronous Mode)

Before transmitting and receiving data, start by writing the initial value 00h to the SCR register and then continue through the procedure for SCI given in Figure 41.28. Whenever the operating mode or transfer format is changed, the SCR register must be initialized before the change is made.

Note that setting the SCR.RE bit to 0 initializes neither the ORER, FER, and PER flags in the SSR register nor the RDR register.

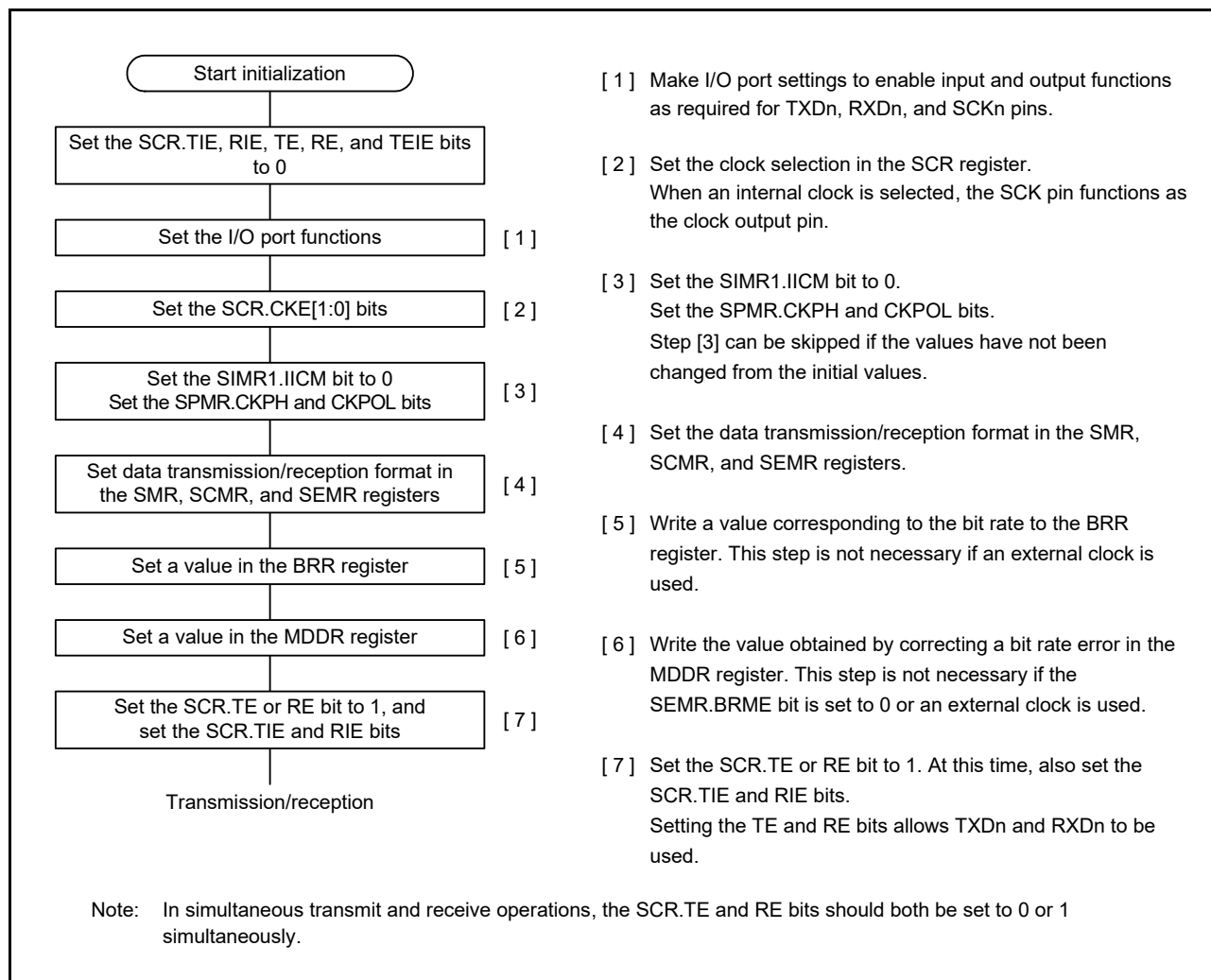


Figure 41.28 Example of SCI Initialization Flowchart (Clock Synchronous Mode)

41.5.4 Serial Data Transmission (Clock Synchronous Mode)

(1) SCI0 to SCI6 and SCI12, and SCI7 to SCI11 when FIFO is Disabled

Figure 41.29, Figure 41.30, and Figure 41.31 show an example of the operation for serial transmission in clock synchronous mode.

In serial data transmission, the SCI operates as described below.

1. The SCI transfers data from the TDR register to the TSR register when data is written to the TDR register in the TXI interrupt handling routine. The TXI interrupt request at the beginning of transmission is generated when the SCR.TE bit is set to 1 after the SCR.TIE bit is set to 1 or when these 2 bits are set to 1 simultaneously by a single instruction.
2. After transferring data from the TDR register to the TSR register, the SCI starts transmission. When the SCR.TIE bit is set to 1 at this time, a TXI interrupt request is generated. Continuous transmission is enabled by writing the next transmit data to the TDR register in this TXI interrupt handling routine before transmission of the current transmit data has finished. When TEI interrupt requests are in use, set the SCR.TIE bit to 0 (a TXI interrupt request is disabled) and the SCR.TEIE bit to 1 (a TEI interrupt request is enabled) after the last of the data to be transmitted are written to the TDR register from the handling routine for TXI requests.
3. 8-bit data is sent from the TXDn pin in synchronization with the output clock when clock output mode has been specified and in synchronization with the input clock when use of an external clock has been specified. Output of the clock signal is suspended until the input CTS signal is at the low level while the SPMR.CTSE bit is 1 (CTS function is enabled).
4. The SCI checks for updating of (writing to) the TDR register at the time of the last bit output.
5. When TDR is updated, the next transmit data is transferred from the TDR register to the TSR register, and serial transmission of the next frame is started.
6. If the TDR register is not updated, set the SSR.TEND flag to 1 and the TXDn pin retains the output state of the last bit. If the SCR.TEIE bit is 1 at this time, a TEI interrupt request is generated. The SCKn pin is held high.

Figure 41.32 shows a sample flowchart of serial data transmission.

Transmission will not start while a receive error flag (ORER, FER, or PER in the SSR register) is set to 1. Be sure to set the receive error flags to 0 before starting transmission. Note that setting the SCR.RE bit to 0 does not clear the receive error flags.

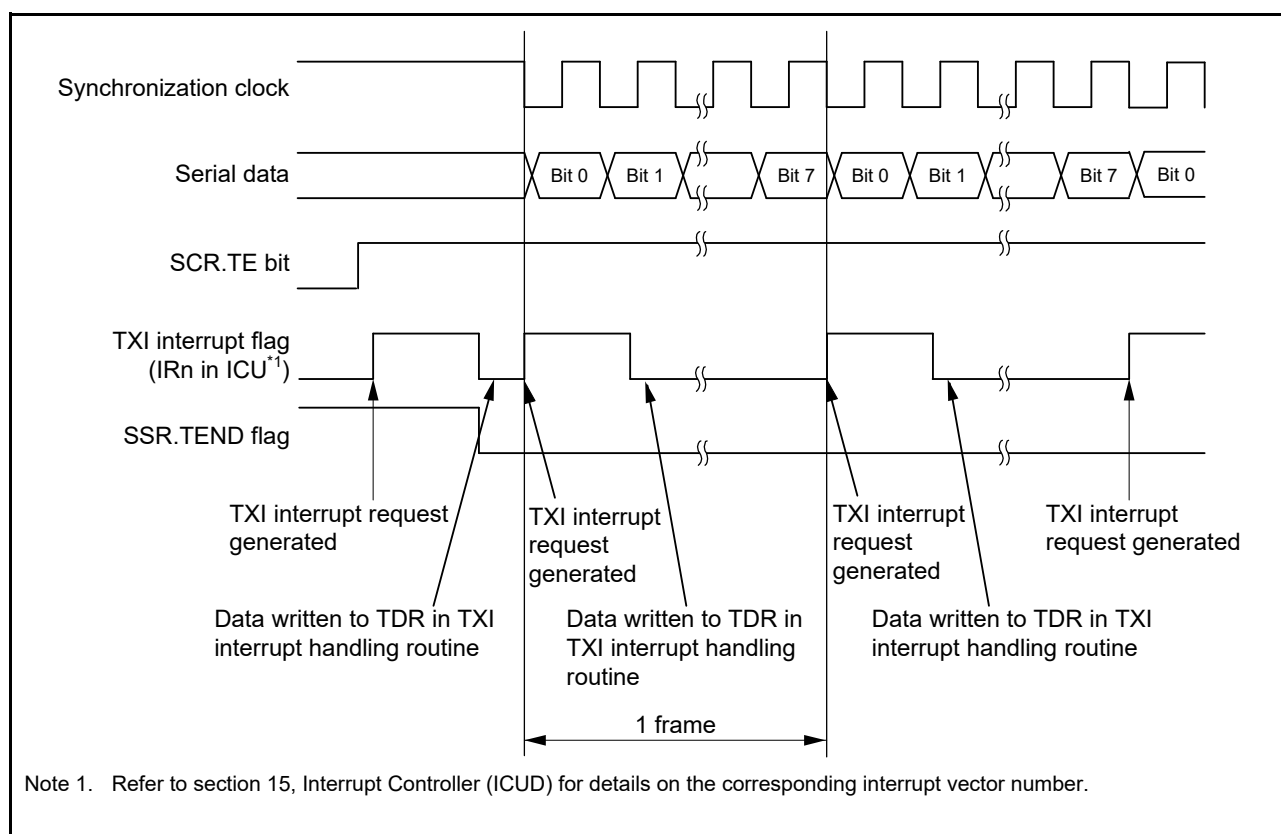


Figure 41.29 Example of Serial Data Transmission in Clock Synchronous Mode When the CTS Function is Not Used at the Beginning of Transmission

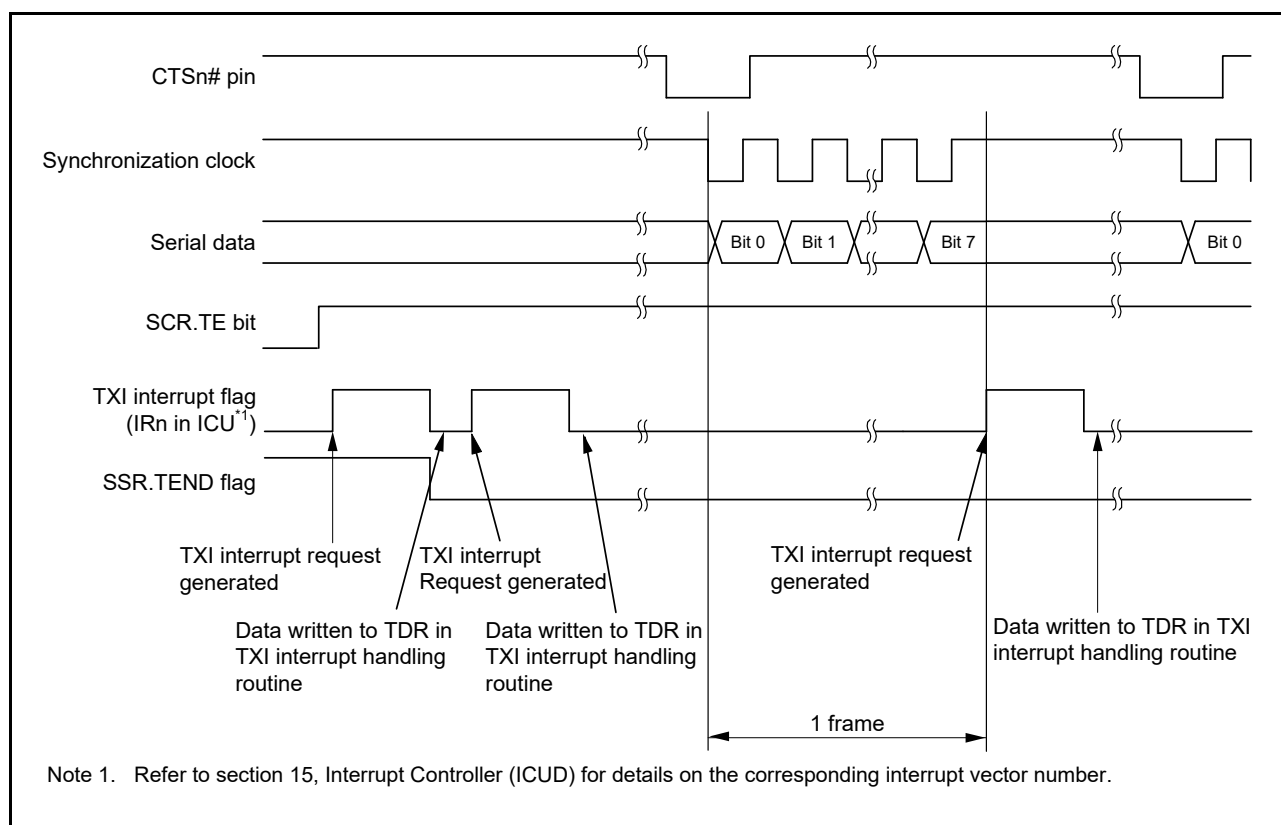


Figure 41.30 Example of Serial Data Transmission in Clock Synchronous Mode When the CTS Function is Used at the Beginning of Transmission

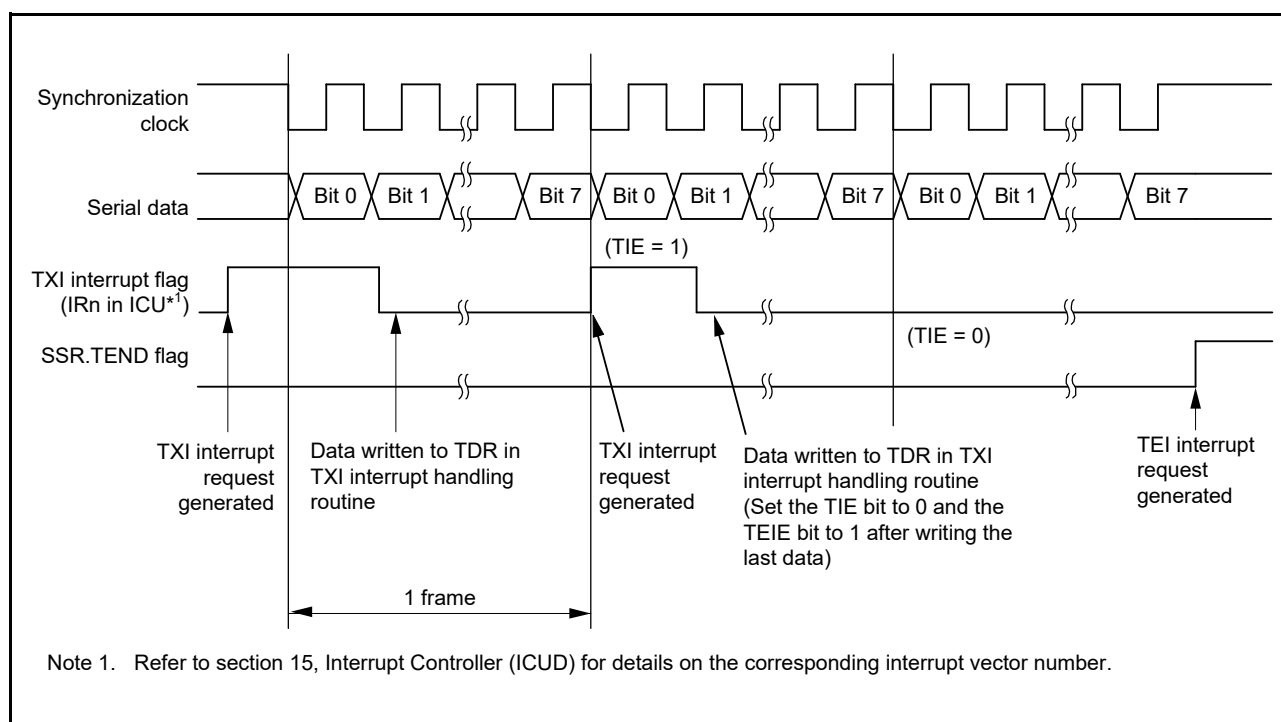


Figure 41.31 Example of Serial Data Transmission in Clock Synchronous Mode from the Middle of Transmission until Transmission Completion

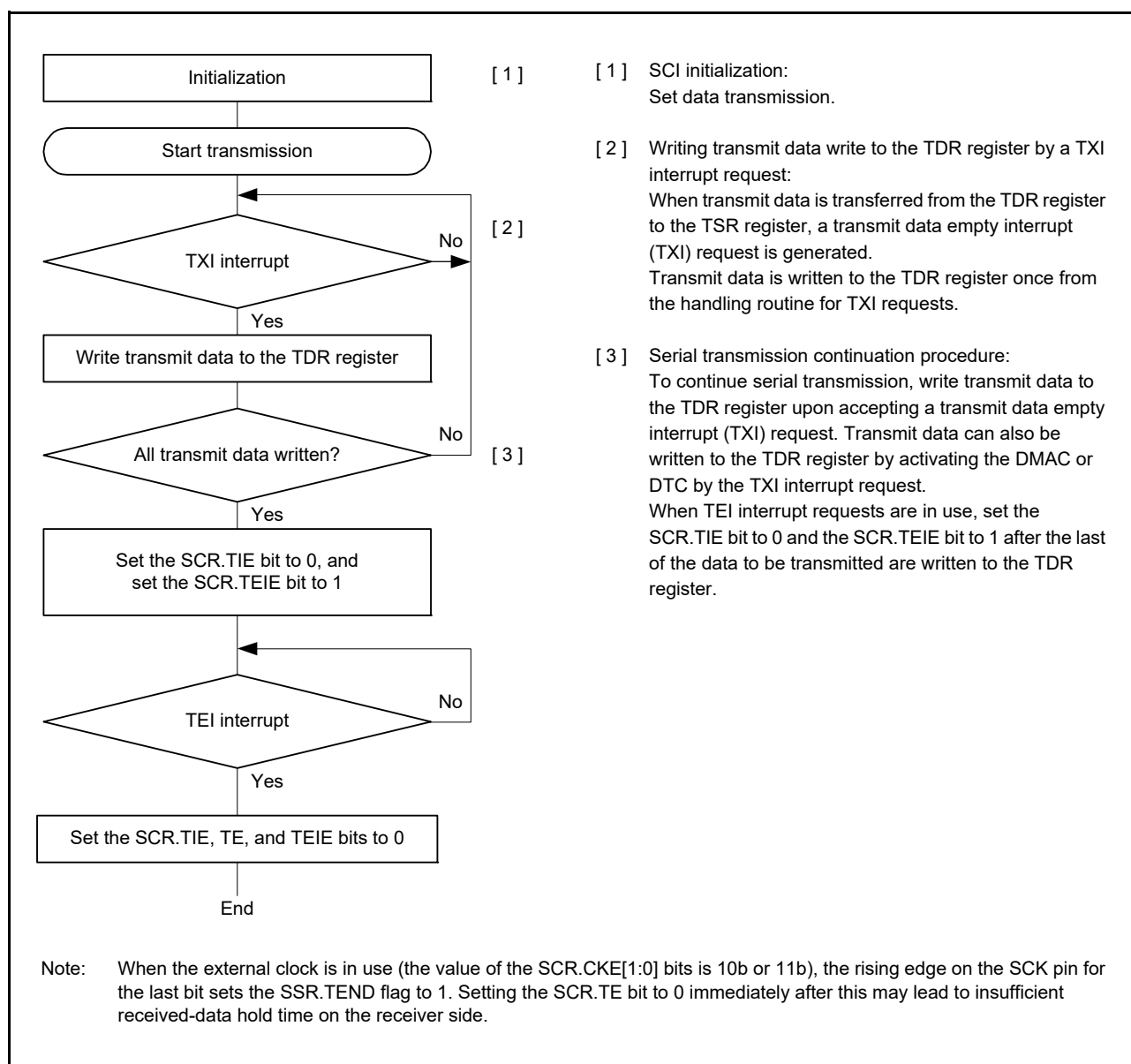


Figure 41.32 Example Flowchart of Serial Transmission in Clock Synchronous Mode

(2) SCI7 to SCI11 When FIFO is Enabled

Set the transmit data in the FTDR register (FTDR.L register) instead of the TDR register. When data is transferred from the transmit FIFO to the TSR register and if the number of data stored in the transmit FIFO is less than or equal to the threshold (FCR.TTRG[3:0] bits), a transmit data empty interrupt (TXI) request is generated.

Up to 16 minus FDR.T[4:0] frames of transmit data can be set within the TXI interrupt processing routine. When the settings of all transmit data are completed, set the SSRFIFO.TDFE flag to 0.

When the transmit data is set by using the DMAC or DTC, the TDFE flag is automatically set to 0.

When transmitting serial data, the SCI operates as follows.

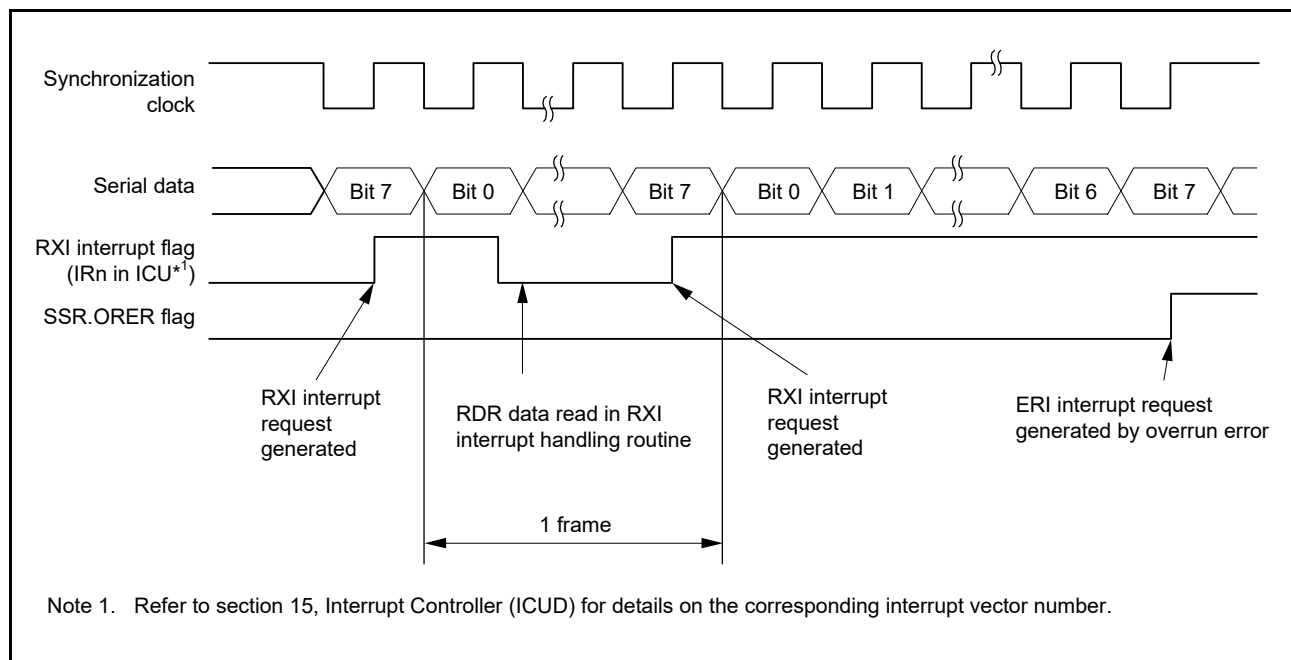
1. When the SCR.TIE and SCR.TE bits are simultaneously set to 1, a TXI interrupt request is generated. Up to 16 minus FDR.T[4:0] bytes of transmit data can be set in the FTDR register (FTDR.L register) within the TXI interrupt processing routine.
2. After the transmit data is written to the FTDR register, the SCI starts transmission by transferring the data to the TSR register in sequence starting from the top of the transmit FIFO. When the number of untransmitted data in the transmit FIFO is less than or equal to the specified threshold (FCR.TTRG[3:0] bits), the SSRFIFO.TDFE flag is set to 1. At this time, if the SCR.TIE bit is 1, a TXI interrupt request is generated. Writing the transmit data to the FTDR register by using the TXI interrupt processing routine before the transmit FIFO becomes empty enables sequence transmission. When a TEI interrupt request is used, the SCI writes the last transmit data to the FTDR register by using the TXI interrupt processing routine, and then sets the SCR.TIE bit to 0 and the SCR.TEIE bit to 1.
3. The SCI outputs 8-bits of data from the TXDn pin in synchronization with the clock for output when the SCR.CKE[1] bit is 0 (internal clock) and in synchronization with the clock for input when the SCR.CKE[1] bit is 1 (external clock). When the SPMR.CTSE bit is 1 (with the CTS function enabled), the SCI waits until the low is input to the CTSn# pin and starts transmission.
4. The SCI checks whether any untransmitted data is in the transmit FIFO when it transmits the last bit.
5. When any untransmitted data is in the transmit FIFO, the SCI transfers the data to the TSR register from the transmit FIFO and starts transmission of the next frame data.
6. When no untransmitted data remains in the transmit FIFO, the SCI sets the SSRFIFO.TEND flag to 1 and stops shifting by the TSR register. At this time, if the SCR.TEIE bit is 1, a TEI interrupt request is generated. The TXDn pin is fixed to the value of the last bit of the last transmitted data and the SCKn pin is fixed to high.

41.5.5 Serial Data Reception (Clock Synchronous Mode)

(1) SCI0 to SCI6 and SCI12, and SCI7 to SCI11 when FIFO is Disabled

Figure 41.33 and Figure 41.34 show an example of SCI operation for serial reception in clock synchronous mode. In serial data reception, the SCI operates as described below.

1. The value of the SCR.RE bit becoming 1 places the signal output on the RTSn# pin at the low level (when the RTS function is in use).
2. The SCI performs internal initialization and starts receiving data in synchronization with a synchronization clock input or output, and stores the receive data in the RSR register.
3. If an overrun error occurs, the SSR.ORER flag is set to 1. If the SCR.RIE bit is 1 at this time, an ERI interrupt request is generated. Receive data is not transferred to the RDR register.
4. When reception finishes successfully, receive data is transferred to the RDR register. If the SCR.RIE bit is 1 at this time, an RXI interrupt request is generated. Continuous reception is enabled by reading the receive data transferred to the RDR register in this RXI interrupt handling routine before reception of the next receive data is completed. Reading out the received data that have been transferred to the RDR register causes the RTSn# pin to output the low level (when the RTS function is in use).



**Figure 41.33 Example of Operation for Serial Reception in Clock Synchronous Mode (1)
(When RTS Function is Not Used)**

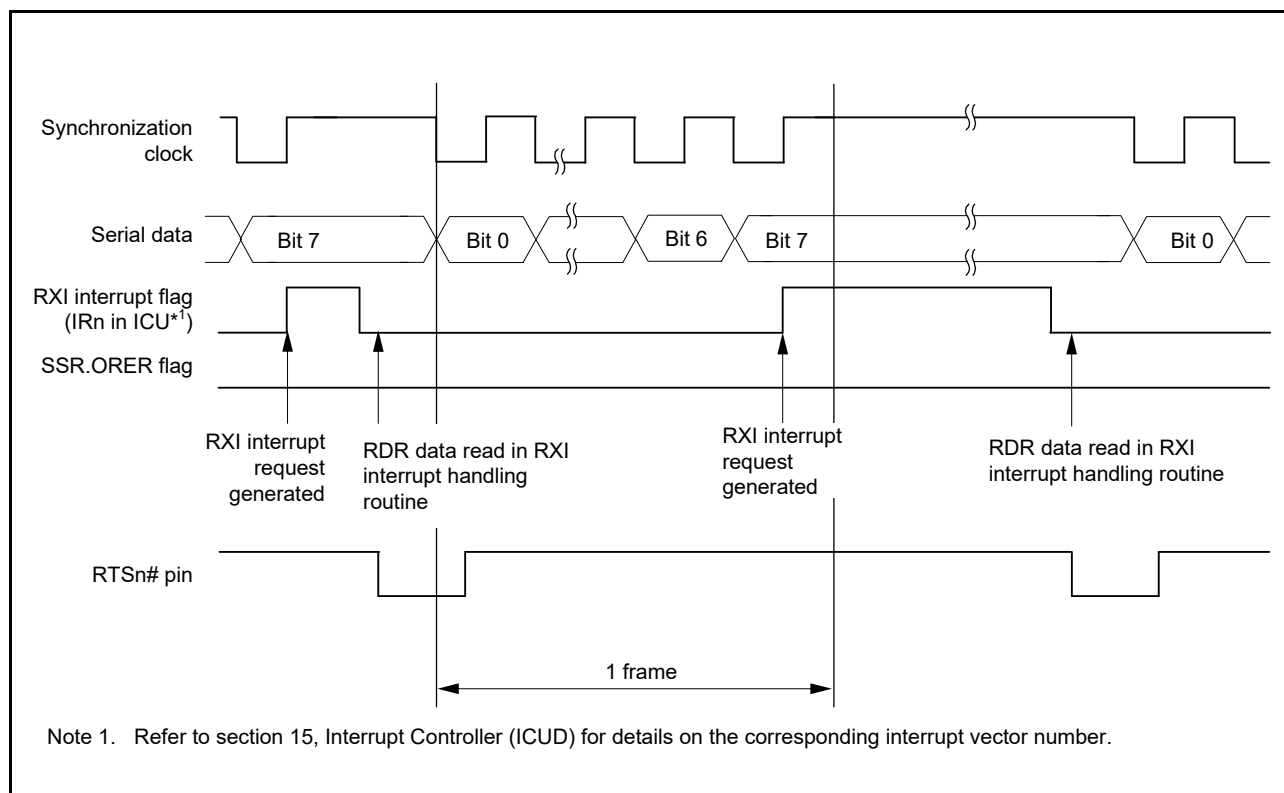


Figure 41.34 Example of Operation for Serial Reception in Clock Synchronous Mode (2) (When RTS Function is Used)

Data transfer cannot be resumed while a receive error flag is 1. Accordingly, clear the ORER, FER, and PER flags in the SSR register before resuming reception. Moreover, be sure to read the RDR register during overrun error processing. When a reception is forcibly terminated by setting the SCR.RE bit to 0 during operation, read the RDR register because received data which has not yet been read may be left in the RDR register.

Figure 41.35 shows a sample flowchart for serial data reception.

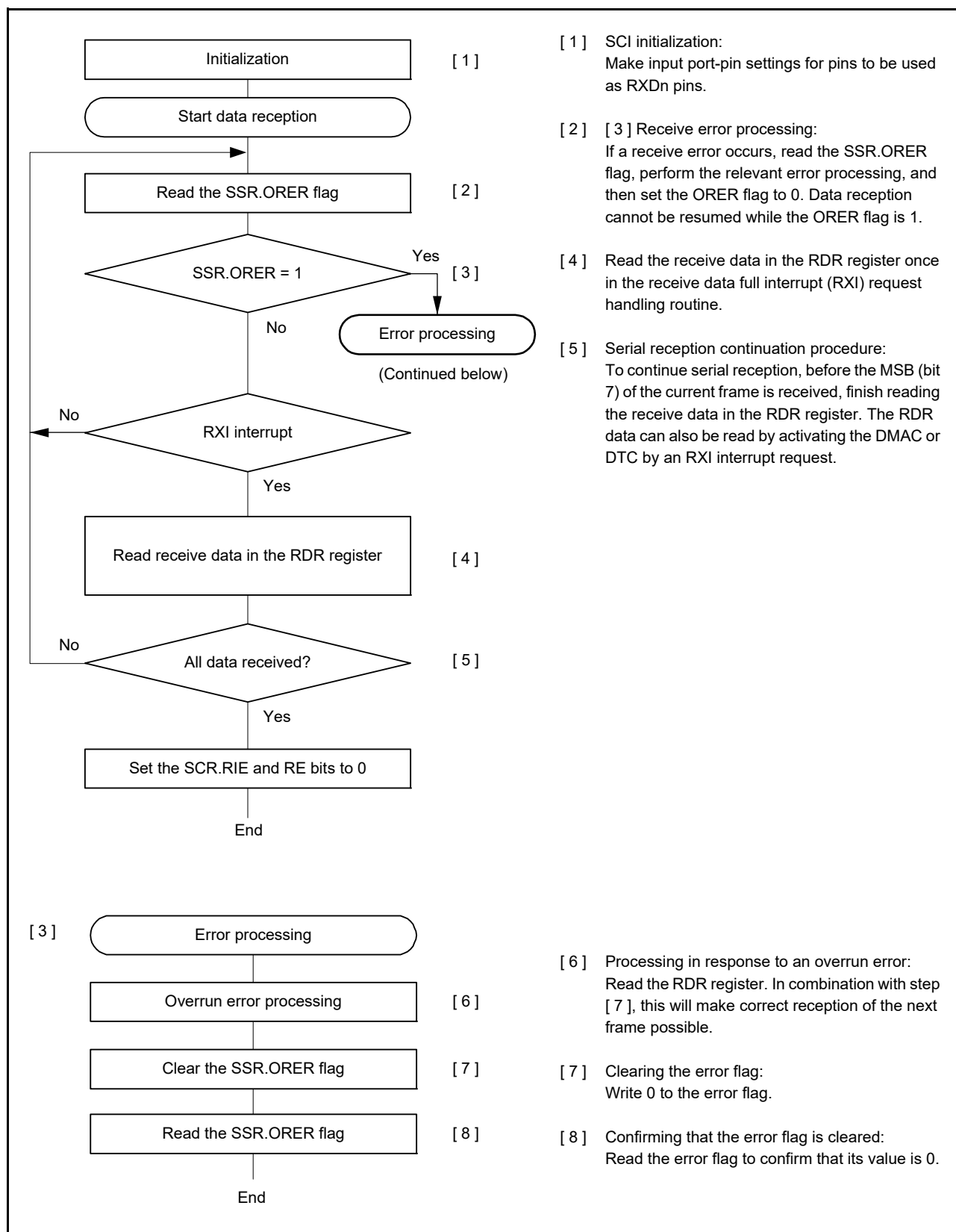


Figure 41.35 Example Flowchart of Serial Reception in Clock Synchronous Mode

(2) SCI7 to SCI11 When FIFO is Enabled

The received data and status flag are read from the FRDR register instead of the RDR register. When reading in byte units, read the FRDR.H register and then the FRDR.L register in this order. Reading the FRDR.L register updates the FRDR.RDAT[7:0] bits. The value of the RDF and ORER flags in the FRDR register is the same as that of the SSRFIFO register.

When receiving serial data, the SCI operates as follows.

1. If the SCR.RE bit is set to 1 and the RTSn# function is in use, the SCI drives the RTSn# pin low.
2. The SCI starts receiving data in synchronization with the clock input or output and stores the received data in the RSR register.
3. When there is no space in the receive FIFO, an overrun error occurs. When an overrun error occurs, the SCI sets the SSRFIFO.ORER flag to 1. At this time, if the SCR.RIE bit is 1, an ERI interrupt request is generated. The received data is not transferred to the receive FIFO.
4. When the data is successfully received, the SCI transfers the received data to the receive FIFO. If the number of data stored in the receive FIFO is equal to or greater than the threshold (FCR.RTRG[3:0] bits), the SCI sets the SSRFIFO.RDF flag to 1. At this time, if the SCR.RIE bit is 1, an RXI interrupt request is generated. Using the RXI interrupt processing routine allows reading of the received data in the RDRF register before an overrun error occurs, thus enabling the sequence reception of data. When the received data that have been transferred to the receive FIFO are read and the number of unread data is less than the value of the FCR.RSTRG[3:0] bits, the SCI drives the RTSn# pin low.

41.5.6 Simultaneous Serial Data Transmission and Reception (Clock Synchronous Mode)

Figure 41.36 shows a sample flowchart for simultaneous serial transmit and receive operations in clock synchronous mode.

After initializing the SCI, the following procedure should be used for simultaneous serial data transmit and receive operations.

To switch from transmit mode to simultaneous transmit and receive mode, check that the SCI has finished transmission by reading that the SSR.TEND flag is 1, and then initialize the SCR register. Then set the TIE, RIE, TE, and RE bits in the SCR register to 1 simultaneously by a single instruction.

To switch from receive mode to simultaneous transmit and receive mode, check that the SCI has finished reception, and then set the RIE and RE bits to 0. Then check that the receive error flags (ORER, FER, and PER in the SSR register) are 0, and then set the TIE, RIE, TE, and RE bits in the SCR register to 1 simultaneously by a single instruction.

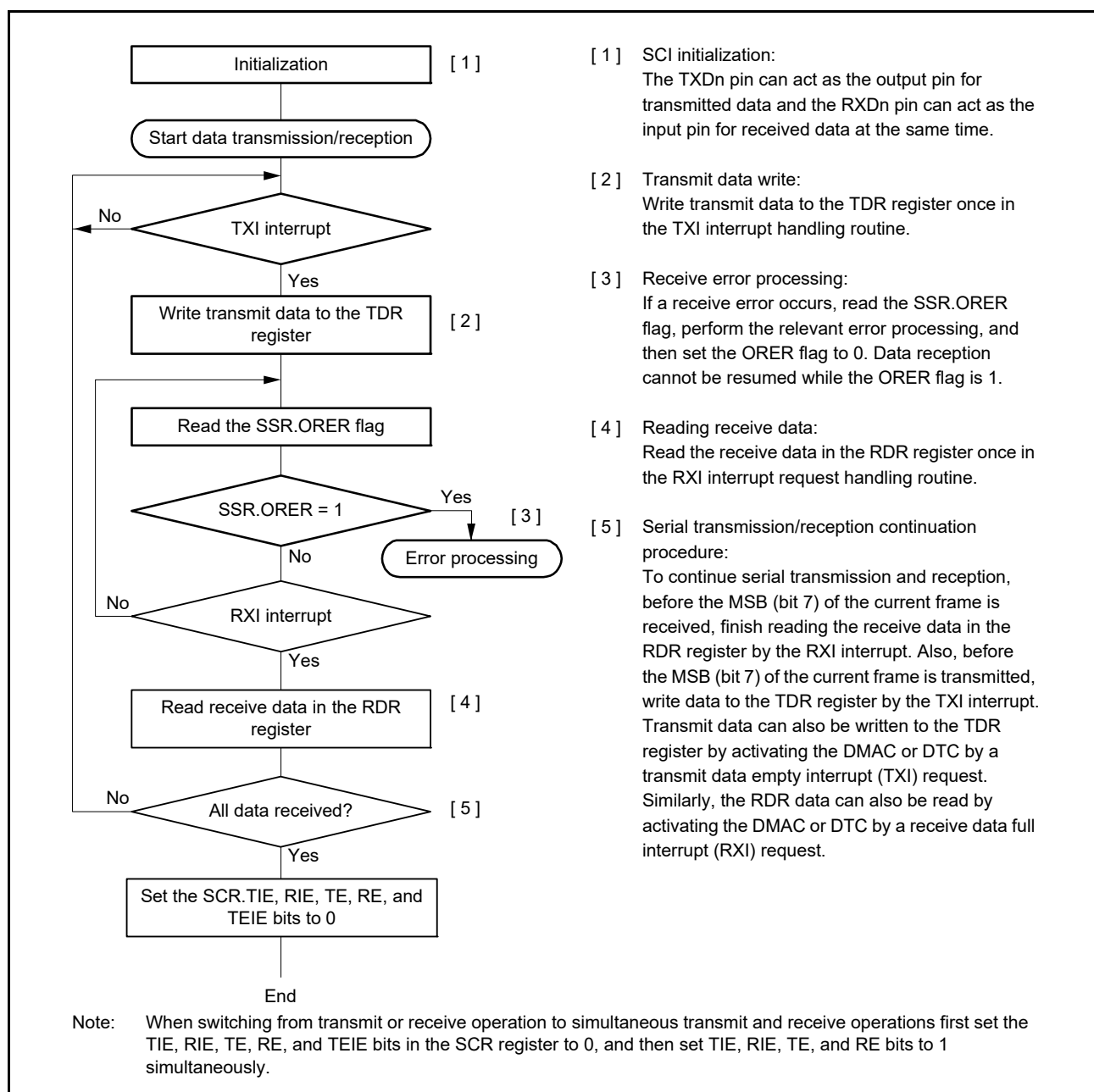


Figure 41.36 Example Flowchart of Simultaneous Serial Transmission and Reception in Clock Synchronous Mode

41.6 Operation in Smart Card Interface Mode

The SCI supports smart card (IC card) interfaces conforming to ISO/IEC 7816-3 (standard for Identification Cards), as an extended function of the SCI.

Smart card interface mode can be selected using the appropriate register.

41.6.1 Sample Connection

Figure 41.37 shows a sample connection between a smart card (IC card) and this MCU.

As in the figure, since this MCU communicates with an IC card using a single transmission line, interconnect the TXDn and RXDn pins and pull up the data transmission line to VCC using a resistor.

Setting the TE and RE bits in the SCR register to 1 with an IC card disconnected enables closed transmission/reception allowing self-diagnosis.

To supply an IC card with the clock pulses generated by the SCI, input the SCKn pin output to the CLK pin of an IC card. The output port of the this MCU can be used to output a reset signal.

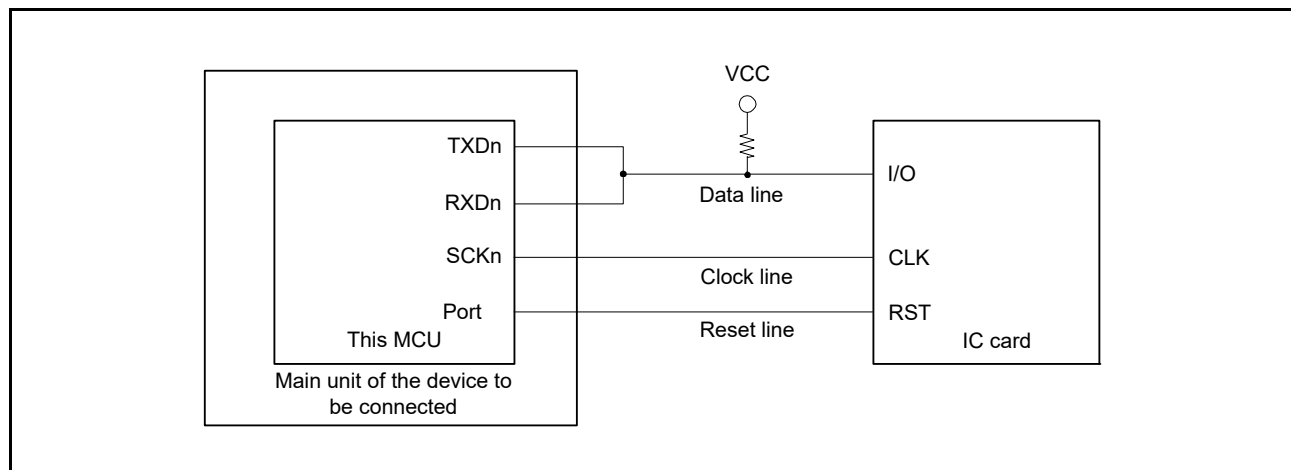


Figure 41.37 Sample Connection with a Smart Card (IC Card)

41.6.2 Data Format (Except in Block Transfer Mode)

Figure 41.38 shows the data transfer formats in smart card interface mode.

- One frame consists of 8-bit data and a parity bit in asynchronous mode.
- During transmission, at least 2 etu (elementary time unit: time required for transferring 1 bit) is secured as a guard time from the end of the parity bit until the start of the next frame.
- If a parity error is detected during reception, a low-level error signal is output for 1 etu after 10.5 etu has passed from the start bit.
- If an error signal is sampled during transmission, the same data is automatically retransmitted after at least 2 etu.

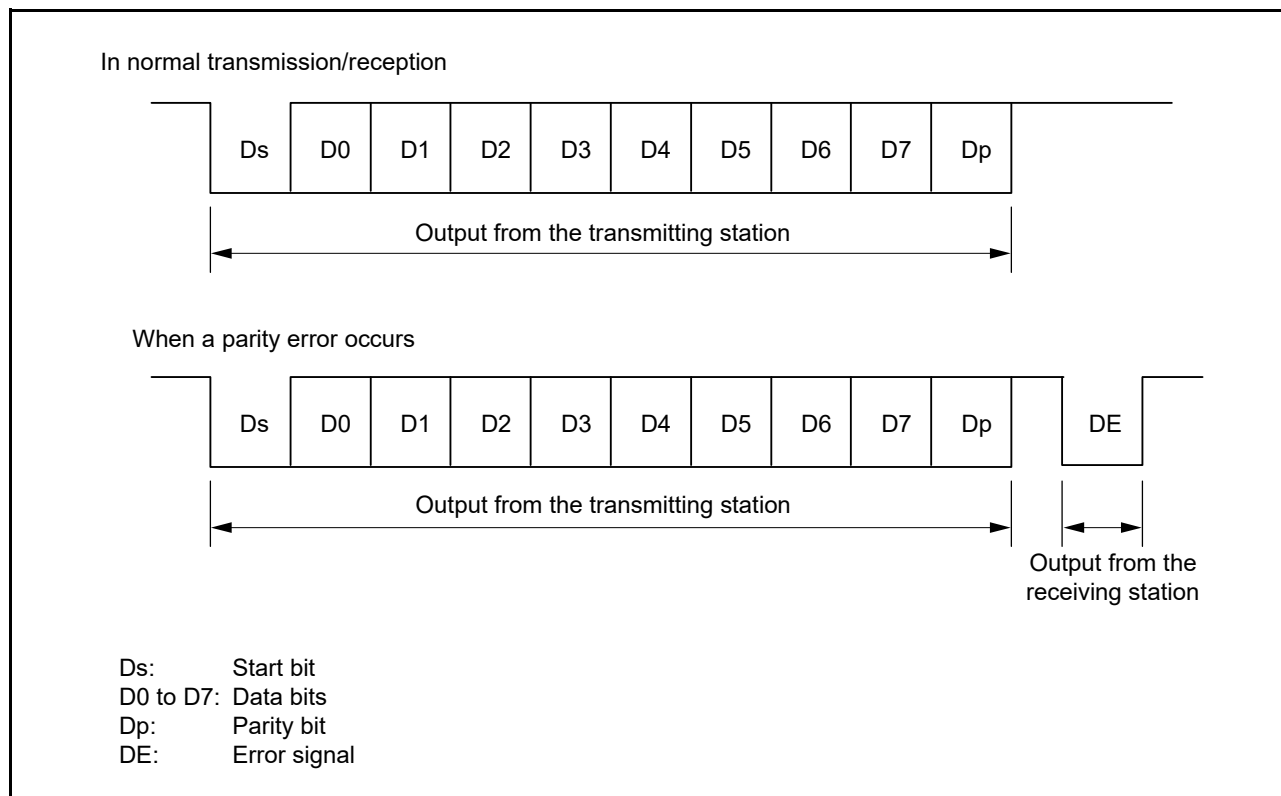


Figure 41.38 Data Formats in Smart Card Interface Mode

For communications with IC cards of the direct convention type and inverse convention type, follow the procedure below.

(1) Direct Convention Type

For the direct convention type, logic levels 1 and 0 correspond to states Z and A, respectively, and data is transferred with LSB first as the start character, as shown in Figure 41.39. Therefore, data in the start character in the figure is 3Bh. When using the direct convention type, write 0 to both the SDIR and SINV bits in the SCMR register. Write 0 to the SMR.PM bit in order to use even parity, which is prescribed by the smart card standard.

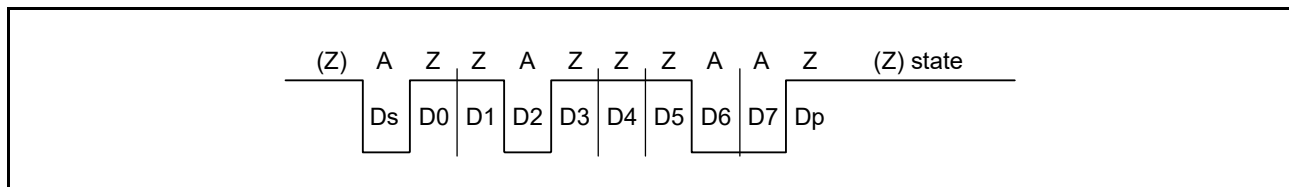


Figure 41.39 Direct Convention (SCMR.SDIR bit = 0, SCMR.SINV bit = 0, SMR.PM bit = 0)

(2) Inverse Convention Type

For the inverse convention type, logic levels 1 and 0 correspond to states A and Z, respectively and data is transferred with MSB first as the start character, as shown in Figure 41.40. Therefore, data in the start character in the figure is 3Fh. When using the inverse convention type, write 1 to both the SDIR and SINV bits in the SCMR register. The parity bit is logic level 0 to produce even parity, which is prescribed by the smart card standard, and corresponds to state Z. Since the SINV bit of this MCU only inverts data bits D7 to D0, write 1 to the SMR.PM bit to invert the parity bit for both transmission and reception.

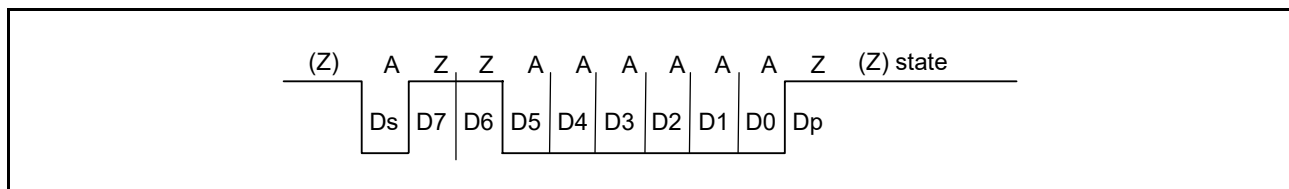


Figure 41.40 Inverse Convention (SCMR.SDIR bit = 1, SCMR.SINV bit = 1, SMR.PM bit = 1)

41.6.3 Block Transfer Mode

Block transfer mode is different from non-block transfer mode in the following respects.

- Even if a parity error is detected during reception, no error signal is output. Since the SSR.PER flag is set by error detection, clear the PER flag before receiving the parity bit of the next frame.
- During transmission, at least 1 etu is secured as a guard time from the end of the parity bit until the start of the next frame.
- Since the same data is not retransmitted during transmission, the SSR.TEND flag is set 11.5 etu after transmission start.
- In block transfer mode, the SSR.ERS flag indicates the error signal status as in non-block transfer mode, but the flag is read as 0 because no error signal is transferred.

41.6.4 Receive Data Sampling Timing and Reception Margin

Only the base clock generated by the on-chip baud rate generator can be used as a transmit/receive clock in smart card interface mode.

In this mode, the SCI can operate on a base clock with a frequency of 32, 64, 372, 256, 93, 128, 186, or 512 times the bit rate according to the settings of the SCMR.BCP2 bit and the SMR.BCP[1:0] bits.

For data reception, the falling edge of the start bit is sampled with the base clock to perform synchronization. Receive data is sampled on the 16th, 32nd, 186th, 128th, 46th, 64th, 93rd, and 256th rising edges of the base clock so that it can be latched at the middle of each bit as shown in Figure 41.41. The reception margin here is determined by the following formula.

$$M = \left| \left(0.5 - \frac{1}{2N} \right) - (L - 0.5)F - \frac{|D - 0.5|}{N}(1 + F) \right| \times 100 (\%)$$

M: Reception margin (%)

N: Ratio of bit rate to clock (N = 32, 64, 372, 256)

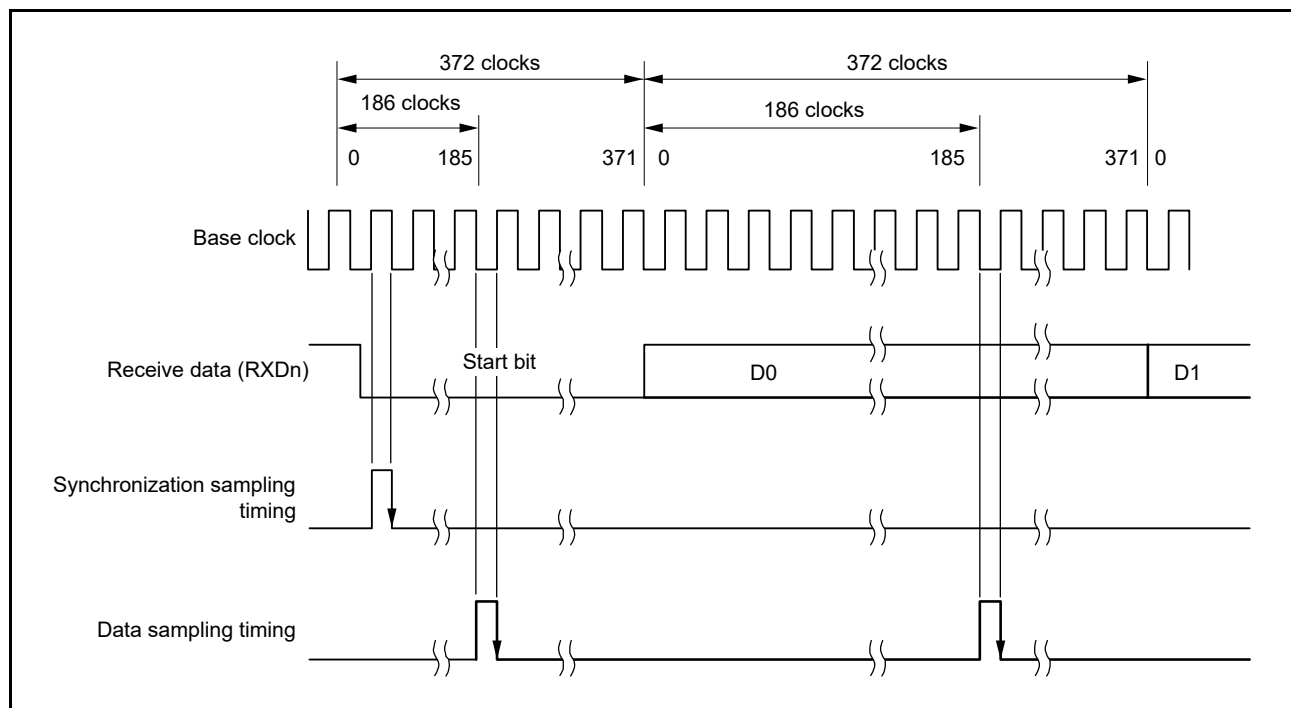
D: Duty cycle of clock (D = 0 to 1.0)

L: Frame length (L = 10)

F: Absolute value of clock frequency deviation

Assuming values of F = 0, D = 0.5, and N = 372 in the above formula, the reception margin is determined by the formula below.

$$M = \{0.5 - 1/(2 \times 372)\} \times 100 (\%) = 49.866 (\%)$$



**Figure 41.41 Receive Data Sampling Timing in Smart Card Interface Mode
(When Clock Frequency is 372 Times the Bit Rate)**

41.6.5 SCI Initialization (Smart Card Interface Mode)

Initialize the SCI following the example of flowchart shown in Figure 41.42.

Initialize the SCR and SSR registers before switching from transmit mode to receive mode and vice versa. When not changing the bit rate, it is not necessary to set the CKE[1:0] bits to 00b. Even if the RE bit is set to 0, the RDR register is not initialized.

To change receive mode to transmit mode, first check that reception has completed, and then execute steps [1] and [3] in Figure 41.42. Set TE = 1 and RE = 0 at step [11]. Reception completion can be verified by reading the RXI request, SSR.ORER, or SSR.PER flag.

To change transmit mode to receive mode, first check that transmission has completed, and then execute steps [1] and [3] in Figure 41.42. Set TE = 0 and RE = 1 at step [11]. Transmission completion can be verified by reading the SSR.TEND flag.

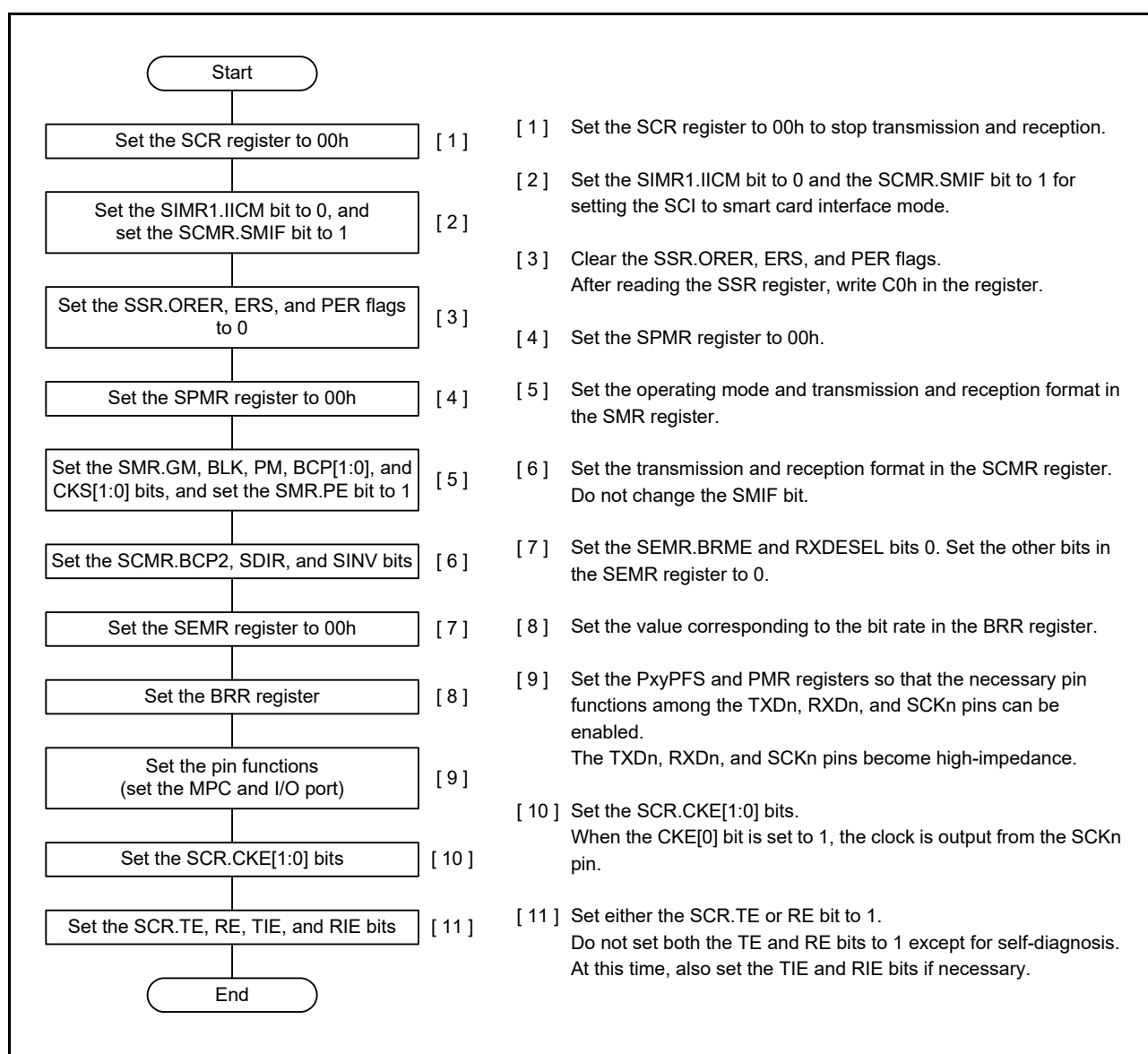


Figure 41.42 Example of SCI Initialization Flowchart (Smart Card Interface Mode)

Figure 41.43 shows an example of data transmission when the SCI is set to smart card interface mode according to the flow described in Figure 41.42 after a reset. When the pin functions are set to the SCK and TXD pins, they are still high-impedance because the SCR.CKE[0] and SCR.TE bits are 0. When the CKE[0] bit is set to 1, clock is output from the SCK pin. When the transmit data is written after setting the TE bit to 1, a data transmission starts. After the TE bit is set to 1, one frame of high-impedance is output from TXD pin (internal wait time) and then the data transmission starts. In smart card interface mode, the clock is continuously output while the CKE[0] bit is set to 1 (clock output) even if both the TE and RE bits are set to 0.

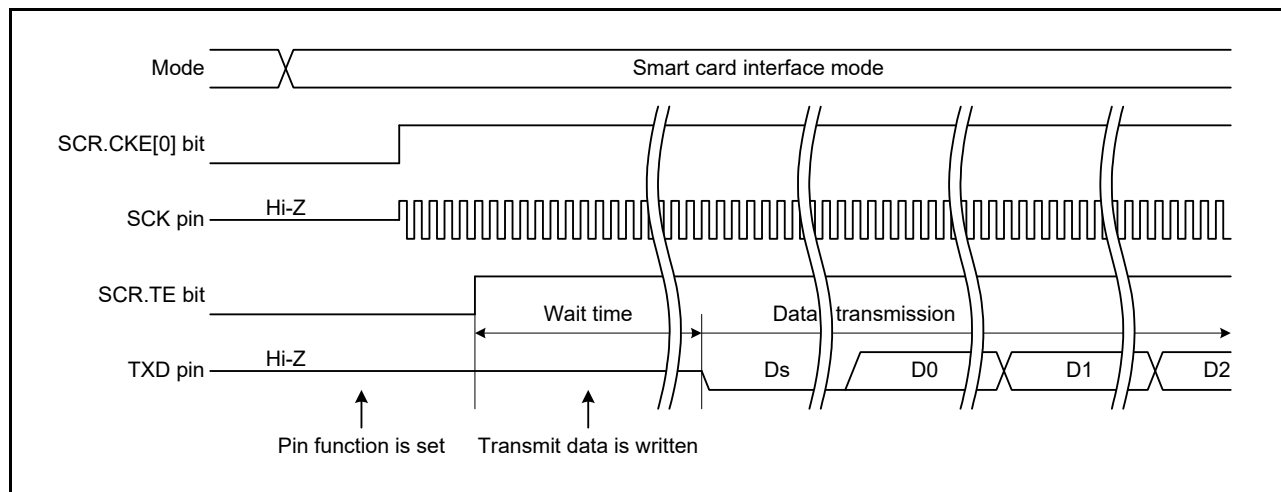


Figure 41.43 Example of Data Transmission Timing in Smart Card Interface Mode

41.6.6 Serial Data Transmission (Except in Block Transfer Mode)

Serial data transmission in smart card interface mode (except in block transfer mode), in that an error signal is sampled and data can be retransmitted, is different from that in non-smart card interface mode. Figure 41.44 shows the data retransmit operation during transmission.

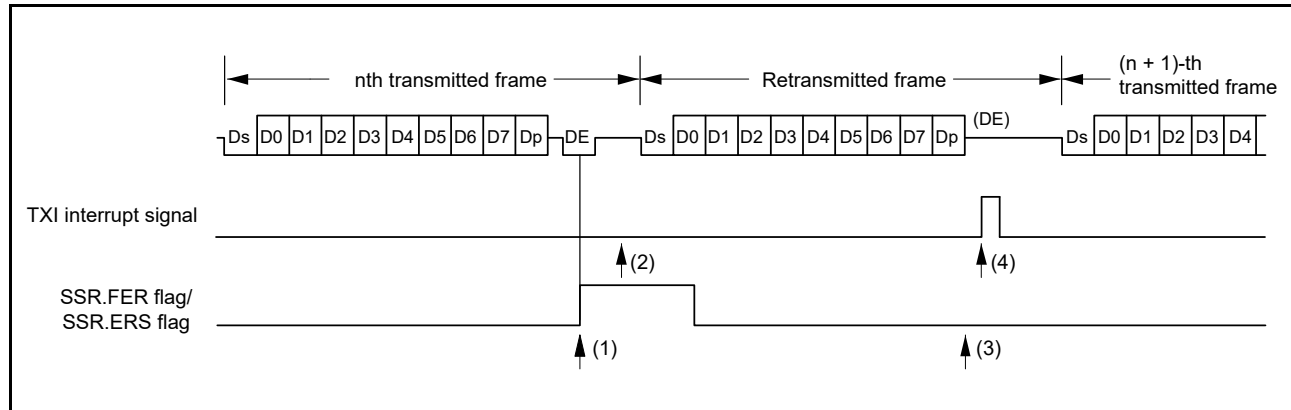


Figure 41.44 Data Retransmit Operation in SCI Transmit Mode

- (1) When an error signal from the receiver end is sampled after one-frame data has been transmitted, the SSR.ERS flag is set to 1. If the SCR.RIE bit is 1 at this time, an ERI interrupt request is generated. Clear the ERS flag before the next parity bit is sampled.
- (2) For a frame in which an error signal is received, the SSR.TEND flag is not set. Data is retransferred from the TDR register to the TSR register allowing automatic data retransmission.
- (3) If no error signal is returned from the receiver, the ERS flag is not set to 1.
- (4) In this case, the SCI judges that transmission of one-frame data (including retransmission) has been completed, and the TEND flag is set. If the SCR.TIE bit is 1 at this time, a TXI interrupt request is generated. Writing transmit data to the TDR register starts transmission of the next data.

Figure 41.45 shows a sample flowchart of serial transmission.

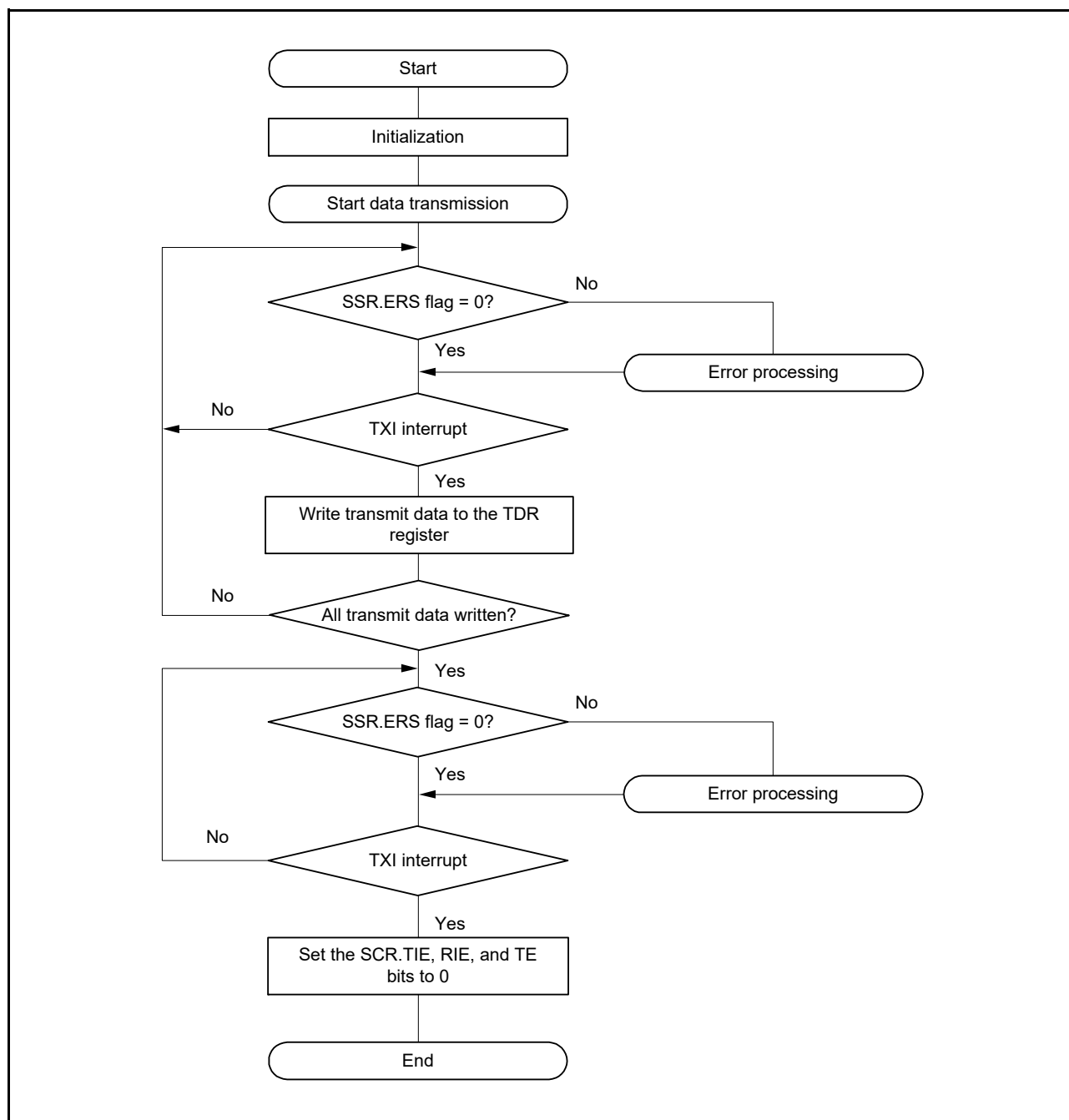


Figure 41.45 Sample Smart Card Interface Transmission Flowchart

All the processing steps are automatically performed using a TXI interrupt request to activate the DTC or DMAC. When the SSR.TEND flag is set to 1 in transmission, if the SCR.TIE bit is 1, a TXI interrupt request is generated. The DTC or DMAC is activated by a TXI interrupt request if the TXI interrupt request is specified as a source of DTC or DMAC activation beforehand, allowing transfer of transmit data. The TEND flag is automatically set to 0 when the DTC or DMAC transfers the data.

If an error occurs, the SCI automatically retransmits the same data. During this retransmission, the TEND flag is kept to 0 and the DTC or DMAC is not activated. Therefore, the SCI and DTC or DMAC automatically transmit the specified number of bytes, including retransmission in the case of error occurrence. However, since the ERS flag is not automatically cleared, set the RIE bit to 1 beforehand to enable an ERI interrupt request to be generated at error occurrence, and clear the ERS flag.

When transmitting/receiving data using the DTC or DMAC, be sure to make settings to enable the DTC or DMAC before making SCI settings.

For DTC or DMAC settings, refer to section 20, Data Transfer Controller (DTCb), section 18, DMA Controller (DMACaA).

Note that the SSR.TEND flag is set in different timings depending on the SMR.GM bit setting. Figure 41.46 shows the TEND flag generation timing.

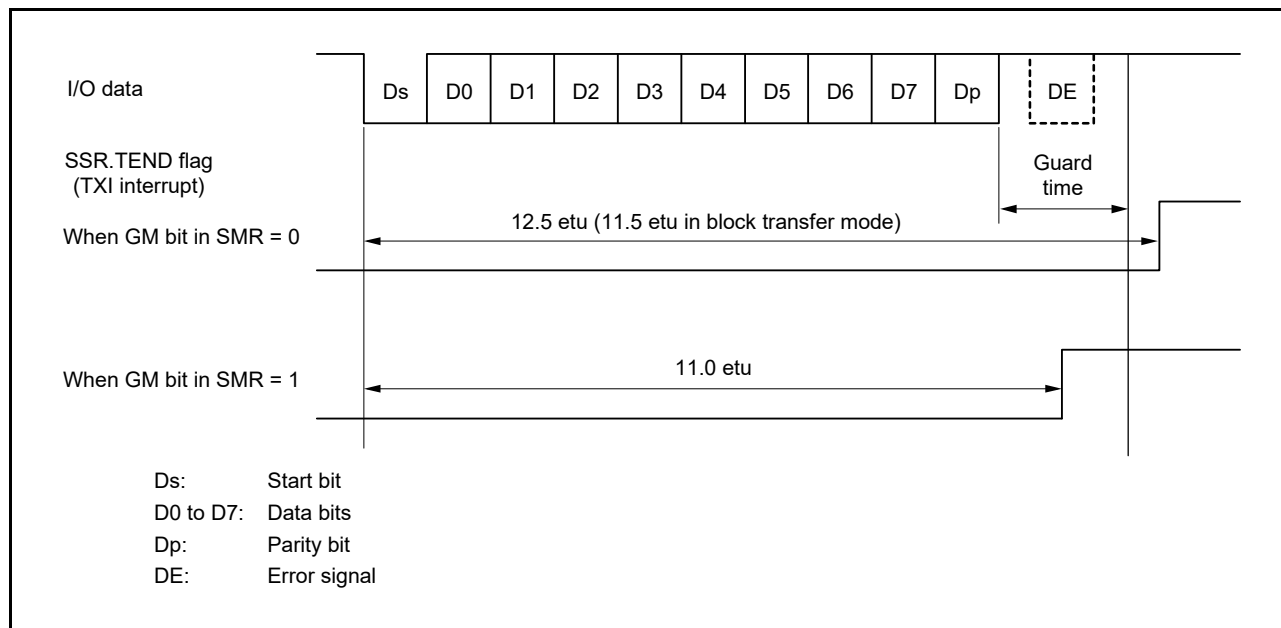


Figure 41.46 SSR.TEND Flag Generation Timing during Transmission

41.6.7 Serial Data Reception (Except in Block Transfer Mode)

Serial data reception in smart card interface mode is similar to that in non-smart card interface mode. Figure 41.47 shows the data retransmit operation in receive mode.

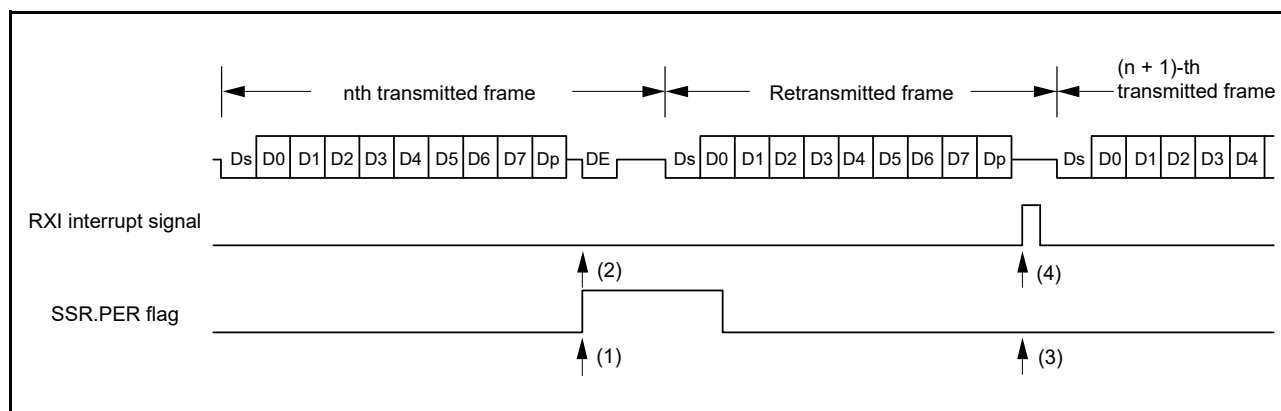


Figure 41.47 Data Retransmit Operation in SCI Receive Mode (Data Retransmit Operation during Reception)

- (1) If a parity error is detected in receive data, the SSR.PER flag is set to 1. When the SCR.RIE bit is 1 at this time, an ERI interrupt request is generated. Clear the PER flag before the next parity bit is sampled.
- (2) For a frame in which a parity error is detected, no RXI interrupt is generated.
- (3) When no parity error is detected, the SSR.PER flag is not set to 1.
- (4) In this case, data is determined to have been received successfully. When the SCR.RIE bit is 1, an RXI interrupt request is generated.

Figure 41.48 shows a sample flowchart for serial data reception.

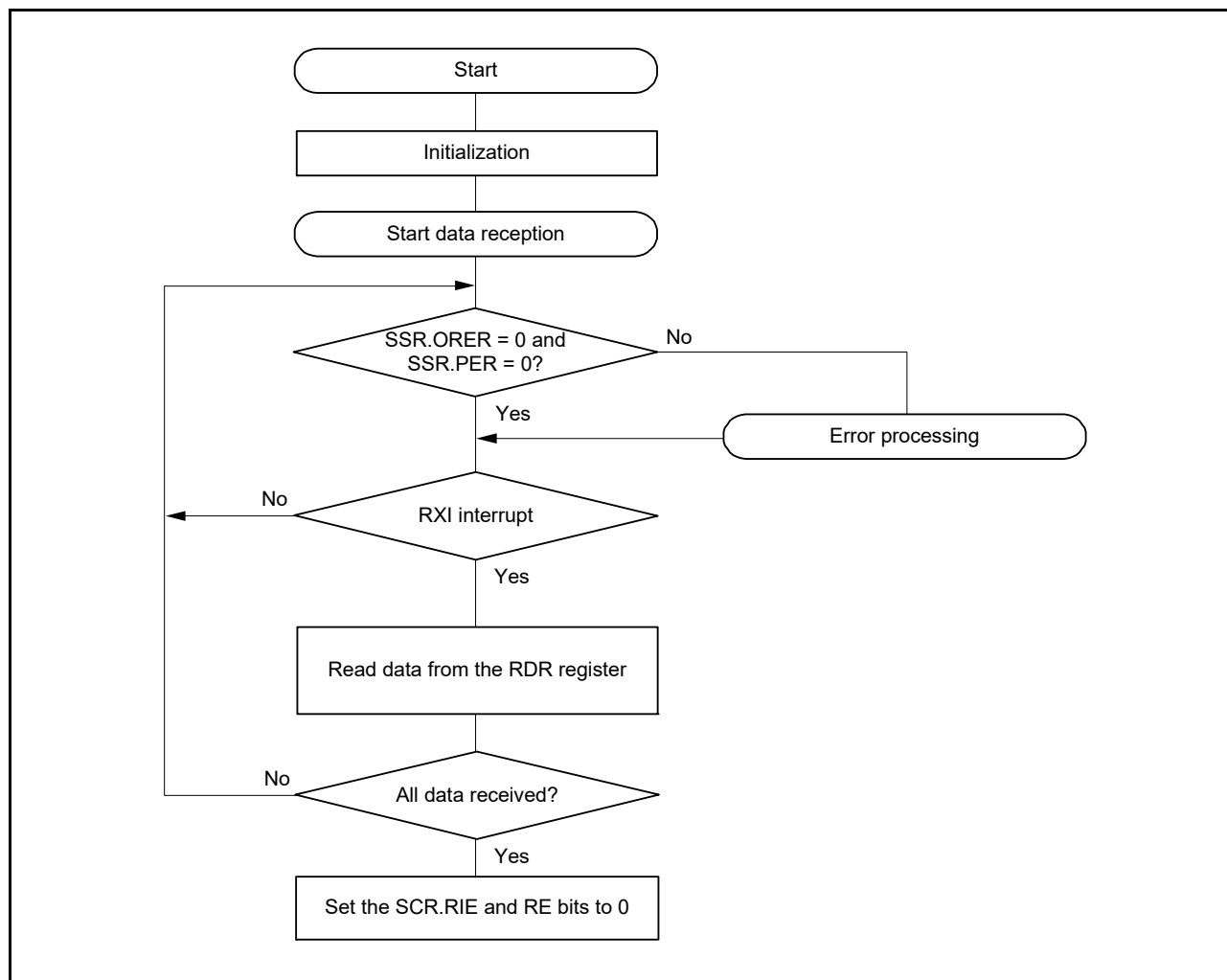


Figure 41.48 Sample Smart Card Interface Reception Flowchart

All the processing steps are automatically performed using an RXI interrupt request to activate the DTC or DMAC.

In reception, setting the RIE bit to 1 allows an RXI interrupt request to be generated. The DTC or DMAC is activated by an RXI interrupt request if the RXI interrupt request is specified as a source of DTC or DMAC activation beforehand, allowing transfer of receive data.

If an error occurs during reception and either the ORER or PER flag in the SSR register is set to 1, a receive error interrupt (ERI) request is generated. Clear the error flag after the error occurrence. If an error occurs, the DTC or DMAC is not activated and receive data is skipped. Therefore, the number of bytes of receive data specified in the DTC or DMAC is transferred.

Even if a parity error occurs and the PER flag is set to 1 during reception, receive data is transferred to the RDR register, thus allowing the data to be read.

When a reception is forcibly terminated by setting the SCR.RE bit to 0 during operation, read the RDR register because the received data which has not yet been read may be left in the RDR register.

Note 1. For operations in block transfer mode, refer to section 41.3, Operation in Asynchronous Mode.

41.6.8 Clock Output Control

Clock output can be fixed to high or low using the SCR.CKE[1:0] bits when the SMR.GM bit is 1. When the CKE[1:0] bits are set to 01b (clock output), the base clock is output from the SCK pin. For the settings of the base clock frequency (bit rate), refer to section 41.2.13, Bit Rate Register (BRR). When the CKE[1:0] bits are set to 00b (output fixed low) or 10b (output fixed to high), the SCK pin can be fixed to low or high.

Figure 41.49 shows a timing chart when the clock output is controlled.

If changing the CKE[1:0] bits while the SMR.GM bit is 0 (non-GSM mode), a pulse of unexpected width may output from SCK pin because the result is immediately reflected to the SCK pin.

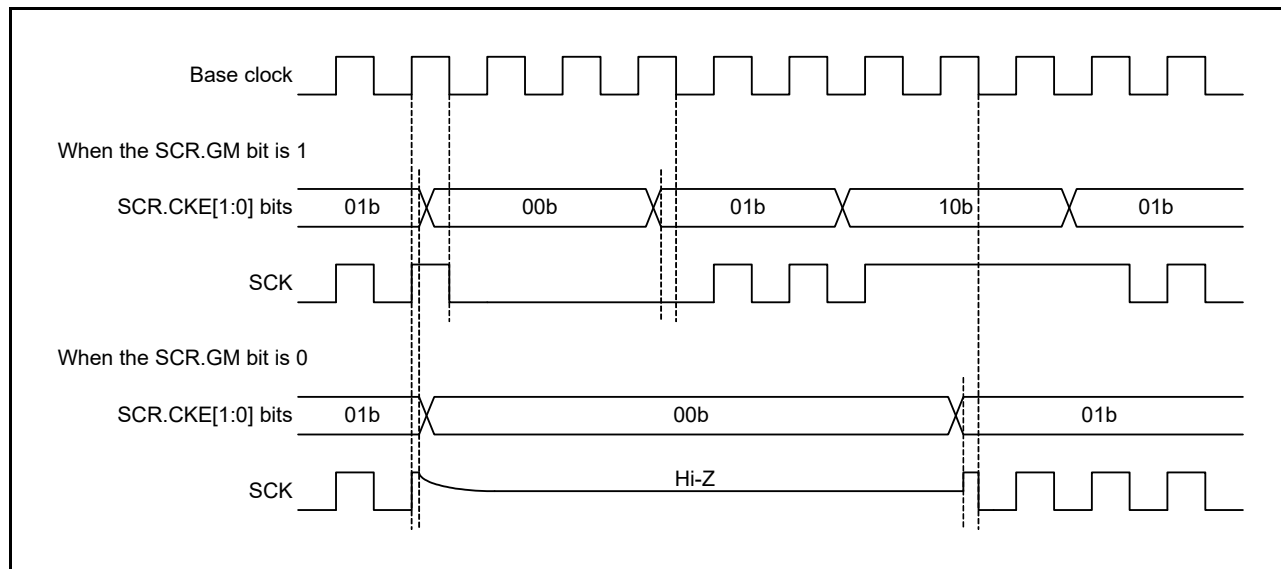


Figure 41.49 Clock Output Control

41.7 Operation in Simple I²C Mode

Simple I²C-bus format is composed of 8 data bits and an acknowledge bit. By continuing into a slave-address frame after a start condition or restart condition, a master device is able to specify a slave device as the partner for communications. The currently specified slave device remains valid until a new slave device is specified or a stop condition is satisfied. The 8 data bits in all frames are transmitted in order from the MSB.

The I²C-bus format and timing of the I²C-bus are shown in Figure 41.50 and Figure 41.51.

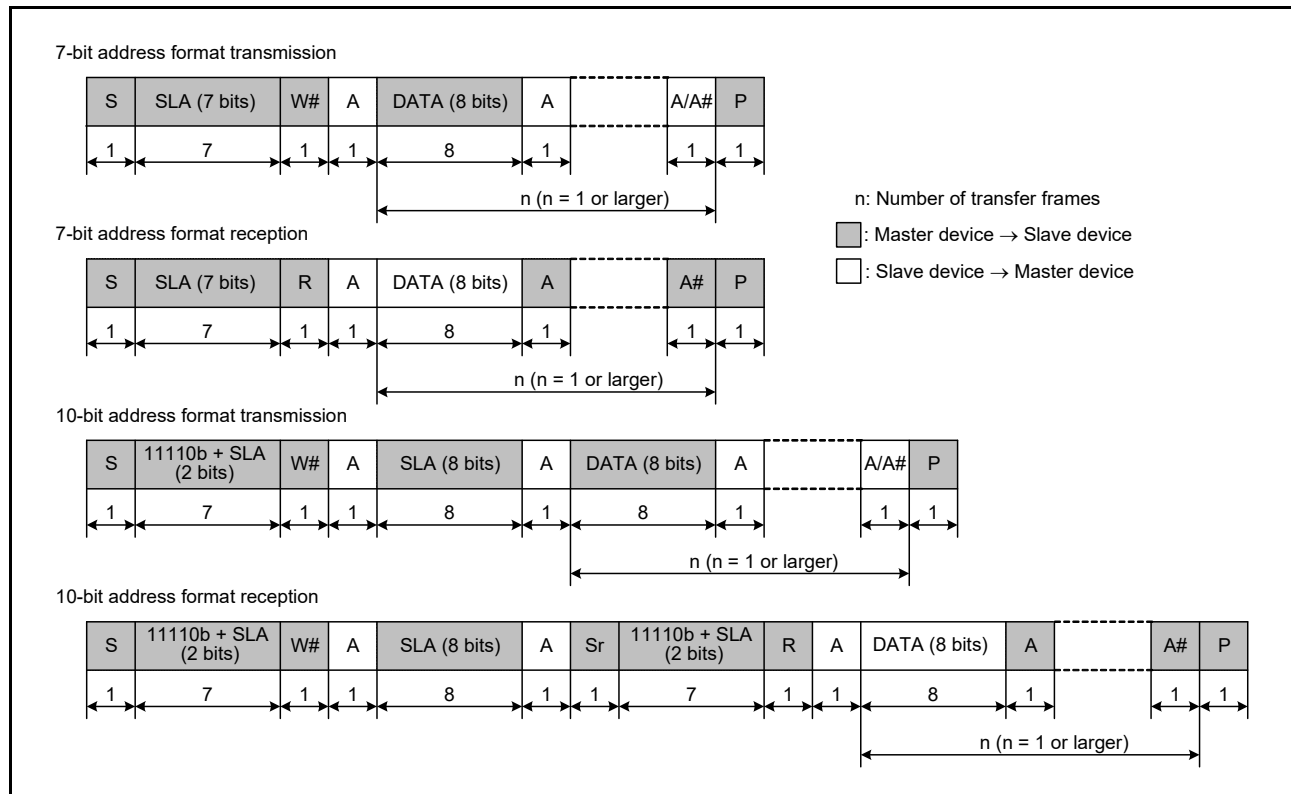


Figure 41.50 I²C-bus Format

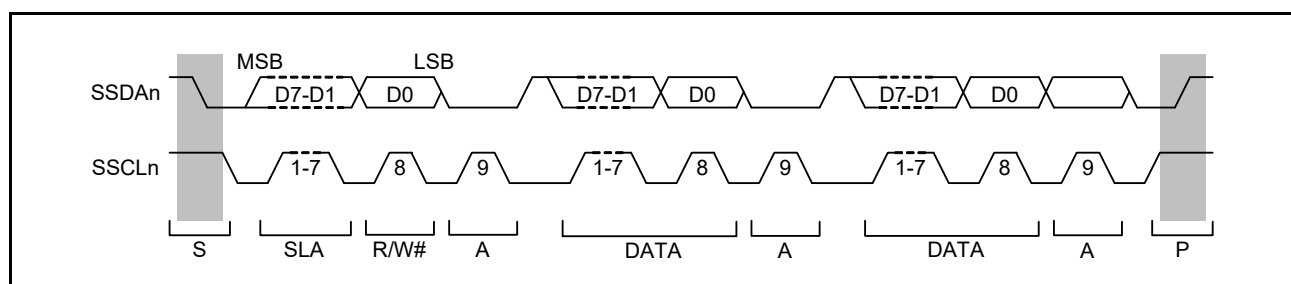


Figure 41.51 I²C-bus Timing (When SLA is 7 Bits)

- S: Indicates a start condition, i.e. the master device changing the level on the SSDAn line from the high to the low level while the SSCLn line is at the high level.
- SLA: Indicates a slave address, by which the master device selects a slave device.
- R/W#: Indicates the direction of transfer (reception or transmission). The value 1 corresponds to transfer from the slave device to the master device and 0 corresponds to transfer from the master device to the slave device.
- A/A#: Indicates an acknowledge bit. This is returned by the slave device for master transmission and by the master device for master reception. Return of the low level indicates ACK and return of the high level indicates NACK.
- Sr: Indicates a restart condition, i.e. the master device changing the level on the SSDAn line from the high to the low level while the SSCLn line is at the high level and after the setup time has elapsed.
- DATA: Indicates the data being received or transmitted.
- P: Indicates a stop condition, i.e. the master device changing the level on the SSDAn line from the low to the high level while the SSCLn line is at the high level.

41.7.1 Generation of Start, Restart, and Stop Conditions

Writing 1 to the SIMR3.IICSTAREQ bit causes the generation of a start condition. The generation of a start condition proceeds through the following operations.

- The level on the SSDAn line falls (from the high level to the low level) and the SSCLn line is kept in the released state.
- The hold time for the start condition is secured as half of a bit period at the bit rate determined by the setting of the BRR.
- The level on the SSCLn line falls (from the high level to the low level), the SIMR3.IICSTAREQ bit is set (to 0), and a start-condition generated interrupt is output.

Writing 1 to the SIMR3.IICRSTAREQ bit causes the generation of a start condition. The generation of a start condition proceeds through the following operations.

- The SSDAn line is released and the SSCLn line is kept at the low level.
- The period at low level for the SSCLn line is secured as half of a bit period at the bit rate determined by the setting of the BRR.
- The SSCLn line is released (transition from the low to the high level).
- Once the high level on the SSCLn line is detected, the setup time for the restart condition is secured as half of a bit period at the bit rate determined by the setting of the BRR.
- The level on the SSDAn line falls (from the high level to the low level).
- The hold time for the restart condition is secured as half of a bit period at the bit rate determined by the setting of the BRR.
- The level on the SSCLn line falls (from the high level to the low level), the SIMR3.IICRSTAREQ bit is set (to 0), and a restart-condition generated interrupt is output.

Writing 1 to the SIMR3.IICSTPREQ bit causes the generation of a stop condition. The generation of a stop condition proceeds through the following operations.

- The level on the SSDAn line falls (from the high level to the low level) and the SSCLn line is kept at the low level.
- The period at low level for the SSCLn line is secured as half of a bit period at the bit rate determined by the setting of the BRR.
- The SSCLn line is released (transition from the low to the high level).
- Once the high level on the SSCLn line is detected, the setup time for the stop condition is secured as half of a bit period at the bit rate determined by the setting of the BRR.
- The SSDAn is released (transition from the low to the high level), the SIMR3.IICSTPREQ bit is set (to 0), and a stop-condition generated interrupt is output.

Figure 41.52 shows the timing of operations in the generation of start, restart, and stop conditions.

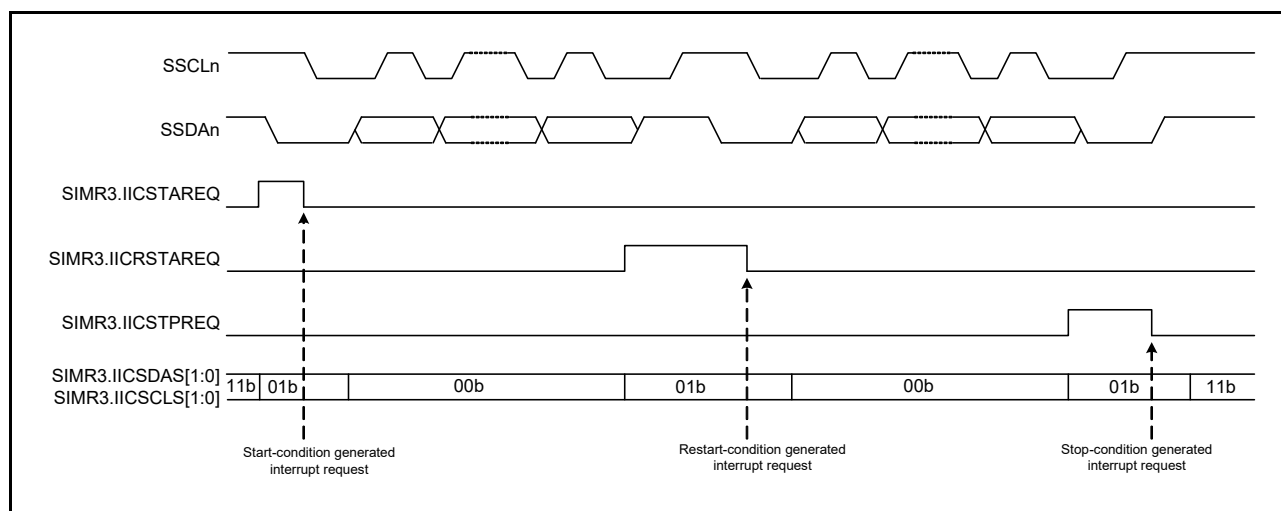


Figure 41.52 Timing of Operations in the Generation of Start, Restart, and Stop Conditions

41.7.2 Clock Synchronization

The SSCLn line may be placed at the low level in the case of a wait inserted by a slave device as the other side of transfer. Setting the SIMR2.IICCSC bit to 1 applies control to obtain synchronization when the levels of the internal SSCLn clock signal and the level being input on the SSCLn pin differ.

When the SIMR2.IICCSC bit is set to 1, the level of the internal SSCLn clock signal changes from low to high, counting to determine the period at high level is stopped while the low level is being input on the SSCLn pin, and counting to determine the period at high level starts after the transition of the input on the SSCLn pin to the high level. The interval from this time until counting to determine the period at high level starts on the transition of the SSCLn pin to the high level is the total of the delay of SSCLn output, delay for noise filtering of the input on the SSCLn pin (2 or 3 cycles of sampling clock for the noise filter), and delay for internal processing (1 or 2 cycles of PCLK). The period at high level of the internal SSCLn clock is extended even if other devices are not placing the low level on the SSCLn line. If the SIMR2.IICCSC bit is 1, synchronization is obtained for the transmission and reception of data by taking the logical AND of the input on the SSCLn pin and the internal SSCLn clock. If the SIMR2.IICCSC bit is 0, synchronization with the internal SSCLn clock is obtained for the transmission and reception of data.

If a slave device inserts a period of waiting into the interval until the transition of the internal SSCLn clock signal from the low to the high level after a request for the generation of a start, restart, or stop condition is issued, the time until generation is prolonged by that period.

If a slave device inserts a period of waiting after the transition of the internal SSCLn clock signal from the low to the high level, although the generation-completed interrupt is issued without stopping the period of waiting, generation of the condition itself is not guaranteed. Figure 41.53 shows an example of operations to synchronize the clocks.

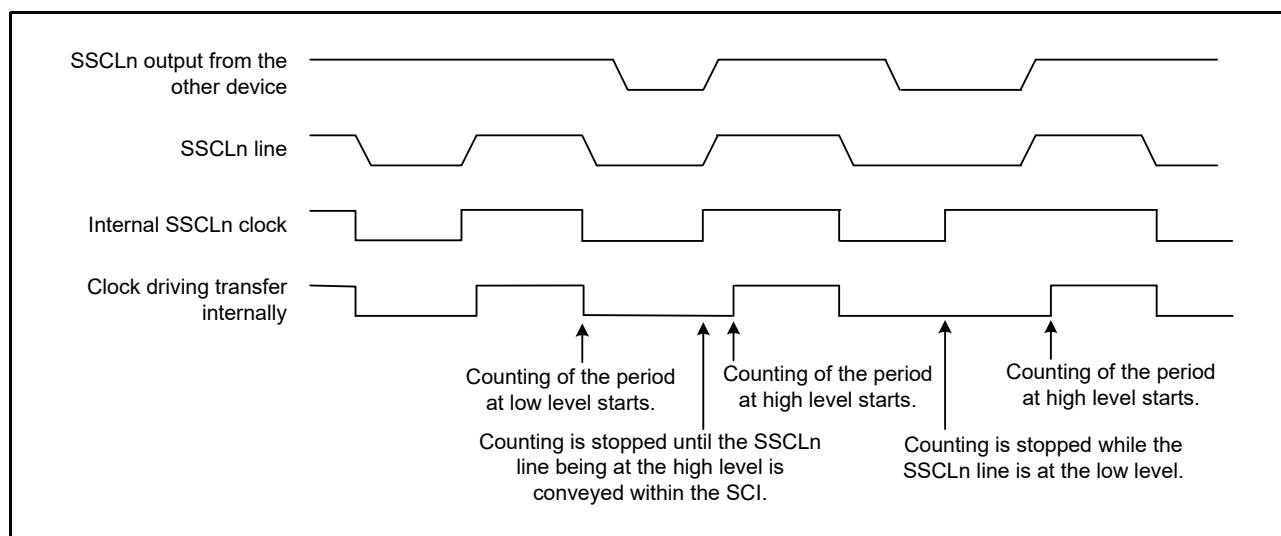


Figure 41.53 Example of Operations for Clock Synchronization

41.7.3 SSDA Output Delay

The SIMR1.IICDL[4:0] bits can be used to set a delay for output on the SSDAn pin relative to falling edges of output on the SSCLn pin. Delay-time settings from 0 to 31 are selectable, representing periods of the corresponding numbers of cycles of the clock signal from the on-chip baud rate generator (derived by frequency-dividing the base clock, PCLK, by the divisor selected by the SMR.CKS[1:0] bits). A delay for output on the SSDAn pin is for the start condition/restart condition/stop condition signal, 8-bit transmit data, and an acknowledge bit.

If the SSDA output delay is shorter than the time for the level on the SSCLn pin to fall, the change of the output on the SSDAn pin will start while the output level on the SSCLn pin is falling, creating a possibility of erroneous operation for slave devices. Ensure that settings for the delay of output on the SSDAn pin are for times greater than the time output on the SSCLn pin takes to fall (300 ns for I²C-bus in normal mode and fast mode).

Figure 41.54 shows the timing of delays in SSDA output.

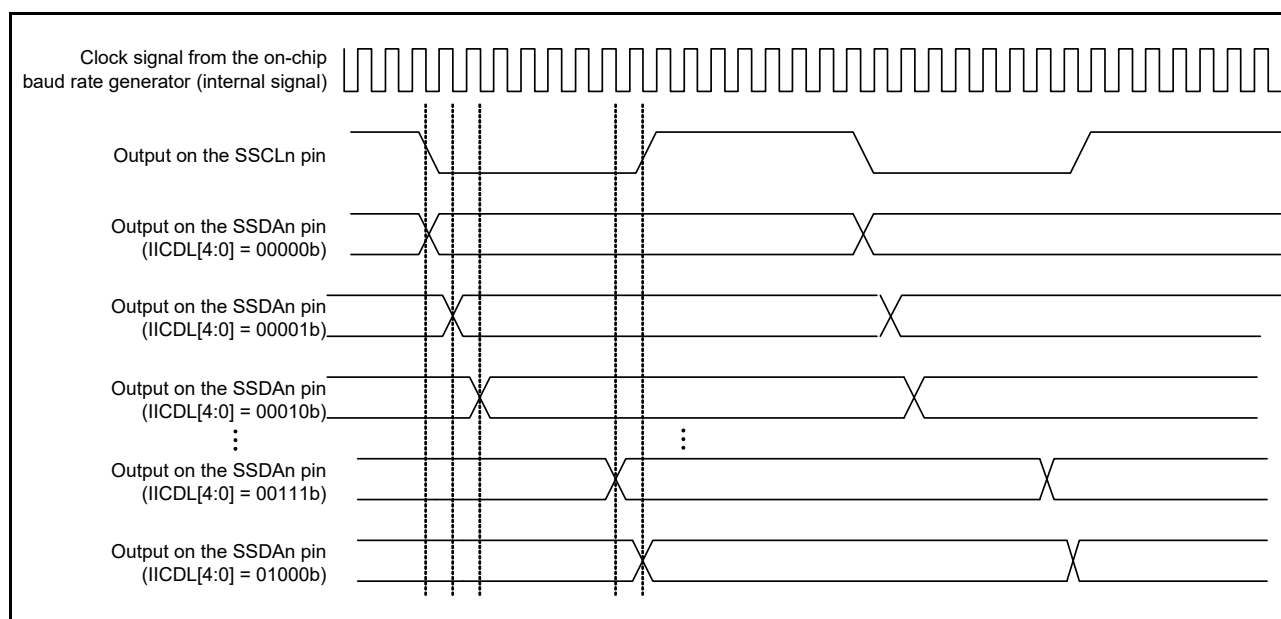


Figure 41.54 Timing of Delays in SSDA Output

41.7.4 SCI Initialization (Simple I²C Mode)

Before transferring data, write the initial value (00h) to the SCR register and initialize the interface following the example shown in Figure 41.55.

When changing the operating mode, transfer format, and so on, be sure to set the SCR register to its initial value before proceeding with the changes.

In simple I²C mode, the open-drain setting for the communication ports should be made on the port side.

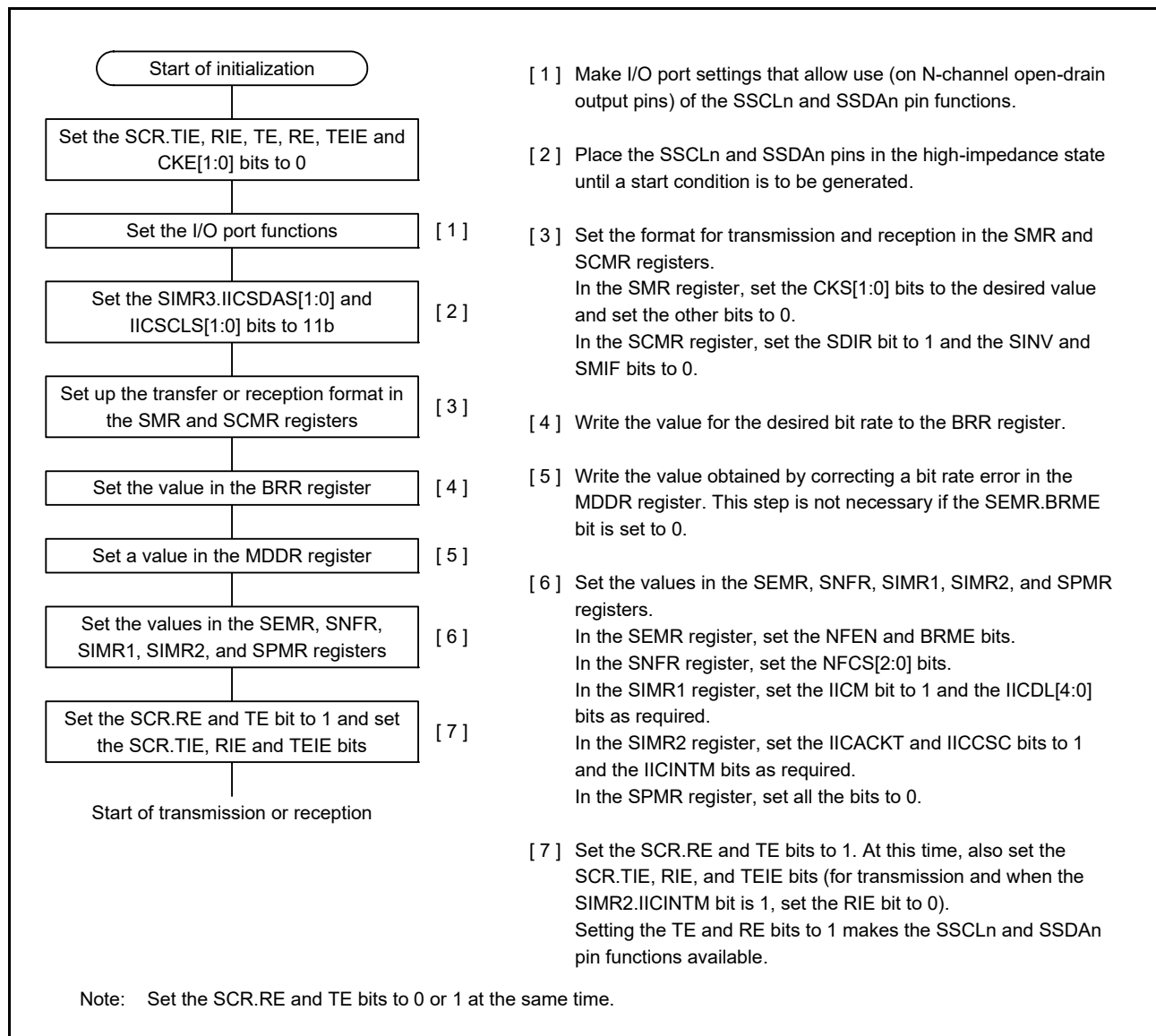


Figure 41.55 Example of the Flowchart of SCI Initialization (for Simple I²C Mode)

41.7.5 Operation in Master Transmission (Simple I²C Mode)

Figure 41.56 and Figure 41.57 show examples of operations in master transmission and Figure 41.58 is a flowchart showing the procedure for data transmission. Refer to Table 41.39 for more information on the STI interrupt.

When 10-bit slave addresses are in use, steps [3] and [4] in Figure 41.58 are repeated twice.

In simple I²C mode, the transmit data empty interrupt (TXI) is generated when communication of one frame is completed, unlike the TXI interrupt request generation timing during clock synchronous transmission.

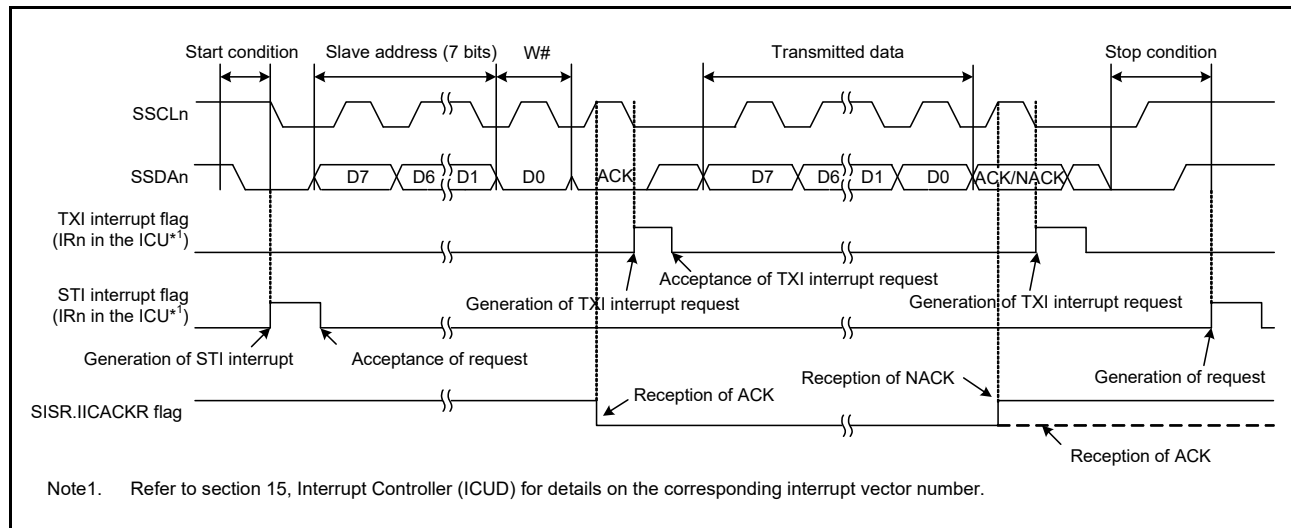


Figure 41.56 Example 1 of Operations for Master Transmission in Simple I²C-bus Mode (with 7-Bit Slave Addresses, Transmission Interrupts, and Reception Interrupts in Use)

When the SIMR2.IICINTM bit is set to 0 (use ACK/NACK interrupts) during master transmission, the DTC or DMAC is activated by the ACK interrupt as the trigger and necessary number of data bytes are transmitted. When the NACK is received, error processing, such as transmission stop and retransmission, is performed by the NACK interrupt as the trigger.

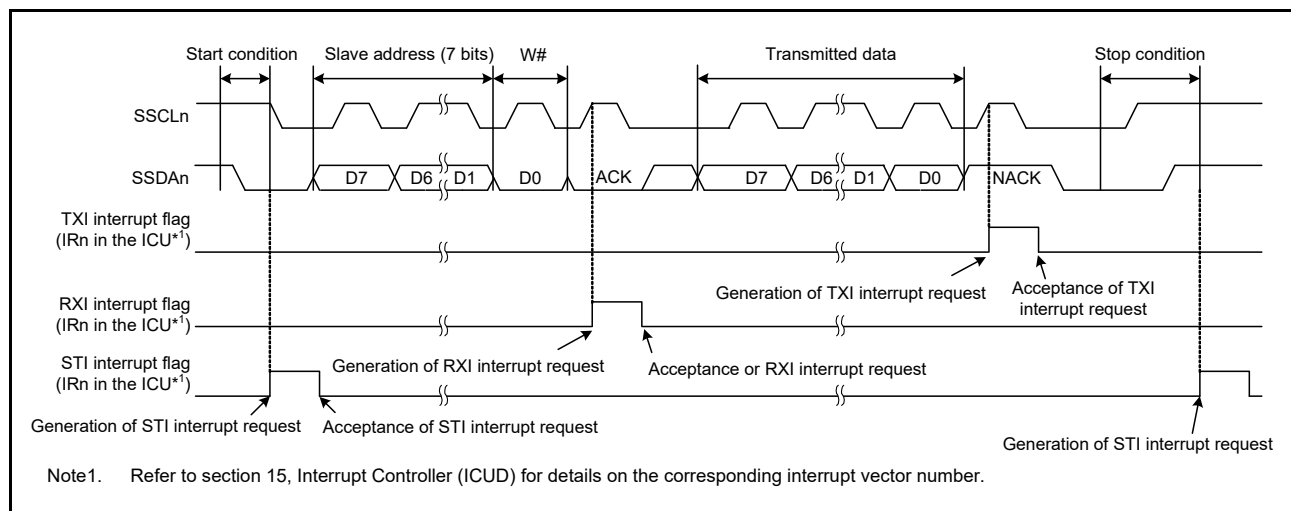


Figure 41.57 Example 2 of Operations for Master Transmission in Simple I²C-bus Mode (with 7-Bit Slave Addresses, ACK Interrupts, and NACK Interrupts in Use)

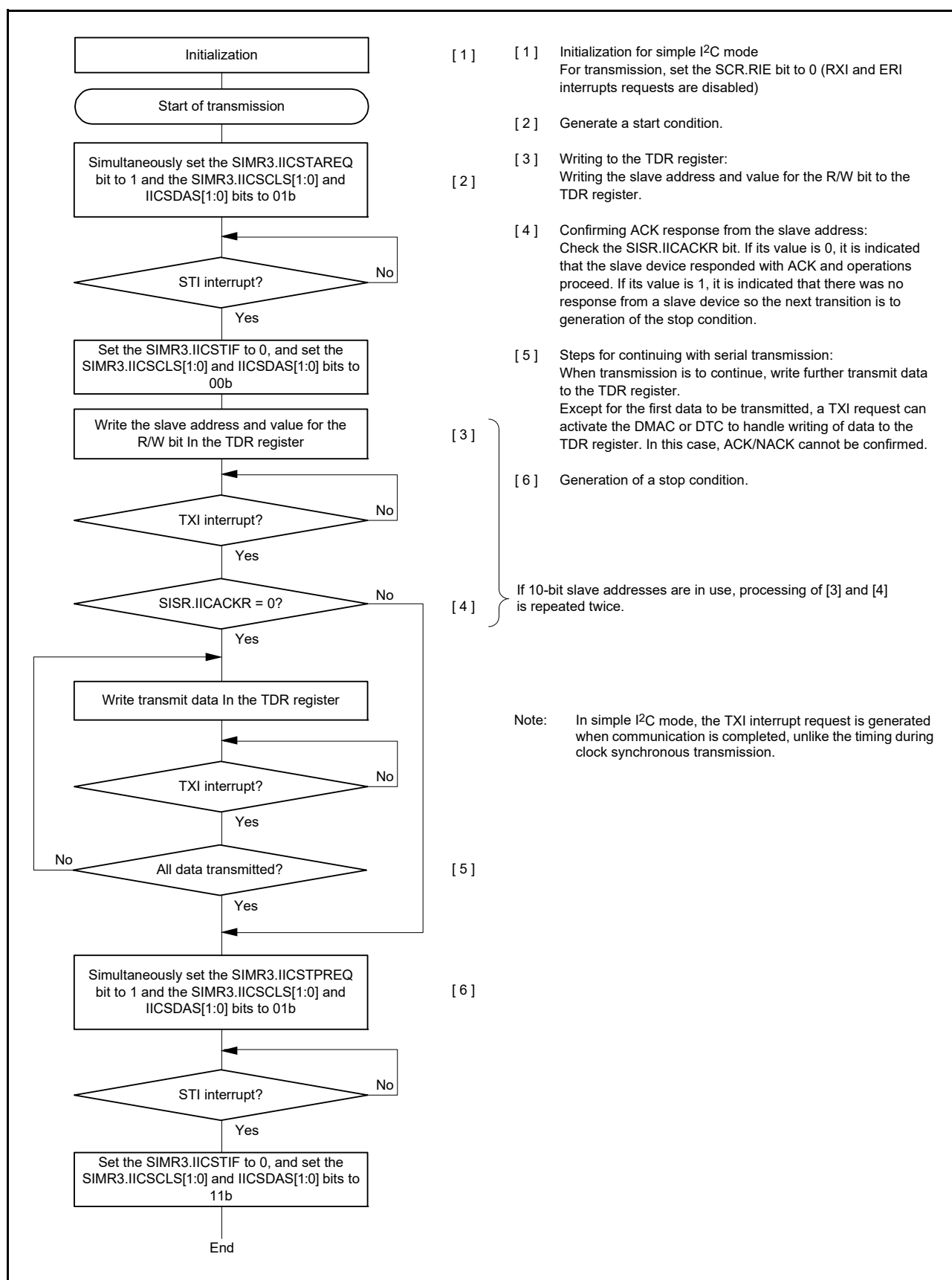


Figure 41.58 Example of the Procedure for Master Transmission Operations in Simple I²C Mode (with Transmission Interrupts and Reception Interrupts in Use)

41.7.6 Master Reception (Simple I²C Mode)

Figure 41.59 shows an example of operations in simple I²C mode master reception and Figure 41.60 is a flowchart showing the procedure for master reception.

The value of the SIMR2.IICINTM bit is assumed to be 1 (use reception and transmission interrupts).

In simple I²C mode, the transmit data empty interrupt (TXI) is generated when communication of one frame is completed, unlike the TXI interrupt request generation timing during clock synchronous transmission.

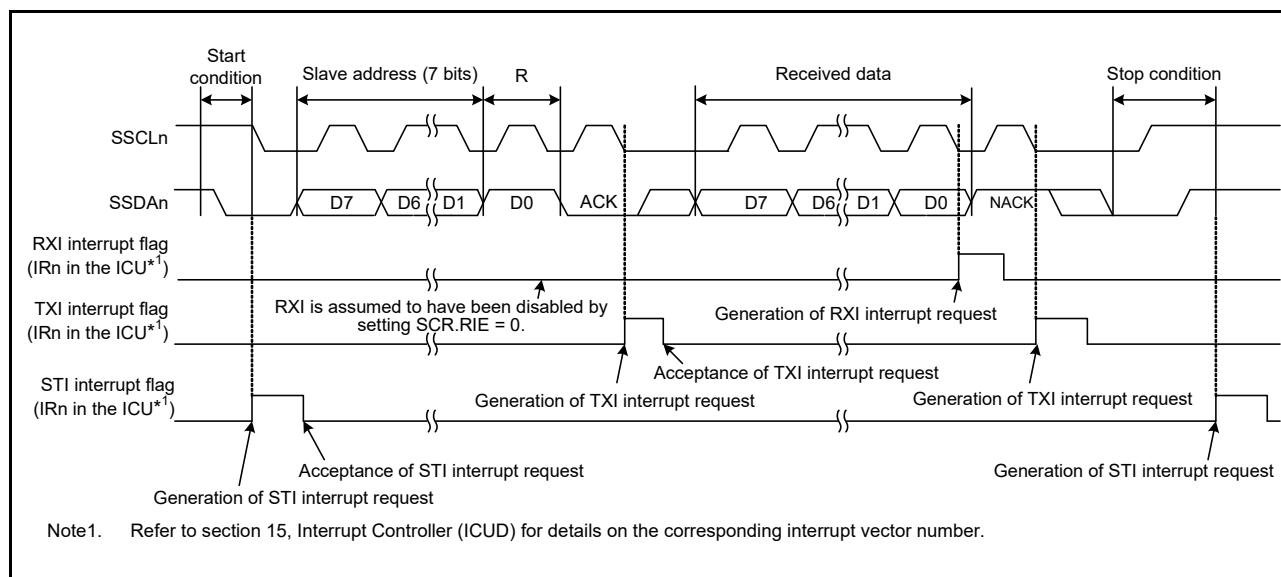


Figure 41.59 Example of Operations for Master Reception in Simple I²C-bus Mode (with 7-Bit Slave Addresses, Transmission Interrupts, and Reception Interrupts in Use)

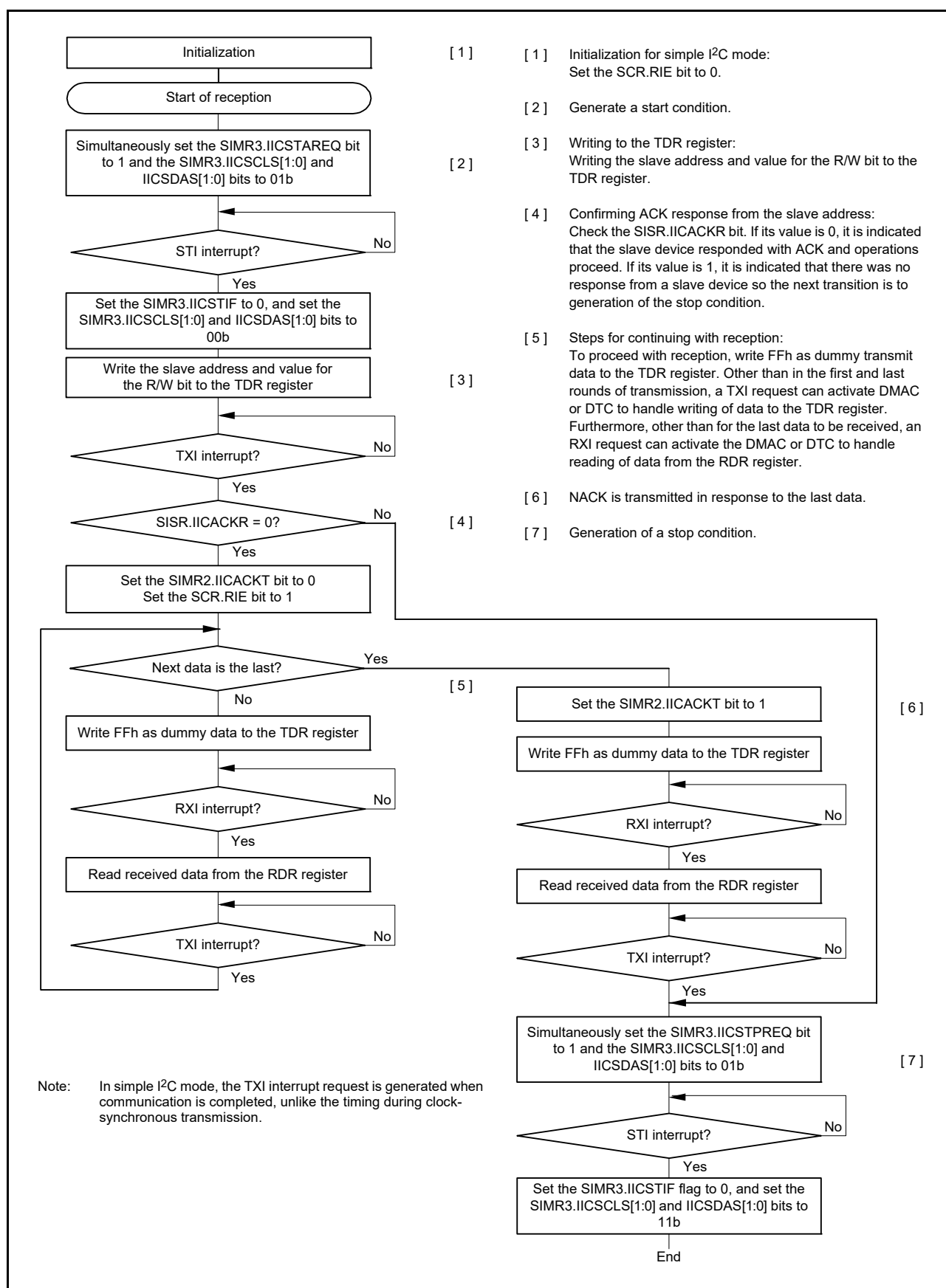


Figure 41.60 Example of the Procedure for Master Reception Operations in Simple I²C Mode (with Transmission Interrupts and Reception Interrupts in Use)

41.7.7 Recovery from Bus Hang-up

If the bus is stuck by an abnormal state in SCI because of the communication error, reset the SCI according to the following steps and release the bus.

- (1) Set the SCR.TE and RE bit to 0 at the same time to reset SCI.
- (2) Set the SIMR3 register to F0h to release the bus.
- (3) If the SSR.RDRF flag is 1, dummy-read the RDR register to clear the flag.
- (4) Set the SCR.TE and RE bit to 1 at the same time.

41.8 Operation in Simple SPI Mode

As an extended function, the SCI supports a simple SPI mode that handles transfer among one or multiple master devices and multiple slave devices.

Making the settings for clock synchronous mode (SCMR.SMIF = 0, SIMR1.IICM = 0, SMR.CM = 1) plus setting the SPMR.SSE bit to 1 places the SCI in simple SPI mode. However, the SS pin function on the master side is unnecessary for connection of the device used as the master in simple SPI mode when the configuration only has a single master, so set the SPMR.SSE bit to 0 in such cases.

Figure 41.61 shows an example of connections for simple SPI mode. Control a general port pin to produce the SS output signal from the master.

In simple SPI mode, data are transferred in synchronization with clock pulses in the same way as in clock synchronous mode. One character of data for transfer consists of 8 bits of data, and parity bits cannot be appended to this. The data can be inverted by setting the SCMR.SINV bit to 1.

Since the receiver and transmitter are independent of each other within the SCI module, full-duplex communications are possible, with a common clock signal. Furthermore, since both the transmitter and receiver have a double-buffered structure, writing of further transmit data while transmission is in progress and reading of previously received data while reception is in progress are both possible. Continuous transfer is thus possible.

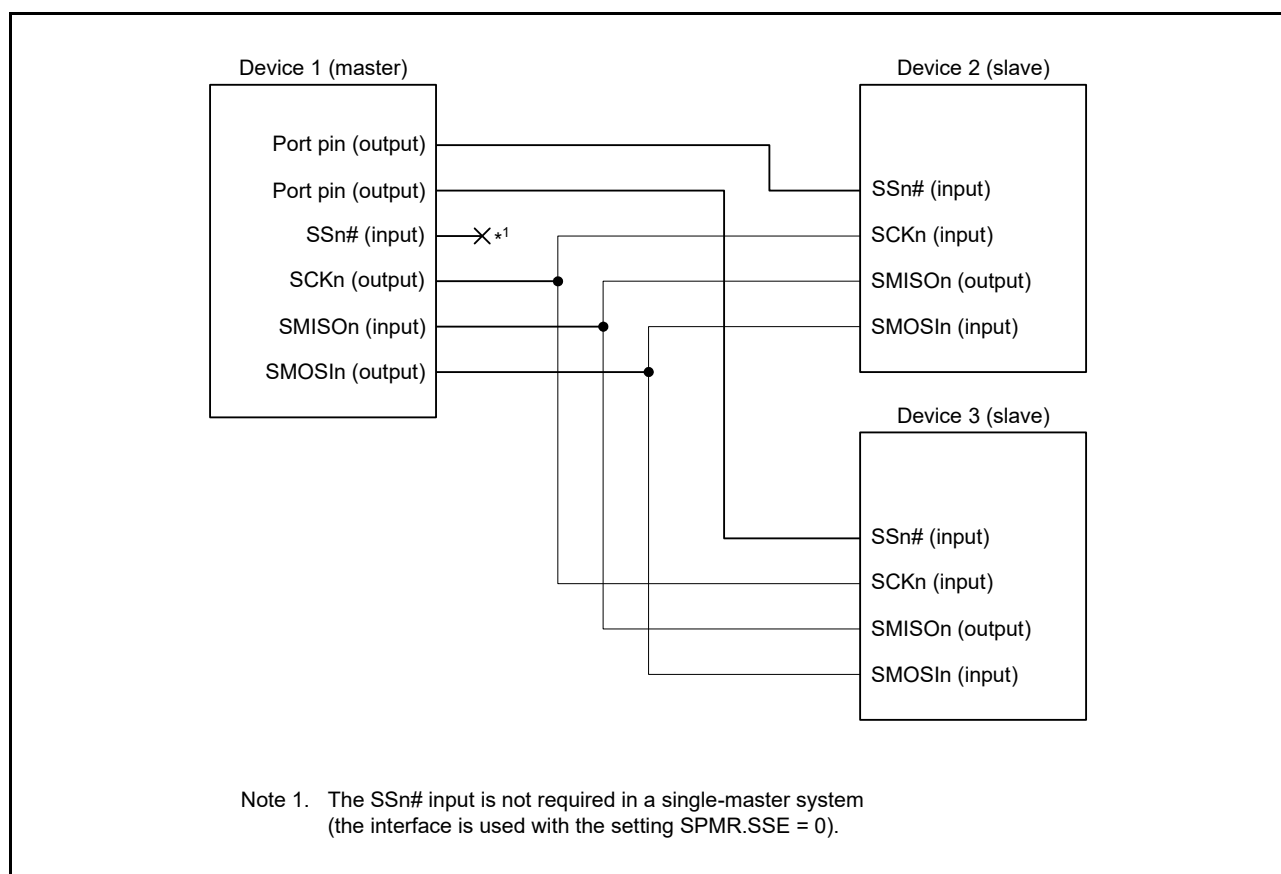


Figure 41.61 Example of Connections via a Simple SPI Mode (In Single Master Mode, SPMR.SSE Bit = 0)

41.8.1 States of Pins in Master and Slave Modes

The direction (input or output) of pins for the simple SPI mode interface differs according to whether the device is a master (SCR.CKE[1:0] = 00b or 01b and SPMR.MSS = 0) or slave (SCR.CKE[1:0] = 10b or 11b and SPMR.MSS = 1).

Table 41.34 lists the states of pins according to the mode and the level on the SSn# pin.

Table 41.34 States of Pins by Mode and Input Level on the SSn# Pin

Mode	Input on SSn# Pin	State of SMOSIn Pin	State of SMISOn Pin	State of SCKn Pin
Master mode*1	High level (transfer can proceed)	Output for data transmission*2	Input for received data	Clock output*3
	Low level (transfer cannot proceed)	High-impedance	Input for received data (but disabled)	High-impedance
Slave mode	High level (transfer cannot proceed)	Input for received data (but disabled)	High-impedance	Clock input (but disabled)
	Low level (transfer can proceed)	Input for received data	Output for data transmission	Clock input

Note 1. When there is only a single master (SPMR.SSE = 0), transfer is possible regardless of the input level on the SSn# pin (this is equivalent to input of a high level on the SSn# pin). Since the SSn# pin function is not required, the pin is available for other purposes.

Note 2. The SMOSIn pin output is in the high-impedance state when serial transmission is disabled (SCR.TE bit = 0).

Note 3. The SCKn pin output is in the high-impedance state when serial transmission is disabled (SCR.TE and RE bits = 00b) in a multi-master configuration (SPMR.SSE = 1).

41.8.2 SS Function in Master Mode

Setting the SCR.CKE[1:0] bits to 00b and the SPMR.MSS bit to 0 selects master operation. The SSn# pin is not used in single-master configurations (SPMR.SSE = 0), so transmission or reception can proceed regardless of the value of the SSn# pin.

When the level on the SSn# pin is high in a multi-master configuration (SPMR.SSE = 1), a master device outputs clock signals from the SCKn pin before starting transmission or reception to indicate that there are no other masters or another master is performing reception or transmission. When the level on the SSn# pin is low in a multi-master configuration (SPMR.SSE = 1), there are other masters, and this indicates that transmission or reception is in progress. At this time the SMOSIn output and SCKn pins will be placed in the high-impedance state and starting transmission or reception will not be possible. Furthermore, the value of the SPMR.MFF bit will be 1, indicating a mode fault error. In a multi-master configuration, start error processing by reading SPMR.MFF flag. Also, even if a mode fault error occurs while transmission or reception is in progress, transmission or reception will not be stopped, but the SMOSIn and SCKn pin output will be placed in the high-impedance state after the completion of the transfer.

Control a general port pin to produce the SS output signal from the master.

41.8.3 SS Function in Slave Mode

Setting the SCR.CKE[1:0] bits to 10b and the SPMR.MSS bit to 1 selects slave operation. When the level on the SSn# pin is high, the SMISOn output pin will be in the high-impedance state and clock input through the SCKn pin will be ignored. When the level on the SSn# pin is low, clock input through the SCKn pin will be effective and transmission or reception can proceed.

If the input on the SSn# pin changes from low to high level during transmission or reception, the SMISOn output pin will be placed in the high-impedance state. Meanwhile, the internal processing for transmission or reception will continue at the rate of the clock input through the SCKn pin until processing for the character currently being transmitted or received is completed, after which it stops. At that time, an interrupt (the appropriate one from among TXI, RXI, and TEI) will be generated.

41.8.4 Relationship between Clock and Transmit/Receive Data

The CKPOL and CKPH bits in the SPMR register can be used to set up the clock for use in transmission and reception in four different ways. The relation between the clock signal and the transmission and reception of data is shown in Figure 41.62. The relation is the same for both master and slave operation.

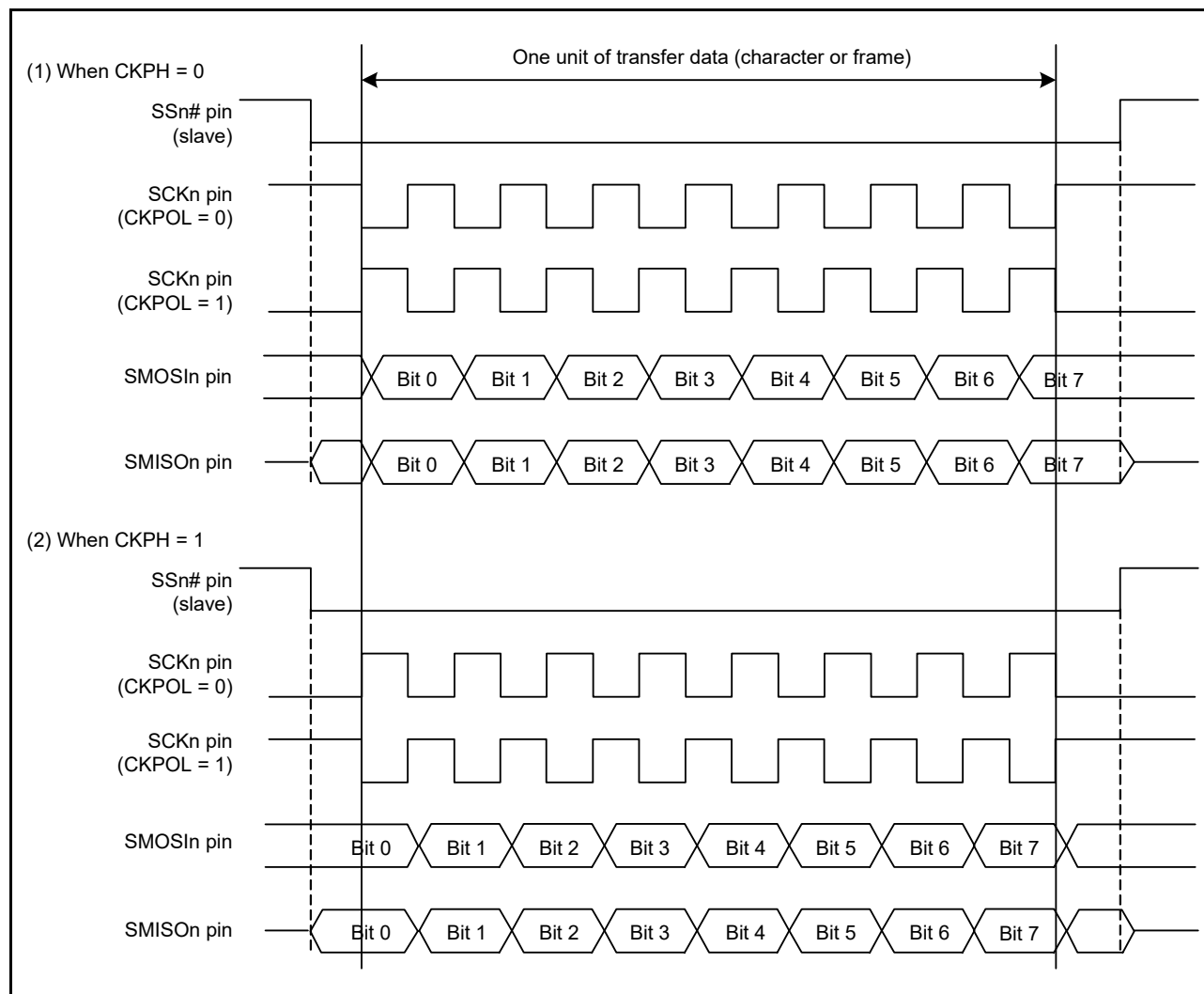


Figure 41.62 Relation between Clock Signal and Transmit/Receive Data in Simple SPI Mode

41.8.5 SCI Initialization (Simple SPI Mode)

The procedure is the same as for initialization in clock synchronous mode Figure 41.28, Sample SCI Initialization Flowchart. The CKPOL and CKPH bits in the SPMR register must be set to ensure that the kind of clock signal they select is suitable for both master and slave devices.

For initialization, changes to the operating mode, changes to the transfer format, and so on, initialize the SCR register before proceeding with changes.

As well as setting the RE bit to 0, note that the SSR. ORER, FER, and PER flags, as well as the RDR register, are not initialized.

41.8.6 Transmission and Reception of Serial Data (Simple SPI Mode)

In master operation, ensure that the SSn# pin of the slave device on the other side of the transfer is at the low level before starting the transfer and at the high level on completion of the transfer. Otherwise, the procedures are the same as in clock synchronous mode.

41.9 Bit Rate Modulation Function

The bit rate modulation function corrects the bit rate by thinning out the specified amount of clocks from those input to the baud rate generator.

When the SEMR.BRME bit is 1, the baud rate generator validates and counts the average interval of the number of clocks set in the MDDR register out of the total 256 clocks input.

Figure 41.63 assumes the SCI is in asynchronous mode, bits SMR.CKS[1:0] are 00b, the BRR register is 00h, and the MDDR register is 160. In this example, the cycle of the base clock is evenly corrected to 256/160, and the bit rate is corrected to 160/256. Note that there is an imbalance in thinning out the internal clock, and expansion and contraction occur in the pulse width of the base clock.

Note: Do not use this function in the highest speed settings (SMR.CKS[1:0] = 00b, SCR.CKE[1] = 0, and BRR = 0) in clock synchronous mode and simple SPI mode.

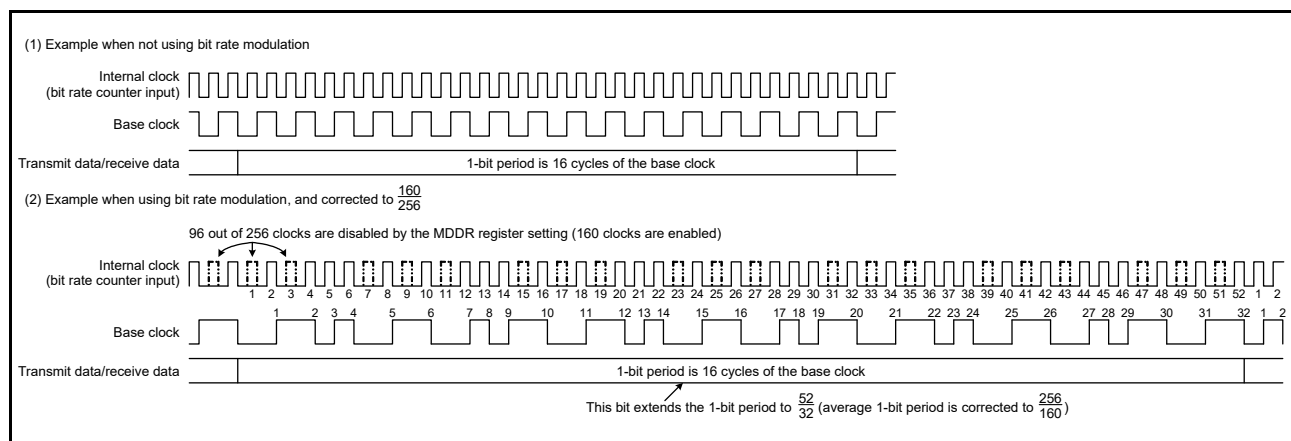


Figure 41.63 Example of the Base Clock When the Bit Rate Modulation Function is Used

The input of a clock signal with a shorter period to the baud rate generator reduces difference in the generated base clock period and, since the division ratio of the baud rate generator also becomes larger, reduces difference in the length of the 1-bit period.

41.10 Extended Serial Mode Control Section: Description of Operation

41.10.1 Serial Transfer Protocol

The extended serial mode control section of the SCI12 can realize the serial transfer protocol composed of Start Frames and Information Frames that is shown in Figure 41.64.

A Start Frame is composed of a Break Field, Control Field 0, and Control Field 1. An Information Frame is composed of a number of Data Fields, a CRC16 Upper Field, and a CRC16 Lower Field.

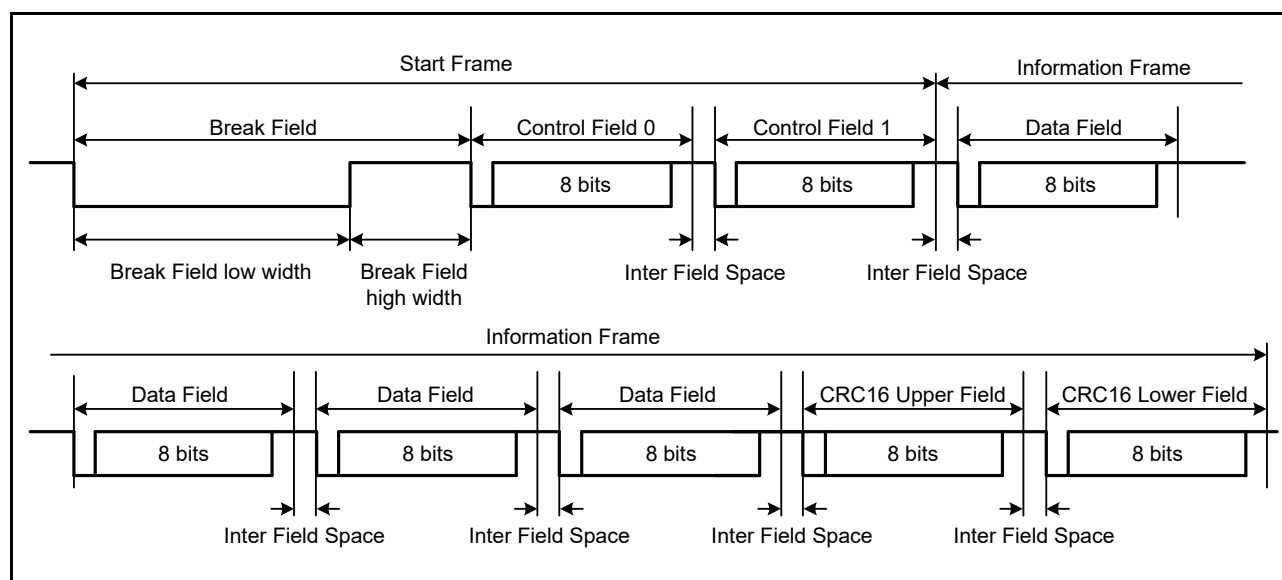


Figure 41.64 Protocol for Serial Transfer by the Extended Serial Mode Control Section

41.10.2 Transmitting a Start Frame

Figure 41.65 shows an example of operations to transmit a Start Frame, which is composed of the Break Field low width, Control Field 0, and Control Field 1. Figure 41.66 and Figure 41.67 are flowcharts for the transmission of a Start Frame.

Operations when the extended serial mode control section is to be used to transmit a Start Frame are as listed below. Be sure to use the SCI12 in asynchronous mode.

- (1) With Break Field low width output mode as the operating mode for the timer, writing 1 to the TCR.TCST bit starts counting by the timer, and the low level will be output from the TXDX12 pin over the period corresponding to registers TCNT and TPRE settings.
- (2) The output on the TXDX12 pin is inverted when the timer counter underflows, and the STR.BFDF flag is set to 1. An SCIX0 interrupt is also generated if the value of the ICR.BFDIE bit is 1.
- (3) Write 0 to the TCR.TCST bit to stop counting by the timer, and send the data for Control Field 0. After the Break Field low width output, stop counting before the next underflow occurs.
- (4) When the data for Control Field 0 have been transmitted, the data for Control Field 1 is transmitted.
- (5) When the data for Control Field 1 have been transmitted, an Information Frame is transmitted.

Omit the Break Field and Control Field 0 to suit the structure of the Start Frame.

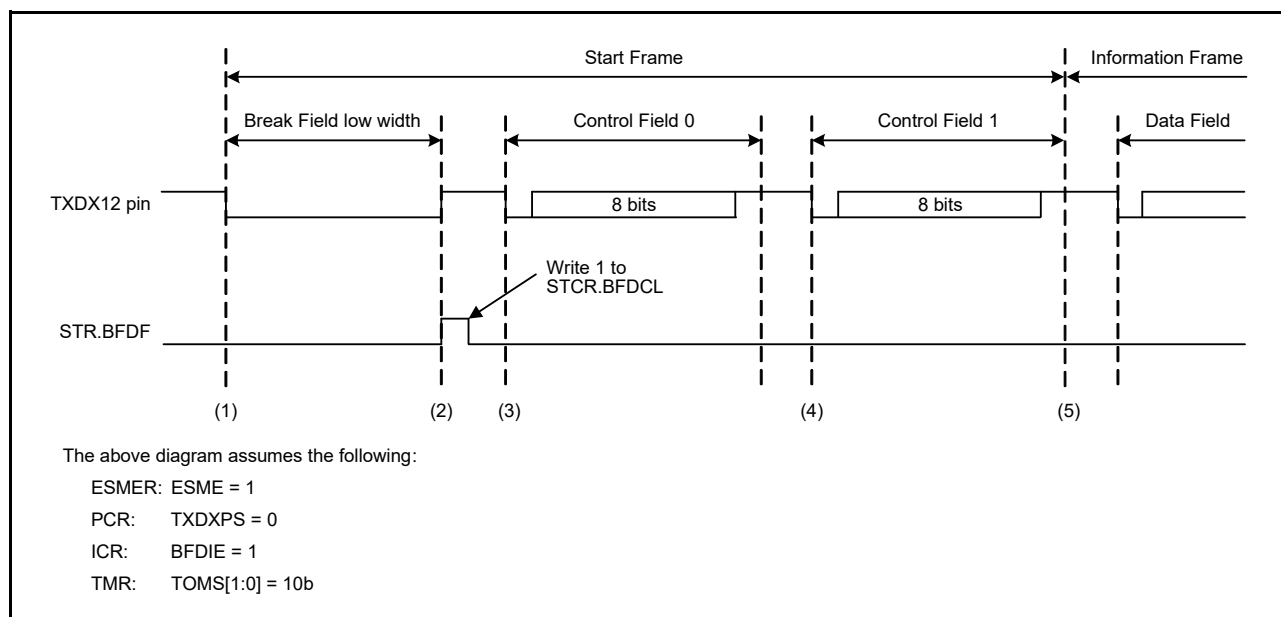


Figure 41.65 Example of Operations When Transmitting a Start Frame

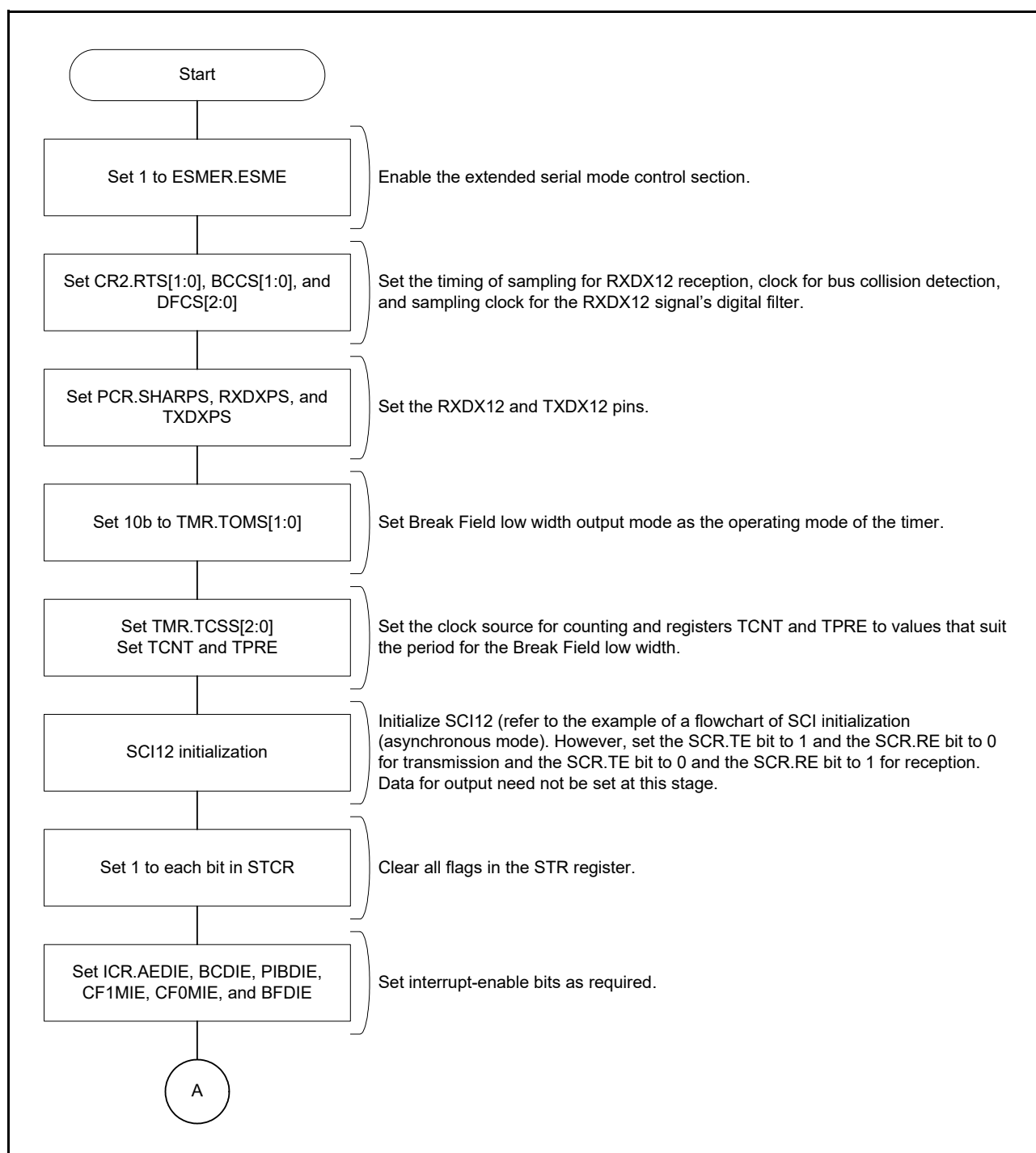


Figure 41.66 Example of Start Frame Transmission (1/2)

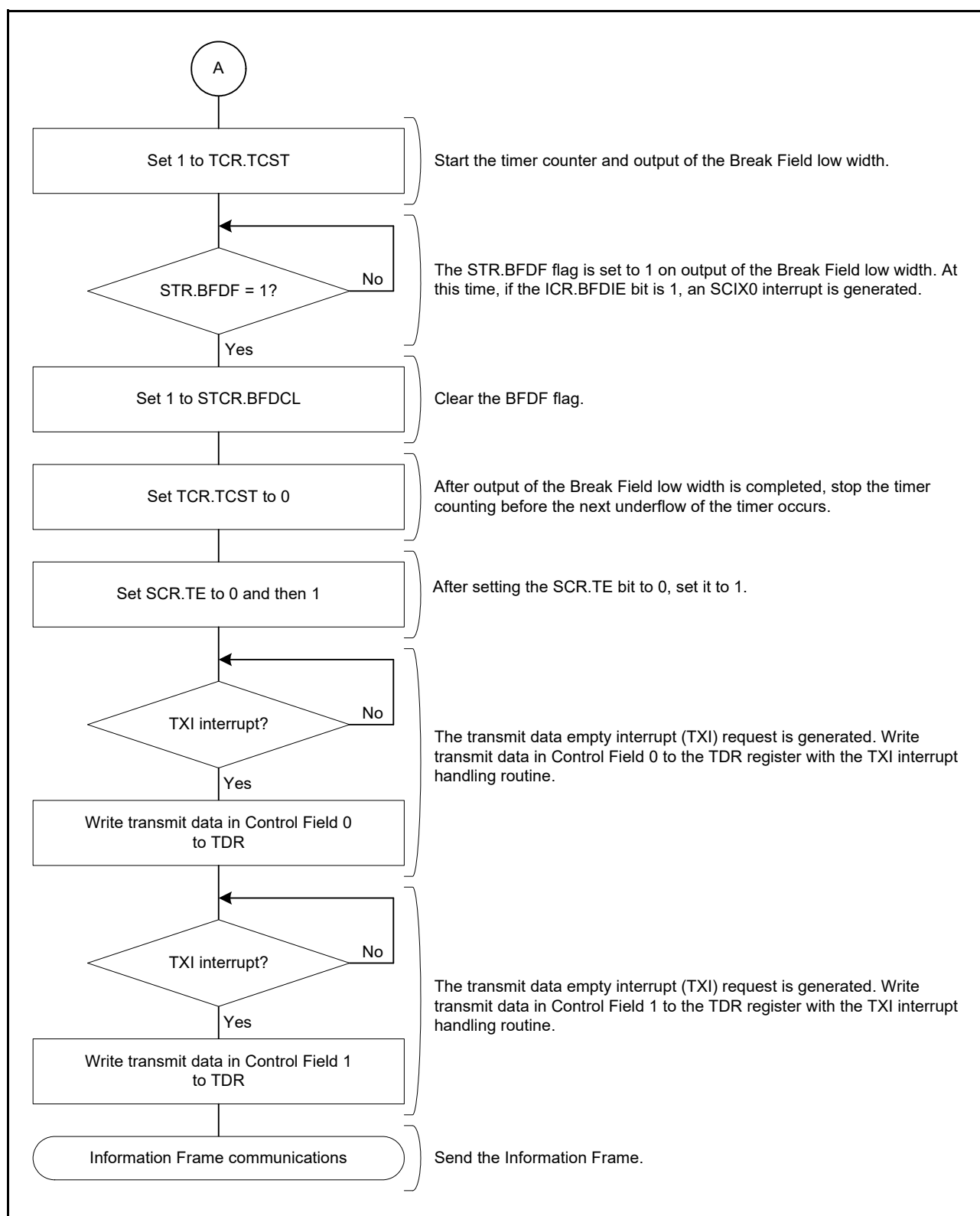


Figure 41.67 Example of Start Frame Transmission (2/2)

41.10.3 Receiving a Start Frame

The extended serial mode control section is capable of receiving Start Frames with the structures listed in Table 41.35.

Table 41.35 Structures of Start Frames

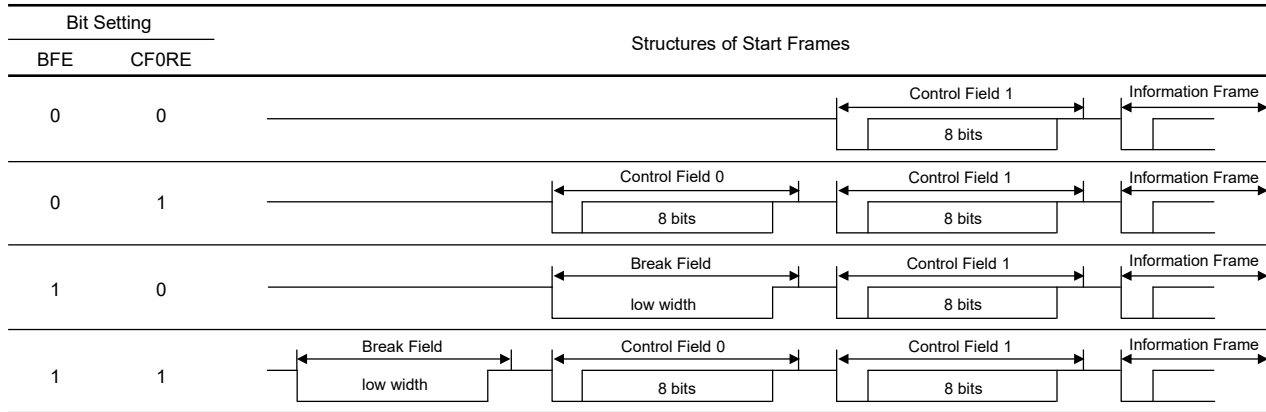


Figure 41.68 shows an example of operations to receive a Start Frame, which is composed of the Break Field low width, Control Field 0, and Control Field 1. Figure 41.69 and Figure 41.70 are flowcharts for the reception of a Start Frame, and Figure 41.71 is a state transition diagram when receiving a Start Frame.

Operations when the extended serial mode control section is to be used to receive a Start Frame are as listed below. Be sure to use the SCI12 in asynchronous mode.

- (1) With Break Field low width detection mode as the operating mode for the timer, writing 1 to the CR3.SDST bit enables detection of the Break Field low width.
- (2) Low-level input on the RXDX12 pin continuing over a period longer than that corresponding to the settings of registers TCNT and TPRE is detected as the Break Field low width. At this time, the STR.BFDF flag is set to 1. An SCIX0 interrupt is also generated if the value of the ICR.BFDIE bit is 1.
- (3) When the input from the RXDX12 pin goes high after the Break Field low width, the CR0.RXDSF flag becomes 0 and reception of Control Field 0 starts.
- (4) If the data received in Control Field 0 match the data set in the CF0DR register, the STR.CF0MF flag is set to 1. An SCIX1 interrupt is also generated if the value of the ICR.CF0MIE bit is 1. Reception of Control Field 1 starts after that. If the data received in Control Field 0 do not match the data set in the CF0DR register, a transition to the state prior to Break Field low width detection proceeds.
- (5) If the data received in Control Field 1 match the data set in registers PCF1DR and SCF1DR, the STR.CF1MF flag is set to 1. An SCIX1 interrupt is also generated if the value of the ICR.CF1MIE bit is 1. Transfer of the Information Frame starts after that. If the data received in Control Field 1 do not match the data set in either or both of registers PCF1DR and SCF1DR, a transition to the state prior to Break Field low width detection proceeds.

Omit the Break Field and Control Field 0 to suit the structure of the Start Frame.

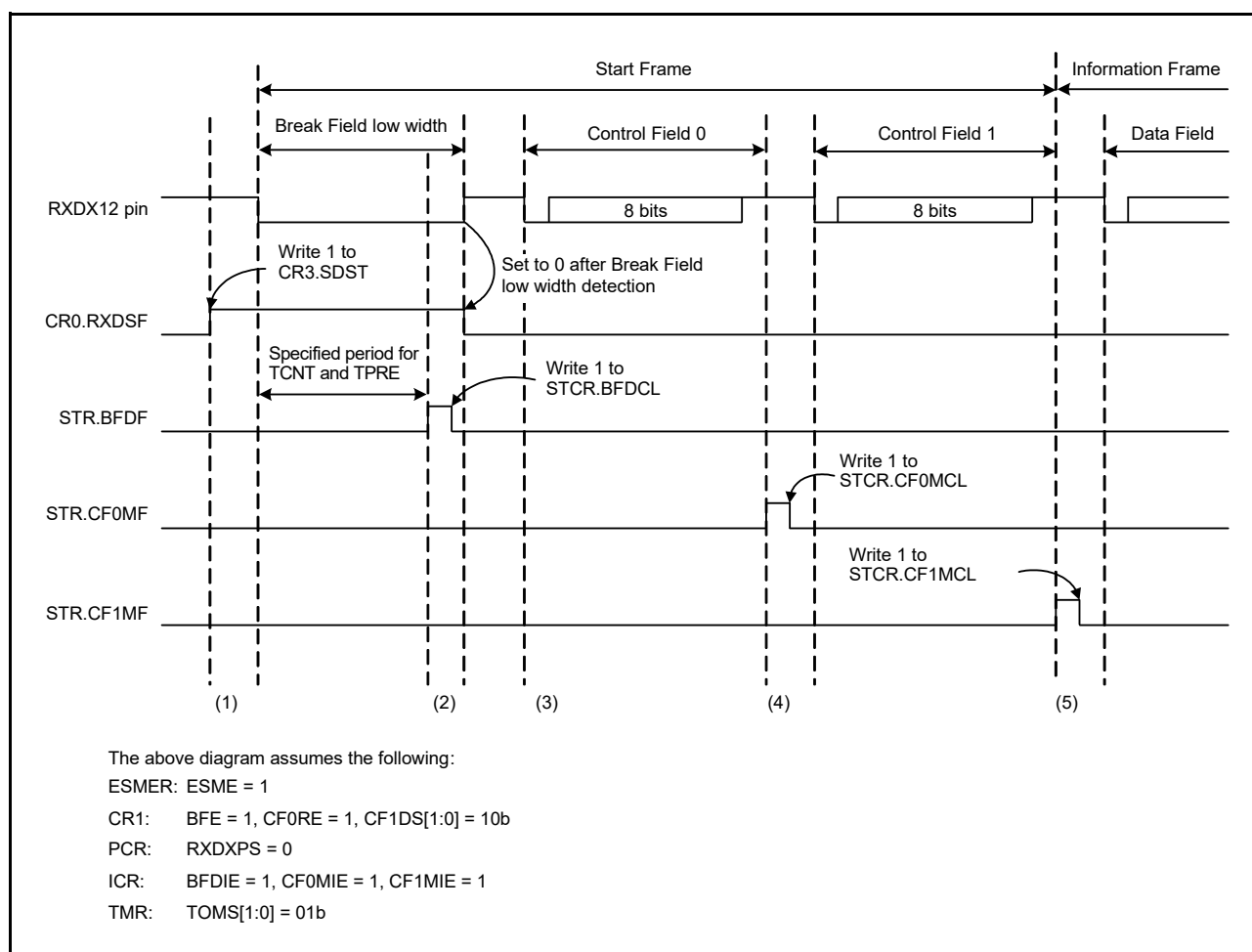


Figure 41.68 Example of Operations at the Time of Start Frame Reception

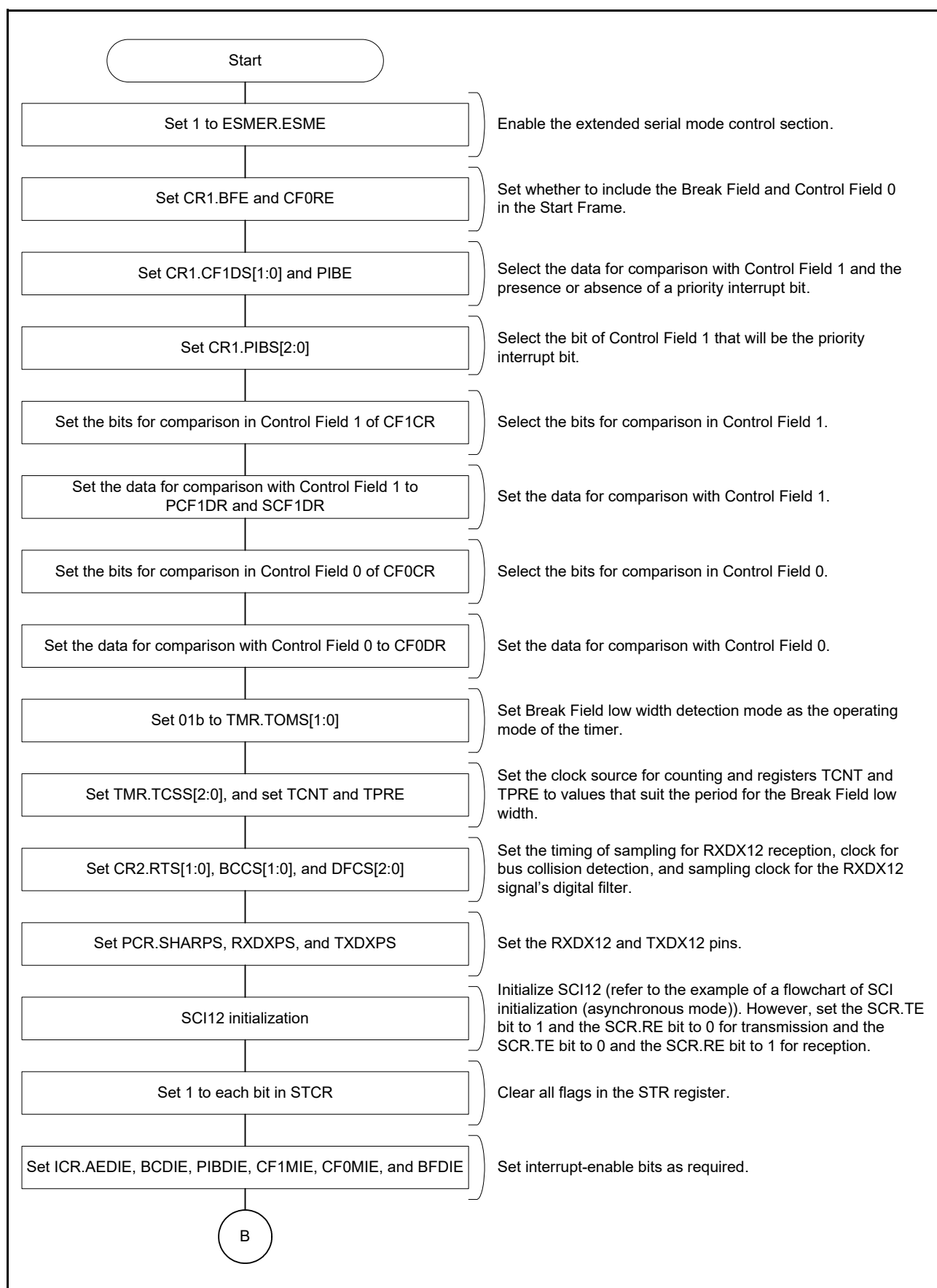


Figure 41.69 Sample Flowchart for Reception of a Start Frame (1)

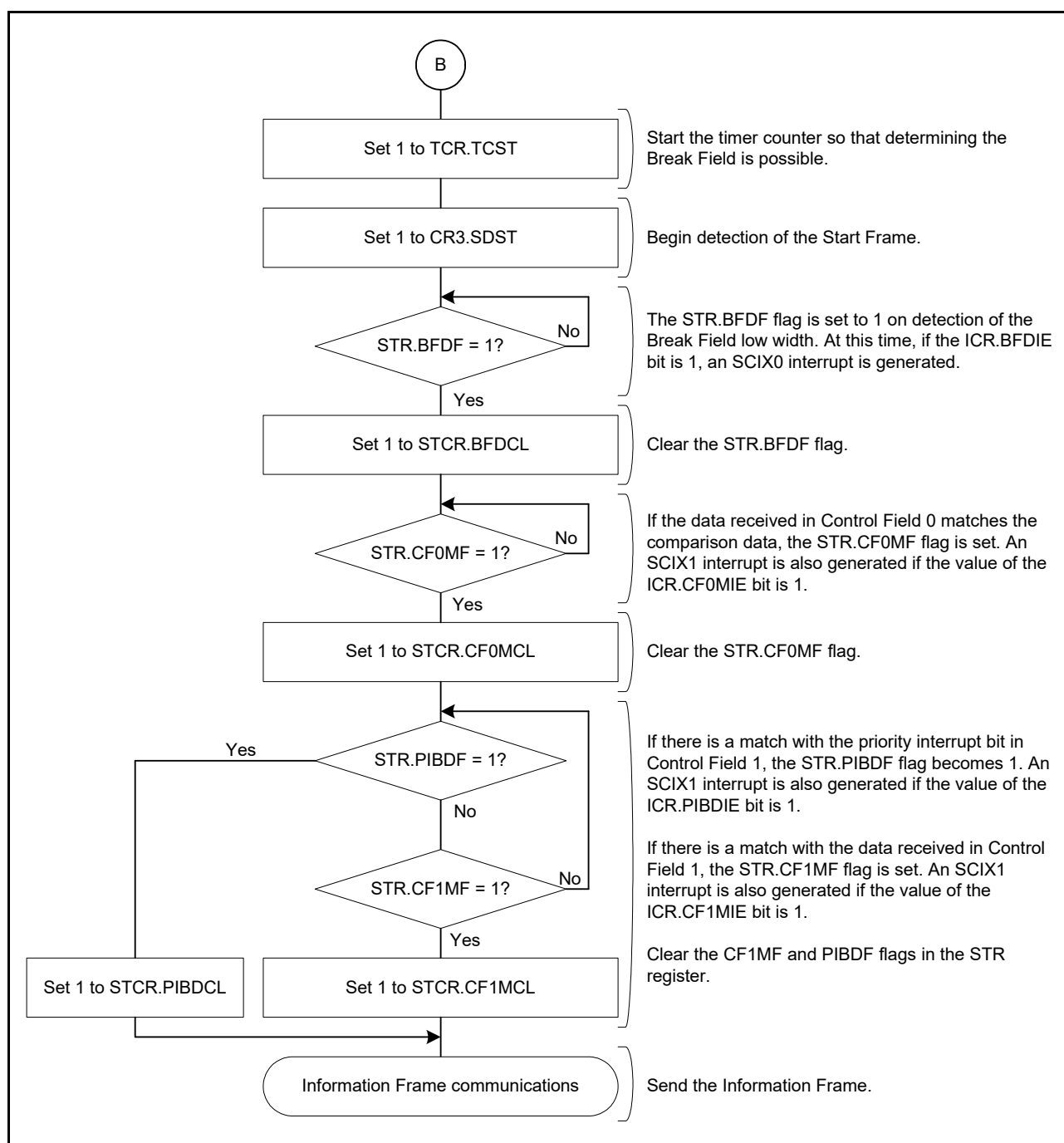
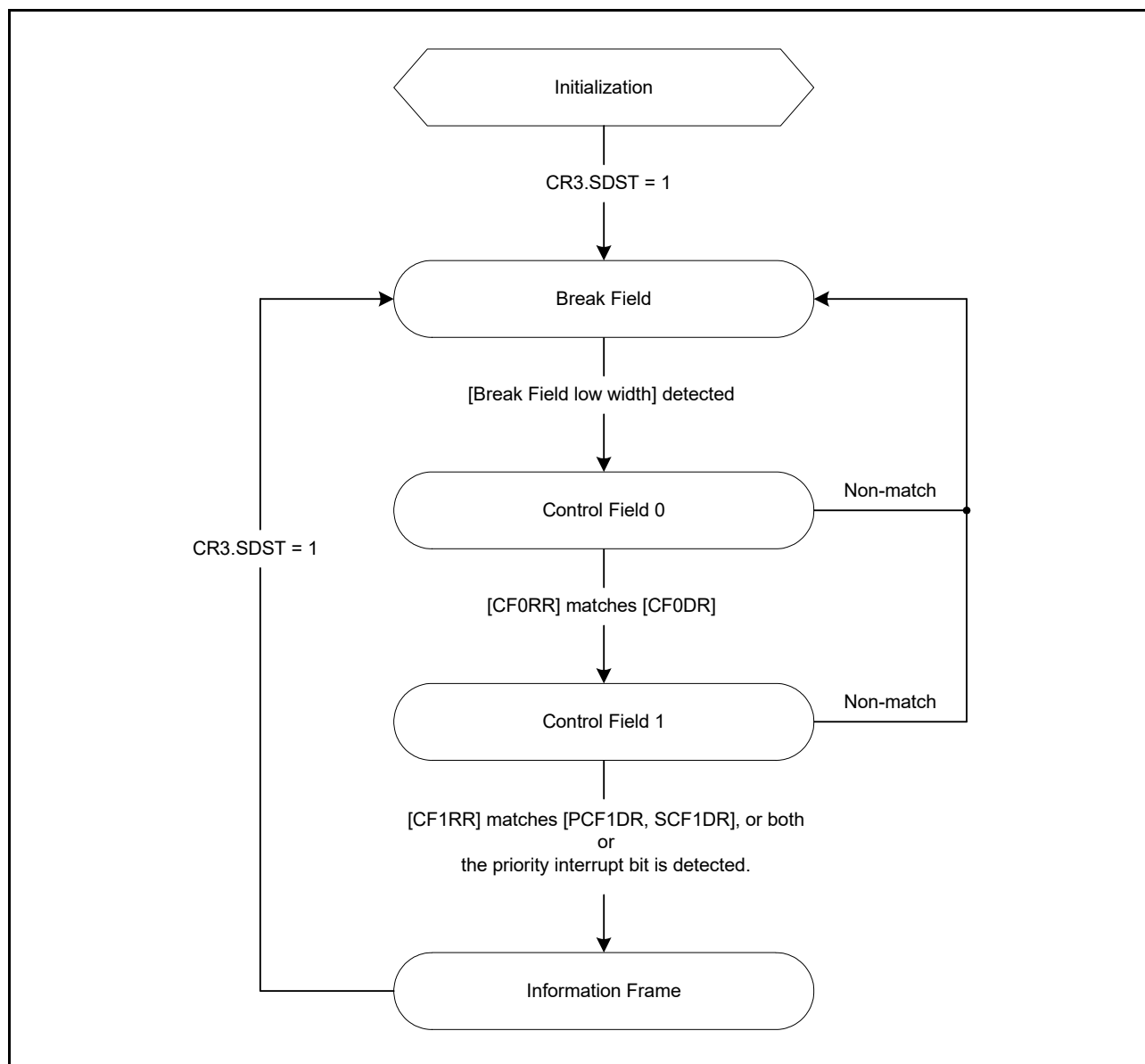


Figure 41.70 Sample Flowchart for Reception of a Start Frame (2)

**Figure 41.71 State Transitions When Receiving a Start Frame**

41.10.3.1 Priority Interrupt Bit

Figure 41.72 shows an example of operation in Start Frame reception where a priority interrupt bit is in use. Setting the CR1.PIBE bit to 1 enables the use of a priority interrupt bit.

Operations of the extended serial mode control section in start Frame reception where a priority interrupt bit is in use are as described below.

Steps (1) to (4) are the same as in Figure 41.68, for Start Frame reception.

- (5) If the value of the bit selected by the CR1.PIBS[2:0] bits matches the corresponding bit in the PCF1DR register, the STR.PIBDF flag is set to 1. An SCIX1 interrupt is also generated if the value of the ICR.PIBDIE bit is 1. Transfer of the Information Frame starts after that. If the data received in Control Field 1 do not match the data set in either or both of registers PCF1DR and SCF1DR and the priority interrupt bit is not detected, a transition to the state prior to Break Field low width detection proceeds.

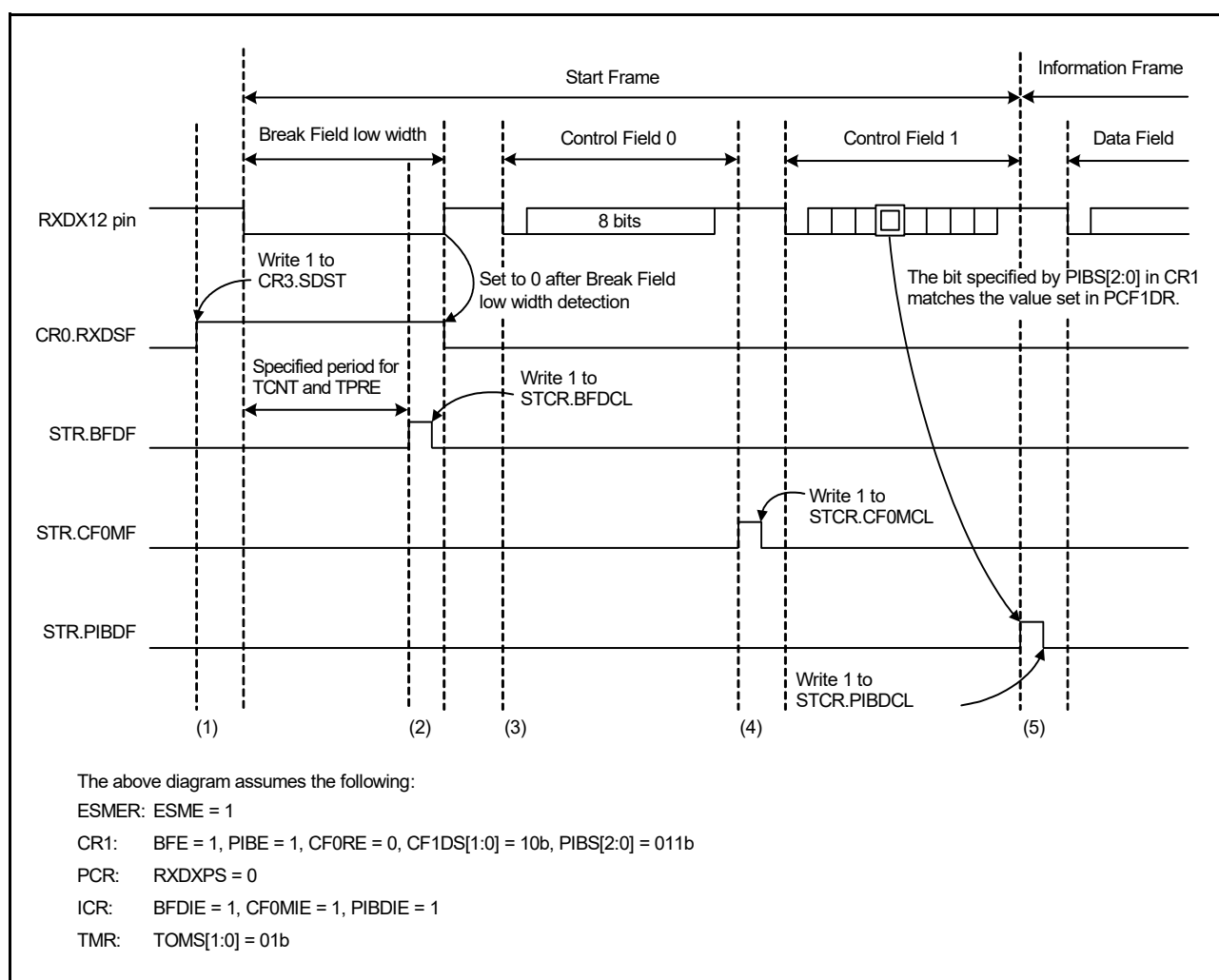


Figure 41.72 Example of Operations When Receiving a Start Frame While the CR1.PIBE Bit is 1

41.10.4 Detection of Bus Collisions

Detection of bus collisions operate for cases where output of the Break Field low width and transmission of data are in progress when the ESMER.ESME bit and the SCR.TE bit are set to 1.

Figure 41.73 shows an example of operations with bus collision detection. Signals output through TXDX12 and input through RXDX12 are sampled with the bus collision detection clock set with the CR2.BCCS[1:0] bits as the sampling clock, and the STR.BCDF flag is set to 1 if the signals fail to match three times in a row. An SCIX2 interrupt is also generated if the value of the ICR.BCDIE bit is 1.

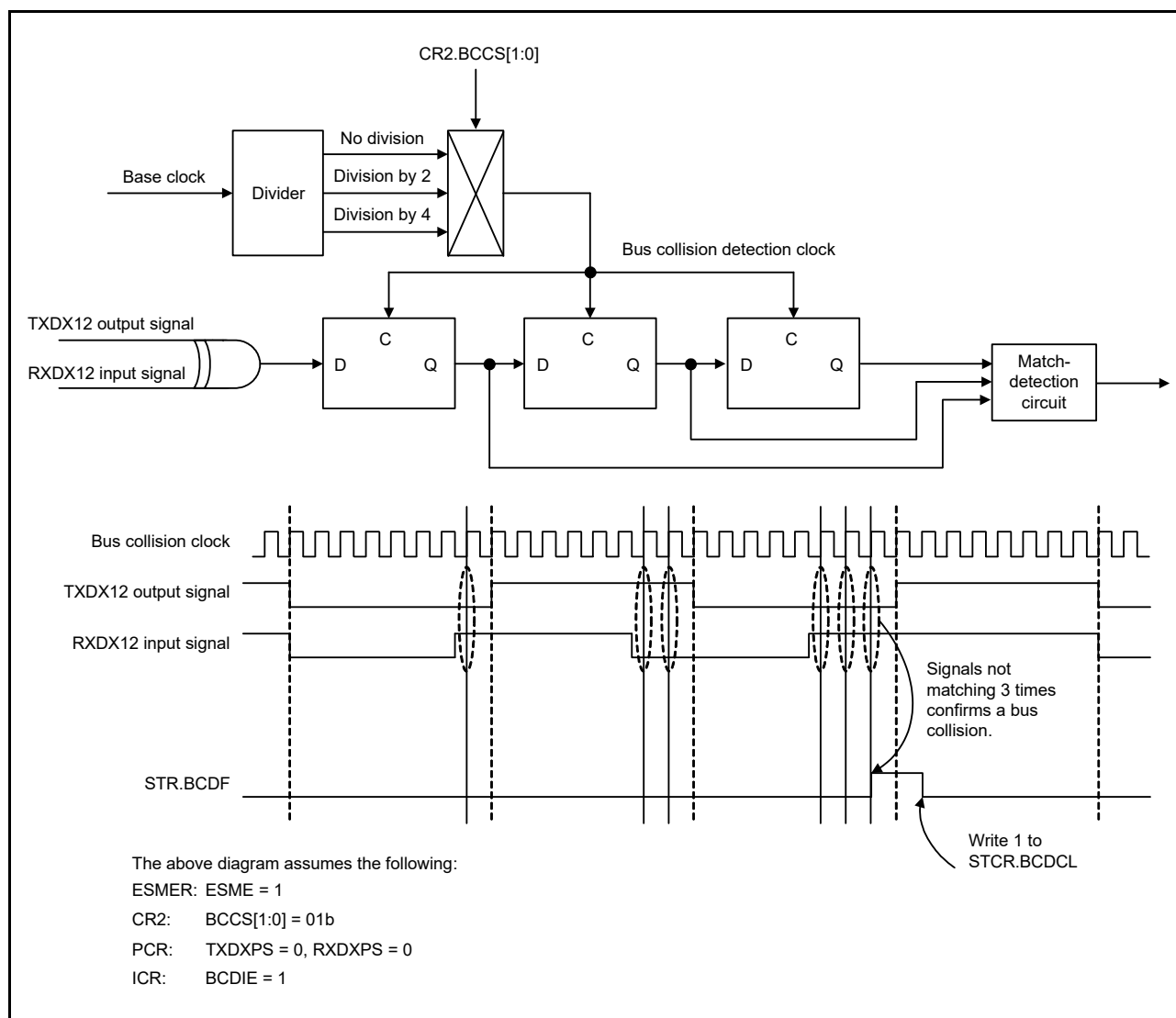


Figure 41.73 Example of Operations with Bus Collision Detection

41.10.5 Digital Filter for Input on the RXDX12 Pin

Signals input through the RXDX12 pin can be passed through a digital filter before they are conveyed to the internal circuits. The digital filter consists of three flip-flop circuit stages connected in series and a match-detecting circuit. The CR2.DFCS[2:0] bits select the sampling clock for the RXDX12 pin input signals. If the outputs of all three latches match, the given level is conveyed to subsequent circuits. If the levels do not match, the previous value is retained. In other words, levels are confirmed as being the signal if they are retained for at least three cycles of the sampling clock but judged to be noise rather than changes in the signal level if they change within three cycles of the sampling clock. Figure 41.74 shows an example of operations with the digital filter.

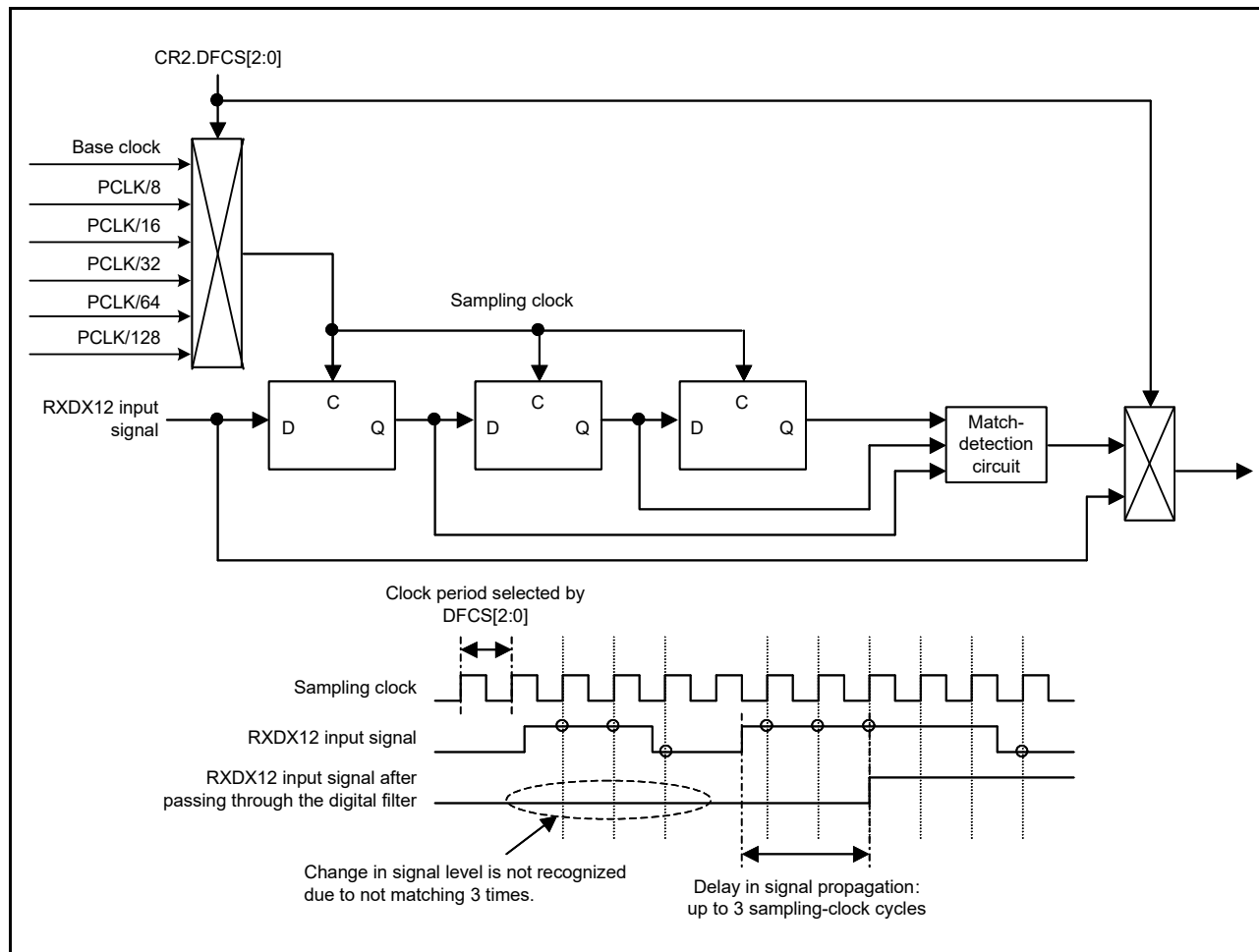


Figure 41.74 Example of Operations with the Digital Filter

41.10.6 Bit Rate Measurement

The bit rate measurement function measures the intervals between rising and falling edges and between falling and rising edges of the signal input from the RXDX12 pin. Figure 41.75 shows an example of operations for bit rate measurement.

- (1) Writing 1 to the CR0.BRME bit enables bit rate measurement. Only set the BRME bit to 1 when you wish to proceed with bit rate measurement. Furthermore, bit rate measurement will not proceed during a Break Field, even if the BRME bit is set to 1.
- (2) After detection of the Break Field low width, bit rate measurement starts when the level input on the RXDX12 pin becomes high.
- (3) Once bit rate measurement has started, counter values from the timer are retained in the read buffers on the input of valid edges from the RXDX12 pin (rising and falling edges) and the counter is reloaded. An SCIX3 interrupt is also generated if the value of the ICR.AEDIE bit is 1. Retention by registers TCNT and TPRES is released by reading these registers.
- (4) The bit rate as calculated from the values counted during intervals between valid edges can be used for adjusting the rate by changing the settings of the BRR register. To disable the bit rate measurement after a match with Control Field 1, write 0 to the CR0.BRME bit.

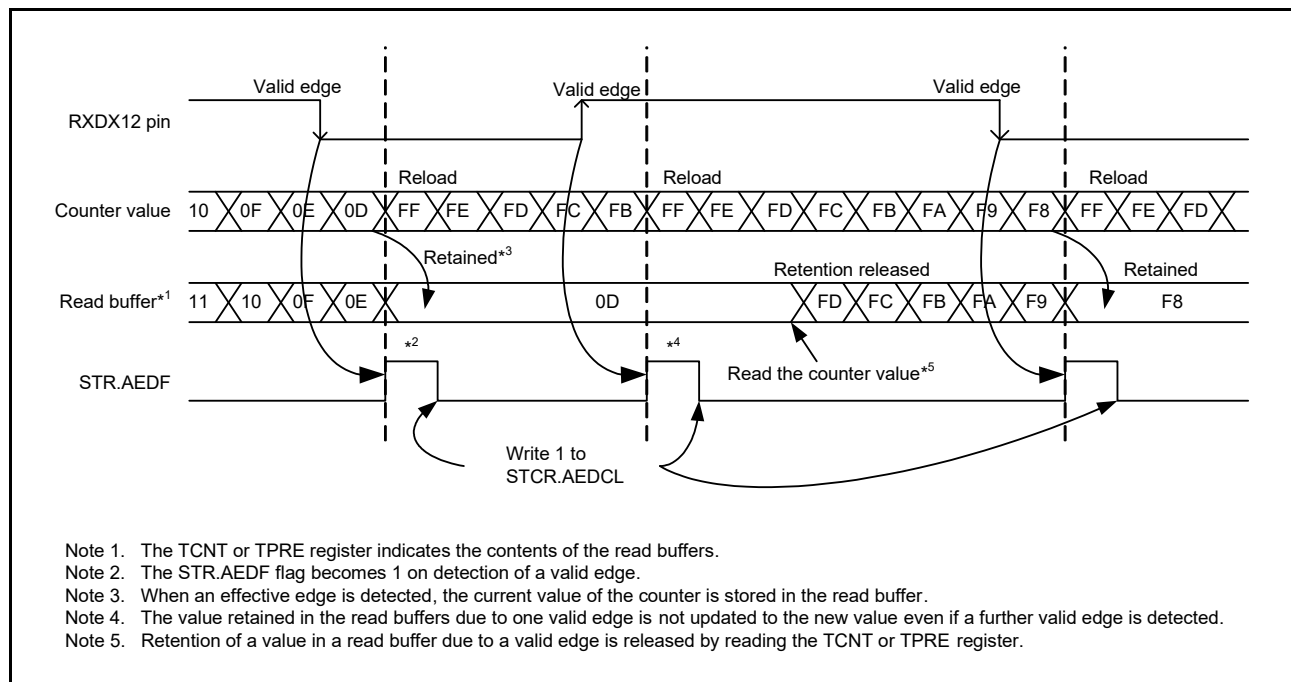


Figure 41.75 Example of Operations for Bit Rate Measurement

41.10.7 Selectable Timing for Sampling Data Received through RXDX12

The extended serial mode control section provides a way of adjusting the timing for the sampling of data received through the RXDX12 pin by setting the CR2.RTS[1:0] bits to select the rising edges of 8th, 10th, 12th, or 14th cycle of the base clock. If the value of the SEMR.ABCS bit is 1, the bits select the rising edges of 4th, 5th, 6th, or 7th cycle of the base clock. Figure 41.76 shows timing for the sampling of data received through RXDX12.

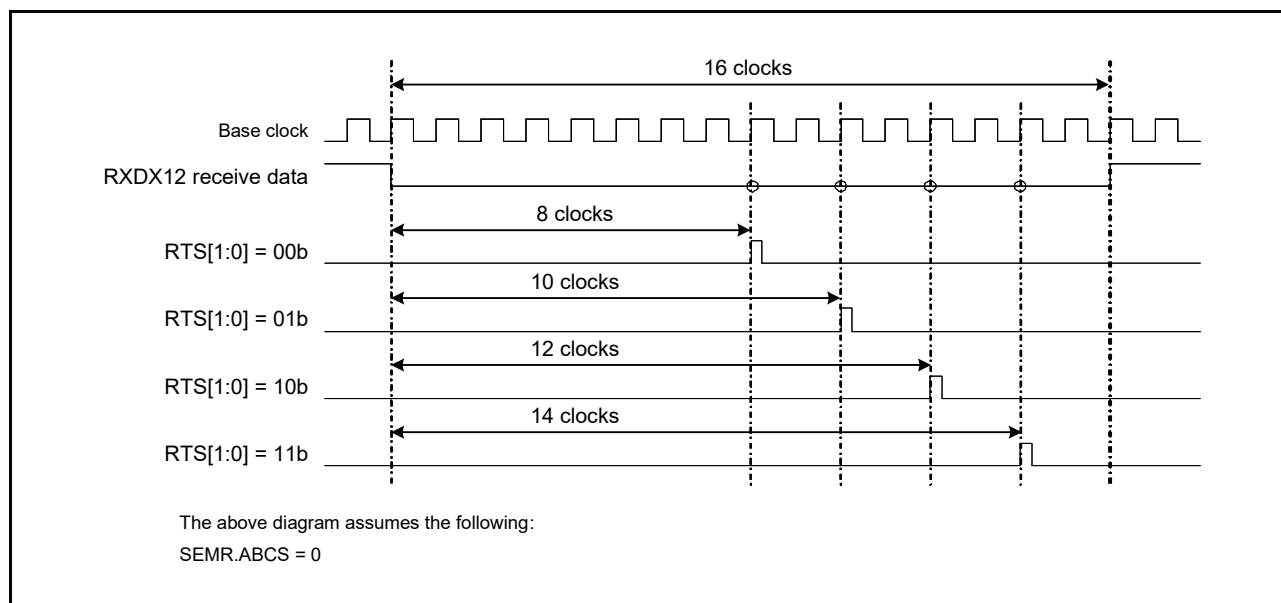


Figure 41.76 Timing for Sampling of Data Received through RXDX12

41.10.8 Timer

The timer has the following operating modes.

(1) Break Field Low Width Output Mode

This mode is for output through the TXDX12 pin of the low level over the Break Field low width at the transmission of a Start Frame. Setting the TMR.TOMS[1:0] bits to 10b switches operation to Break Field low width output mode. The TMR.TCSS[2:0] bits select the clock source for the counter. When the TCR.TCST bit is set to 1, the output on the TXDX12 pin goes to the low level and counting starts. When the timer underflows, the output on the TXDX12 pin goes to the high level and the STR.BFDF flag is set to 1. An SCIX0 interrupt is also generated if the value of the ICR.BFDIE bit is 1. When 0 is written to the TCR.TCST bit, counting stops after reloading of registers TPRES and TCNT. After output of the Break Field low width is completed, stop the timer before it underflows again. Figure 41.77 shows an example of operations in Break Field low width output mode.

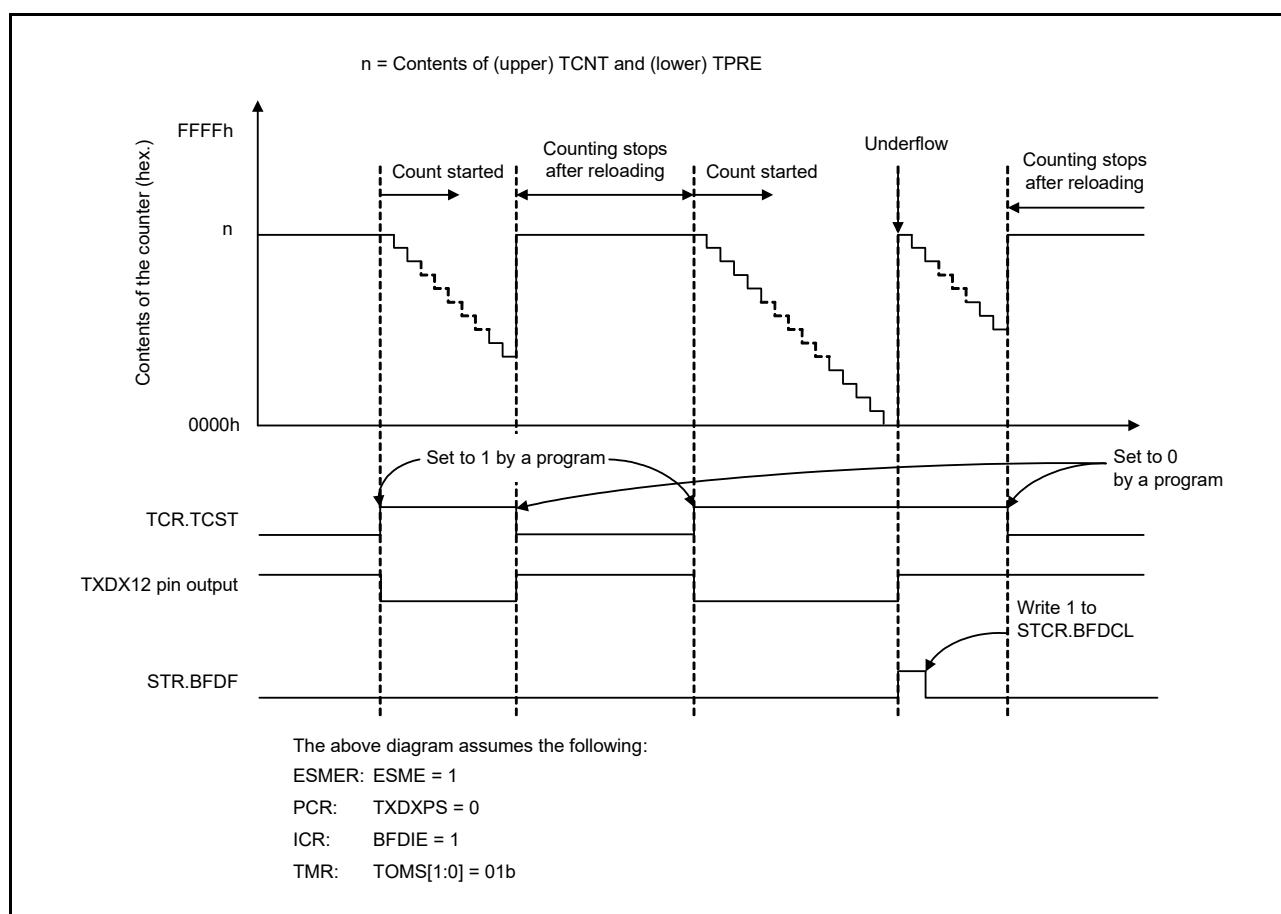


Figure 41.77 Example of Operations in Break Field Low Width Output Mode

(2) Break Field Low Width Determination Mode

This mode is for determining the Break Field low width in the input signal on the RXDX12 pin at the reception of a Start Frame. Setting the TMR.TOMS[1:0] bits to 01b switches operation to Break Field low width determination mode. The TMR.TCSS[2:0] bits select the clock source for the counter. When the TCR.TCST bit is set to 1, the interface enters the Break Field low width determinable state. Determination starts when a low level is input from the RXDX12 pin. When a high level is then input on the RXDX12 pin, registers TPRE and TCNT are reloaded and the interface enters the Break Field low width determinable state. When the timer underflows during Break Field low width determination, the STR.BFDF flag is set to 1. An SCIX0 interrupt is also generated if the value of the ICR.BFDIE bit is 1. If an underflow of the timer during data transfer cause a problem in the form of interrupt generation, stop the timer after Break Field low width determination. Figure 41.78 shows an example of operations in Break Field low width output mode.

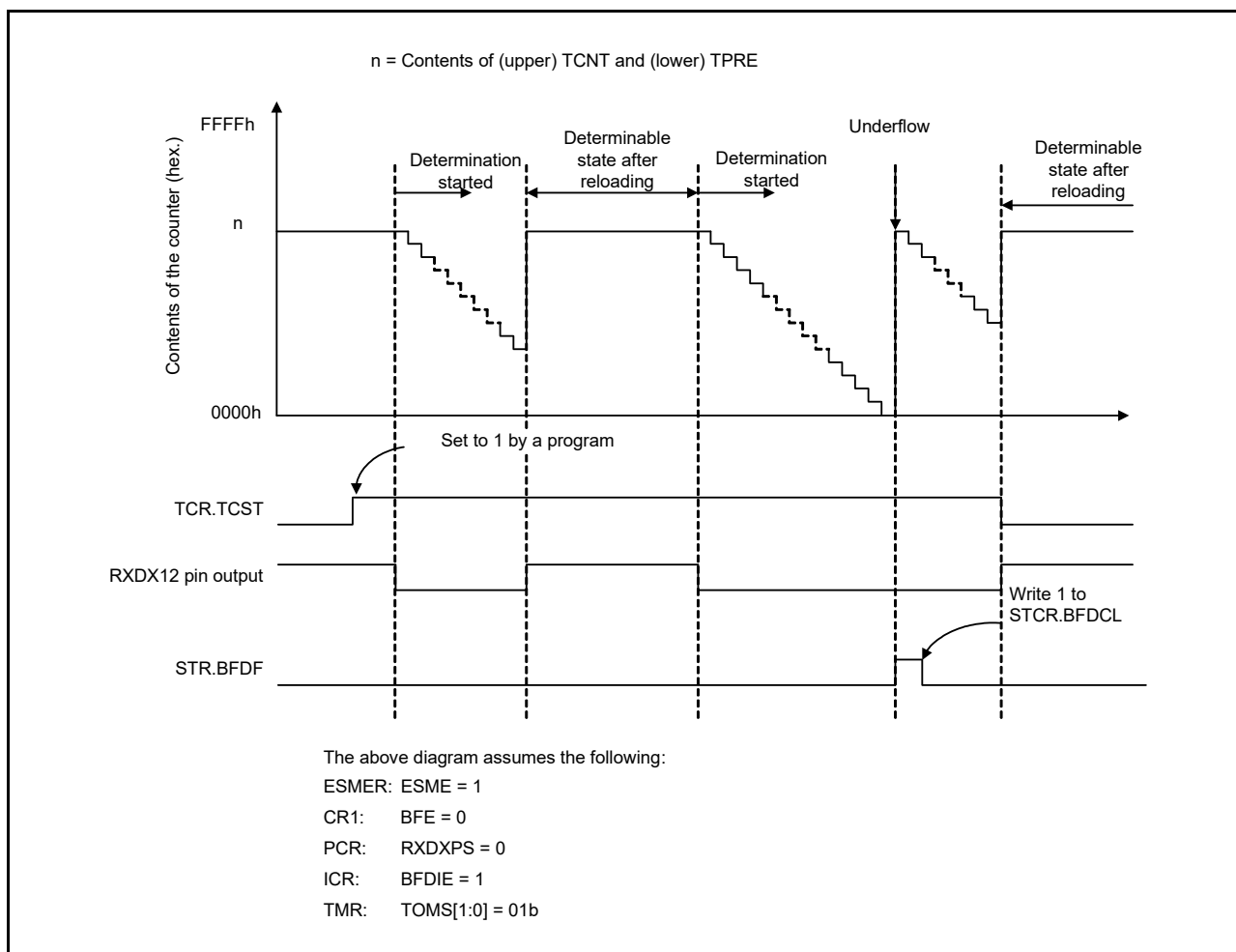


Figure 41.78 Example of Operations in Break Field Low Width Determination Mode

(3) Timer Mode

This mode is for counting cycles of the internal clock as the clock source. Setting the TMR.TOMS[1:0] bits to 00b switches operation to timer mode. The TMR.TCSS[2:0] bits select the clock source for the counter. Counting starts when 1 is written to the TCR.TCST bit and stops when 0 is written to the TCST bit. Registers TPRE and TCNT both count down. The TPRE register counts cycles of the clock source for counting, and underflows of the TPRE register provide the clock source for counting by the TCNT register. When the timer underflows, the STR.BFDF flag is set to 1. An SCIX0 interrupt is also generated if the value of the ICR.BFDIE bit is 1.

41.11 Noise Cancellation Function

Figure 41.79 shows the configuration of the noise filter used for noise cancellation. The noise filter consists of two stages of flip-flop circuits and a match-detection circuit. When the level on the pin matches in three consecutive samples taken at the set sampling interval, the matching level continues to be conveyed internally until the level on the pin again matches in three consecutive samples.

In asynchronous mode, the noise cancellation function can be applied on the RXDn input signal. The period of the base clock ($1/16$ th of a bit-period when SEMR.ABCSE = 0 and SEMR.ABCS = 0, $1/8$ th of a bit-period when SEMR.ABCSE = 0 and SEMR.ABCS = 1, and $1/6$ th of a bit-period when SEMR.ABCSE = 1) is the sampling interval.

In simple I²C mode, the noise cancellation function can be applied on the SSDAn and SSCLn input signals. The sampling clock is the clock signal produced by frequency-dividing the signal from the clock source for the internal baud-rate generator by one, two, four, or eight as selected by the setting of the SNFR.NFCS[2:0] bits.

If the base clock is stopped with the noise filter enabled and then the clock input is started again, the noise filter operation resumes from where the clock was stopped. If SCR.TE and SCR.RE are set to 0 during base clock input, all of the noise filter flip-flop values are initialized to 1. Accordingly, if the input data is 1 when reception operation resumes, it is determined that a level match is detected and is conveyed to the internal signal. When the level being input corresponds to 0, the initial output of the noise filter is retained until the level matches in three consecutive samples.

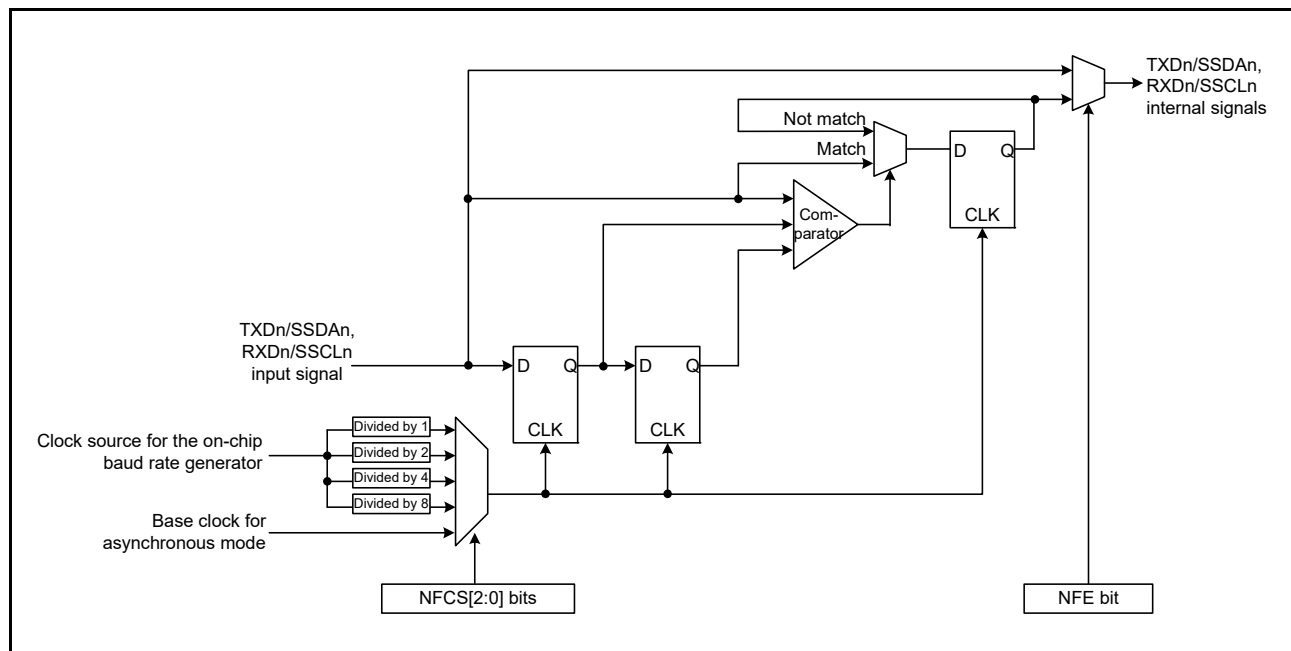


Figure 41.79 Block Diagram of Digital Noise Filter

41.12 Interrupt Sources

41.12.1 Buffer Operations for TXI and RXI Interrupts

(1) SCI0 to SCI6 and SCI12, and SCI7 to SCI11 when FIFO is Disabled

If the conditions for a TXI and RXI interrupt are satisfied while the interrupt status flag in the interrupt controller is 1, the SCI does not output the interrupt request but retains it internally (with a capacity for retention of one request per source). When the value of the interrupt status flag in the interrupt controller becomes 0, the interrupt request retained within the SCI is output. The internally retained interrupt request is automatically discarded once the actual interrupt is output. Clearing of the corresponding interrupt enable bit (the TIE or RIE bit in the SCR register) can also be used to discard an internally retained interrupt request.

(2) SCI7 to SCI11 When FIFO is Enabled

When the FCR.FM bit is 1, the SCI does not retain the interrupt request internally. When the interrupt status flag in the interrupt controller is 1 and the condition for interrupt generation is satisfied, the status flag is updated, but an interrupt request is not generated.

41.12.2 Interrupts in Asynchronous Mode, Clock Synchronous Mode, and Simple SPI Mode

(1) SCI0 to SCI6 and SCI12, and SCI7 to SCI11 when FIFO is Disabled

Table 41.36 lists interrupt sources in asynchronous mode, clock synchronous mode, and simple SPI mode. Individual interrupt sources can be enabled or disabled with the enable bits in the SCR register.

If the SCR.TIE bit is 1, a TXI interrupt request is generated when transmit data is transferred from the TDR or TDRL register*¹ to the TSR register. A TXI interrupt request can also be generated by setting the SCR.TE bit to 1 after setting the SCR.TIE bit to 1 or by using a single instruction to set the SCR.TE and SCR.TIE bit to 1 at the same time. A TXI interrupt request can activate the DTC or DMAC to handle data transfer.

A TXI interrupt request is not generated by setting the SCR.TE bit to 1 while the setting of the SCR.TIE bit is 0 or by setting the SCR.TIE bit to 1 while the setting of the SCR.TE bit is 1.*²

Note that setting the SCR.TE bit to 0 while the SCR.TIE bit is 1 leads to the generation of a TXI interrupt.

When new data is not written by the time of transmission of the last bit of the current transmit data and the setting of the SCR.TEIE bit is 1, the SSR.TEND flag becomes 1 and a TEI interrupt request is generated. Furthermore, when the setting of the SCR.TE bit is 1, the SSR.TEND flag retains the value 1 until further transmit data are written to the TDR or TDRL register*¹, and setting the SCR.TEIE bit to 1 leads to the generation of a TEI interrupt request.

Writing data to the TDR or TDRL register*¹ leads to clearing of the SSR.TEND flag and, after a certain time, discarding of the TEI interrupt request.

If the SCR.RIE bit is 1, an RXI interrupt request is generated when received data is stored in the RDR register. An RXI interrupt request can activate the DTC or DMAC to handle data transfer.

Setting of any from among the ORER, FER, and PER flags in the SSR register to 1 while the SCR.RIE bit is 1 leads to the generation of an ERI interrupt request. An RXI interrupt request is not generated at this time. Clearing all three flags (ORER, FER, and PER) leads to discarding of the ERI interrupt request.

Note 1. In the case where asynchronous mode and 9-bit data length are selected

Note 2. To temporarily disable TXI interrupts at the time of transmission of the last of the data and so on when you wish a new round of transmission to start after handling of the transmission-completed interrupt, control disabling and enabling of the interrupt by using the interrupt request enable bit in the interrupt controller rather than using the SCR.TIE bit. This can prevent the suppression of TXI interrupt requests in the transfer of new data.

Table 41.36 Interrupt Sources

Name	Interrupt Source	Interrupt Flag	DTC Activation	DMAC Activation
ERI	Receive error	ORER, FER, PER, DFER* ¹ , or DPER* ¹	Not possible	Not possible
RXI	Receive data full	RDRF	Possible	Possible
	Data match* ¹	DCMF* ¹		
TXI	Transmit data empty	TDRE	Possible	Possible
TEI	Transmit end	TEND	Not possible	Not possible

Note 1. Available in SCI0 to SCI11 only.

(2) SCI7 to SCI11 When FIFO is Enabled

Table 41.37 lists the interrupt sources in asynchronous mode and clock synchronization mode when the FIFO is enabled. Each interrupt source is enabled independently by using the enable bit in the SCR register.

When the SCR.TIE bit is 1, the number of data to be transmitted in the transmit FIFO is less than or equal to the threshold (FCR.TTRG[3:0] bits), a TXI interrupt request is generated. A TXI interrupt request is generated when the SCR.TE bit is set to 1 and then SCR.TIE bit is 1, or when the SCR.TIE and SCR.TE bit are simultaneously set to 1. A TXI interrupt request is not generated when the SCR.TE bit is set to 1 while the SCR.TIE bit is 0 or the SCR.TE bit is set to 1 while the SCR.TIE bit is 1.

Note that setting the SCR.TIE bit to 1 while the SCR.TE bit is 0 leads to the generation of a TXI interrupt.

When the SCR.TEIE bit is 1 and if the next data is not yet written to the FTDR register by the timing when the last bit of the transmit data is transmitted, the SSRFIFO.TEND flag is 1 and a TEI interrupt request is generated.

When the SCR.RIE bit is 1, the number of the data stored in the receive FIFO is equal to or greater than the threshold (FCR.RTRG[3:0] bits), an RXI interrupt request is generated.

When the SCR.RIE bit is 1 and the SSRFIFO.ORER flag is 1 or the data with a framing or parity error is stored in the receive FIFO, an ERI interrupt request is generated. At this time, if the number of the data stored in the receive FIFO is equal to or greater than the threshold (FCR.RTRG[3:0] bits), an RXI interrupt request is generated. Clearing the ORER, FER, and PER flags in the SSRFIFO register cancels an ERI interrupt request.

Table 41.37 Interrupt Source

Name	Interrupt Source	Interrupt Flag	DTC Activation	DMAC Activation
ERI	Receive error	ORER, FER, PER, DFER, or DPER DR (when FCR.DRES = 1)	Not possible	Not possible
RXI	Receive FIFO full	RDF	Possible	Possible
	Receive data ready	DR (when FCR.DRES = 0)		
	Data match	DCMF		
TXI	Transmit FIFO empty	TDRE	Possible	Possible
TEI	Transmit end	TEND	Not possible	Not possible

41.12.3 Interrupts in Smart Card Interface Mode

Table 41.38 lists interrupt sources in smart card interface mode. A transmit end interrupt (TEI) request cannot be used in this mode.

Table 41.38 SCI Interrupt Sources

Name	Interrupt Source	Interrupt Flag	DTC Activation	DMAC Activation
ERI	Receive error or error signal detection	ORER, PER, or ERS	Not possible	Not possible
RXI	Receive data full	—	Possible	Possible
TXI	Transmit data empty	TEND	Possible	Possible

Data transmission/reception using the DTC or DMAC is also possible in smart card interface mode, similar to in the normal SCI mode. In transmission, when the SSR.TEND flag is set to 1, a TXI interrupt request is generated. This TXI interrupt request activates the DTC or DMAC allowing transfer of transmit data if the TXI request is specified beforehand as a source of DTC or DMAC activation. The TEND flag is automatically set to 0 when the DTC or DMAC transfers the data.

If an error occurs, the SCI automatically retransmits the same data. During the retransmission, the TEND flag is kept to 0 and the DTC or DMAC is not activated. Therefore, the SCI and DTC or DMAC automatically transmit the specified number of bytes, including retransmission in the case of error occurrence. However, the SSR.ERS flag is not automatically cleared to 0 at error occurrence. Therefore, the ERS flag must be cleared by previously setting the SCR.RIE bit to 1 to enable an ERI interrupt request to be generated at error occurrence.

When transmitting/receiving data using the DTC or DMAC, be sure to make settings to enable the DTC or DMAC before making SCI settings. For DTC or DMAC settings, refer to [section 20, Data Transfer Controller \(DTCb\)](#) and [section 18, DMA Controller \(DMACa\)](#).

In reception, an RXI interrupt request is generated when receive data is set to the RDR register. This RXI interrupt request activates the DTC or DMAC allowing transfer of receive data if the RXI request is specified beforehand as a source of DTC or DMAC activation. If an error occurs, the error flag is set. Therefore, the DTC or DMAC is not activated and an ERI interrupt request is issued to the CPU instead; the error flag must be cleared.

41.12.4 Interrupts in Simple I²C Mode

The interrupt sources in simple I²C mode are listed in Table 41.39. The STI interrupt is allocated to the transmit end interrupt (TEI) request. The receive error interrupt (ERI) request cannot be used.

The DTC or DMAC can also be used to handle transfer in simple I²C mode.

When the value of the SIMR2.IICINTM bit is 1, an RXI request will be generated on the falling edge of the SSCLn signal for the eighth bit. If the RXI has been set up as an activating request for the DTC or DMAC beforehand, the RXI request will activate the DTC or DMAC to handle transfer of the received data. Furthermore, a TXI request is generated on the falling edge of the SSCLn signal for the ninth bit (acknowledge bit). If the TXI has been set up as an activating request for the DTC or DMAC beforehand, the TXI request will activate the DTC or DMAC to handle transfer of the transmit data.

When the value of the SIMR2.IICINTM bit is 0, an RXI request (ACK detection) if the input on the SSDAn pin is at the low level or a TXI request (NACK detection) if the input on the SSDAn pin is at the high level will be generated on the rising edge of the SSCLn signal for the ninth bit (acknowledge bit). If the RXI has been set up as an activating request for the DTC or DMAC beforehand, the RXI request will activate the DTC or DMAC to handle transfer of the received data. Also, if the DTC or DMAC is used for data transfer in reception or transmission, be sure to set up and enable the DTC or DMAC before setting up the SCI.

When the IICSTAREQ, IICRSTAREQ, and IICSTPREQ bits in the SIMR3 register are used to generate a start condition, restart condition, or stop condition, the STI request is issued when generation is complete.

Table 41.39 SCI Interrupt Sources

Name	Interrupt Source		Interrupt Flag	DTC Activation	DMAC Activation
	IICINTM bit = 0	IICINTM bit = 1			
RXI	ACK detection	Reception	—	Possible	Possible
TXI	NACK detection	Transmission	—	Possible*1	Possible*1
STI	Completion of generation of a start, restart, or stop condition		IICSTIF	Not possible	Not possible

Note 1. Activation of the DTC or DMAC is only possible when the SIMR2.IICINTM bit is 1 (use reception and transmission interrupts).

41.12.5 Interrupt Requests from the Extended Serial Mode Control Section

The extended serial mode control section has a total of six types of interrupt request for generating the SCIX0 interrupt (Break Field low width detected), SCIX1 interrupt (Control Field 0 match, Control Field 1 match, priority interrupt bit detected), SCIX2 interrupt (bus collision detected), and SCIX3 interrupt (valid edge detected). When any of the interrupt factors is generated, the corresponding status flag is set to 1. Details of all of the interrupt requests are listed in Table 41.40.

Table 41.40 Interrupt Sources of the Extended Serial Mode Control Section

Interrupt Request	Status Flag	Interrupt Factors
SCIX0 interrupt (Break Field low width detected)	BFDF	• Detection of a Break Field low width longer than the interval corresponding to the timer setting • Completion of the output of a Break Field low width over the interval corresponding to the timer setting • Underflow of the timer
SCIX1 interrupt (Control Field 0 match)	CF0MF	The data received in Control Field 0 matching the value set in the CF0DR register
SCIX1 interrupt (Control Field 1 match)	CF1MF	The data received in Control Field 1 matching the value set in the PCF1DR or SCF1DR register
SCIX1 interrupt (priority interrupt bit detected)	PIBDF	The value of the bit specified as the priority interrupt bit matching the value set in the PCF1DR register
SCIX2 interrupt (bus collision detected)	BCDF	The output level on the TXDX12 pin and the input level on the RXDX12 pin not matching on three consecutive cycles of the bus collision detection clock
SCIX3 interrupt (valid edge detected)	AEDF	Detection of a valid edge during bit rate measurement

41.13 Event Linking

By employing interrupt request signals as event signals, SCI5 is able to provide linked operation through the event link controller (ELC) for modules selected in advance.

Event signals can be output regardless of the values of the corresponding interrupt request enable bits.

- (1) Error (receive error, error signal detected) event output
 - Indicates abnormal termination due to a parity error during reception in asynchronous mode.
 - Indicates abnormal termination due to a framing error during reception in asynchronous mode.
 - Indicates abnormal termination due to an overrun error during reception.
 - Indicates detection of the error signal during transmission in smart card interface mode.
- (2) Receive data full event output
 - Indicates that received data have been set in the receive data register (RDR or RDRL).
 - Indicates that ACK has been detected if the SIMR2.IICINTM bit is 0 in simple I²C mode.
 - Indicates that the 8th-bit SSCL5 falling edge has been detected if the SIMR2.IICINTM bit is 1 in simple I²C mode.
 - When the SIMR2.IICINTM bit is 1 during master transmission in simple I²C mode, set the event link controller (ELC) so that receive data full events are not used.
- (3) Transmit data empty event output
 - Indicates that the SCR.TE bit has been changed from 0 to 1.
 - Indicates that transmit data have been transferred from the transmit data register (TDR or TDRL) to the transmit shift register (TSR).
 - Indicates that transmission has been completed in smart card interface mode.
 - Indicates that NACK has been detected if the SIMR2.IICINTM bit is 0 in simple I²C mode.
 - Indicates that the ninth-bit SSCL5 falling edge has been detected if the SIMR2.IICINTM bit is 1 in simple I²C mode.
- (4) Transmit end event output
 - Indicates the completion of transmission.
 - Indicates that the starting condition, resumption condition, or termination condition has been generated in simple I²C mode.

41.14 Usage Notes

41.14.1 Setting the Module Stop Function

Module stop control register B (MSTPCRB) and module stop control register C (MSTPCRC) are used to stop and start SCI operations. With the value after a reset, SCI operations are stopped. Register access is enabled by releasing the module stop state. For details, refer to section 11, Low Power Consumption.

41.14.2 Break Detection and Processing

(1) SCI0 to SCI6 and SCI12, and SCI7 to SCI11 when FIFO is Disabled

When a framing error is detected, a break can be detected by reading the RXDn pin value directly or reading the value of the SPTR.RXDMON flag (only for SCI0 to SCI11). In a break, the input from the RXDn pin becomes all 0s, and so the SSR.FER flag is set to 1 (framing error has occurred), and the SSR.PER flag may also be set to 1 (parity error has occurred). When the SEMR.RXDESEL bit is 0, the SCI continues the receive operation even after a break is received. Therefore, note that even if the FER flag is set to 0 (no framing error occurred), it will be set to 1 again. When the SEMR.RXDESEL bit is 1, the SCI sets the SSR.FER flag to 1 and stops receiving operation until a start bit of the next data frame is detected. If the SSR.FER flag is set to 0 at this time, the SSR.FER flag retains 0 during the break. When the RXDn pin becomes high and the break ends, detecting the beginning of the start bit at the first falling edge of the RXDn pin allows the SCI to start the receiving operation.

(2) SCI7 to SCI11 When FIFO is Enabled

When a framing error is detected and the following one-frame data received is all 0s, the SCI stops receiving operation. Upon the detection of a framing error, reading the value of the SPTR.RXDMON flag detects a break. After the RXDn pin becomes high and the break ends, the SCI resumes receiving operation.

41.14.3 Mark State and Sending Breaks

When the SCR.TE bit is 0 (serial transmission is disabled), the TXDn pin becomes high-impedance. To forcibly set the TXDn pin to mark or space state while the TE bit is 0, set the I/O port associated registers and switch the TXDn pin to general output port.

For holding the communication line in the mark ("1") state until the TE bit is set to 1 (serial transmission is enabled), set the corresponding bit in the PODR register to 1 for high output from general output port. To start communications, set the TE bit to 1 and then the corresponding bit in the PMR register to 1.

To send a break (the space state for longer than a certain period of time) while data transmission, set the corresponding bit in the PODR register to 0 (low output), and set the corresponding bit in the PMR register to 0 (general I/O port). Then set the TE bit to 0 if necessary. When the TE bit is set to 0, the transmitter is initialized regardless of the current transmit status.

The SPTR register, if it is included, can set the TXDn pin in mark/space state without switching the pin function to general output port. Set the SPTR.SPB2IO bit to 1 (output) and the SPB2DT bit to 1 (mark) or 0 (space), and then set the TE bit to 0.

41.14.4 Receive Error Flags and Transmit Operations (Clock Synchronous Mode and Simple SPI Mode)

Transmission cannot be started when a receive error flag (SSR.ORER) is set to 1, even if data is written to the TDR register (FTDR register). Be sure to set the receive error flags to 0 before starting transmission. Note also that the receive error flags cannot be set to 0 even if the SCR.RE bit is set to 0 (serial reception is disabled).

41.14.5 Writing Data to the TDR Register

Data can be written to registers TDR, TDRH, and TDRL. However, if new data is written to registers TDR, TDRH, and TDRL when transmit data is remaining in registers TDR, TDRH, and TDRL, the previous data in registers TDR, TDRH, and TDRL is lost because it has not been transferred to the TSR register yet. Be sure to write transmit data to registers TDR, TDRH, and TDRL in the TXI interrupt request handling routine.

41.14.6 Restrictions on Clock Synchronous Transmission (Clock Synchronous Mode and Simple SPI Mode)

When the external clock source is used as a synchronization clock, the following restrictions apply.

(1) Start of transmission

Update the TDR register by the CPU, DMAC, or DTC and wait for at least five PCLK cycles before allowing the transmit clock to be input (refer to Figure 41.80).

(2) Continuous transmission

- (a) Write the next transmit data to the TDR or TDRL register before the falling edge of the transmit clock (bit 7) (refer to Figure 41.80).
- (b) When updating the TDR register after bit 7 has started to transmit, update the TDR register while the synchronization clock is in the low-level period, and set the high-level width of the transmit clock (bit 7) to four PCLK cycles or longer (refer to Figure 41.80).

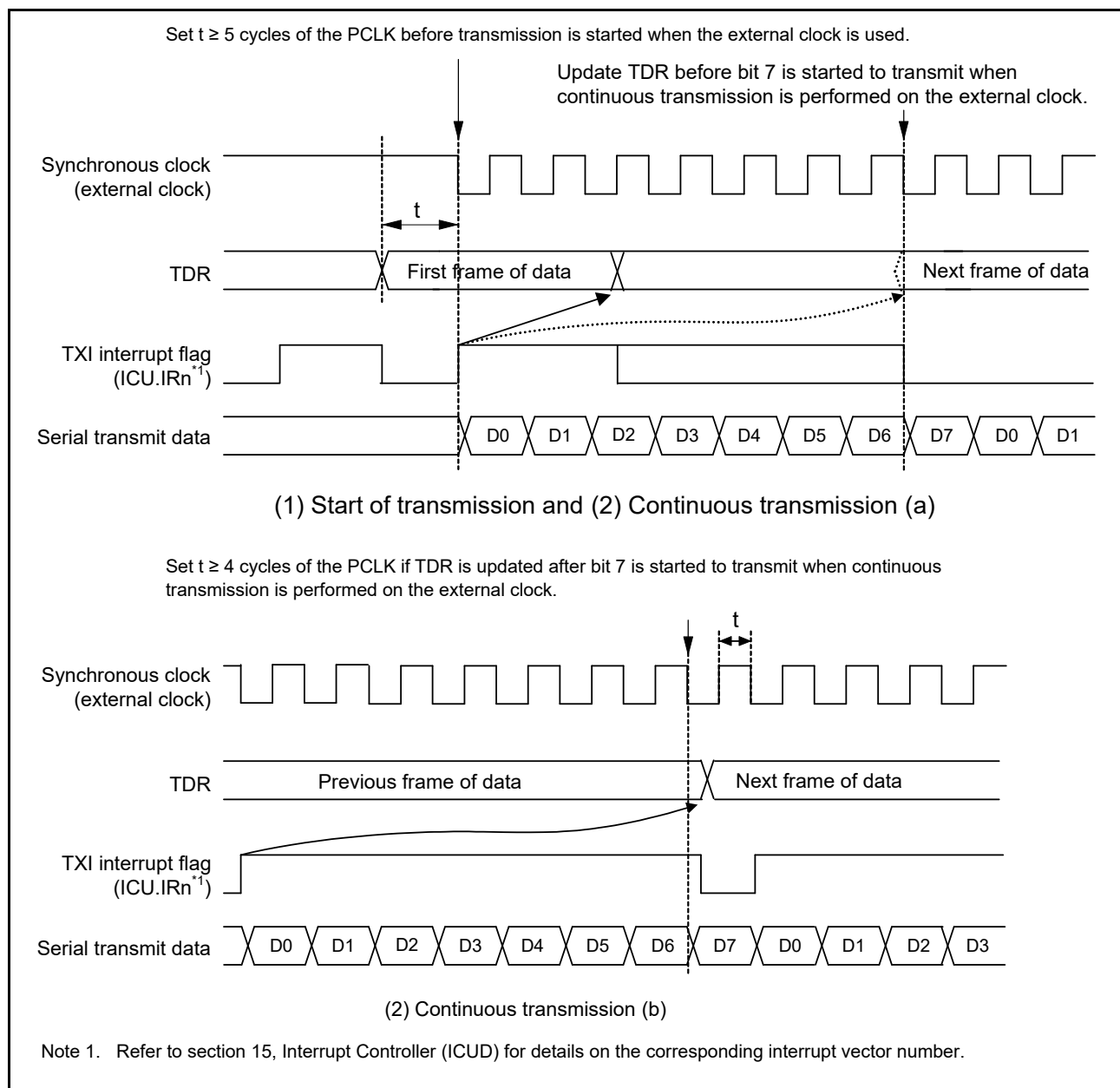


Figure 41.80 Restrictions on Use of External Clock in Clock Synchronous Transmission

41.14.7 Restrictions on Using DMAC or DTC

When using the DMAC or DTC to read the RDR, RDRH, and RDRL registers, be sure to set the receive data full interrupt (RXI) as the activation source of the relevant SCI.

41.14.8 Notes on Starting Transfer

At the point where transfer starts when the interrupt status flag (IRn.IR flag) in the interrupt controller is 1, follow the procedure below to clear interrupt requests before permitting operations (by setting the SCR.TE or SCR.RE bit to 1). For details on the interrupt status flag, refer to **section 15, Interrupt Controller (ICUD)**.

- Confirm that transfer has stopped (the setting of the SCR.TE or SCR.RE bits is 0).
- Set the corresponding interrupt enable bit (SCR.TIE or SCR.RIE) to 0.
- Read the corresponding interrupt enable bit (SCR.TIE or SCR.RIE bit) to check that it has become 0.
- Set the interrupt status flag (IRn.IR flag) in the interrupt controller to 0.

41.14.9 SCI Operations during Low Power Consumption State

(1) Transmission

When making settings for the module stopped state or in transitions to software standby, stop operations (by setting the TIE, TE, and TEIE bits in the SCR register to 0) after switching the TXDn pin to the general I/O port pin function or fix the output level of the TXDn pin by the SPTR register (only for SCI0 to SCI11). Setting the TE bit to 0 resets the TSR register and the SSR.TEND flag. The SSRFIFO.TEND flag in the SCI7 to SCI11 is not initialized. Depending on the port settings or SPTR register setting (only for SCI0 to SCI11), output pins may output the level before a transition to the low power consumption state is made after release from the module stopped state or software standby mode. When transitions to these states are made during transmission, the data being transmitted become indeterminate.

To transmit data in the same transmit mode after cancellation of the low power consumption state, set the TE bit to 1, read the SSR register, and write data to the TDR register sequentially to start data transmission. To transmit data with a different transmit mode, initialize the SCI first.

Figure 41.81 shows a sample flowchart for transition to software standby mode during transmission. Figure 41.82 and Figure 41.83 show the port pin states during transition to software standby mode.

Before specifying the module stop state or making a transition to software standby mode from the transmit mode using DTC/DMA transfer, stop the transmit operations (TE = 0). To start transmission after cancellation using the DTC/DMAC, set the TE and TIE bits to 1. The TXI interrupt flag is set to 1 and transmission starts using the DTC/DMAC.

(2) Reception

(a) When the data match function is not used for release from the low-power consumption state

Before specifying the module stop state or making a transition to software standby mode, stop the receive operations (SCR.RE = 0). If transition is made during data reception, the data being received will be invalid.

To receive data in the same receive mode after cancellation of the low power consumption state, set the RE bit to 1, and then start reception. To receive data in a different receive mode, initialize the SCI first.

Figure 41.84 shows a sample flowchart for transition to software standby mode during reception.

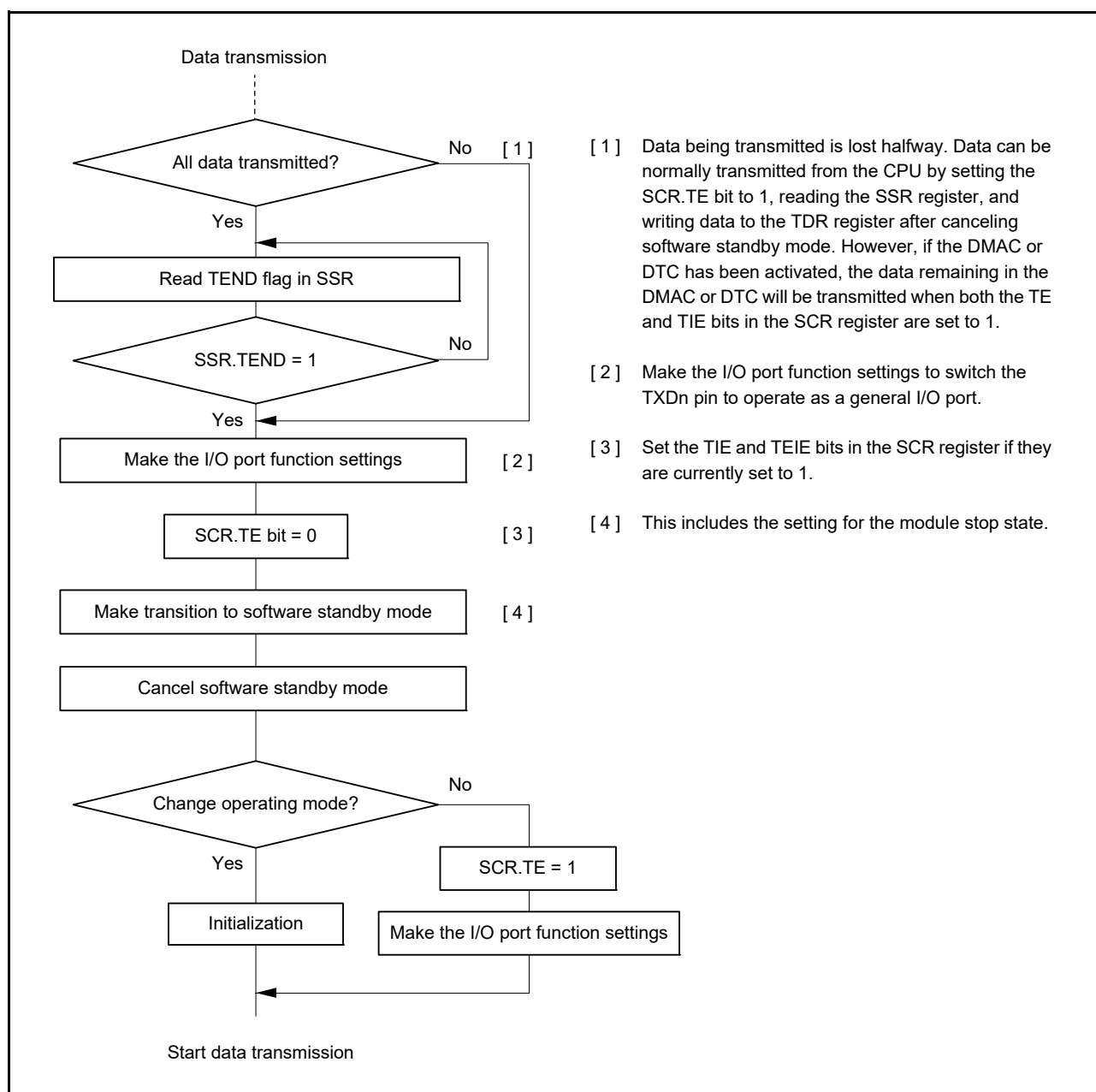


Figure 41.81 Example of Flowchart for Transition to Software Standby Mode during Transmission

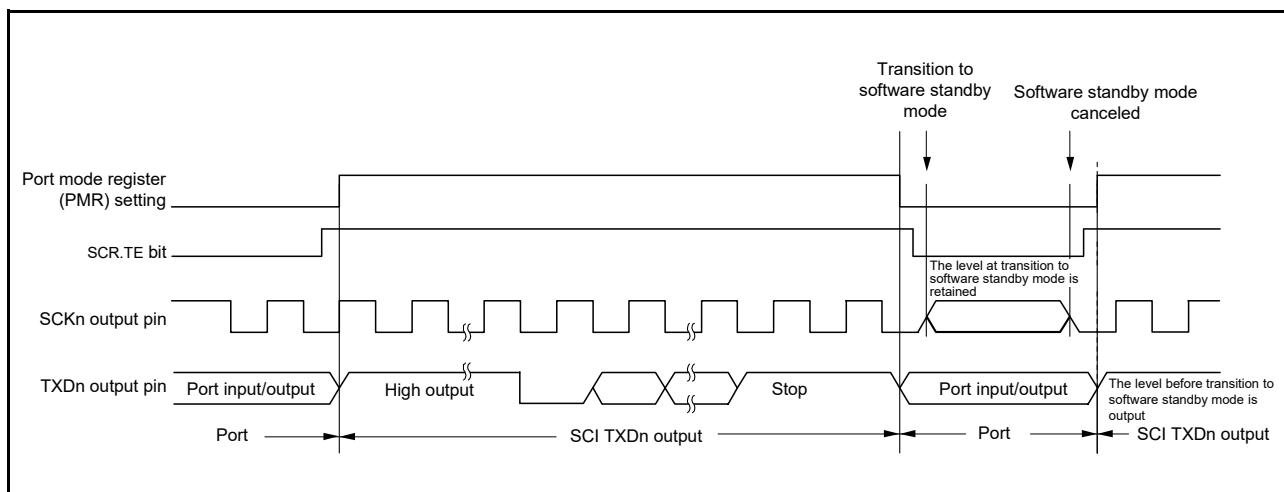


Figure 41.82 Port Pin States during Transition to Software Standby Mode (Internal Clock, Asynchronous Transmission)

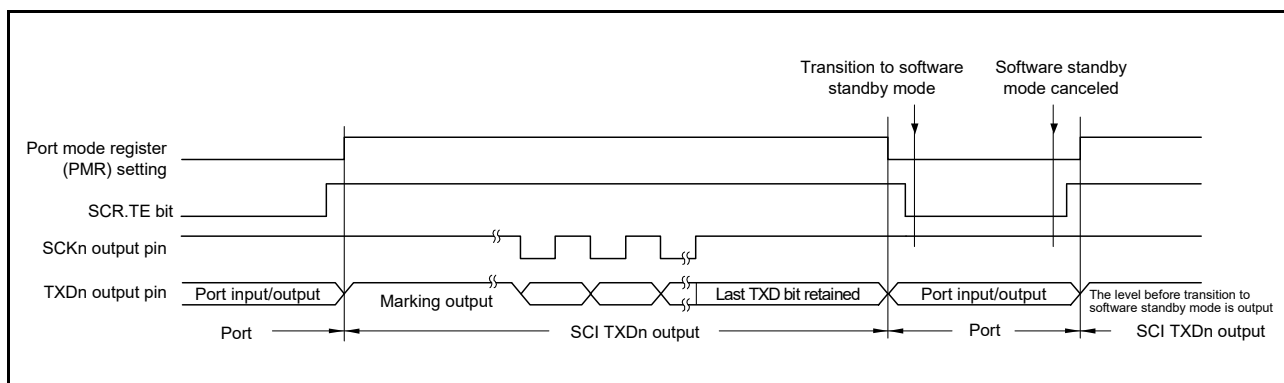


Figure 41.83 Port Pin States during Transition to Software Standby Mode (Internal Clock, Clock Synchronous Transmission)

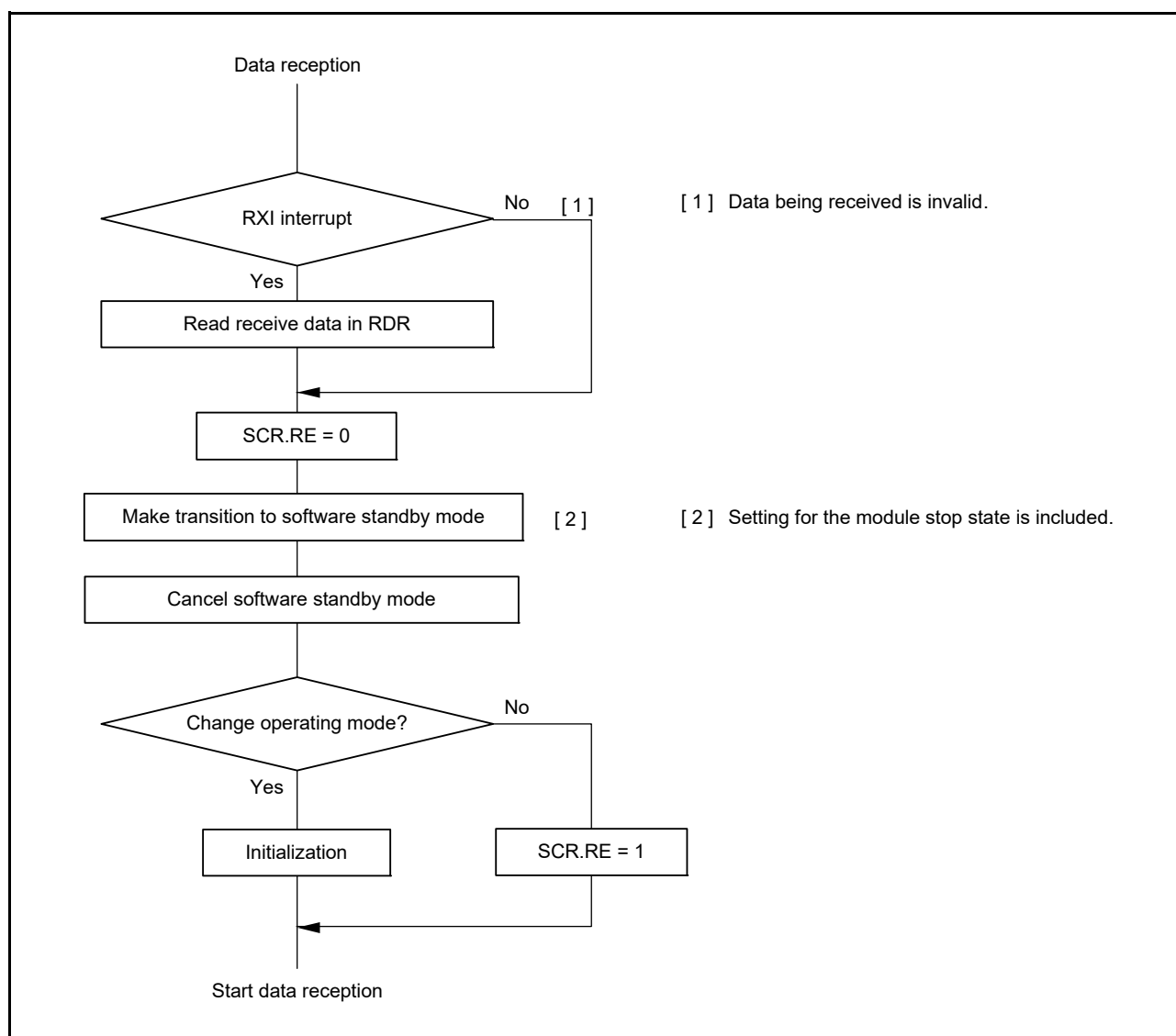


Figure 41.84 Example of Flowchart for Transition to Software Standby Mode during Reception

(b) When the data match function is used for release from the low-power consumption state

Set an operating mode to be set after release from the low-power consumption state before reducing power consumed by the SCI by using the function of reducing power consumption. Afterwards, set the comparison data in the CDR.CMPD[8:0] bits and set the DCCR.DCME bit to 1. While the SCR.RE bit is 1, enter to the low-power consumption state.

If transition to the low-power consumption state may take place while the RXDn pin is driven low, set the SEMR.RXDESEL bit to 0. When the SEMR.RXDESEL bit is 1, the start bit may not be detected at the time of release from the low-power consumption state.

41.14.10 External Clock Input in Clock Synchronous Mode and Simple SPI Mode

In clock synchronous mode and simple SPI mode, the external clock SCKn must be input as follows:

High-pulse period, low-pulse period = 2 PCLK cycles or more, period = 6 PCLK cycles or more

41.14.11 Limitations on Simple SPI Mode

(1) Master Mode

- Use a resistor to pull up or pull down the clock line matching the initial settings for the transfer clock set by the SPMR.CKPH and CKPOL bits when the SPMR.SSE bit is 1.
This prevents the clock line from being placed in the high-impedance state when the SCR.TE bit is set to 0 or unexpected edges from being generated on the clock line when the SCR.TE bit is changed from 0 to 1. When the SPMR.SSE bit is 0 in single master mode, pulling up or pulling down the clock line is not necessary because the clock line is not placed in the high-impedance state even when the SCR.TE bit is set to 0.
- In the case of the setting for clock delay (SPMR.CKPH bit is 1), the receive data full interrupt (RXI) is generated before the final clock edge on the SCKn pin as indicated in Figure 41.85. If the TE and RE bits in the SCR register become 0 at this time before the final edge of the clock signal on the SCKn pin, the SCKn pin is placed in the high-impedance state, so the width of the last clock pulse of the transfer clock is shortened. Furthermore, an RXI interrupt may lead to the input signal on the SSn# pin of a connected slave going to the high level before the final edge of the clock signal on the SCKn pin, leading to incorrect operation of the slave.
- In a multi-master configuration, take care because the SCKn pin output becomes high-impedance while the input on the SSn# pin is at the low level if a mode fault error occurs as the current character is being transferred, stopping supply of the clock signal to the connected slave. Remake the settings for the connected slave to avoid misaligned bits when transfer is restarted.

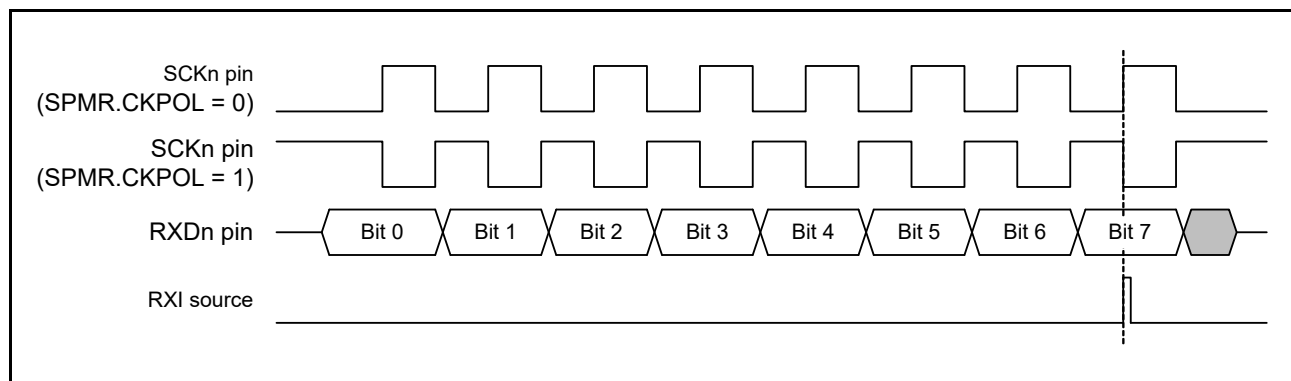


Figure 41.85 Timing of the RXI Interrupt in Simple SPI Mode (with Clock Delay)

(2) Slave Mode

- Secure at least five cycles of the PCLK from writing transmit data in the TDR register to start of the external clock input. Also secure at least five cycles of the PCLK from input of low level on the SSn# pin to start of the external clock input.
- Provide an external clock signal from the master the same as the transmit/receive data length.
- Control the input on the SSn# pin before the start and after the end of data transfer.
- When the level being input on the SSn# pin is to be changed from low to high while the current character is being transferred, set the TE and RE bits in the SCR register to 0 and, after remaking the settings, restart transfer of the first byte.

41.14.12 Limitation 1 on Usage of the Extended Serial Mode Control Section

When the PCR.SHARPS bit is set to 1, output on the TXDX12/RXDX12 pin is only possible when the following conditions apply.

- The timer is in Break Field low width output mode and the value of the TCR.TCST bit is 1 (when the TCST bit is set to 1, the high level continues to be output for up to one cycle of the clock source for counting by the timer counter before output of the low level)
- The value of the SCR.TE bit is 1.

41.14.13 Limitation 2 on Usage of the Extended Serial Mode Control Section

The TXI, RXI, ERI, and TEI interrupt requests are generated even if the extended serial mode is enabled. However, the RXI interrupt should not be enabled during reception of a Start Frame because the extended serial mode control section uses the receive data full signal.

To use the RXI interrupts during a reception of the Information Frame, use it in accordance with one of the following procedures. When a receive error is detected, clear the receive error flag and initialize the extended serial mode control section.

- (1) Set the SCR.RIE bit to 0 to disable the output of interrupt requests. Check the error flags in the SSR register on completion of the reception of a Start Frame, because an ERI interrupt is not generated if a receive error occurs. After reception of the Start Frame is completed, set the SCR.RIE bit to 1 by the time the first byte of the Information Frame is received.
- (2) Set the SCR.RIE bit to 1 to disable RXI interrupts and enable ERI interrupts for ICU. Clear the IRn.IR flag to enable the acceptance of RXI interrupts by ICU by the time the first byte of the Information Frame is received after the completion of Start Frame reception.

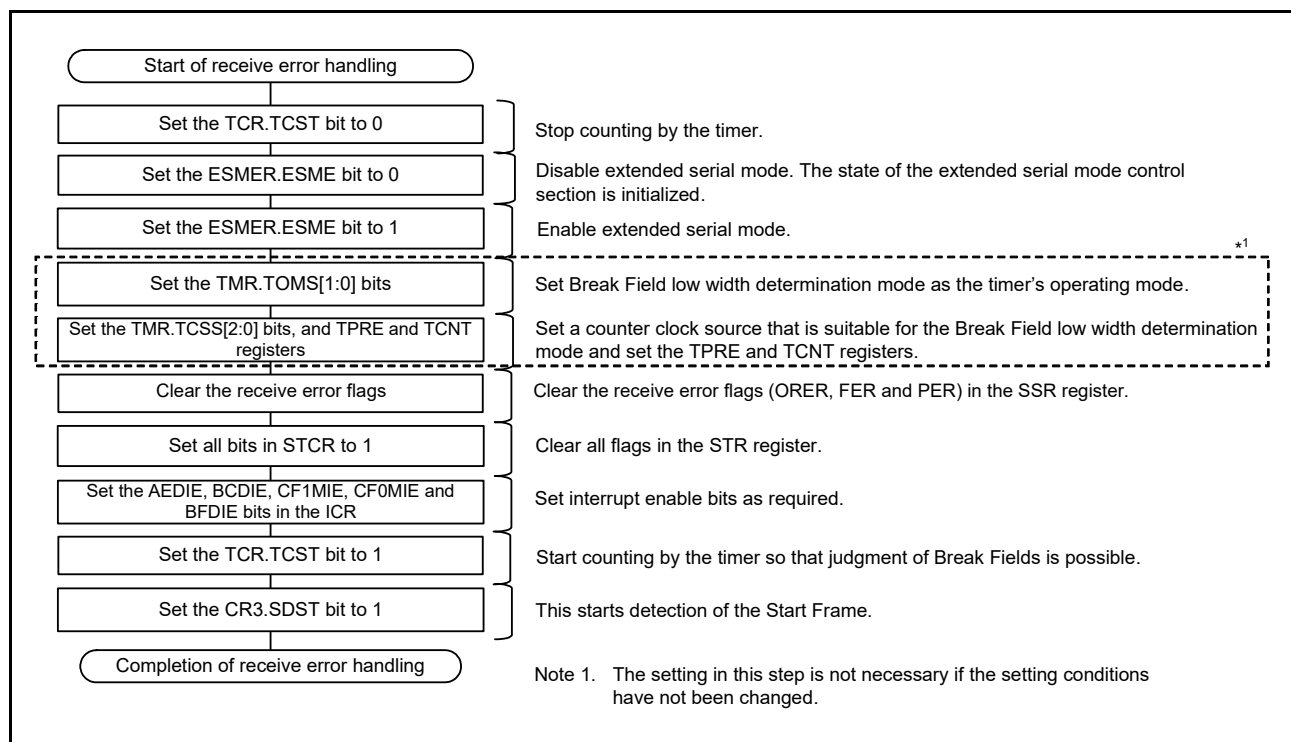


Figure 41.86 Example of Flowchart for Receive Error Handling (during Reception of the Start Frame)

41.14.14 Note on Transmit Enable Bit (TE Bit)

When setting the pin function to “TXDn” while the SCR.TE bit is 0 (serial transmission is disabled) or setting the TE bit to 0 while the pin function is “TXDn”, output of the TXDn pin becomes high-impedance.

Prevent the TXDn line from becoming high-impedance by any of the following ways:

- (1) Connect a pull-up resistor to the TXDn line.
- (2) Set the TE bit to 1^{*1} before changing the pin function to “TXDn”. Before setting the TE bit to 0, change the pin function to “general-purpose I/O port” and drive the pin high or low.
- (3) Set the SPTR.SPB2IO bit to 1 first, and change the pin function to “TXDn”. Leave the value of the SPB2IO bit as 1 after that (for SCI0 to SCI11).

Note 1. An interrupt is generated when the TE bit is set to 1 while the TXI interrupt is enabled (SCI0 to SCI6 and SCI12, and SCI7 to SCI11 when FIFO is Disabled). If this creates a problem, change the pin function to “TXDn” first, and then set the corresponding ICU.IERm.IENj bit to 1.

41.14.15 Note on Stopping Reception When Using the RTS Function in Asynchronous Mode

One clock cycle of PCLK is required for the time from setting the SCR.RE bit to 0 to stopping the RTS signal generator in asynchronous mode.

When reading the RDR (or RDRL) register after setting the SCR.RE bit to 0, confirm that the RE bit has been set to 0 before reading the RDR (or RDRL) register to prevent these two processes from being performed consecutively.